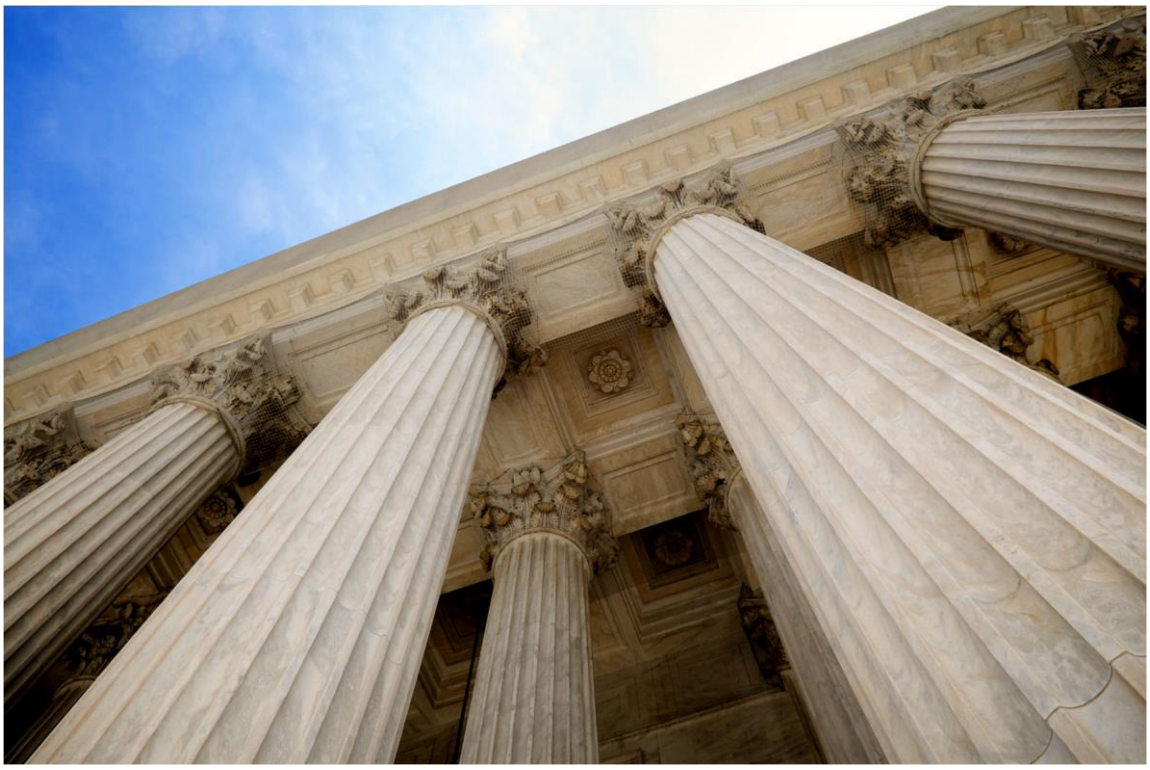
SAP MASTER GOVERNANCE (MDG) Overview of the integration into the business processes for financial (MDG-F) customer (MDG-C) supplier (MDG-S) business partner (BP) - ARIBA

Dipl.-Wirtsch.-Ing. Hans-G. Emrich

SAPCon

SAP MDG



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1 Preface

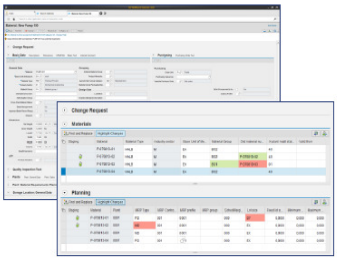
SAP Master Data Governance is a Master Data Management solution and controls, harmonizes the master data process in the company. Master data is the beating heart of your organization. Without it, you're dead in the water.

This book is a description of the use of SAP-MDG. It is designed to provide project managers, IT managers and departmental managers with an overview of the application of MDG. MDG is a software tool that directly intervenes horizontally and vertically in business processes. The SAP MDG requires a system and business integration in the company. The book provides an overview of the MDG architecture, integration and implementation into the business process. Examples of implementation and project structures are presented. The book will serve the project manager as a compass. Due to the large number of different company structures and different handling of business processes, not all possible variants of business processes can be mapped. The company, which wants to introduce MDG must be clear that not only an SAP module is implemented, but rather established structures are changed. The MDG stands and falls with the degree of integration or integration. The audience of this document can be business analysts, techno-functional analysts, business process experts, master data experts, solution architects, technical developers, data scientists and customers.

2 Introduction SAP Master Data Governance (MDG)[1]

SAP Master Data Governance: the efficient way towards consistent and high quality master data – the foundation of your Digital Business

SAP MDG is the one single application that provides out-of-the-box, domain-specific master data applications as well as a flexible framework for custom data models and processes. This allows to de-centrally manage and consolidate, or centrally manage and distribute all master data for your company's complete system landscape.



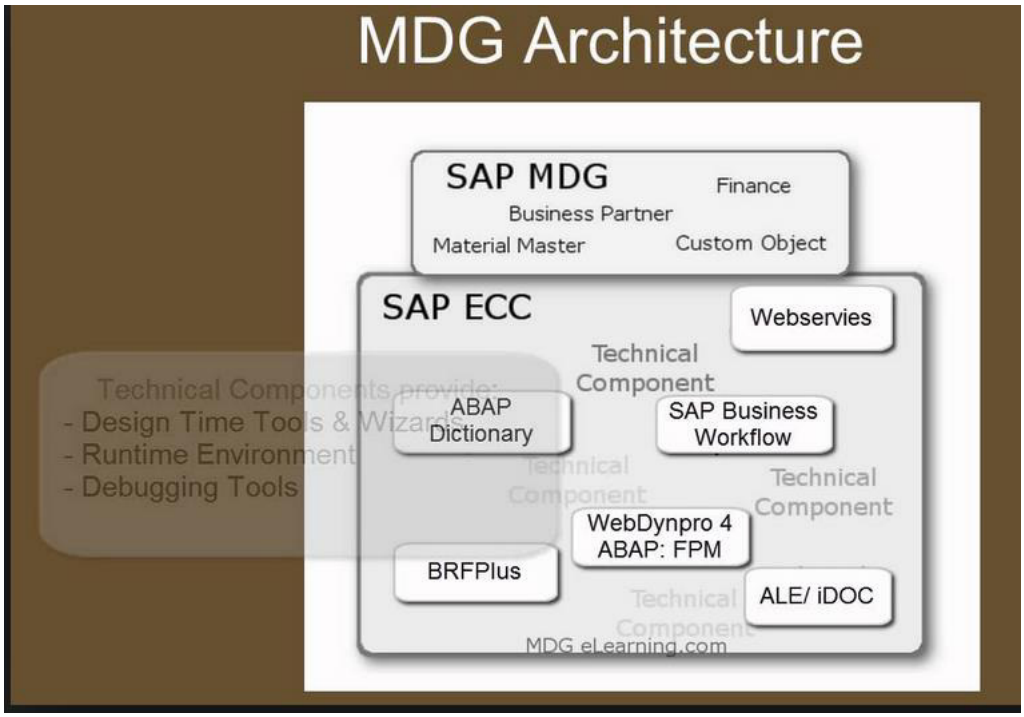
The clean master data from SAP MDG enables you to accelerate your enterprise software journey, and to enable new digital business models. For example…

- …SAP MDG supports your system landscape simplification and move to the cloud with synchronized master data across on-premise and on-demand applications
- …Leverage SAP MDG's harmonized master data and SAP's platform capabilities to connect transactional and IoT or other big data, and derive real business value
- …Mobilize your business with simplified master data access and efficient scenarios for request, approval, or to monitor KPIs and trends
- …MDG helps preparing a SAP S/4HANA move: get experience with a controlled subset, ensure master data quality before the implementation and maintain quality after go live

[1] SAP : Product overview 2018

SAP Master Data Governance (MDG) is a state-of-the-art master data management solution, providing out-of-the-box, domain-specific master data governance to centrally create, change, and distribute, or to consolidate master data across your complete enterprise system landscape.
In this context, MDG can either run as a separate master data hub or be co-deployed on an operational system. In both cases MDG can run on:

- ECC on any DB
- ECC on HANA
- S/4HANA
- ARIBA
- Non - SAP



SAP Master Data Governance (MDG) is a **state-of-the-art master data management solution**, providing **out-of-the-box**, domain-specific master data governance to **centrally** create, change, and distribute, or to **consolidate** master data across your **complete enterprise system landscape**.
There are two versions of the product: **SAP MDG** (which technically runs on SAP ERP 6.0) and **SAP MDG on SAP S/4HANA**.

2.1 Central governance or consolidation for consistent master data

Enables centralized governance, compliance and transparency of master data during creation and change through integrated staging, approval and central audit trail

Delivers consistent definition, authorization and replication of key master data entities

Eliminates error-prone manual maintenance processes for master data in multiple systems

Supports consolidation of de-central master data in any enterprise system landscape in order to create best records and key mapping betweenduplicates

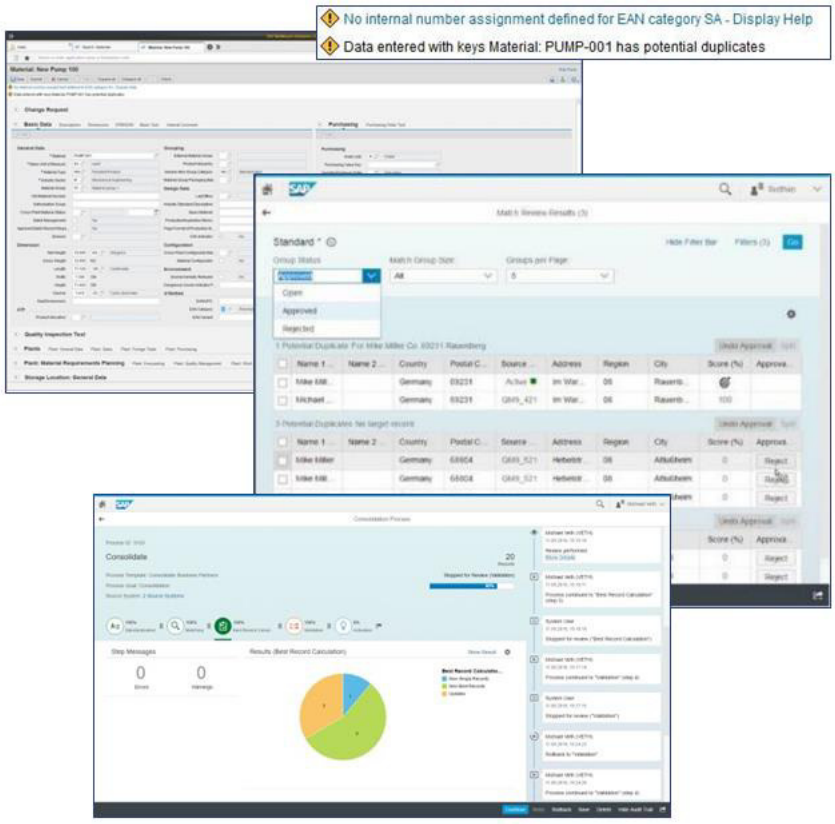
Optionally combines consolidation with central governance for sustained master data quality

2.2 Integration and re-use

Provides native integration and openness to integrate 3rd partyservices

Re-use of SAP data model, existing business logic and configuration for data validation

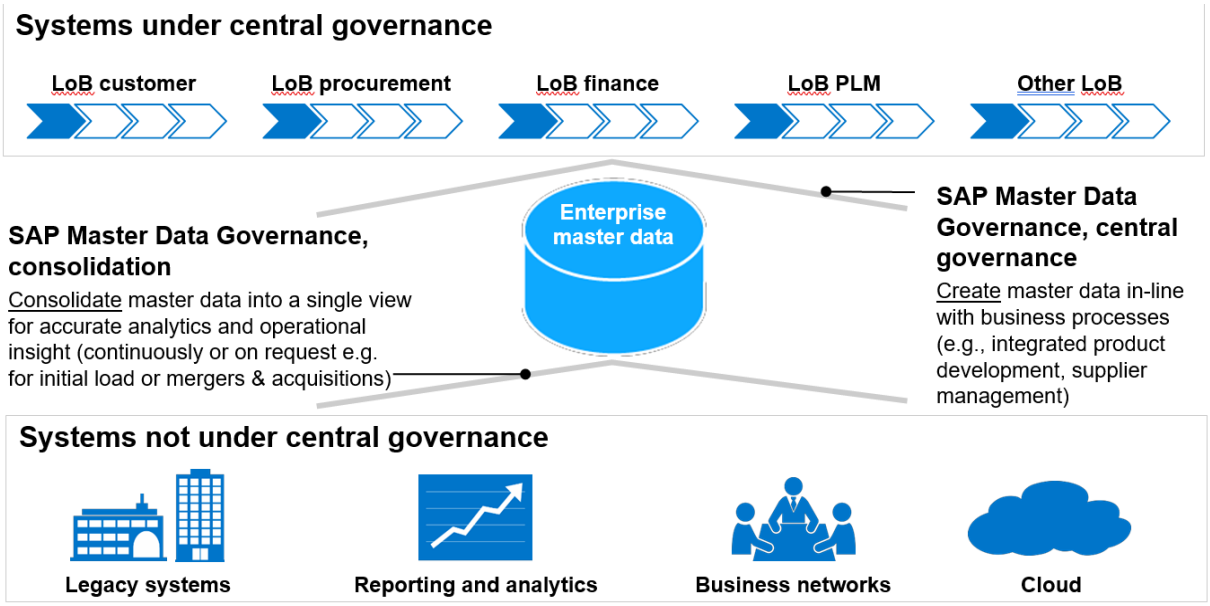
Open to manage your own master data flexibly also for non-SAPenvironments



2.3 Data quality

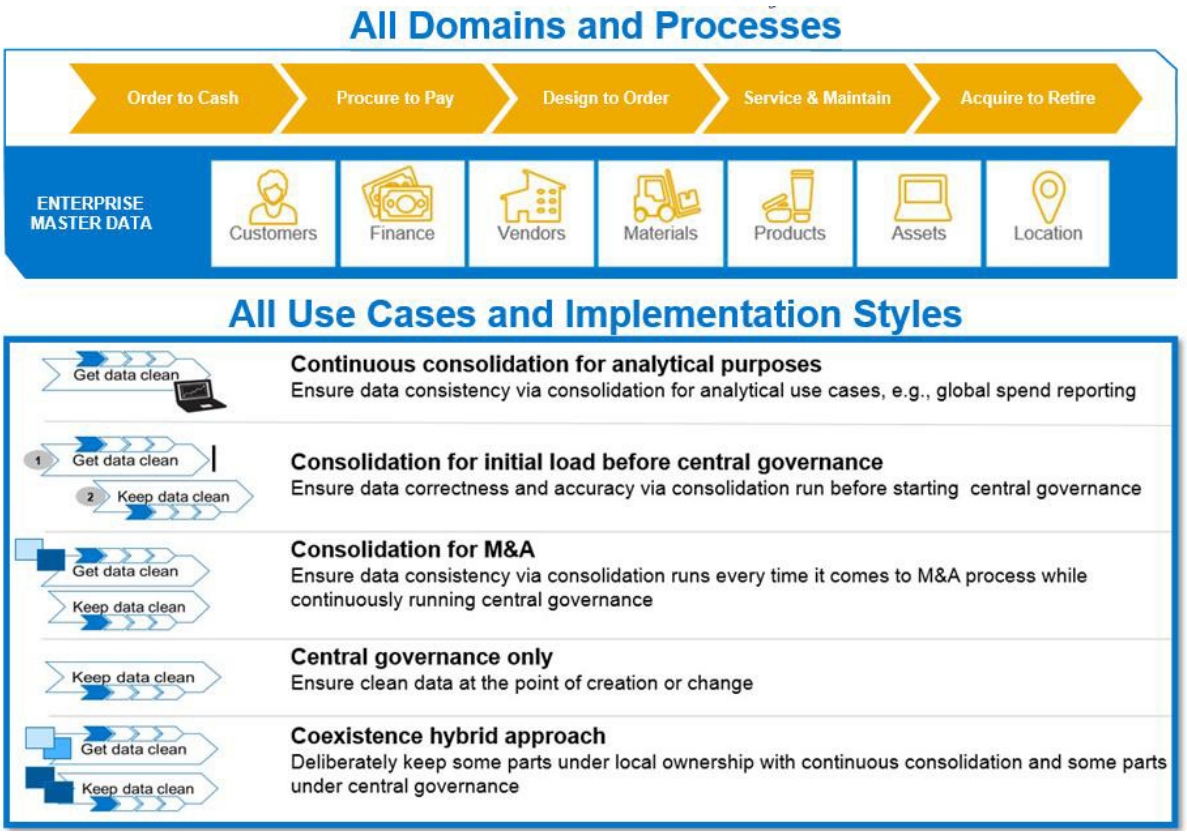
Analyze process quality and integrate with SAP HANA smart data quality and SAP Data Services / SAP Information Steward for quality, enrichment, and remediation

2.4 Central governance & distribution / de-central ownership & consolidation[2]

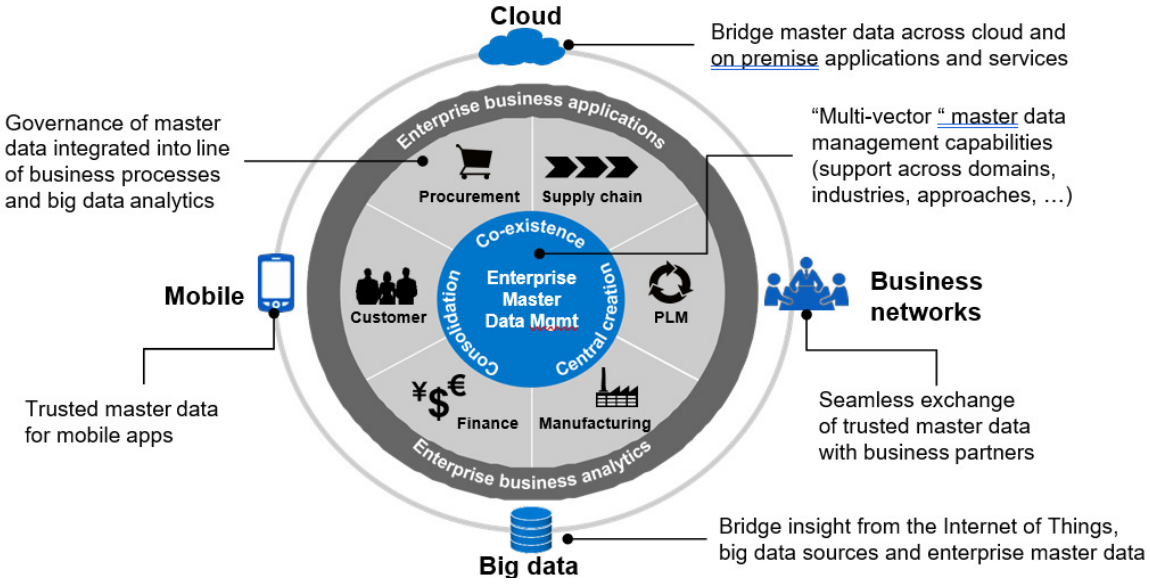


[2] SAP Master Data Governance on SAP S/4HANA 1709 / SAP Master Data Governance 9.1

2.5 One single application for Enterprise master data management



2.6 Key trends and impacts in business and technology



2.7 Master data governance capatilities[3]

Master Data Governance for all business data offers a central hub-based maintenance of material , customer, supplier , financial master data.

To address the key business issues related to such a governance solution for master data, MDG for material follows some basic principles that are shared across the domains of MDG

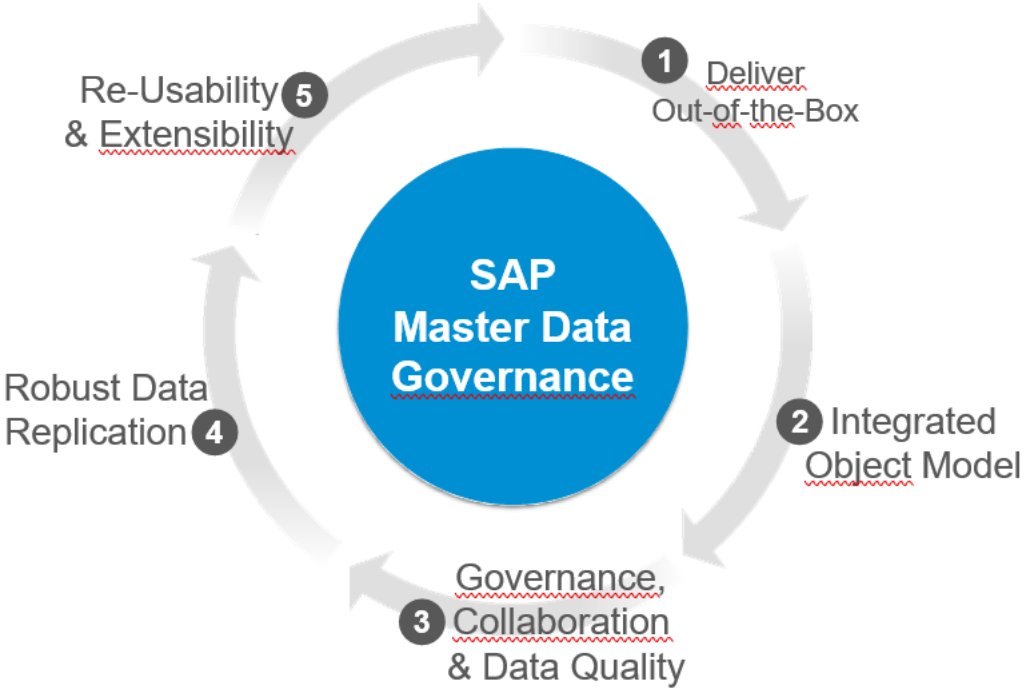
- Maintain once, use many
- One standardized process to be used by everybody
- One standardized user interface for everybody
- One standardized set of business logic

In order to fulfill these principles, SAP has built a Master Data Governance solution that offers

- Central based maintenance of master data
- Workflow-driven processes that are flexible while ensuring governance and standardization
- Harmonized user interface, based upon Enterprise Portal, Business Client or FIORI Launchpad
- Data model and business logic follow the best practices of SAP and reuse existing data structures, business validations and various services

[3] SAP Master Data Governance on SAP S/4HANA 1709 / SAP Master Data Governance 9.1

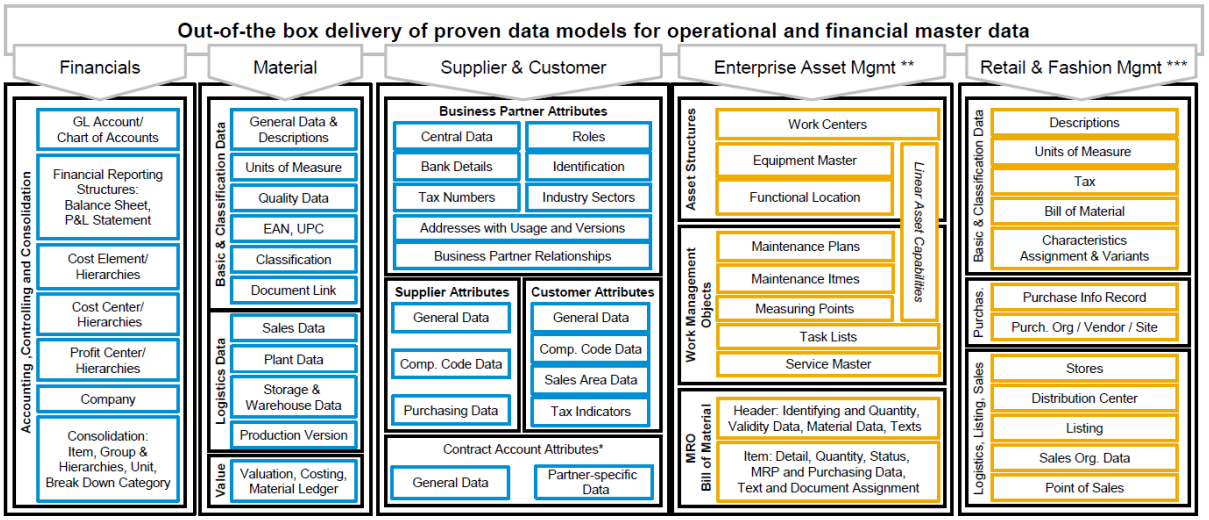
2.7.1 Establish Best Quality Master Data to Optimize Business Processes



2.7.2 Deliver Out-of-the-Box

Low TCO	Full Integration to SAP systems as a client Offer integration services for non-SAP client systems
Flexibility	Workflow flexibility closely integrated to a framework for defining business rules
Ease of Consumption	Deliver a process-centric solution, with roles and a web-based user interface Deliver predefined data quality services
Best Practice	Deliver a data model proven by several applications & industries
Process Transparency	Reuse of existing processing logic
Integration	Out-of-the box data replication mechanisms, using stable mechanisms

2.7.3 Integrated Object Model[4]



Identifying Material Data ~ 10 fields

- Both SAP internal (material number) as well as external standards (GTIN)

Descriptive Material Data ~ 100 fields

- Help to understand what a material is but do not control processes
- Are as different as the products of our customers
- Typically realized via Classification (rather than Field Extension)

Process Controlling Material Data ~ 1.000 fields

- Control the behavior of SAP business processes
- Proprietary to SAP applications

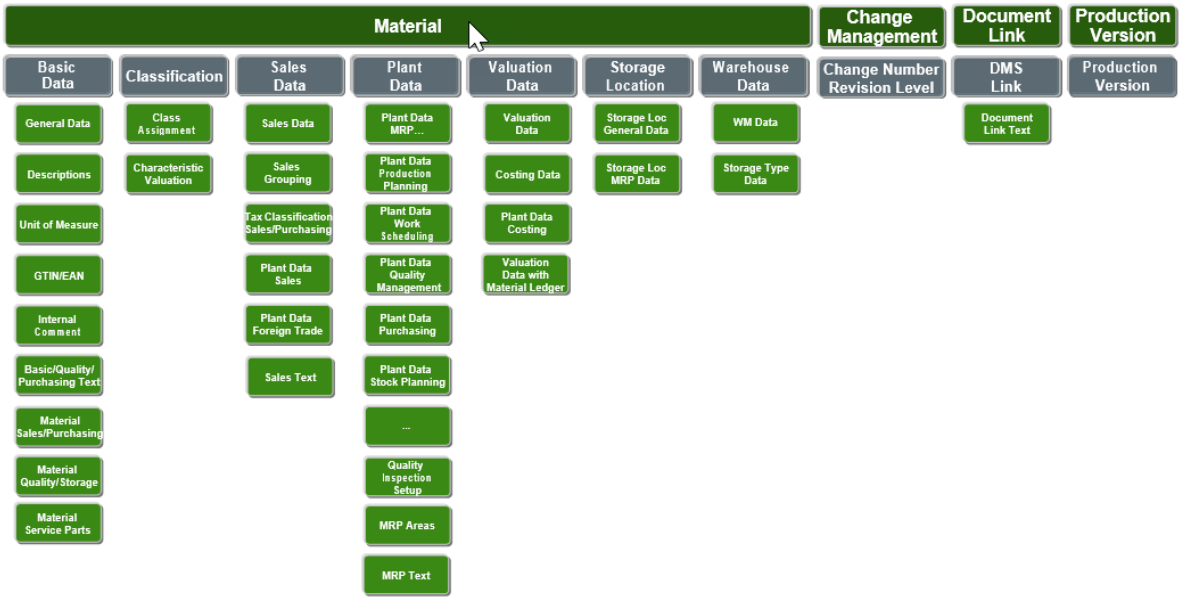
SAP MDG 9.1

SAP MDG Scope includes (~ 600 fields)

- Basic Data (Purchasing, Sales, Storage, Service Parts) and Long texts /notes
- Dimensions/ alternative UOMs / GTIN/EAN handling
- Classification / Lean Classification
- Sales Organization / Distribution Channel specific data / TaxClassification
- Plant specific data (with for Inspection Setup, MRP Area, Production Version)
- Storage Location Data
- Valuation data (with Material Ledger)
- Document Assignment
- Warehouse Data

Customers can enhance the data model

[4] SAP Master Data Governance on SAP S/4HANA 1709 / SAP Master Data Governance 9.1



2.7.4 Master data governance in hybrid landscapes

All systems always in synch

- Master data creation can happen in on-demand systems, on-premise systems, or in SAP MDG on SAP S/4HANA
- De-central creation triggers a process in SAP MDG on SAP S/4HANA
- After approval, enriched high-quality master data is replicated to all relevant cloud and on-premise systems

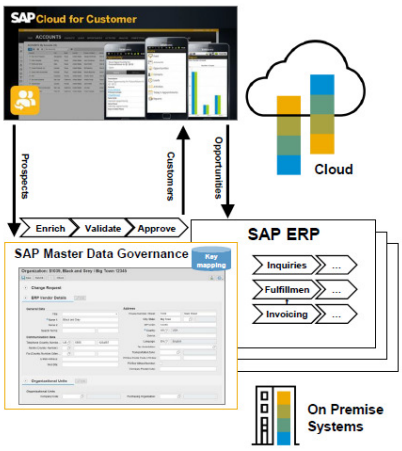
SAP MDG provides adaptable integration

- Integration scenarios can be implemented based on standard capabilities, like for example using extendable SOA services
- Any system can be involved as source and/or target
- Pre-built integration for selected SAP systems

Move to the cloud at your own speed

- A master data hub (like MDG) that synchronizes master data across on-premise and on-demand solutions is a prerequisite for your system landscape simplification and your step-by-step move to the cloud

Real-life customer example: SAP MDG managing prospect and customer data between cloud and on premise systems



2.7.5 End-to-end integration in heterogeneous landscapes [5]

Business-led integration

SAP Master Data Governance provides interfaces based on a process choreography for easy end-to-end integration…

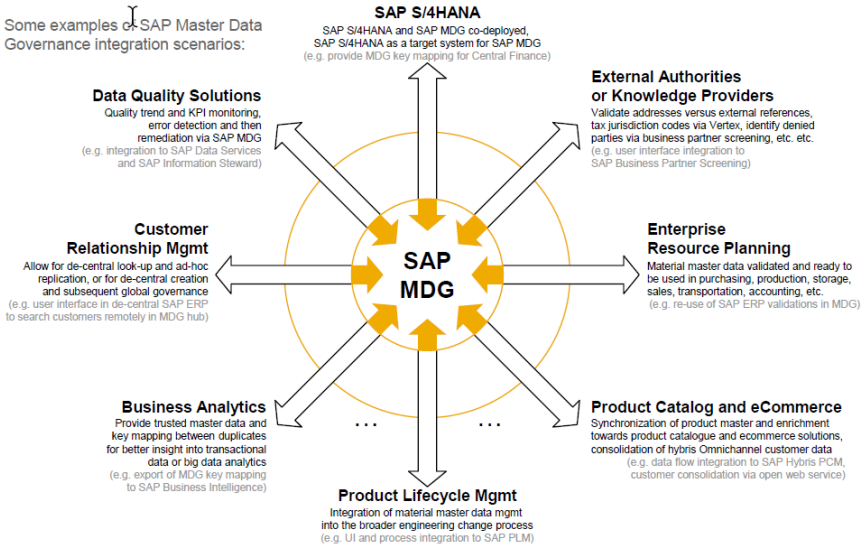
… not just technical interfaces

Not only for SAP

SAP Master Data Governance provides open interfaces to integrate with every solution as needed

Out-of-the-box for SAP

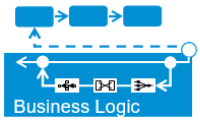
In addition, for SAP to SAP integration, we provide out-of-the-box integration content



[5] SAP Master Data Governance on SAP S/4HANA 1709 / SAP Master Data Governance 9.1

2.7.6 Governance, Collaboration

Linear or distributed and workflow based



Possibility to extend the process by adding business logic and thereby bringing in data quality

Has roles and responsibilities and also task authority (e.g., create, change, approve)

Data in process stored in a separated repository, will be transferred to operational database after final approval

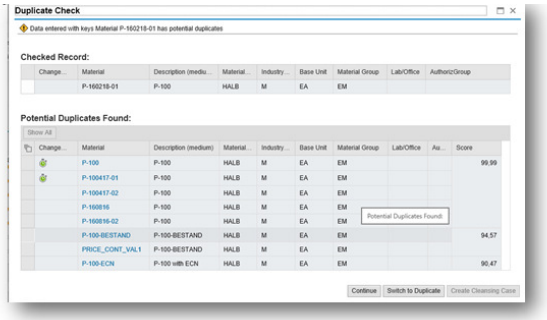
Can be adapted and tailored based on customer needs

Flexible enough to respect the distributed responsibilities existing in a company across various business units

2.7.7 Data quality

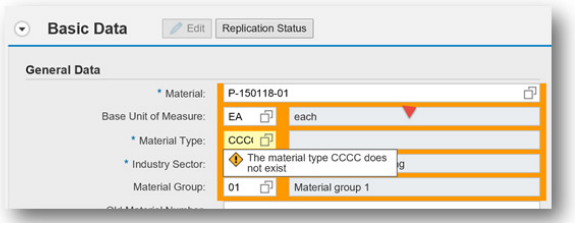
Prevent creation of duplicates for increased effectiveness and efficiency

- Checked early and embedded in the process
- High detection quality of matching using SAP HANA or Enterprise Search



Validations

- Re-use of existing validation logic
- Custom validations can be modeled and programmed (For example, code lists, simple checks, or rules using BRFplus)



Showing what was changed

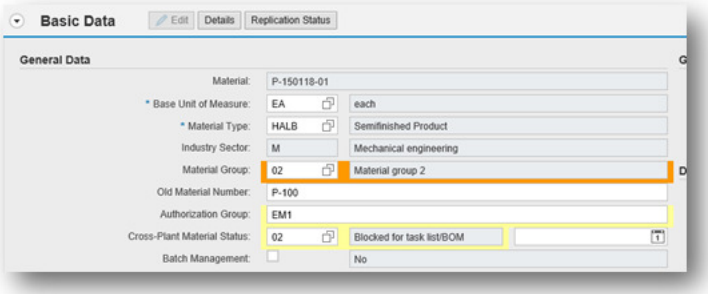
- Changes by earlier processors and own (unsaved) changes in two colors
- Previous, changed, and last saved value per attribute
- Available for all fields, including table cells, rows, navigation elements, etc.

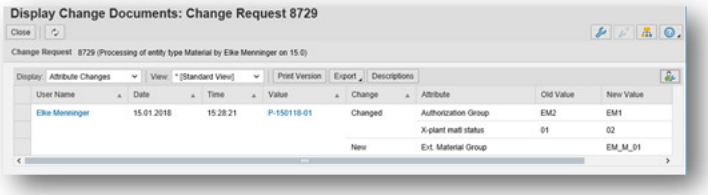
Helps people to do their job

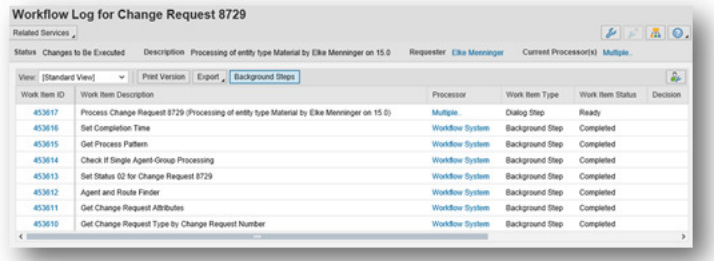
- Supports the processor to decide on their next action
- Supports approvers to quickly spot what they will approve
- Supports auditors to easily see what was changed

Transparency on what has happened earlier

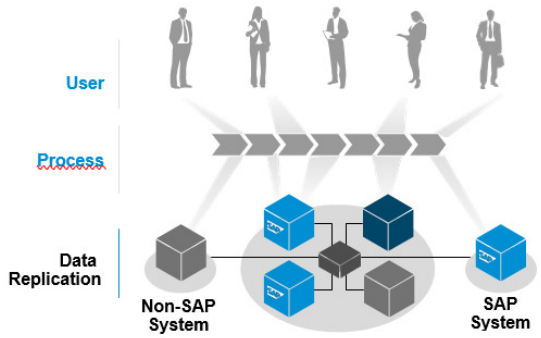
- Change documents provide information who changed which attribute from what value to what other value by when
- Workflow logs tell who was involved in the change request process and who approved which data change







2.7.8 Robust Data Replication



Replication

- Direct, pooled or manual replication to SAP and non-SAP systems
- Peer-to-peer or mediated

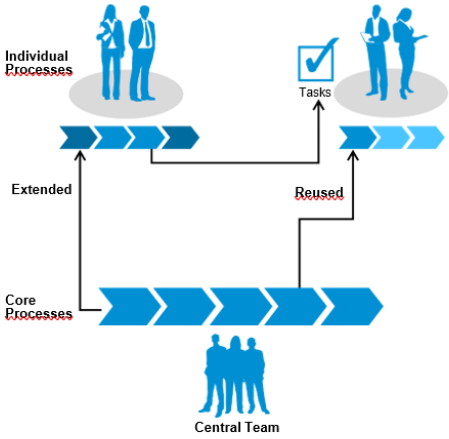
Replication Techniques

- DRF controlling replication mechanisms like
  - ALE for SAP S/HANA, SAP ECC, R/3 and non SAP systems
  - SOA service (only in S/4) for SAP and non SAP systems
- CIF for SAP SCM and MW for SAP CRM/SRM

Replication Features

- Flexibility to replicate selected data only to a specific client system by defining filters (controlled by DRF)
- Support of different object IDs or customizing keys between hub and client systems
  (via mapping in MDG or PI)
- Monitoring and error handling

2.7.9 Reusability & Extensibility



Reuse standard content

- Reuse existing data structures and functionality of SAP Business Suite

Enhance functionality

- Be able to enhance MDG functionality based on predefined content (e.g. plant or sales data) or customer-specific extensions

Extend data structures

- Provide easy-to-use extensibility capabilities for enhancing nodes and fields
  - data models
  - user interfaces
  - processes
  - data replication

2.7.10 MDG and S/4HANA[6]

As more customers move from the classic SAP ERP to SAP S/4HANA to transform their digital cores and make their enterprise landscape ready for the future, master data owners and IT teams responsible for master data must ask many tough questions: Should we also migrate our SAP Master Data Governance (SAP MDG) system to SAP S/4HANA? Should we upgrade our SAP MDG on SAP ERP or SAP MDG on the SAP Business Suite on SAP HANA to SAP MDG on SAP S/4HANA? What is the value of running SAP MDG on SAP S/4HANA? What functions and features are available in SAP MDG for SAP S/4HANA 1709 that aren't available in SAP MDG on SAP ERP?

The role of SAP MDG in the context of SAP S/4HANA

MDG can accelerate your S/4HANA roadmap

SAP S/4HANA does not change the role of SAP Master Data Governance

- Distribute or consolidate master data across all systems in the landscape: for example SAP Business Suite systems, SAP S/4HANA systems, other SAP and non-SAP systems…

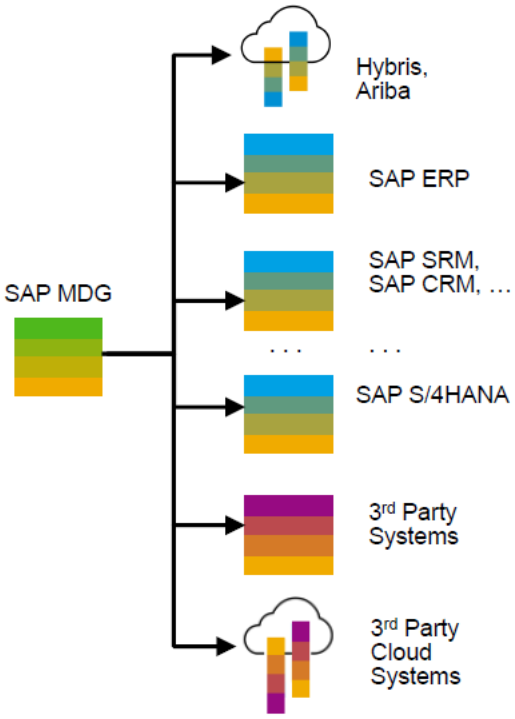
Integration with SAP S/4HANA is available today

- SAP S/4HANA can be a distribution target for SAP Master Data Governance
- Central Finance can leverage MDG's centralized master data & key mapping for aligned postings
- SAP Master Data Governance can be co-deployed on SAP S/4HANA on premise
- Simplified versions of Consolidation, Mass Processing and Master Data Quality Remediation available for public cloud consumption in SAP S/4HANA Cloud

Preparing for a move to SAP S/4HANA

- SAP Master Data Governance is a good way to get experience with a controlled subset of SAP S/4HANA before a full blown implementation
- SAP Master Data Governance is a good way to ensure data quality and readiness before SAP S/4HANA implementation and reduce deployment risk.
- SAP Master Data Governance is a good way to establish strong governance processes that will help you maintain master data quality after SAP S/4HANA go live

[6] Rheinwerk Verlag SAP Master Data Governance (SAP MDG) on SAP S/4HANA



Remark:
SAP expects innovations in SAP MDG on S/4HANA to outpace innovations in SAP MDG on ERP

2.8 Master data excellence for the digital business

Benefit in performing enterprise master data management with MDG innovations across key disciplines such as **master data consolidation, mass processing, master data remediation**, and **central governance.**

Master data consolidation and mass processing

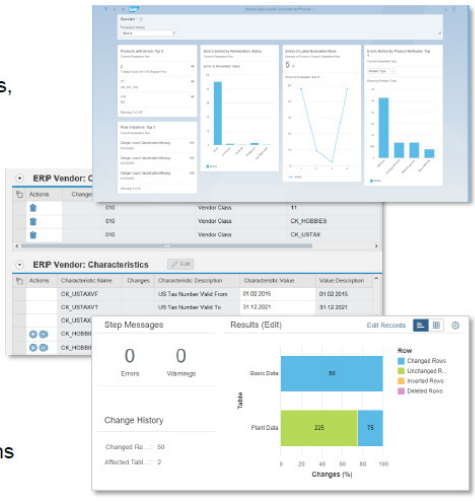
- Consolidation of active data; handling of classification data, business partner relationships, mass creation or deletion scenarios, file upload, and transparency of changes – for business partner, material, and custom objects

Master data remediation

- Master data quality analytics for simple quality issue detection and correction of product data (only available on SAP MDG on SAP S/4HANA)

Central governance

- Improve the user experience for classification data handling for central governance of material data and enable classification data handling for central governance of customer and supplier data and within multiple records processing
- Capabilities related to the European Union General Data Protection Regulation (GDPR) like business partner blocking and deletion as well as read access logging
- Customer-driven optimization, simplification and round-offs across all master data domains



*) Innovations provided in SAP MDG on SAP S/4HANA 1709 and SAP MDG 9.1

End-to-end processes for the entire master data lifecycle, covering central governance and consolidation
Empowering business agility with data quality and process analytics and consumer-grade user experience
Strong foundation for improved business efficiency and process flexibility enabling simplification

Consolidation and mass processing

- Multiple vendor and customer assignments of business partners
- Classification data of materials, customers, vendors in mass processing

Central master data governance

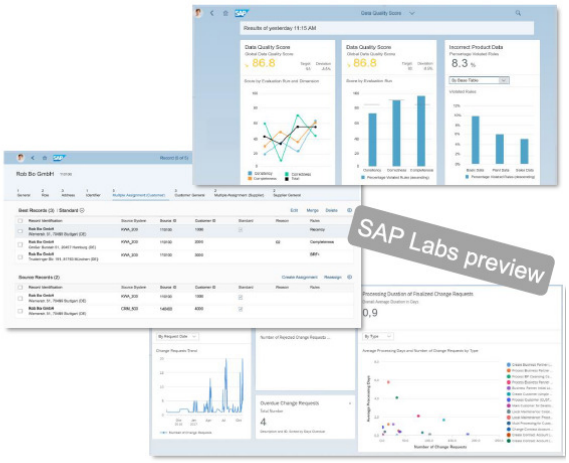
- Financial Data: Adopt S/4HANA universal journal, i.e. G/L accounts as basis for cost element; support for new domain: internal order
- Product: Adopt S/4HANA simplifications and industry-specific enhancements
- Business Partner: Multiple vendor and customer assignments; integration with SAP Credit Management

Integration scenarios

- e.g. SAP Ariba Supplier SLP, SAP Hybris C4C

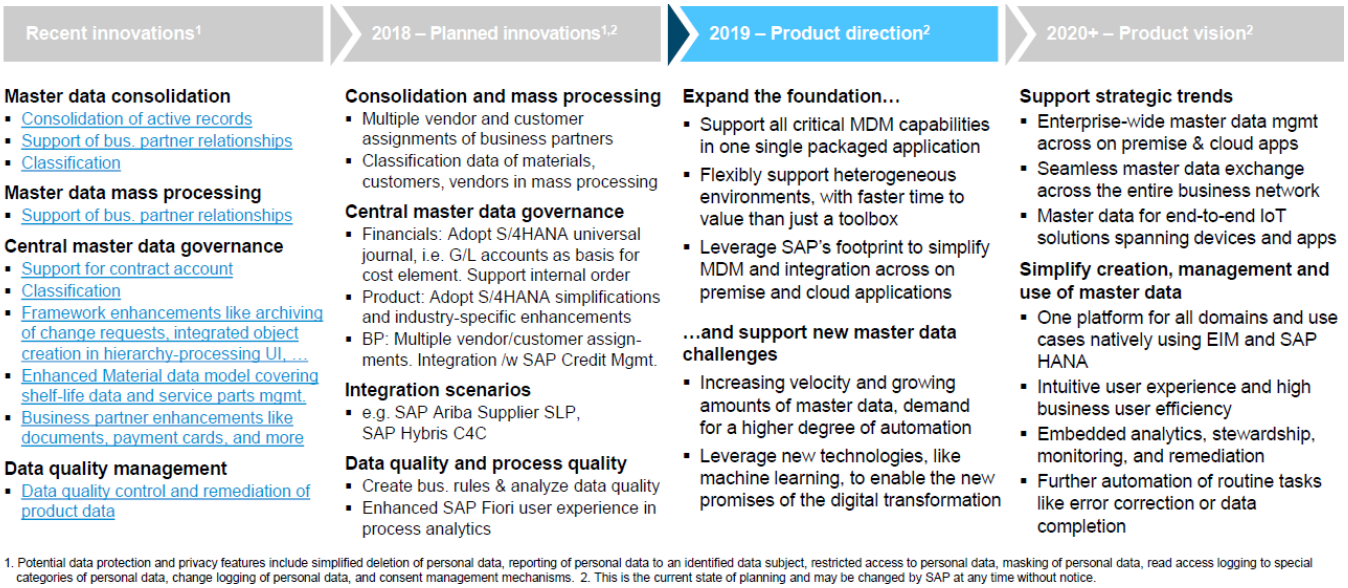
Data quality and process quality

- Create business rules and analyze data quality
- Enhanced SAP Fiori user experience in process analytics



This is the current state of planning and may be changed by SAP at any time.

2.9 MDG road map overview – key innovations[7]



1. Potential data protection and privacy features include simplified deletion of personal data, reporting of personal data to an identified data subject, restricted access to personal data, masking of personal data, read access logging to special categories of personal data, change logging of personal data, and consent management mechanisms. 2. This is the current state of planning and may be changed by SAP at any time without notice.

2.10 New MDG functions

Integration scenarios

- SAP Ariba Supplier SLP
- SAP Hybris C4C

Central Master Data Governance for Customer / Supplier

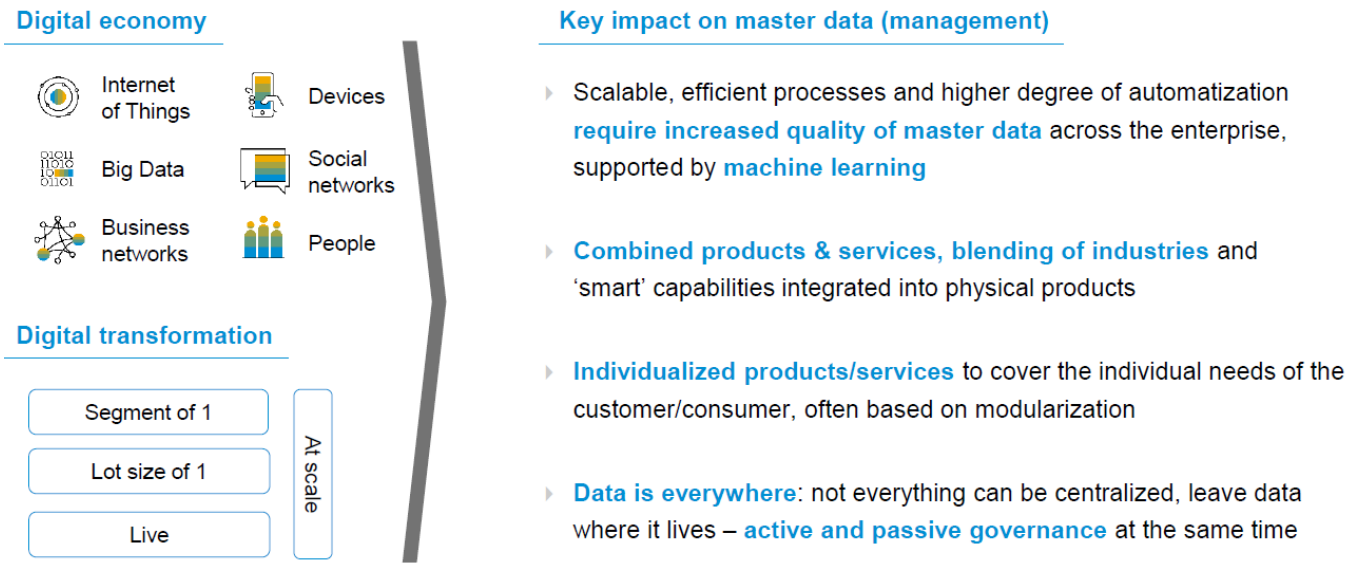
- Roundoff Multiple vendor and customer assignments of business partners (creation process within one Change Request)
- Global search variants (MDG on S/4HANA only)
- Enhancement by SAP Credit Management data

Central Master Data Governance for FI Contract Account

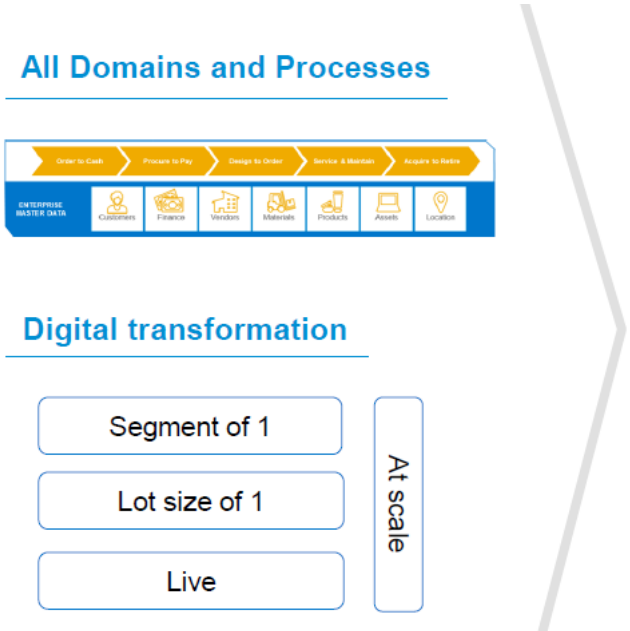
- Data Load
- Data Replication via SOA-Service
- Enable Client Maintenance
- Time Dependency

[7] SAP Master Data Governance on SAP S/4HANA 1709 / SAP Master Data Governance 9.1

2.10.1 The impact of digitization on master data



Expand the foundation …
Further strengthen SAP MDG's ability to support all critical capabilities for master data management in one single application. Continue to deliver a packaged solution and out of the box content, with all flexibility to support heterogeneous environments, but with a faster time to value than just a toolbox
Leverage SAP's footprint in applications, analytics, and business networks; and simplify master data management and integration across on premise and cloud applications
… and support new master data challenges . Increasing velocity and growing amounts of master data, demand for a higher degree of automation in handling of master data
Start to leverage new technologies, like machine learning, to enable the new promises of the digital transformation, such as customized offers to the individual customer in real time.



2.10.2 MDG > 2020+ Product vision - High quality master data for every use case

Support current and future strategic trends

Enterprise-wide master data management across on premise and cloud deployed applications

Seamless exchange of master data across the entire business network value chain

Support master data for end-to-end IoT solutions spanning devices, platforms & applications

Mobile capabilities that are fully integrated into backend systems

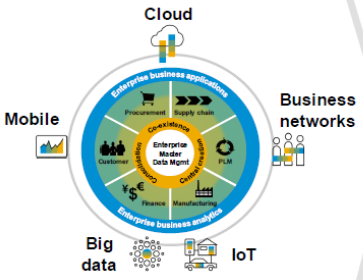
Simplify the creation, management and use of master data

One platform supporting all domains, styles and use cases available on premise or in the cloud, natively using SAP's Information Management portfolio and SAP HANA

Intuitive user experience and high efficiency through business oriented interfaces, flexible workflow, and integration of content and context

Embedded analytics and stewardship capabilities for monitoring, root cause analysis and remediation

Automation using machine learning for things like error detection and correction, completion of missing attributes, entity resolution, rule detection, and more



2.10.3 SAP Master Data Governance and cloud consumption models

Enterprise master data is part of companies' Digital Core

SAP MDG is offered on premise and in the private cloud, today

In the future, we plan to leverage SAP S/4HANA Cloud and SAP Cloud Platform as the platform to extend MDM scenarios for consumption in the public cloud

Master Data Management support for cloud applications

SAP MDG customers synchronize master data between all cloud and on premise applications, today

SAP provides cloud-based data integration and data quality capabilities

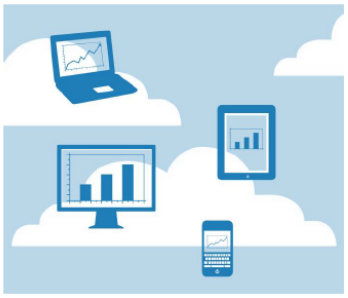
Advanced local master data maintenance in SAP S/4HANA Cloud, like e.g. business partner data enrichment, or re-use of MDG code for mass maintenance…

Leveraging Micro Services and Business Networks

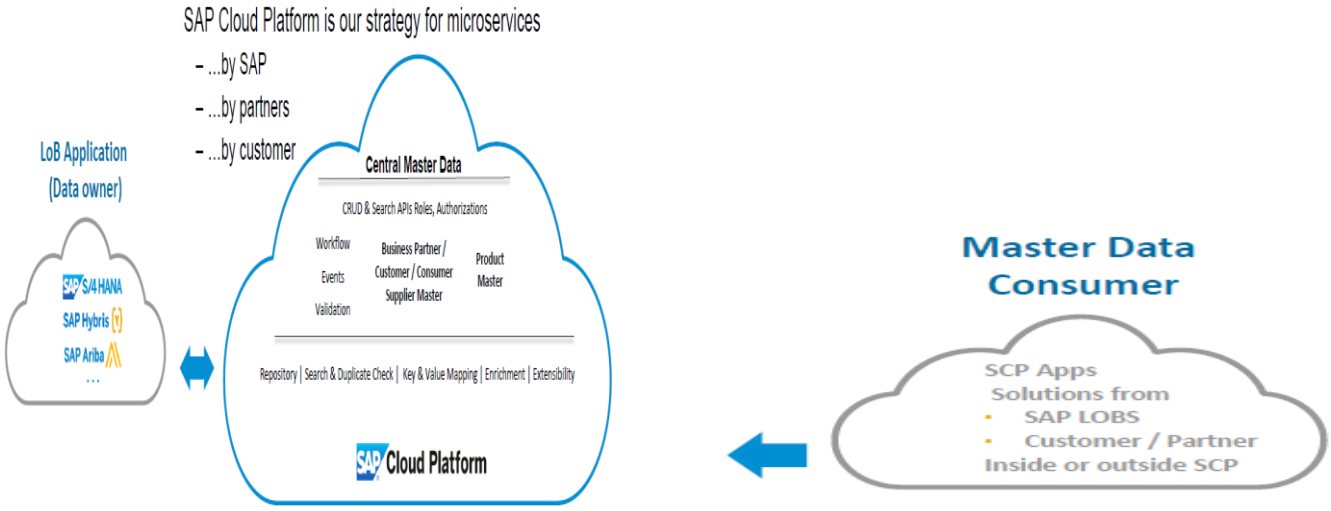
SAP Data Quality Management, microservice, Identity-as-a-Service, …

SAP Asset Intelligence Network, SAP Ariba, SAP Hybris, …

SCP-based microservices planned as an alternative way to synchronize between SAP MDG and any public cloud application (SOAP & ODATA)



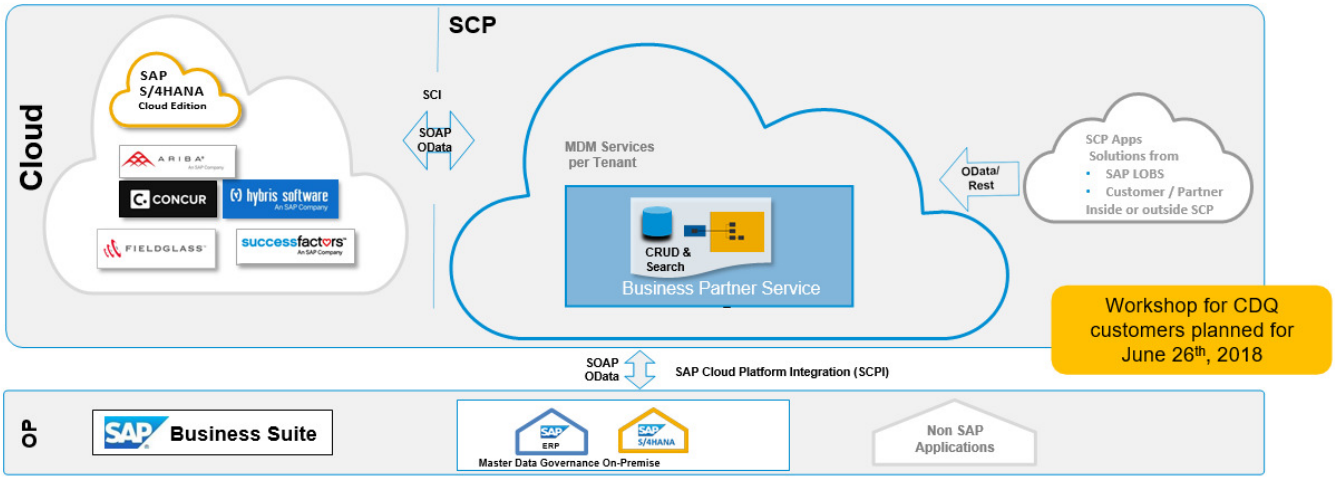
2.10.4 Master data services on SAP Cloud Platform



Guiding Principles

1. Shared core data model between applications
2. Local enhancements by specific attribute-sets in local apps
3. Start with Business partner/customer re-use service, extend to Supplier and Product Master
4. Built on SAP Cloud Platform, services published in API Hub

2.10.5 MDG – Business Partner Service Consumption



2.11 Strategic direction, using connectivity in business networks

Connect to suppliers with SAP Ariba

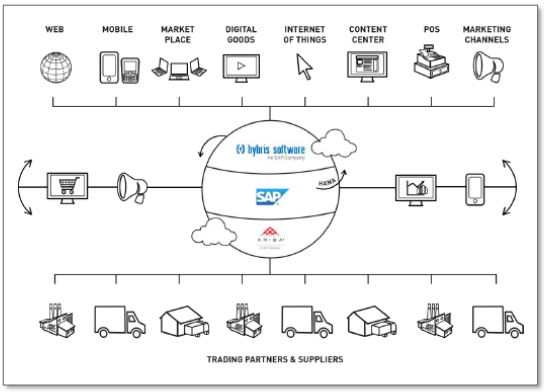
Exchange master data between business partners via the SAP Ariba network

Manage data internally in SAP MDG

Governed provisioning of master data with SAP Master Data Governance, ready to be used in all internal business applications

Provide product data via SAP Hybris PCM

Enrich product data for use in catalog, eCommerce and other customer-facing channels



3 SAP MDG for Financials (MDG-F)[8]

Master Data Governance for Financials (MDG-F) is an application to centrally govern the maintenance of financial master data.

•MDG-F provides central steering and control mechanisms independently of local or centralized execution.

•In MDG-F, master data records can be created, completed or changed in a staging area, following a distributed step-by-step way, before making the record available to connected business applications.

SAP Master Data Governance provides out-of-the box solutions for the central management of master data objects. Domain-specific solutions include business partner (MDG-BP), customer (MDG-C), supplier (MDG-S) governance, material governance (MDG-M), and financials governance (MDG-F).
This guide provides you with the foundation knowledge you need to know about financial data and its related governance solution financial governance (MDG-F).

Inconsistencies in financial master data arise from local, non-standardized maintenance processes (e.g. different charts of accounts) with, in some cases, manual workflow steps in a heterogeneous global system landscape. This not only affects the operational efficiency, but also the input required for the respective financial statements when preparing evaluations and the subsequent manual adjustments.

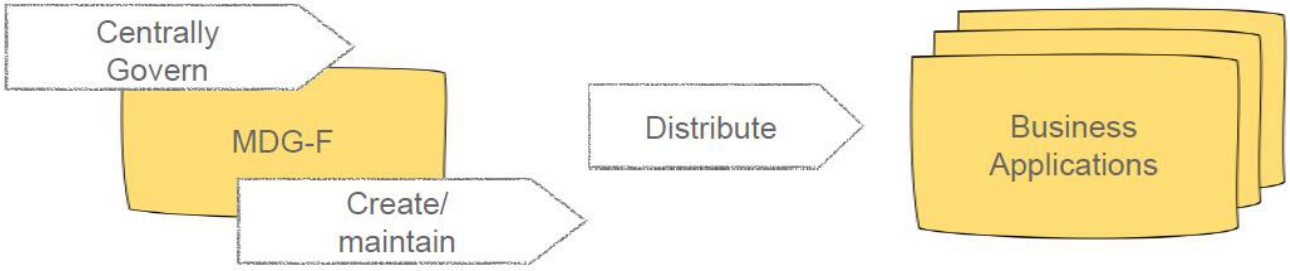
In addition to this, increasing complex regulations require organizations to have change transparency and audit trails.

Master Data Governance for Financials (MDG-F) is an application to centrally govern the maintenance of financial master data.

•MDG-F provides central steering and control mechanisms independently of local or centralized execution.

[8] SAP Master Data Governance on SAP S/4HANA 1709 / SAP Master Data Governance 9.1

•In MDG-F, master data records can be created, completed or changed in a staging area, following a distributed step-by-step way, before making the record available to connected business applications.



SAP MDG for Financials (MDG-F) offers central maintenance and governance solutions for a wide range of financial master data. These include, for example, accounting objects such as standard and group charts of accounts, general ledger accounts, companies and Controlling objects such as profit centers, cost centers and their hierarchies. In addition, customer-specific objects can also be implemented with MDG-F.

3.1 Specifics of SAP MDG-F:[9]

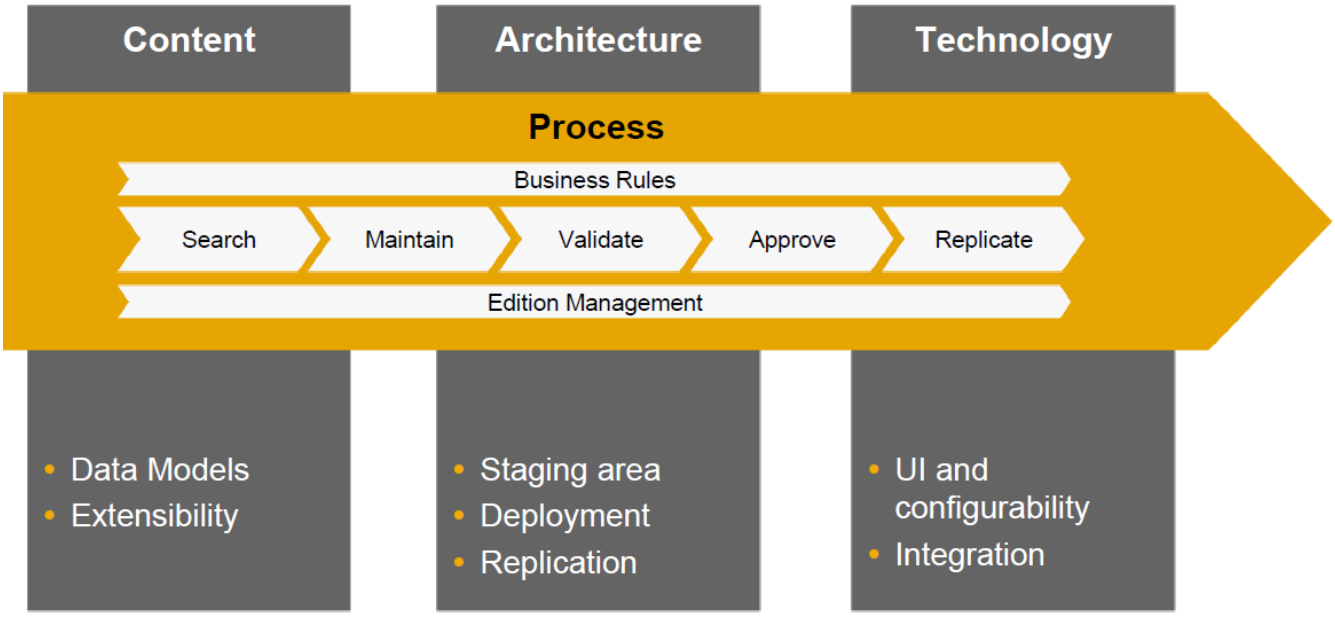
- Consolidation of company financial data with the financial data of the subsidiaries
- Mapping between corporate and operative standard charts of accounts
- Tracking of the change processes and process setup possible within the scope of IFRS, Sarbanes-Oxley, GAAP, etc.
- Time-based and period-based versioning: Transparency, traceability, edition (summary of multiple work steps for the final release)

The centralized MDG-F system permits the user-friendly management of your financial data and specific customizing options in order to comply with legal requirements with the aid of holistic integration into your SAP ERP system. In particular, SAP MDG-F is notable for allowing existing SAP ERP data structures and objects to be reused, which in turn allows checks and validations to be used analogously. The user interface is based on ABAP WebDynpro programming to respond flexibly to changes in requirements.

[9] SAP SE

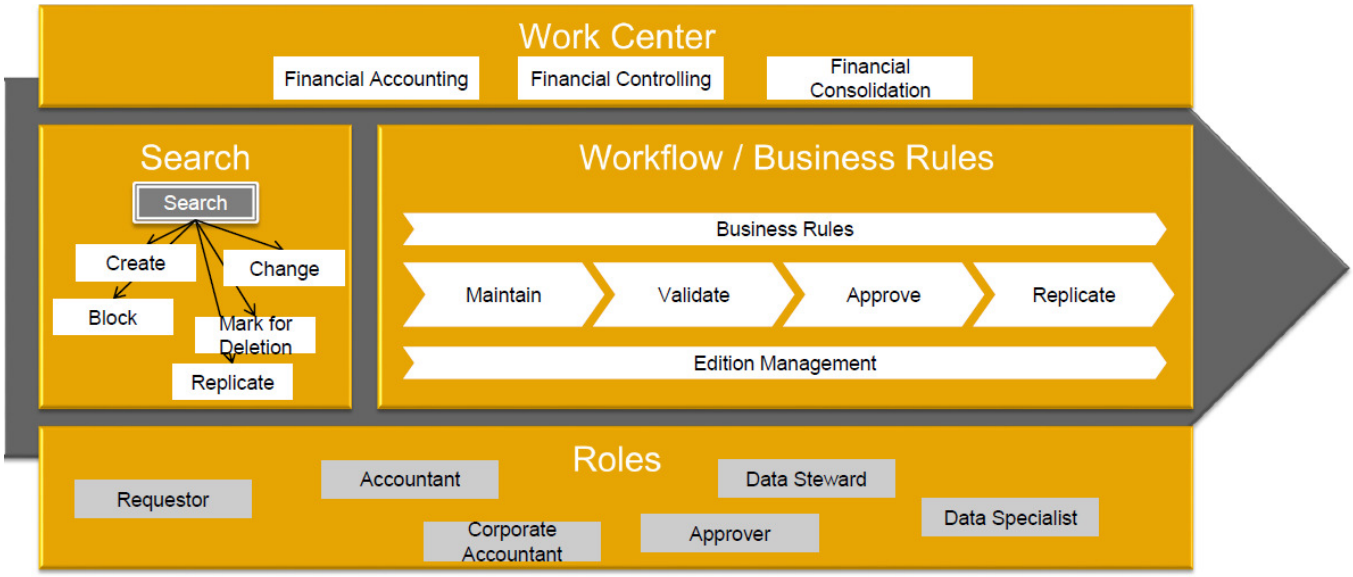
3.2 MDG for Financials Addresses Main Action Areas[10]

3.3 The Solution Building Block Overview



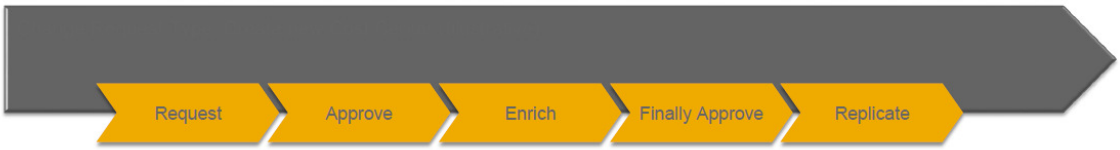
[10] SAP SE

3.4 MDG-F Process Framework



3.5 The Change Request

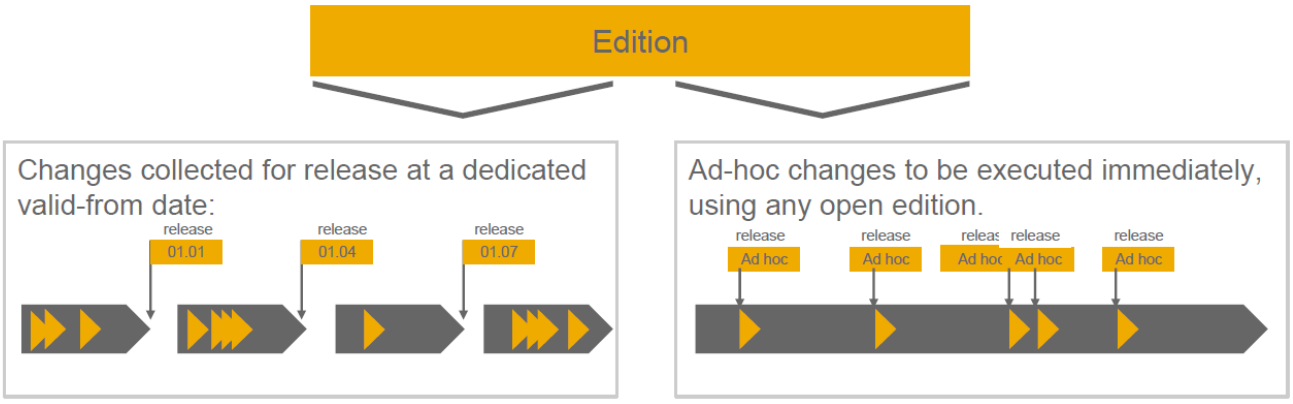
- **The Change Request is an MDG-specific object that:**
- **acts as the container for new and changed data**
  - Master data content
  - Comments
  - Approval history
- **defines and controls the process flow and processors according to defined Change Request type**
- **optionally holds multiple changes**



3.6 Editions

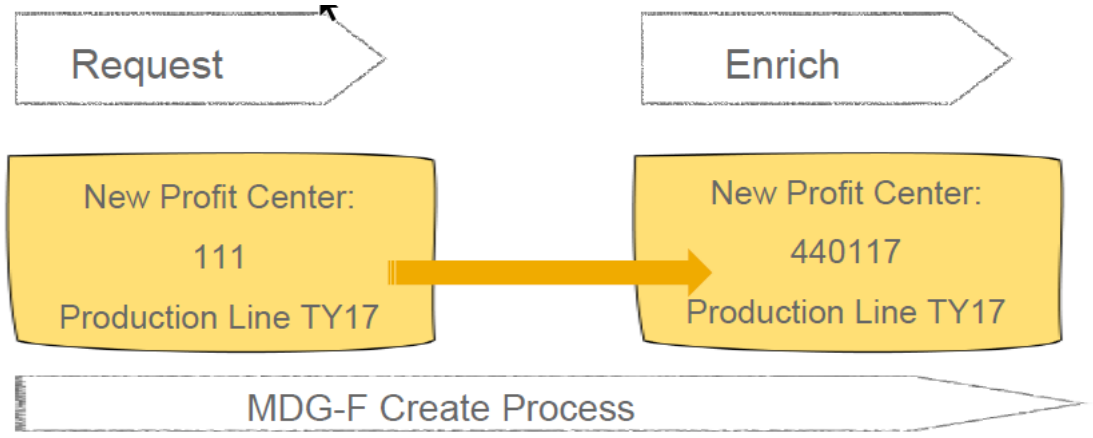
Editions are used to control the validity of a change. In the edition, planned change requests for the creation of new data or for changes to existing data are collected and released collectively at a certain

date. Time-dependent objects (Profit/Cost Center) automatically inherit the valid-from date from the edition's valid-from date.



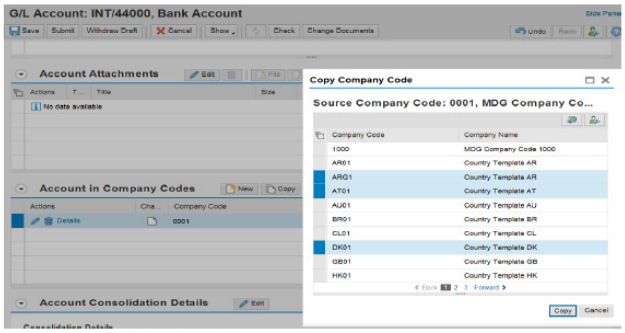
3.7 Work with Changeable Keys

When requesting a new financial master data item, often the requesting user does not yet know the number to be used in Business Suite Applications. To provide flexibility to the requestor whilst adhering to corporate numbering standards, the single-object maintenance in SAP MDG-F offers the feature to use a generated key first, and changing it to the desired final number in a later step of the creation process.

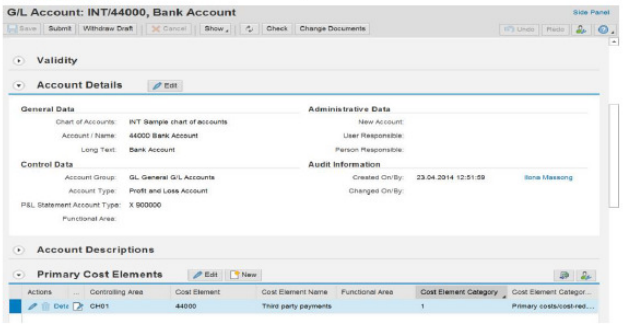


3.8 Process Features

- Copy multiple company codes: create multiple company code-specific G/L accounts by simple copying



- One-step creation of primary cost element: automatically create the primary cost element connected to a new account in the same maintenance step



3.9 Hierarchy Assignment in Single-Object Processing

Master Data and Hierarchy Assignment in the same Main Financial Objects UI

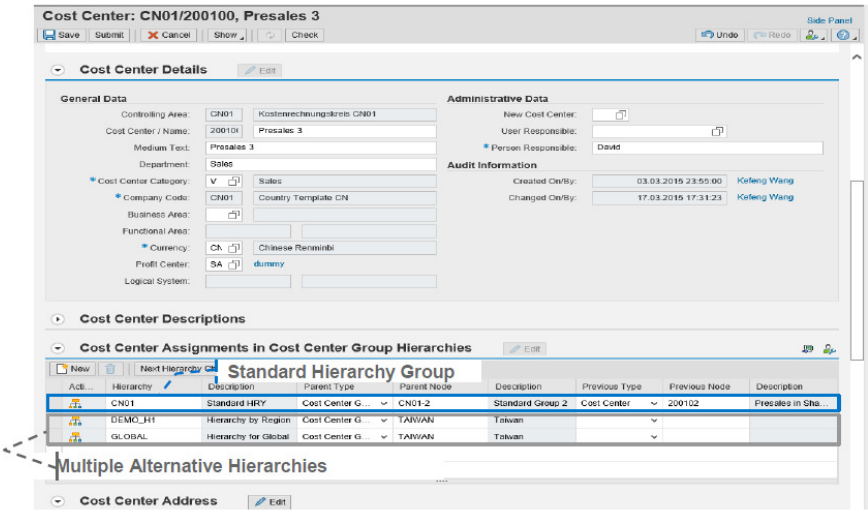
- Cost Center
- Profit Center
- Cost Element
- Cost Center Group
- Profit Center Group
- Cost Element Group

And also:

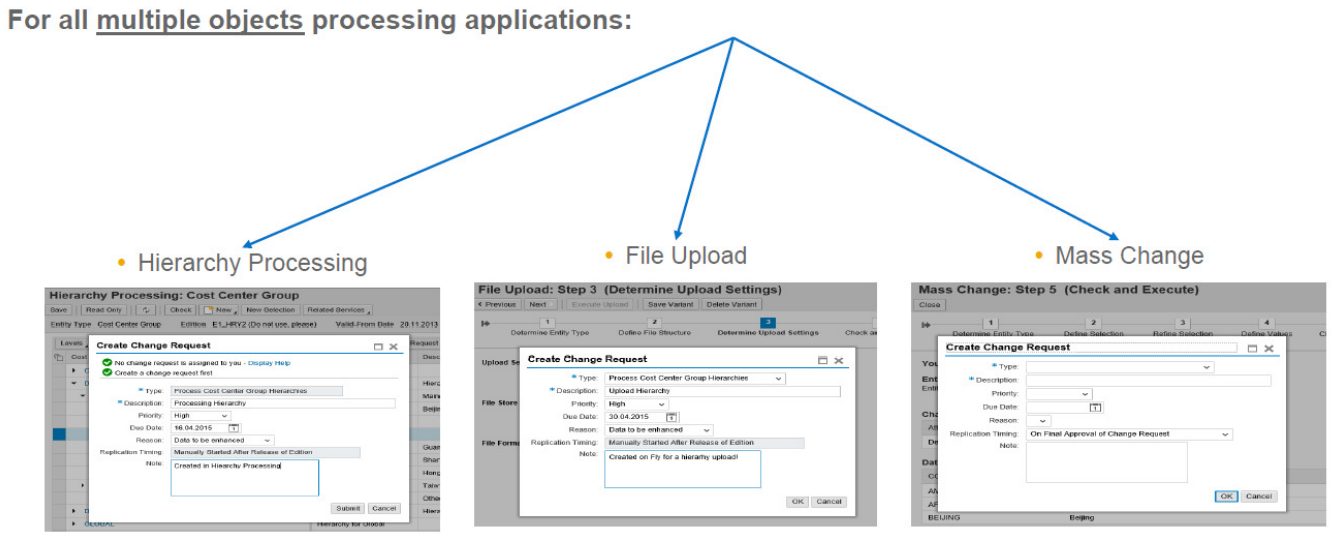
- G/L Account
- Financial Reporting Structure Item

In addition to ERP:

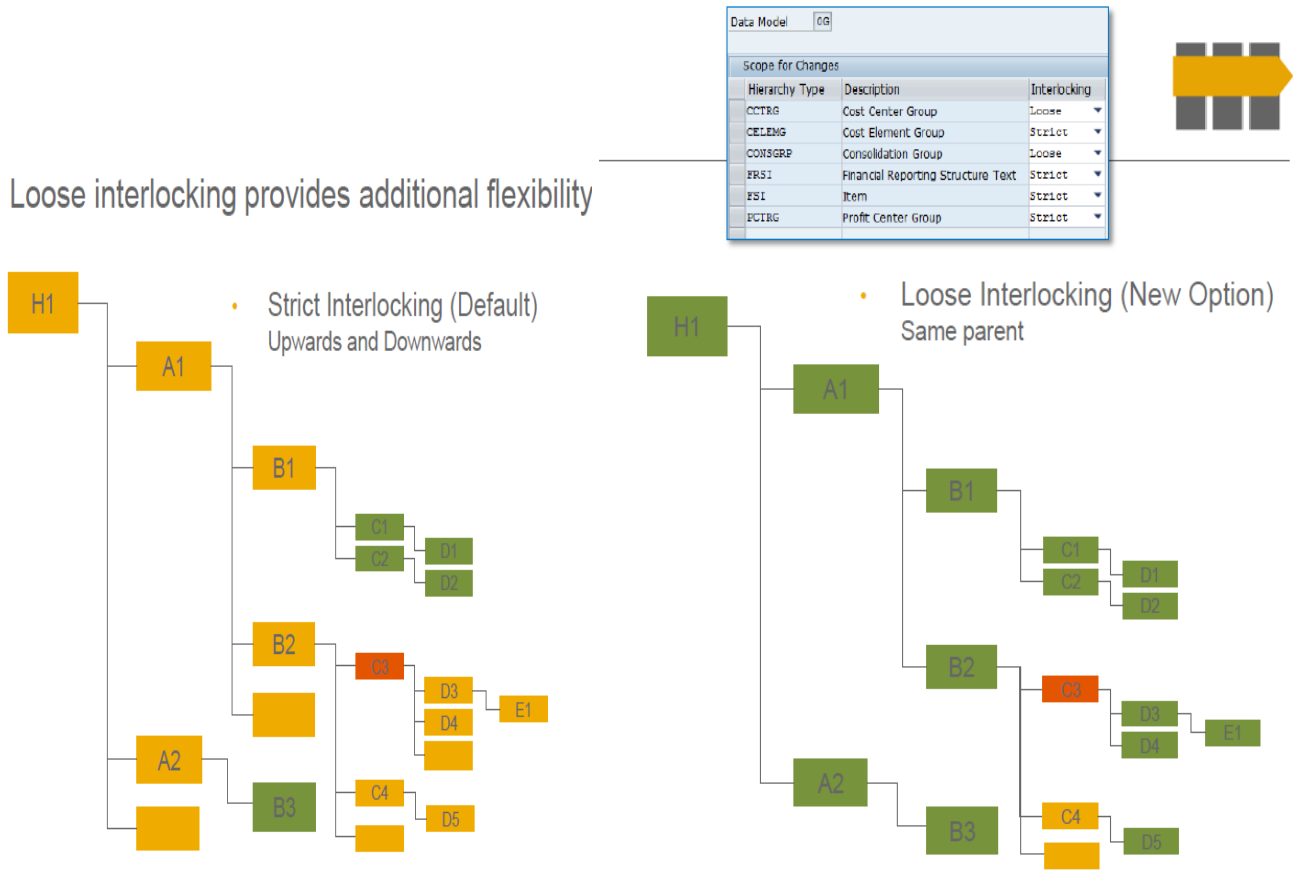
- Overview of all assignments in all hierarchies
- Display and maintain not only standard group, but also alternative hierarchies on master data screen



3.10 Integrated Change Request



3.11 Hierarchy Interlocking

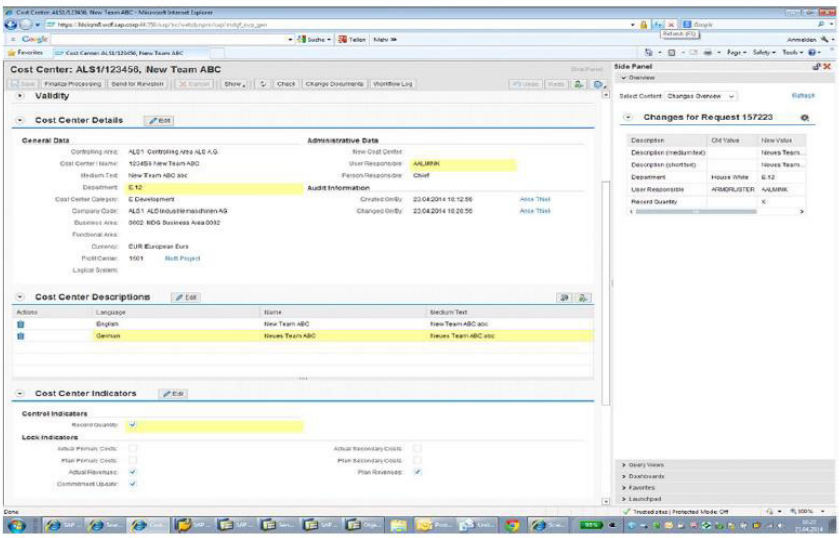


Node currently maintained in a pending change request **Interlocked**: You must use the pending change request **Not Interlocked:** You can use a separate change request

3.12 Usability Features in Single-Object Maintenance UI

Side panel / Business Context Viewer:
Context information can be displayed in the side panel to provide further related information about the current data record, e.g. the change history

Highlight Changes: Fields that were changed in the previous step are highlighted in color

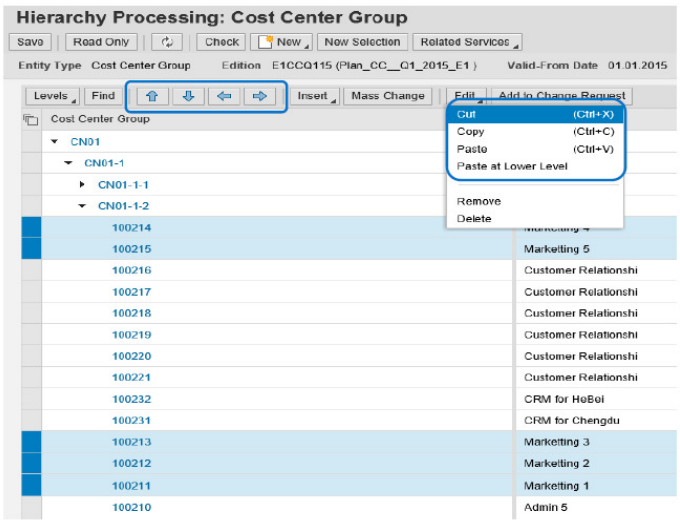


3.13 Usability Features in Hierarchy Processing

Easy Actions supported for multiple objects:

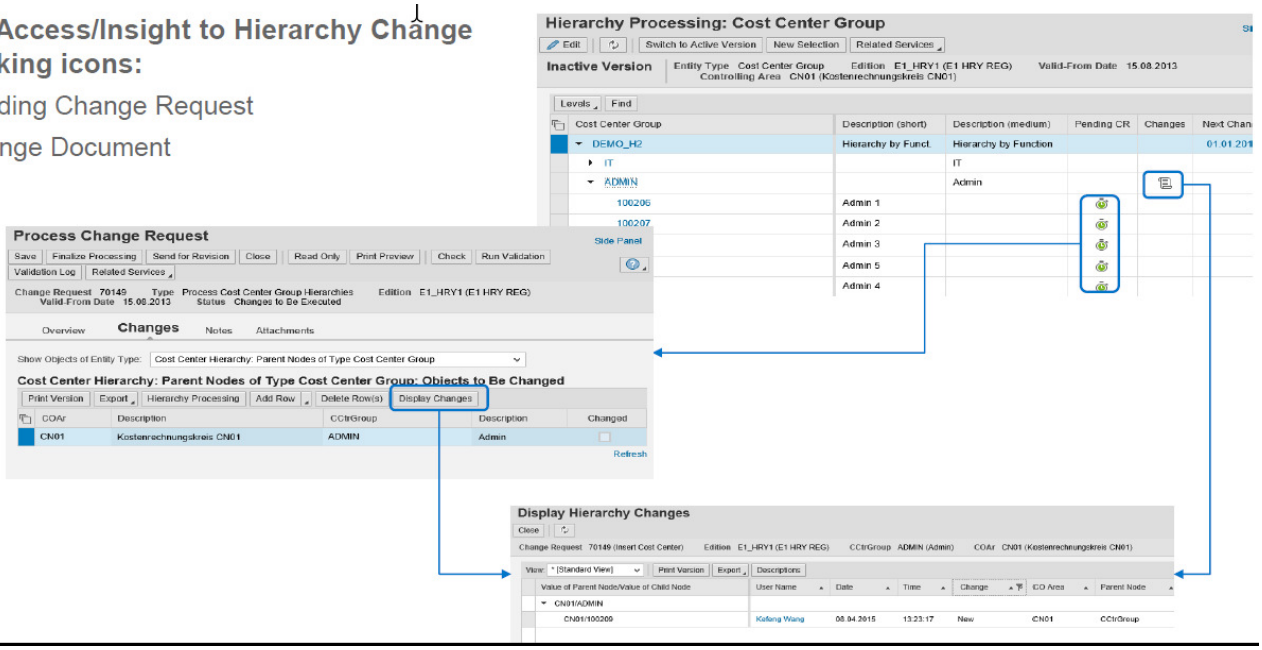
- Drag & Drop
- Move Up, Down
- Rank Higher, Rank Lower
- Copy & Paste or Paste at Lower Level
- Cut & Paste or Paste at Lower Level

Easy to use:
Select multiple subnodes of one node and carry out any of the actions above.

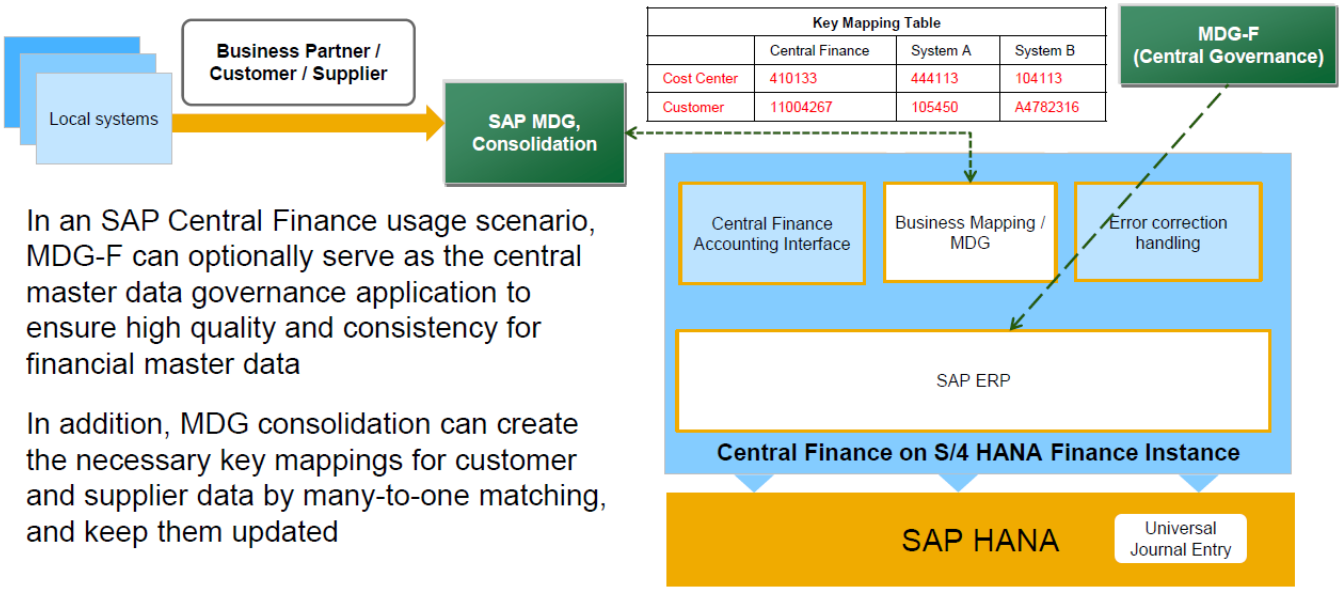


Direct Access/Insight to Hierarchy Change by clicking icons:

- Pending Change Request
- Change Document



3.14 SAP MDG Supporting Central Finance (S/4 HANA Finance)[11]

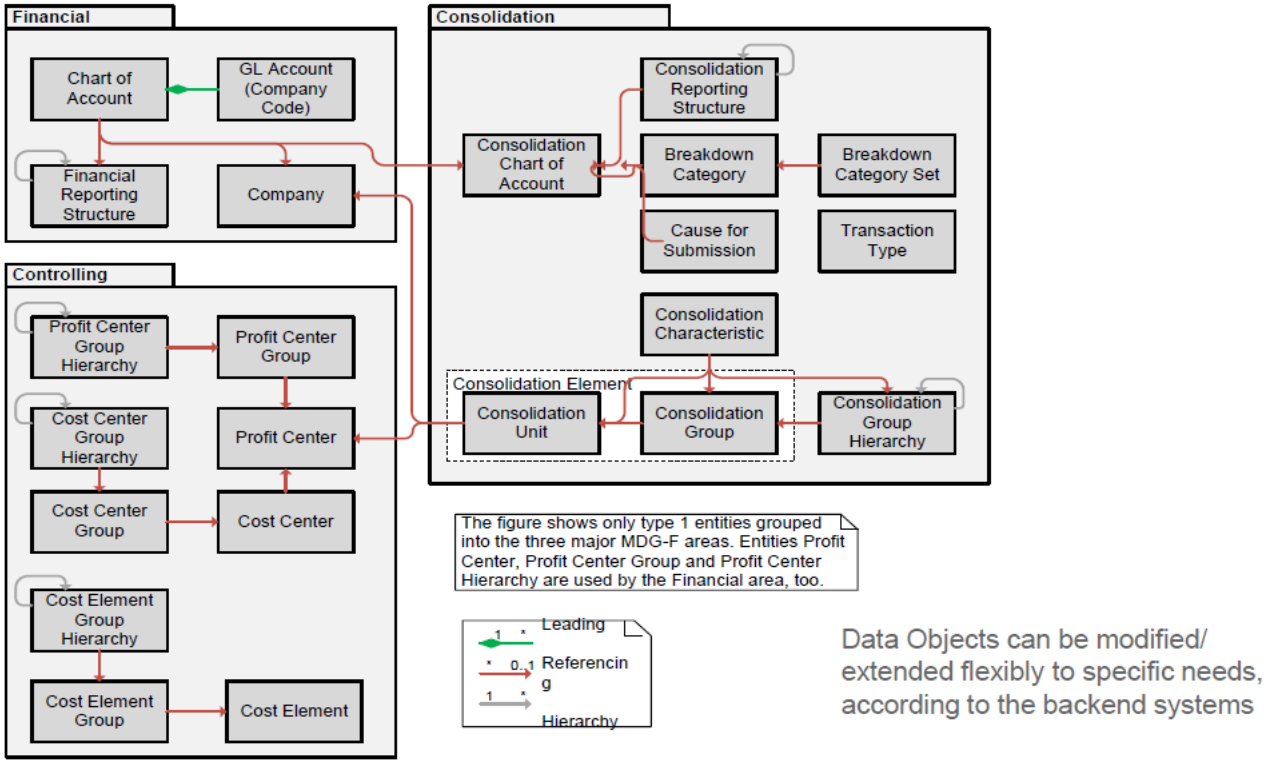


In an SAP Central Finance usage scenario, MDG-F can optionally serve as the central master data governance application to ensure high quality and consistency for financial master data

In addition, MDG consolidation can create the necessary key mappings for customer and supplier data by many-to-one matching, and keep them updated

[11] SAP SE

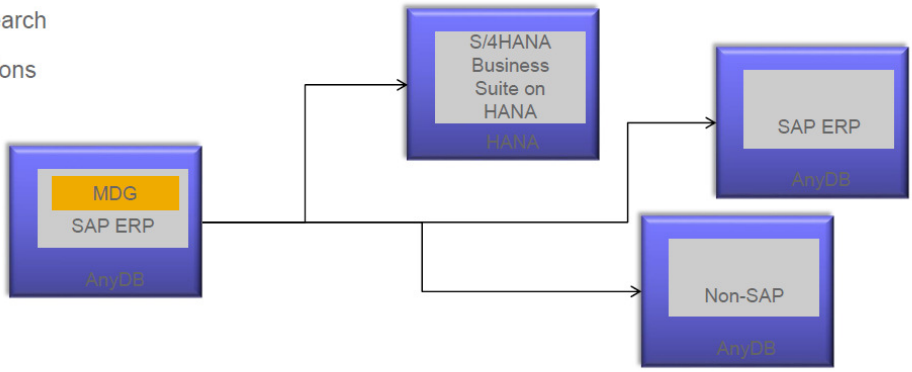
3.15 Standard Objects in Data Model 0G



3.16 Deployment

- MDG is delivered as a business function within SAP ECC
- Deployment as standalone hub or on top of existing ERP instance
- Optional embedding of HANA search
- Replication into various applications including S/4HANA

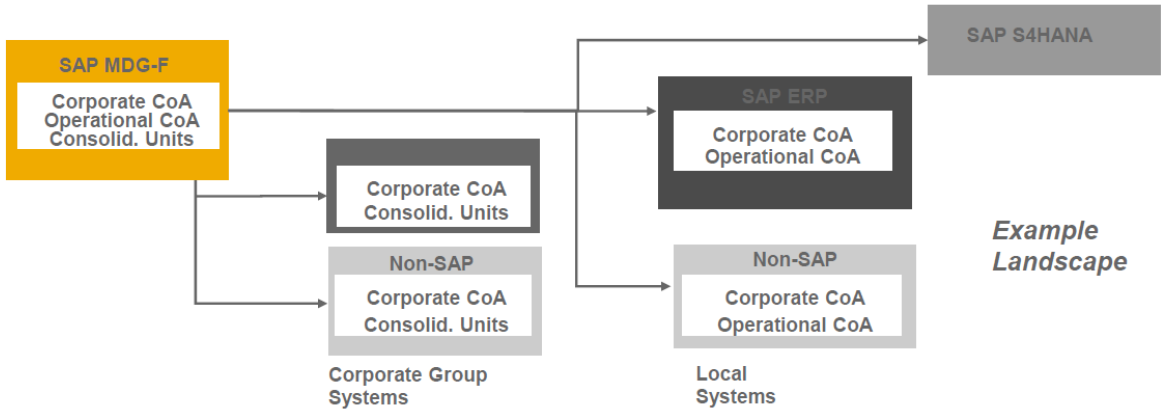
Example



> MDG is pre-integrated with SAP ECC, and provides publish/subscribe and master/slave implementation modes with automatic distribution to SAP and 3rd party applications.

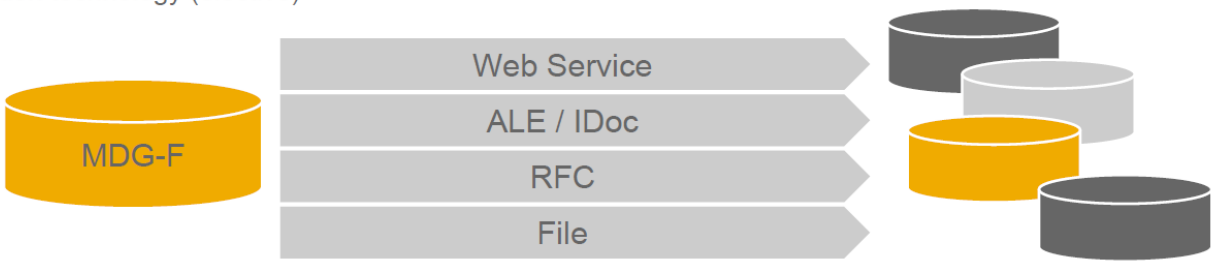
3.17 Replication

MDG-F replicates the data according to the needs of the connected business systems. This is independent of the deployment option, if MDG is used on top of one of the operational ERP systems, or as a standalone MDG data hub.



3.18 Integration

Integration technology (elective)



SOA

Replication Model	Outbound Impl.	Description
SOA_ACC	1010	Replication model for Account
SOA_CCTRH	1110	Replication model for Cost Center Group
SOA_CELE	1180	Replication model for Cost Element (SOA)
SOA_CELEH	1190	Replication model for Cost Element Group Hierarchy (SOA)
SOA_COA	1000_V1	Replication model for Chart of Account (SOA)
SOA_COMP	1140	Replication model for Company (SOA)
SOA_COST	1100	Replication model for Cost Centre (SOA)
SOA_FRS	1020	Replication model for Financial Reporting Structure (SOA)
SOA_PCTH	1130	Replication model for Profit Center Group Hierarchy (SOA)
SOA_PCTR	1120	Replication model for Profit Center (SOA)

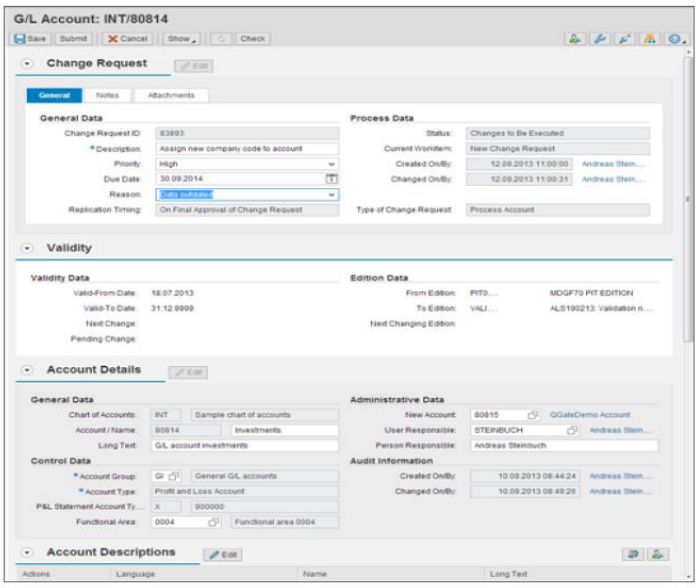
IDoc

Replication Model	Outbound Impl.	Description
IDOC_ACC	1012	Replication model for Account
IDOC_COST	1102	Replication model for Cost Center (IDoc)
IDOC_CCTRH	1112	Replication model for Cost Center Group Hierarchy
IDOC_CELE	1182	Replication model for Cost Element (IDoc)
IDOC_CELEH	1192	Replication model for Cost Element Group Hierarchy (IDoc)
IDOC_PCTR	1122	Replication model for Profit Center (IDoc)
IDOC_PCTH	1132	Replication model for Profit Center Group Hierarchy (IDoc)

3.19 User Interface

Technology

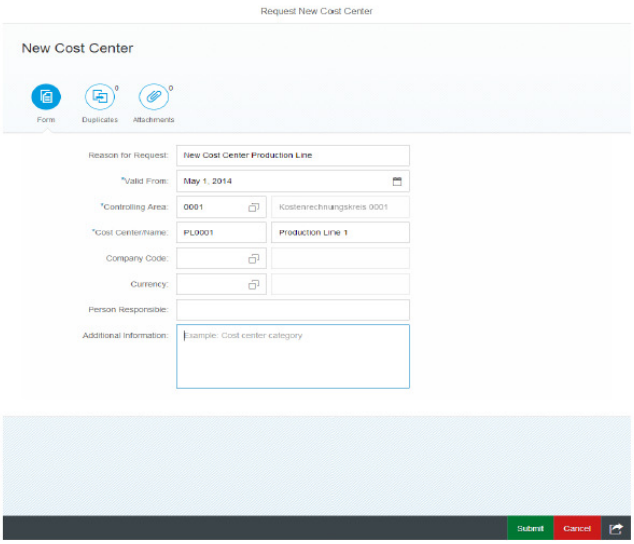
- Browser-based UI (Web Dynpro ABAP) to allow decentral task execution
- Use of Floorplan Manager for easy screen configuration and personalization
- Pre-delivered workcenters provide a structured entry point for individual roles



3.20 Request Cost Center or Profit Center With FioriApp[12]

You can also use Fiori apps to request a new cost center or a new profit center.

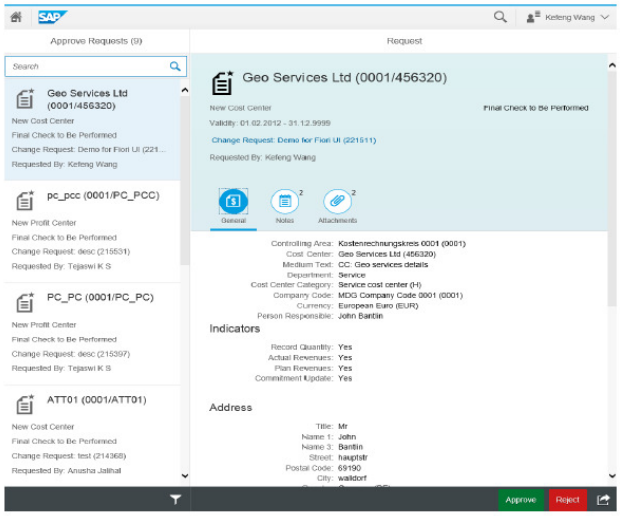
These apps can be flexibly used on a variety of devices including tablets and smartphones.



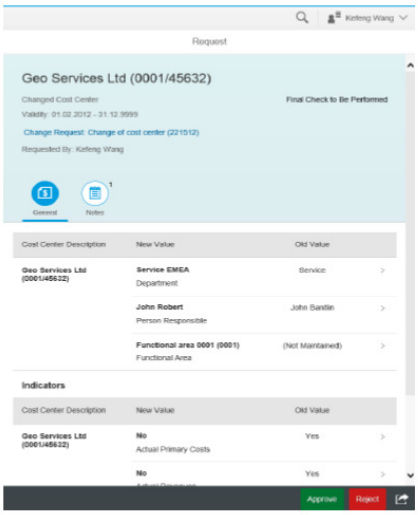
[12] SAP SE

3.21 Approve Cost Center, Profit Center or G/L Account With Fiori App

Approve new object: Only entered fields are displayed



Approve change: Only changed fields (with old values) are displayed



3.22 Copying of accounts

When creating a G/L account, you can now save time and effort by copying the

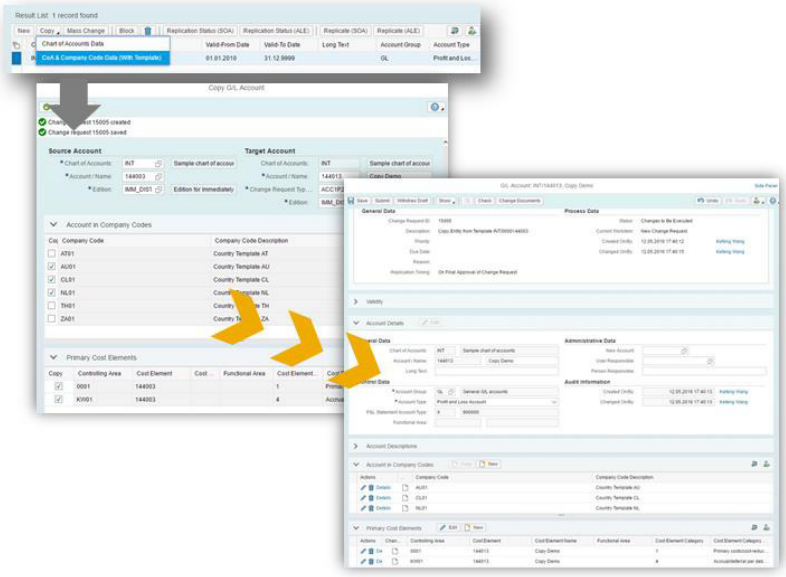
Chart of Accounts and Company Code Data (With Template)

from an existing G/L account using the new Copy function on the Search screen. Alternatively, you can copy just the Chart of Accounts Data.

Copy relevant G/L Account data in one step, including:

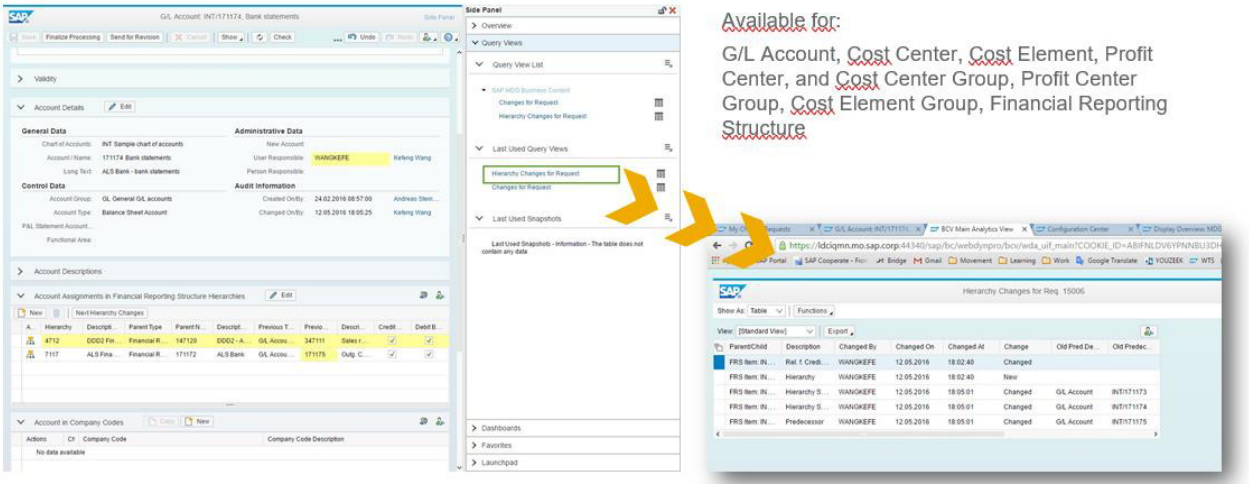
- Chart of Accounts data
- Company Code data
  (selectable on Company Code level)
- Primary Cost Element
  (selectable on Controlling Area).

- New G/L Account can be changed on the populated change request drafts and submitted directly from there.



3.23 Enhanced BCV content to display hierarchy changes

A new display option was added to the Query View. In addition to displaying changes per change request, you can now also display only hierarchy-specific changes for a change request for a detailed overview.



Available for:

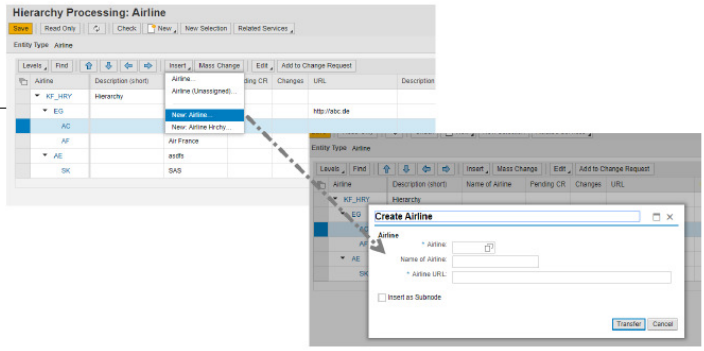
G/L Account, Cost Center, Cost Element, Profit Center, and Cost Center Group, Profit Center Group, Cost Element Group, Financial Reporting Structure

3.24 New functions (planned Enhancements in MDG Application Framework)

3.24.1 Integrated Object creation in Hierarchy Processing UI

Feature description

- In hierarchy processing UI, create and insert object into hierarchy within one step.

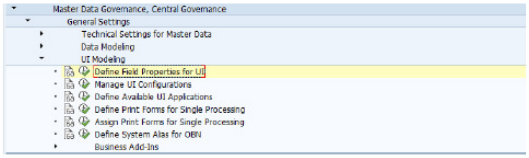


Business value

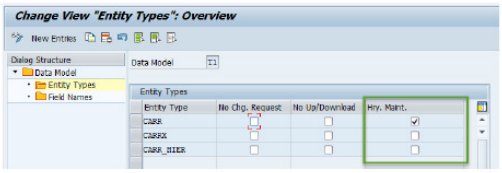
- Simplified maintenance process for hierarchy management
- Improved user usability for hierarchy management

Valid in MDG-F hierarchy

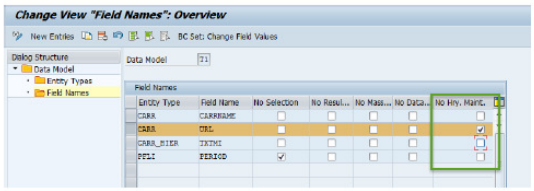
Integrated Object Creation feature is configured under MDG Customizing -> UI Modelling.



You can configure whether Integrated Object Creation in Hierarchy Processing UI is allowed for hierarchy related entity type.



You can configure entity attribute level whether it should NOT be maintained on the creation Popup in Hierarchy Process UI.



Valid in MDG-F hierarchies

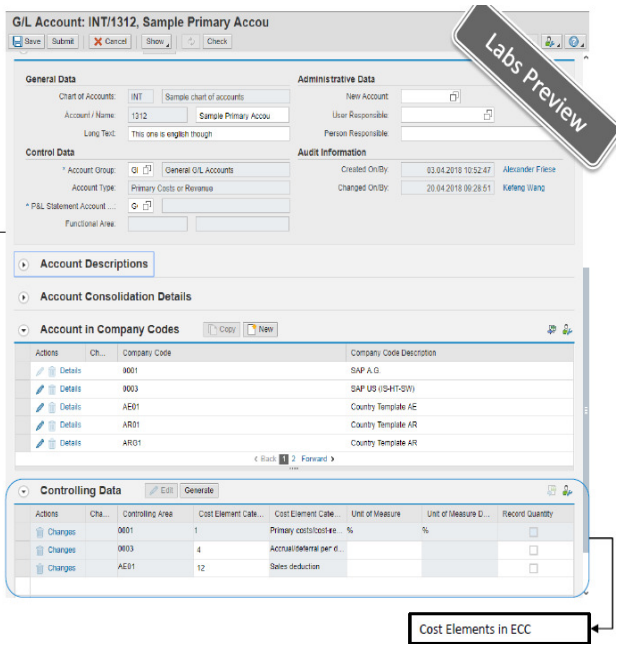
3.24.2 Merge Cost Element into G/L Account

Feature description

- Maintain cost element data in G/L Account UI
- Only for MDG on S/4HANA

Business value

- Simplified financial master data maintenance
- Harmonized behavior with S/4 HANA environment



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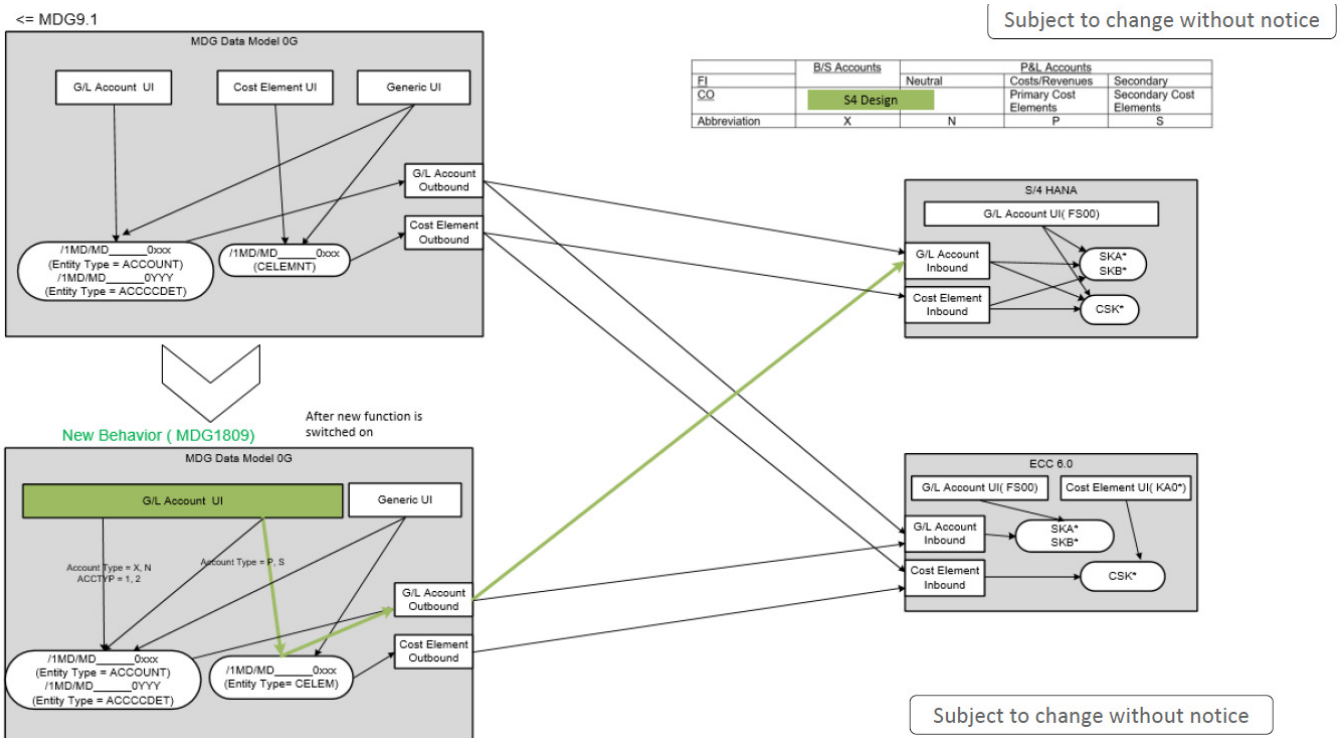


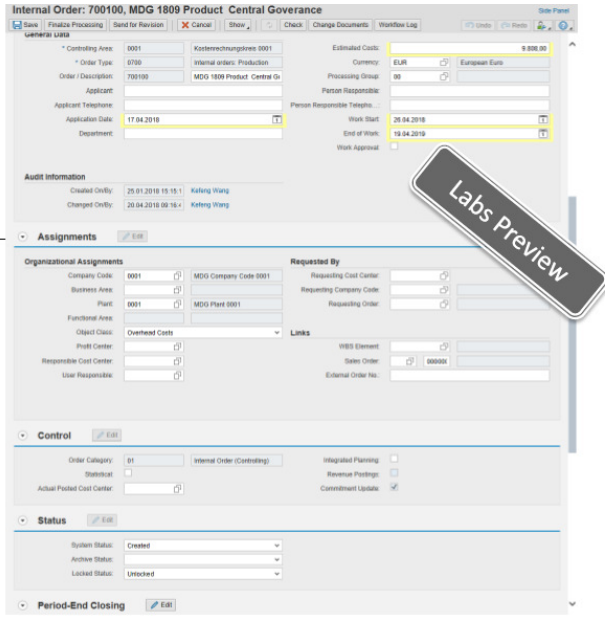
3.24.3 Internal Order Central Governance

Feature description

- Central maintenance and governance of Internal Order
- Replicate Internal Order to ECC or S/4HANA system (Both SOA and Idoc)
- Initial Load Internal Order from ECC or S/4HANA into MDG

Business value

- Centralized governance, compliance and transparency of master data during creation and change through integrated staging, approval and central audit trail
- Delivers consistent definition, authorization and replication of internal order master data, keeps data consistent cross systems
- Eliminates error-prone manual maintenance processes for internal order master data in multiple systems



3.25 MDG-F road map overview – key innovations[13]

Recent innovations[1]	2018 – Planned innovations[1,2]	2019 – Product direction[2]	2020+ – Product vision[2]
Master data consolidation ▪ Consolidation of active records ▪ Support of bus. partner relationships ▪ Classification **Master data mass processing** ▪ Support of bus. partner relationships **Central master data governance** ▪ Support for contract account ▪ Classification ▪ Framework enhancements like archiving of change requests, integrated object creation in hierarchy-processing UI, … ▪ Enhanced Material data model covering shelf-life data and service parts mgmt. ▪ Business partner enhancements like documents, payment cards, and more **Data quality management** ▪ Data quality control and remediation of product data	**Consolidation and mass processing** ▪ Multiple vendor and customer assignments of business partners ▪ Classification data of materials, customers, vendors in mass processing **Central master data governance** ▪ Financials: Adopt S/4HANA universal journal, i.e. G/L accounts as basis for cost element. Support internal order ▪ Product: Adopt S/4HANA simplifications and industry-specific enhancements ▪ BP: Multiple vendor/customer assign-ments. Integration /w SAP Credit Mgmt. **Integration scenarios** ▪ e.g. SAP Ariba Supplier SLP, SAP Hybris C4C **Data quality and process quality** ▪ Create bus. rules & analyze data quality ▪ Enhanced SAP Fiori user experience in process analytics	**Expand the foundation…** ▪ Support all critical MDM capabilities in one single packaged application ▪ Flexibly support heterogeneous environments, with faster time to value than just a toolbox ▪ Leverage SAP's footprint to simplify MDM and integration across on premise and cloud applications **…and support new master data challenges** ▪ Increasing velocity and growing amounts of master data, demand for a higher degree of automation ▪ Leverage new technologies, like machine learning, to enable the new promises of the digital transformation	**Support strategic trends** ▪ Enterprise-wide master data mgmt across on premise & cloud apps ▪ Seamless master data exchange across the entire business network ▪ Master data for end-to-end IoT solutions spanning devices and apps **Simplify creation, management and use of master data** ▪ One platform for all domains and use cases natively using EIM and SAP HANA ▪ Intuitive user experience and high business user efficiency ▪ Embedded analytics, stewardship, monitoring, and remediation ▪ Further automation of routine tasks like error correction or data completion

1. Potential data protection and privacy features include simplified deletion of personal data, reporting of personal data to an identified data subject, restricted access to personal data, masking of personal data, read access logging to special categories of personal data, change logging of personal data, and consent management mechanisms. 2. This is the current state of planning and may be changed by SAP at any time without notice.

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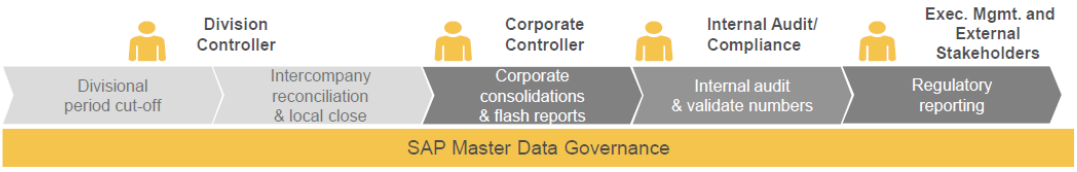
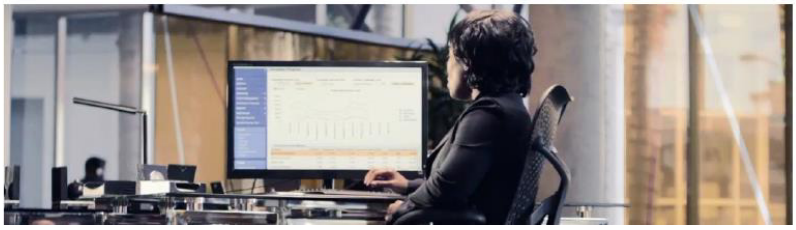
3.26 Benefits of SAP Master Data Governance for Financials

What can be improved?	How can SAP Master Data Governance help?	What benefits can be achieved?
Process efficiency • Intercompany reconciliations • Period closing • Corporate controlling • Financial consolidation	**Greater governance across local and corporate systems** • Automation of data maintenance through integrated validations, derivations and role-based transactional guidance • Reduction of manual correction work due to "First Time Right" data entry • More automation in reconciliation of charts of accounts across multiple ledgers	• Faster financial close • Increased coordination across operating units • Increased trust in reporting data • Reduced cost of finance as a percentage of revenues



What can be improved?	How can SAP Master Data Governance help?	What benefits can be achieved?
Compliance Management • Transparency • Traceability • Risk exposure	**Greater governance across local and corporate systems** • Separation of duties between change and approval authority for financial master data • Standardized workflow and automated routing • Complete history and audit trail of changes to financial master data	• Increased visibility into change impact on financial statements • Reduced audit workloads and costs • Greater collaboration in support of internal policies and government/industry regulations • Fewer non-compliance penalties
Quality and Consistency • Master data • Business rules • Governance processes	**Greater governance across local and corporate systems** • Single Source of Truth allowing for creating and storing the right data at the right place, even in distributed and changing landscapes • Standardized data models, validation rules and workflows • Easier Cross-System Data Synchronization through configurable and selective distribution logic	• Reduced effort to find information • Increased trust in data by business users • Greater consistency of financial statements and less need for restatements • Improved modeling, planning and analysis



SAP Master Data Governance (MDG) is used for embedded Master Data Management (MDM), that is, centralized, out-of-the-box, domain-specific creation, modification, and distribution of master data with a focus on SAP Business Suite.

For MDG-F, however, customers might want to run the MDG hub in the same physical client as the ERP client due to the nature of the MDG-F data model. Since all MDG-F entity types use the flexible mode their data is stored only in the MDG generated tables and not within ERP tables.

This document explains how-to setup an ALE scenario for data replication where MDG hub and ERP client share the same physical client of the same system.

We recommend that you study the following how-to guide before working with the current guide:

Extensibility Options for SAP Master Data Governance for Financial Data Overview about MDG-F

Strong ALE configuration expertise is required, too.

SAP Master Data Governance (MDG) is used for embedded Master Data Management (MDM), that is, centralized, out-of-the-box, domain-specific creation, modification, and distribution of master data with a focus on SAP Business Suite.

Domain-specific content (data models, user interfaces, workflows) is provided as part of the standard for several application areas. It is a common requirement from customers to adapt the MDG data models to their specific needs.

This document explains the SAP-delivered solution for financial master data within MDG (MDG-F). It describes the key concepts and implementation details and links to guides that provide real-life use cases of MDG-F. All software artifacts of the MDG hub that are provided or used by MDG-F belong to the **MDG_FND** software layer.

If you use either the accounting components or the controlling components of MDG-F, we recommend that the MDG hub includes the **SAP_APPL** layer. This layer owns customizing tables that the relevant accounting entity types or controlling entity types use.

If you use the consolidation components of MDG-F only, the **MDG_APPL** and **SAP_APPL** software layers are optional.

Master Data Governance for Financials (MDG-F) is an application to centrally govern the maintenance of financial master data.

•MDG-F provides central steering and control mechanisms independently of local or centralized execution.

•In MDG-F, master data records can be created, completed or changed in a staging area, following a distributed step-by-step way, before making the record available to connected business applications.

3.27 Financial Master Data[14]

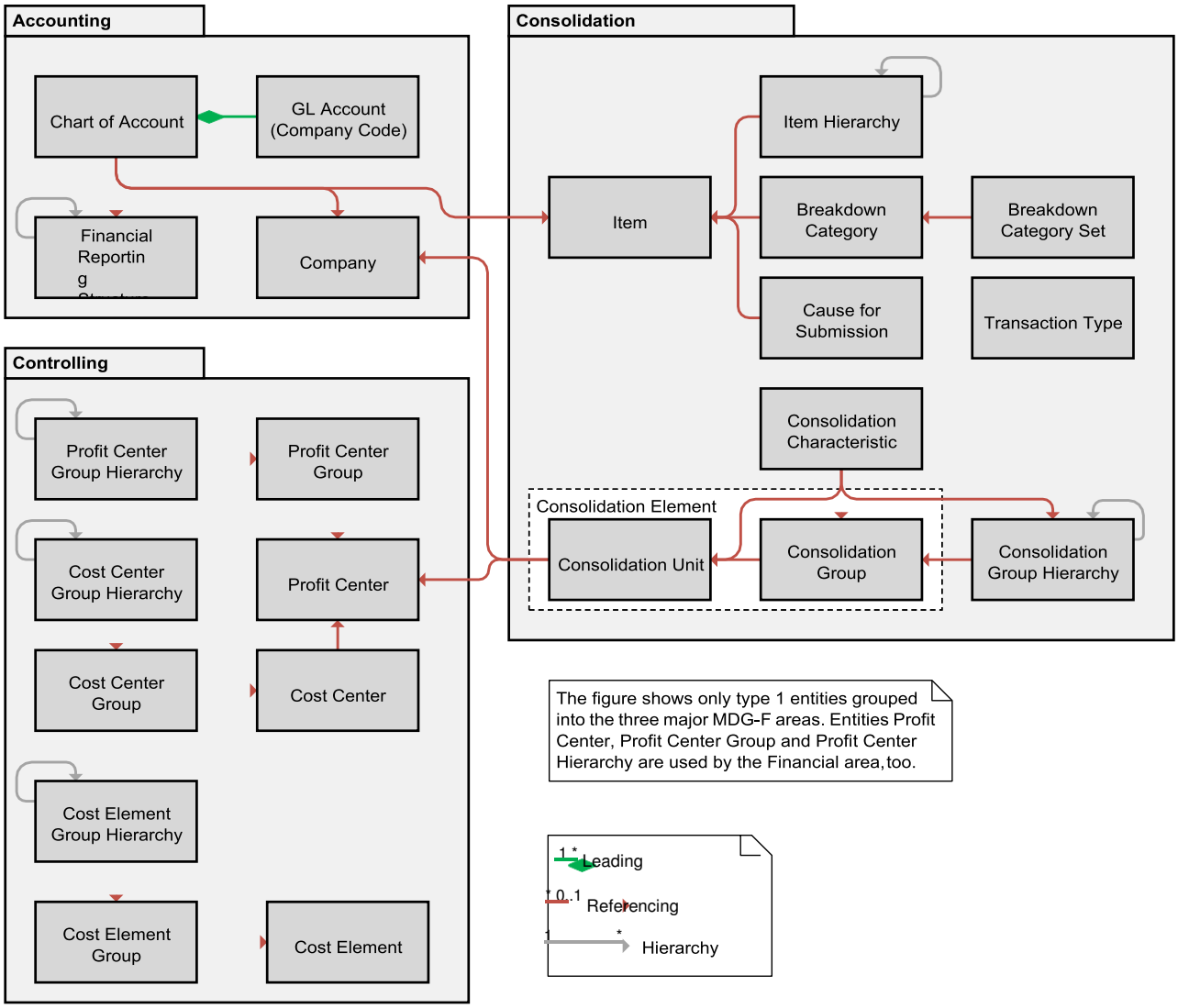
3.27.1 MDG for Financials

MDG offers a domain specific solution for financial governance (MDG-F). The current MDG-F data model is called **0G**. It covers entity types of the accounting, controlling and consolidation components of financial master data as indicated by the examples below:

- Accounting: G/L Account **(ACCOUNT & ACCCCDET)**, Company **(COMPANY)**
- Controlling: Cost Center **(CCTR**), Cost Element **(CELEM)** and Profit Center **(PCTR)**
- Consolidation: Consolidation Unit **(CONSUNIT)**, Item **(FSI)**

MDG offers a single data model combining several entity types of financial master data. The entity types can be separated roughly in three areas: accounting, controlling and consolidation. Nevertheless most of the entities are connected with each other. The figure below shows the currently supported entity types and their relations.

[14] SAP SE



For an excel file providing a detailed overview of all fields in MDG-F, see MDG Financials – (Un-) Supported Segments and Fields.

3.28 Data Model 0G

The MDG-F entity types, their attributes, and relations are shipped within the data model **0G**.

The data model uses only **flexible entity types**. Both the actual (active) data as well as the staging (inactive) data that exists only within a governance process, is stored in generated MDG tables. Although some entity types refer to existing objects of SAP systems, the active master data is not stored directly in the related backend tables. It requires a data replication to send the MDG active data into the related SAP system tables.

Some of the entity types with *SU Type* 1 **reference** other entity types with *SU Type* 1 as attributes (for example the entity type “account” uses the entity type “company” as an attribute “trading partner”). Using a reference requires that the record to be referenced already exists as an active one within MDG-F:

- Keep the validity of an object according to the chosen edition in mind. If you have created a company that is valid from the 1st of January 2015, the same company cannot be used for an account that is valid from 1st of January 2013 since the company does not exist on the 1st of January 2013.
- Keep the sequence of objects to be created in mind. You must create the company that an account uses before creating the account. It is not sufficient to create the change request for the company only. The change request must be processed completely so that the company record is activated.
- Keep the edition and change request type in mind. If you want to create dependent objects at the same time within a single change request, you have to define both the edition and change request type accordingly. They must contain both entity types.

Each of the MDG-F entity types with *SU Type* 1 uses the **MDG Edition functionality**. Editions enable the grouping of different entity types to ensure the data consistency across all entity types. Furthermore, the edition enables the time-dependent maintenance of master data. This is required for some of the related SAP ECC objects like cost or profit centers.

Using editions for the master data maintenance requires that you configure **Edition Types** to indicate which of the entity types with *SU Type* 1 can be grouped within an edition. It is recommended to use the SAP pre- defined edition type **0G_ALL**. The edition type uses all entity types with *SU Type* 1. If you want to create your own edition types, keep in mind that:

- Many entity types with *SU Type* 1 reference other entity types with *SU Type* 1.
- You cannot assign the same entity type with *SU Type* 1 to multiple edition types as long as the edition types are not released.

- If objects are not part of the same edition, you will not be able to create cross references.

Additionally **Change Request Types** are needed for master data maintenance.

SAP pre-defines both Edition Types and **Change Request Types** in release dependent BC-Sets:

Release	BC-Sets	SAP Note
MDG 7.0 SP00	**CA-MDG-APP-FIN_EDITION_CR_04**	1908432
MDG 7.0 SP02	**CA-MDG-APP-FIN_EDITION_05** **CA-MDG-APP-FIN_CR_TYPES_05**	2003344
MDG 8.0 SP00 MDG 9.0 SP00	**CA-MDG-APP-FIN_EDITION_05** **CA-MDG-APP-FIN_CR_TYPES_06**	2138778

Each MDG-F entity type with *SU Type* 1 uses **external and non-changeable keys**. The entity's key must be defined manually during the creation of the entity. It cannot be changed. With MDG 7.0 SP02 MDG-F offers more flexibility with regard to changeable keys during the creation process of an object (see the related how- to guide for more details).

Most of the MDG-F entity types with *SU Type* 1 support **attachments**. This enables adding documents and/or links to the master data. The attachments are a MDG specific functionality. They are not related to the SAP ECC document features and thus not part of data replication.

All **hierarchies** that are defined in data model **0G**use the **MDG Edition functionality**. It is possible to create the same hierarchy in different, multiple points in time. This allows a different structure, for example one for the active business year and a second one for planning the next business year. Note that each hierarchy is an independent object. Following the example above, if you change something in the hierarchy for the active business year, the same change is not applied automatically to the planning hierarchy of the next business year. You have to repeat the change in the planning hierarchy.

In **MDG 7.0** it is only possible to maintain hierarchies using the hierarchy processing user interface.

MDG 8.0 introduces the hierarchy maintenance within the single object maintenance user interfaces for accounting-related and controlling-related entity types.

MDG 9.0 does not introduce any new change request types.

3.29 Accounting

Entity types of the accounting area relate to common SAP ECC objects:

MDG Entity Type	Description
Chart of Accounts (**ACCOUNT**)	Reflects the so called "***A-Segment***" of the SAP ECC ***Account*** (tables **SKA1** and **SKAT**). It is the chart of accounts dependent data of an account.
G/L Account in Company Code (**ACCCCDET**)	Reflects the so called ***B-Segment*** of the SAP ECC ***Account*** (table **SKB1**). It is the company code dependent data of an account.
Entity types **ACCOUNT** *and* **ACCCCDET** *are usually combined within MDG-F to the term* ***G/L Account****.*	
Company (**COMPANY**)	Reflects the SAP ECC ***Company*** customizing (table **T880**).
Financial Reporting Structure (**FRS**)	Reflects the SAP ECC ***Financial Statement Version*** / ***Financial Reporting Structure*** (table **T011**).
FRS Item (**FRSI**)	Reflects the SAP ECC ***Financial Statement Version*** / ***Financial Reporting Structure Items*** (tables **FAGL_011***).
Entity types **FRS** *and* **FRSI** *are usually combined within MDG-F to the term Financial Reporting Structure. The financial reporting structure supports the creation of a hierarchy, too. You can assign accounts to this hierarchy.*	

3.30 Controlling

Entity types of the controlling area relate to common SAP ECC objects:

MDG Entity Type	Description
Cost Center (**CCTR**)	Reflects the SAP ECC ***Cost Center*** (tables **CSKS**and **CSKT**).
Cost Center Group (**CCTRG**) ***Cost Center Group Hierarchy*** (**CCTRH**)	Reflect the SAP ECC ***Cost Center Group Hierarchy***. In MDG-F the entity ***Cost Center Group Hierarchy*** defines the root node of the hierarchy. Both entities are required to build the full hierarchy. Cost Centers are added as leafs.
Cost Element (**CELEM**)	Reflects the SAP ECC ***Cost Element*** (tables **CSKA**, **CSKB**, and **CSKU**). In MDG-F a ***Cost Element*** does not differentiate between common and controlling area dependent data as done in SAP ECC.
Cost Element Group (**CELEMG**) ***Cost Element Group Hierarchy*** (**CELEMH**)	Reflect the ***Cost Element Group Hierarchy***. In MDG-F the entity ***Cost Element Group Hierarchy*** defines the root node of the hierarchy. Both entities are required to build the full hierarchy. Cost Elements are added as leafs.
Profit Center (**PCTR**)	Reflects the SAP ECC ***Profit Center*** (tables **CEPC**, **CEPC_BUKRS**and **CEPCT**).
Profit Center Group (**PCTRG**) ***Profit Center Group Hierarchy*** (**PCTRH**)	Reflect the ***Profit Center Group Hierarchy***. In MDG-F the entity ***Profit Center Group Hierarchy*** defines the root node of the hierarchy. Both entities are required to build the full hierarchy. ***Profit Centers*** are added as leafs.

3.31 Consolidation

Entity types of the consolidation area relate to the SAP solution for Business Consolidation Services of Strategic Enterprise Management (SAP SEM BCS):

MDG Entity Type	Description
Consolidation Characteristic (**CONSCHAR**)	Defines the basis for consolidation groups and consolidation elements. The characteristic's scope either relates to a company or a profit center.
Consolidation Unit (**CONSUNIT**)	Defines the consolidation unit of a consolidation element. Depending on the used consolidation characteristics, either company or profit center specific data can be maintained.
Consolidation Group (**CONSGRP**)	Defines the consolidation group of a consolidation element.
Entity types **CONSUNIT***and* **CONSGRP***are usually combined within SAP SEM BCS Consolidation Element. In MDG-F both are treated as separated entity types.*	
Consolidation Group Hierarchy (**CONSGRPH**)	Defines the root node of the consolidation group hierarchy. Consolidation groups and units can be added to the hierarchy.
Item (**FSI**)	Defines the item. MDG-F uses Items for the maintenance of Group Accounts, too. Therefore entity type **ACCOUNT**has a cross reference to **FSI**.
Item Hierarchy (**FSIH**& **FSIT**)	Reflect the item hierarchy resp. consolidation reporting structure of SAP SEM BCS. Entity type **FSIH** defines the root node of the hierarchy. Entity type **FSIT**defines text items usable as sub-nodes of the hierarchy. Item can be assigned as leafs.
Breakdown Category (**BDC**)	Defines the breakdown category. Each category uses sets. A category is usually assigned to an item.
Breakdown Category Set (**BDCSET**)	Defines the breakdown category sets.
Cause for Submission (**SUBMPACK**)	The cause for submission does not exist in SAP SEM BCS. The entity type serves as an example how-to extend MDG-F with additional information.

Transaction Type (**TRANSTYPE**)	Defines the transaction type.

All entity types are created according to the **SOFEX** example model that is shipped as a template model within SAP SEM BCS. Any changes performed in the SAP SEM BCS model have to be applied to MDG-F, too.

3.32 User Interface

The user interfaces for MDG-F are based on ABAP Web Dynpro. They are built with the floor plan manager (FPM) using its specific Business Object Layer (BOL) / Generic Integration Layer (genIL) technology. Some advantages to be mentioned are:

- Loose coupling of the user interfaces to the MDG-specific processes.
- High flexibility for the creation of the user interfaces. The huge amount of fields to be displayed is split into small User Interface Building Blocks (UIBBs). UIBBs support lists, forms and special kinds like pop-ups, search input and search results.
- Possibility to create object-specific user interfaces to create a common look and feel and/or a similarity of the MDG user interfaces compared to the SAPGUI maintenance transactions.
- Reuse of the MDG-F generated tables, structures and fields (including naming) during the user interface creation.

3.33 Generic Interaction Layer (genIL)

The generic interaction layer (genIL) is required for the MDG-F user interfaces. It consists of a genIL model and one or more genIL implementation classes for the specific model. Transaction **GENIL_MODEL_BROWSER** can be called in the SAP backend to view the model. A genIL model basically consists of objects and relations.

- Objects consist of attributes. Each attribute reflects a usable field for the user interface.
- Relations connect one object to another. They define the cardinality of objects in a relation, too. Relations are reflected in the user interface by the wires (connections) from one UIBB to another. It is mandatory that the UIBB hierarchy in the overview page is consistent to the genIL object hierarchy as defined by the relations.

SAP provides the following genIL models for MDG-F:

- **MDG 7.0**: Model **MDGF** and its implementation class **CL_MDGF_GENIL_ADAPTER.**
- **MDG 8.0**: Model **MDGFHP** and its implementation class **CL_MDGF_GENIL_ADAPTER_HRY**.

Both models are dynamic genIL models. It is strictly forbidden to change the models manually. The models are generated by their implementation classes according to the given run-time information of the MDG-F data model **0G**.

Model **MDGF** is used by the single object maintenance user interface for the common entity types and their attributes.

- Each entity type with *SU Type* 1 of the data model is transferred to a genIL root object.
- Each entity type with *SU Type* 4 of the data model is transferred to a genIL dependent object.

- Relations (for example, between entity types with *SU Type* 1 and other entity types with *SU Type* 1; or between entity types with *SU Type* 1 and entity types with *SU Type* 4) are determined and transferred into genIL relations.
- Each entity type with *SU Type* 1 retrieves additional genIL query and query result objects to support the search.
- If an entity type with *SU Type* 1 supports multi-lingual texts, a dependent object is created in genIL to enable the text maintenance within a table.
- If an entity type with *SU Type* 1 supports attachments, two dependent objects are created in genIL to enable the attachment maintenance within a table and related pop-ups.
- The generated structures belonging to an entity are used for the genIL key and attribute structures. This ensures that all fields of the MDG data model **0G** are available for the creation of the related user interfaces. Attribute structures are used by FPM to build the field catalog being available during the user interface creation

Model **MDGFHP** is used by the single object maintenance user interface for the entity types that are used within hierarchies.

- Each entity type with *SU Type* 1 that is used by hierarchies as a root node is transferred into a genIL root object. Those objects are used to build the hierarchy tree in the user interface.
- Each entity type with *SU Type* 1 that defines a hierarchy is transferred into a genIL dependent object. Those objects are used to build the hierarchy tree in the user interface.
- Each entity type with *SU Type* 1 that is used by hierarchies as a root node, a grouping node, or leaf node is transferred into a genIL access object. Those objects are used to build the hierarchy assignment lists in the user interface.
- Each entity type with *SU Type* 1 that is used by hierarchies as a root node, a grouping node, or a leaf node retrieves an additional genIL query objects to support the search. The search result is displayed as hierarchy tree.
- If a hierarchy defines hierarchy attributes for a specific entity type with *SU Type* 1, a corresponding genIL dependent object is created additionally.

3.34 Caution

Enhancements of the MDG-F data model 0G are reflected immediately after activation of the data model in the genIL components. A manual change or enhancement of the genIL components is strictly forbidden.

If enhancements in genIL are required, all changes have to be implemented in a related genIL implementation class. It is mandatory that this class inherits from the SAP classes **CL_MDGF_GENIL_ADAPTER**respectively **CL_MDGF_GENIL_ADAPTER_HRY**.

3.35 Context Based Adaptations

A context based adaptation (CBA) is an FPM concept that allows changing the user interface in a flexible way based upon given values (for example application parameters, user input, and others). A CBA consists of an **Adaptation Schema** that consists of one or more **Adaptation Dimensions.**

- The pre-defined adaptation schema for MDG-F is **MDG_FIN**.
- The adaptation schema includes the following adaptation dimensions:
  - **USMD_OTC**: usable for adaptations according to the current business object type code.
  - **ACTION**: usable for adaptations according to the current logical action like “create” or “mark for deletion”.
  - **CRTYPE**: usable for adaptations according to the current change request type.
  - **WFSTEP**: usable for adaptations according to the current workflow step.

Using both the adaptation schema and its dimensions, you can create various adaptations of the user interface (for example changing the layout of an overview page (OVP), or either adding rows to or removing rows from a list UIBB, and so on). You can combine several dimensions to create a very specific adaptation.

The CBA concept is based on the common FPM event handling. You can trigger one or more CBA events that are handled by FPM's event loop processing. Unfortunately FPM handles multiple CBA events one after the other. It does not cumulate the dimension information given within each event (which basically means that the final CBA event wins).

3.36 Work Centers, Roles and Landing Pages

Work centers and roles are used within MDG for the creation of so called landing pages. A landing page is considered as a single point of entry for an end user to the MDG-F specific user interface.

Work centers define a grouping of the different MDG-F entity types into a single menu role. Additionally each role uses the common MDG components such as data replication, change request, editions, and so on.

Three work center menu roles are pre-defined according to the MDG-F data model setup:

- **SAP_MDGF_ACC_MENU_04**: reflects all accounting entity types.
- **SAP_MDGF_CTR_MENU_04**: reflects all controlling entity types.
- **SAP_MDGF_CO_MENU_04**: reflects all consolidation entity types.

Next to the menu roles as mentioned above, authorization roles are pre-defined, too. The authorization roles follow the general role setup of all MDG based applications:

- The **Display Role** contains all authorizations to display master data, change requests, change documents, replication status, and other information for the respective work center data.
  The role is assigned to an auditor, who is then authorized to display all information, but not to change anything.

- The **Requester Role** contains all authorizations to create and display change requests within the respective work center. The role is assigned to a business user, who is then able to request changes for master data.
- The **Specialist Role** contains all authorizations to process and approve change requests in the respective work center.
- The **Steward Role** contains all authorizations needed for special tasks that involve the processing and replication of master data within the respective work center. The role is assigned to the user responsible for the quality of the master data. It is recommended to combine this role with the specialist.

The combination of the work center and the explained general authorization roles results in the following pre- defined roles:

Release	Role	Description
MDG 7.0	SAP_MDGF_ACC_DISP_04	MDG-F: Accounting Display
	SAP_MDGF_ACC_REQ_04	MDG-F: Accounting Requester
	SAP_MDGF_ACC_SPEC_04	MDG-F: Accounting Specialist
	SAP_MDGF_ACC_STEW_04	MDG-F: Accounting Data Steward
	SAP_MDGF_CTR_DISP_04	MDG-F: Controlling Display
	SAP_MDGF_CTR_REQ_04	MDG-F: Controlling Requester
	SAP_MDGF_CTR_SPEC_04	MDG-F: Controlling Specialist
	SAP_MDGF_CTR_STEW_04	MDG-F: Controlling Data Steward
	SAP_MDGF_CO_DISP_	MDG-F: Consolidation Display
		MDG-F: Consolidation Requester
		MDG-F: Consolidation Specialist
		MDG-F: Consolidation Data Steward

	04 SAP_MDGF_CO_REQ_ 04 SAP_MDGF_CO_SPEC _04 SAP_MDGF_CO_STEW_0 4	
MDG 8.0[1]	SAP_MDGF_ACC_REQ_0 6 SAP_MDGF_CO_REQ_06 SAP_MDGF_CTR_REQ_06	MDG-F: Accounting Requester MDG-F: Consolidation Requester MDG-F: Controlling Requester

Each pre-defined role must be considered as a template. The roles must not be assigned directly to user. An SAP template role must be copied and adjusted according to the customer project requirements.

3.37 MDG-F User Interfaces

All software artifacts related to the new MDG-F user interfaces for single object maintenance are stored in package **USMDZ10**.

A complete user interface consists of the following elements:

- The application configuration that defines the general settings of a user interface including application parameters.
- The communicator settings configuration that defines the search of each overview page. The communicator settings configuration must have the same name as the application configuration.
- The Overview Page (OVP) that combines one or more form UIBBs and/or list UIBBs for the actual data maintenance. Forms, lists and their related feeder classes can be identified by their naming.
  - Web Dynpro Component **MDGF_0G_CCTR** belongs to the cost center.
  - Feeder class **CL_MDGF_GUIBB_FI_ACCOUNT** belongs to the account.

The new user interface uses the business object type code (OTC) as a key parameter. The OTC links the data model and the entity type with the user interface application. It is mandatory that each user interface uses an OTC. Pre-defined combinations are listed in the next chapters.

3.37.1 Accounting

Application / OVP / OTC	Entity Type(s)	Description
MDGF_0G_OVP_FI_ACCOUNT **MDGF_0G_FI_ACCOUNT_OVP 892**	**ACCOUNT** **ACCCCDET**	The OVP combines two entity types with SU Type 1 for account master data maintenance in a single user interface. Accounts can be assigned to financial reporting structure hierarchies in collective processing.
MDGF_0G_OVP_COMPANY **MDGF_0G_COMPANY_OVP 154**	**COMPANY**	The OVP represents the company master data.

MDGF_0G_OVP_FI_REPORT **MDGF_0G_FI_REPORT_OVP 901**	**FRS** **FRSI**	The OVP combines two entity types with SU Type 1 for financial reporting structure maintenance in a single user interface. The user interface creates single entities that can be used to define the complete financial reporting structure hierarchy in collective processing.

3.37.2 Controlling

Application / OVP / OTC	Entity Type(s)	Description
MDGF_0G_OVP_CCTR **MDGF_0G_CCTR_** **OVP 158**	**CCTR**	The OVP represents the cost center master data. The assignment of the cost center to its (standard) hierarchy must be done in collective processing.
MDGF_0G_OVP_CCTRG **MDGF_0G_CCTRG** **_OVP 895**	**CCTRG**	The OVP represents cost center groups. The assignment of the cost center groups to hierarchy nodes must be done in collective processing.
MDGF_0G_OVP_CCTRH **MDGF_0G_CCTRH** **_OVP 897**	**CCTRH**	The OVP represents the root node of a cost center group hierarchy. Hierarchy maintenance must be done in collective processing.
MDGF_0G_OVP_CELEM **MDGF_0G_CELEM** **_OVP 983**	**CELEM**	The OVP represents the cost element master data. The assignment of the cost center to its hierarchy must be done in collective processing.
MDGF_0G_OVP_CELEMG **MDGF_0G_CELEM** **G_OVP 984**	**CELEMG**	The OVP represents cost element groups. The assignment of the cost element groups to hierarchy nodes must be done in collective processing.
MDGF_0G_OVP_CELEMH **MDGF_0G_CELEM** **H_OVP 985**	**CELEMH**	The OVP represents the root node of a cost element group hierarchy. Hierarchy maintenance must be done in collective processing.
MDGF_0G_OVP_PCTR **MDGF_0G_PCTR_** **OVP 229**	**PCTR**	The OVP represents the profit center master data. The assignment of the profit center to its (standard) hierarchy must be done in collective processing.
MDGF_0G_OVP_PCTRG **MDGF_0G_PCTRG** **_OVP 896**	**PCTRG**	The OVP represents profit center groups. The assignment of the profit center groups to hierarchy nodes must be done in collective

processing.

Application / OVP / OTC	Entity Type(s)	Description
MDGF_0G_OVP_PCTRH **MDGF_0G_PCTRH_** **OVP 898**	**PCTRH**	The OVP represents the root node of a profit center group hierarchy. Hierarchy maintenance must be done in collective processing.

3.37.3 Consolidation

Application / OVP / OTC	Entity Type(s)	Description
MDGF_0G_OVP_BDC **MDGF_0G_BDC_OVP** **MDGF_BCG**	**BDC**	The OVP represents the breakdown category master data. Breakdown categories include breakdown category sets. They can be used to refine items.
MDGF_0G_OVP_BDCSET **MDGF_0G_BDCSET_O** **VP MDGF_BCS**	**BDCSET**	The OVP represents breakdown category sets.
MDGF_0G_OVP_CONSCH AR **MDGF_0G_CONSCHAR_OV** **P MDGF_CCC**	**CONSCHA R**	The OVP represents the consolidation characteristic master data.
MDGF_0G_OVP_CONSGR P **MDGF_0G_CONSGRP** **_OVP 904**	**CONSGRP**	The OVP represents the consolidation group. The assignment of the consolidation groups to hierarchy nodes must be done in collective processing.
MDGF_0G_OVP_CONSGR P **MDGF_0G_CONSGRP** **_OVP 894**	**CONSGRP H**	The OVP represents the root node of a consolidation group hierarchy. Hierarchy maintenance must be done in collective processing.
MDGF_0G_OVP_CONSUNI T **MDGF_0G_CONSUNIT** **_OVP 905**	**CONSUNIT**	The OVP represents the consolidation unit master data. Consolidation units can be assigned to consolidation group hierarchies in collective processing.

MDGF_0G_OVP_CO_ACCOUNT **MDGF_0G_CO_ACCOUNT_OVP 900**	**FSI**	The OVP represent the item. Items can be assigned to item hierarchies in collective processing.

Application / OVP / OTC	Entity Type(s)	Description
MDGF_0G_OVP_CO_REPORT **MDGF_0G_CO_REPORT_OVP 902**	**FSIH** **FSIT**	The OVP combines two entity types with SU Type 1 for item hierarchy maintenance in a single user interface. The user interface creates single entities that can be used to define the complete item hierarchy in collective processing.
MDGF_0G_OVP_SUBMPACK **MDGF_0G_SUBMPACK_OVP MDGF_SMP**	**SUBMPACK**	The OVP represents the cause for submission. A cause for submission is used to refine items.
MDGF_0G_OVP_TRANSTYPE **MDGF_0G_TRANSTYPE_OVP MDGF_TTP**	**TRANSTYPE**	The OVP represents the transaction types.

3.37.4 User Interfaces for Collective Processing

The previous chapters Accounting, Controlling and Consolidation listed user interfaces for single object maintenance.

Collective processing summarizes the user interfaces for both mass change and for hierarchy maintenance. The related user interfaces are not refactored. They are accessible through links in the related work centers.

3.37.5 Search and Value Helps for Single Object Maintenance User Interfaces

MDG-F uses the **generic text helper** functionality provided by the MDG framework with class **CL_USMD_GENERIC_GENIL_TEXT**. The text helper is able to provide values of domains and check tables that have only a single key field. This functionality is combined with the **OVS search** technology of the user interface:

- Fields having domain fixed values are usually displayed as drop-down list boxes in the user interface. The values of the drop-down list box are determined by the text helper.
- Fields having simple check tables are usually displayed as input fields including a related read-only field for the description. MDG-F uses the generic OVS search implementation

provided by FPM (see class **CL_MDGF_GUIBB_FORM**). The text helper is called indirectly by the MDG-F specific genIL implementation.

- Fields having complex check tables or depending on other values are usually displayed as input fields including a related read-only field for the description. MDG-F implements specific OVS search helps in the related feeder classes since this cannot be done with a generic approach.

There are two options for filtering search help results:

- The text helper includes the enhancement spot **USMD_GENERIC_GENIL_TEXT** respectively BAdI **USMD_GENERIC_GENIL_TEXT_BADI**. They are usable to restrict the values being shown on the user interface for domains or simple check tables.
- The generic form feeder class **CL_MDGF_GUIBB_FORM**provides a rather generic OVS filter method called **OVS_OUTPUT_FILTER**. You can enhance this logic (for example, using an implicit enhancement at the end of the method) with your custom filters.

3.38 Data Transfer

Data Transfer in MDG-F covers both the initial load of master data into and MDG system and the replication of created or modified data to client systems.

The initial load is used for preparing the MDG system with the currently existing data of client systems. The initial load can be accomplished using a file-based approach or using the MDM Generic Extractor. When setting up the system it is important to load all objects that shall be put under governance completely. For time-dependent objects such as Profit Centers or Cost Centers, it is ensured that all existing time slices of a single object are loaded to MDG.

Data replication is possible using either SOA Services or ALE IDocs. Combining both for the same receiving system is not recommended. IDocs are only available for accounting and controlling entity types.

Consolidation entity types do not exist within SAP ECC systems and thus do not offer IDocs.

SAP Note (Un-) Supported Segments & Fields offers a detailed overview about all fields covered by MDG-F and how they are used in the different replication technologies.

Data replication supports different modes that are explained in the next chapters.

3.39 Object-based Replication

This is a new feature provided by MDG 7.0. It enables the immediate replication of an existing object, for example by selecting the same in the search result list and choosing the **Replicate** button. It is possible to select one or more target systems for the current replication run. The edition cannot be selected. The outbound implementation always uses the currently valid data for non-time-dependent objects respectively all valid time slice for time-dependent objects. It is not possible to replicate outdated objects.

3.40 Change-Request based Replication

The change-request based replication has not been changed. It is triggered once the change request has been activated. If the changed object is non-time-dependent and the change happened for a time slice that is not valid (for example, in an outdated or future edition), the object is not replicated. This prevents overwriting non-time-dependent objects with invalid data in the target systems.

3.41 Edition-based Replication

The edition-based replication has not been changed. It still requires a released edition to replicate all objects of the same edition. For non-time-dependent objects the same rules as defined by the change-request based replication apply. The system prevents the replication of invalid time slices.

3.42 Replication of Deletions

MDG-F supports the replication of deletion only using SOA messages. It is not possible to use ALE IDocs. A “Deletion” means both the actual deletion of a complete object and the deletion of a single time slice to set

the “valid to date” to a specific value.

3.43 Replication of Accounts

MDG-F provides two entity types with **SUT1**for the account.

- The entity type **ACCOUNT**represents the so-called A-Segment of an account. It relates to the SAP ECC table **SKA1***Accounts*.
- The entity type **ACCCCDET** represents the so-called B-Segment of an account. It relates to the SAP ECC table **SKB1***Accounts in Company Code*.

The predefined replication models for the account reflect this setup.

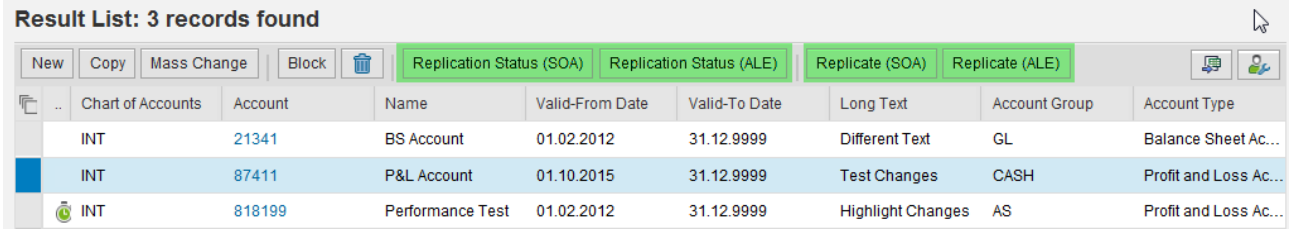
The IDoc based replication uses the common **GLMAST**IDoc type that combines both the A and B Segments. The replication is triggered using the business object type for the general ledger accounts master (892). It is possible to replicate A segments without B segments. If the object list to be replicated contains only B segments, related A segments are added automatically.

The SOA service-based replication is more complex since there are two separate service messages used for the replication.

- Service **ChartOfAccountsReplicationRequest_V1**contains only the A segment without any attributes of the B segment. The replication is triggered using the object type code for the financial accounting chart of accounts (899).
- Service **GeneralLedgerAccountMasterReplicationBulkRequest** contains additional attributes of the A Segment and the complete list of attributes of the B segment. The replication is triggered using the object type code for the general ledger accounts master (892).

The service definition requires that any replication of A segments has to send both of the abovementioned services to ensure data consistency in the receiving system. This is implemented by the corresponding outbound implementations. B segment replication is possible without the chart of accounts service.

A side-effect of the different outbound implementations that are required to ensure data consistency is the design of the search result list for accounts as shown in the screenshot below.



The button **Replication Status (SOA)** shows the status for any replication of accounts using the chart of accounts replication request. The replication itself is triggered with the button **Replicate (SOA)**.

The button **Replication Status (ALE)** shows both the status for any IDoc-based replication of accounts and the status for the SOA-based replication using the general ledger account master service. The replication itself is triggered with the button **Replicate (ALE)**. You choose this button for IDoc based replication only.

3.44 Replication Monitoring

Data replication offers several monitors to check the current replication status.

- The search result list of MDG-F user interfaces offers a link to the DRF replication status. The button
  Replication Status is active for all entity types that support data replication in general.
- ALE IDoc replication offers a specific monitor. You can use backend transaction **WE02** to display both outbound IDocs (on the MDG hub) as well as inbound IDocs (on the MDG client). You can use the monitor for detailed troubleshooting. Post-processing of IDocs is possible using transaction **BD87**.
- SOA Service replication offers a specific monitor. You can use backend transaction **SXMB_MONI** (available until SAP Basis 7.32) to display outbound services (on the MDG hub). You can use backend transaction and **SRT_MONI** (available in SAP Basis 7.40 and newer) to display inbound services (on the MDG client). You can use the monitor for detailed troubleshooting. In the forward error handling, you can post-process services, using backend transaction /**SAPPO**/**PPO2**.

3.45 Validations and Data Derivation

As mentioned before, the data model **0G**uses only **flexible entity types**. This means that there is no access class available that could be used either for validations or for data derivation, or for both. Nevertheless validations and data derivation are needed by the MDG-F entities, too.

MDG-F implements the common MDG rule service BAdI **USMD_RULE_SERVICE**for both validations and derivations. All software artifacts are stored in package **USMDZ7**.

You can enhance the existing validations with custom checks, too. The recommended approach is a custom class that inherits from the pre-defined SAP class to ensure, that all SAP pre-defined validations are executed. SAP Note 2337685 - MDG-F: Standard Validations of Data Model 0G provides a detailed overview of all pre-defined validations in MDG-F.

MDG supports different types of derivation:

- BRF+ based derivation
- Single entity derivation of the rule service BAdI **USMD_RULE_SERVICE**
- Cross entity derivation of the rule service BADI **USMD_RULE_SERVICE_CROSS_ET**

The default implementations provided by SAP use only the cross entity derivation. There is a specific how to guide “Cross Entity Derivation in MDG-F” available that explains all details.

3.46 Data Modeling

3.46.1 Q: Why is there an active data model 0F in the MDG hub?

Data model 0F has been introduced with the very first shipment of MDG-F for SAP ECC 6.0 EhP4. As of today, the data model is outdated and must not be used anymore. Delete the data model in your system.

3.46.2 Q: Is it recommended to copy data model 0G?

It is **not** recommended to create a copy of the data model

0G. The existing data model can be easily enhanced in

various ways:

- Adding new fields to an existing entity type.
- Adding new tables as type 4 entity to an existing entity type.
- Adding new entity types with *SU Type* 1 to the data model.

The generic implementation of the genIL layer simplifies the integration of the enhancements in the user interface.

3.46.3 Q: How do I add custom Data Dictionary search helps to attributes of data model 0G?

The new user interfaces for MDG-F consider the following data dictionary-based search helps:

1. Search helps that are defined within the related data element or domain.
2. Search helps that are defined within the UI configuration, for example using a re-definition of the feeder class method GET_DEFINITION. The UI configuration overrules a definition on data element or domain level.

The new user interfaces **ignore** all data dictionary-based search help that are maintained in the MDG data model customizing views. It is recommended to use the UI configuration instead.

3.47 User Interface Design

3.47.1 Q: Is it recommended to copy existing user interfaces?

It is **not** recommended to copy existing user interfaces.

The existing user interfaces can be easily enhanced in various ways without the need of a copy or a modification of the pre-defined user interfaces:

- You can customize the existing user interface by adding fields or removing fields.
- You can adapt the user interface adapted dynamically using context-based adaptations.
- You can enhance the user interface with custom UIBBs using the enhancement functionality of FPM.
- How-to guide "Extending the Data Model by New Fields in MDG-F" provides a real-life example how the user interface can be enhanced without copying.

A copy might destroy the binding to the improvements and corrections made by SAP.

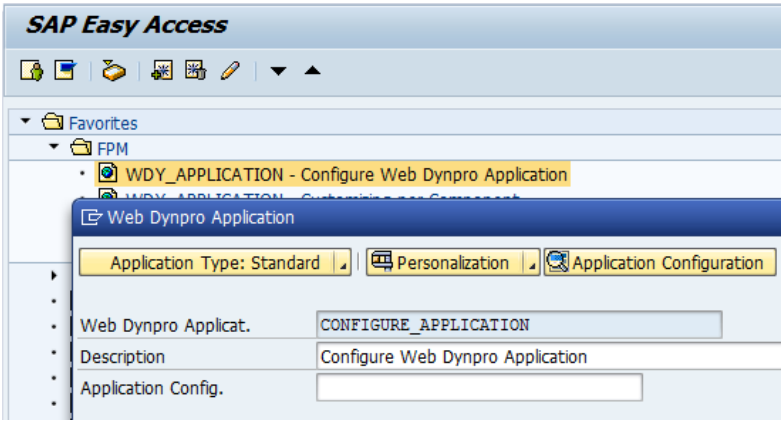
If, in spite of the recommendation, you copy the user interface anyway, ensure that you copy it completely. Remember that each user interface consists of an application configuration, a UI configuration and communicator settings.

3.47.2 Q: How can I start the user Web Dynpro application configurator for user interfaces?

You use the Web Dynpro application configurator to define your Web Dynpro based applications. It defines the main components of the user interfaces (for example the overview pages to be used) as well as application specific parameters.

The most efficient way to start the Web Dynpro application configurator for user interfaces is using a favorite in the SAP Menu.

1. Logon to your MDG Hub system.
2. Create a new entry in you SAP Menu Favorite List.
3. Choose *Web Dynpro Application*.
4. Define the *Web Dynpro Application* as **CONFIGURE_APPLICATION**.
5. Define any *Description*.

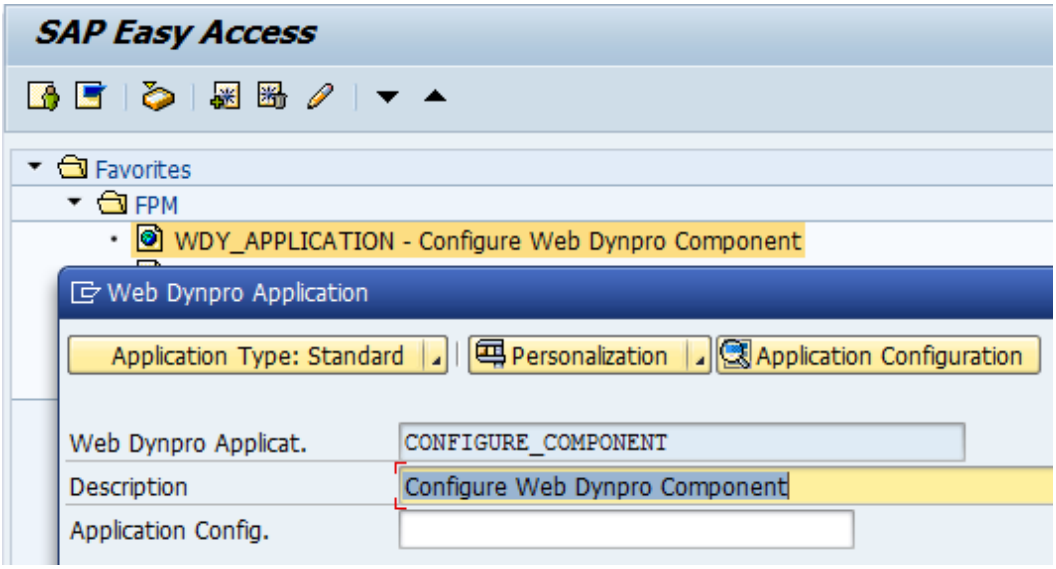

SAP Easy Access
Favorites
FPM
WDY_APPLICATION - Configure Web Dynpro Application
Web Dynpro Application
Application Type: Standard
Personalization
Application Configuration
Web Dynpro Applicat.
CONFIGURE_APPLICATION
Description
Configure Web Dynpro Application
Application Config.

3.47.3 Q: How can I start the user Web Dynpro component configurator for user interfaces?

You use the Web Dynpro component configurator to create your own Web Dynpro components, for example for MDG Communicator settings.

The most efficient way to start the Web Dynpro component configurator for user interfaces is using a favorite in the SAP Menu.

1. Favorite in the SAP Menu
    a. Logon to your MDG Hub system.
    b. Create a new entry in you SAP Menu Favorite List.
    c. Choose *Web Dynpro Application*.
    d. Define the *Web Dynpro Application* as **CONFIGURE_COMPONENT**.
    e. Define any *Description*.

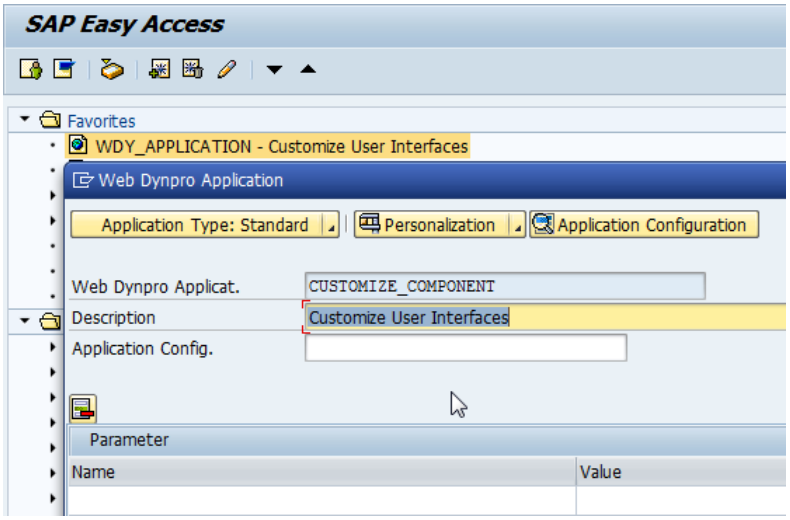


3.47.4 Q: How can I start the user Web Dynpro customizing configurator for user interfaces?

You use the Web Dynpro component customizing configurator to customize SAP defined user interface components like complete overview pages or single forms or lists.

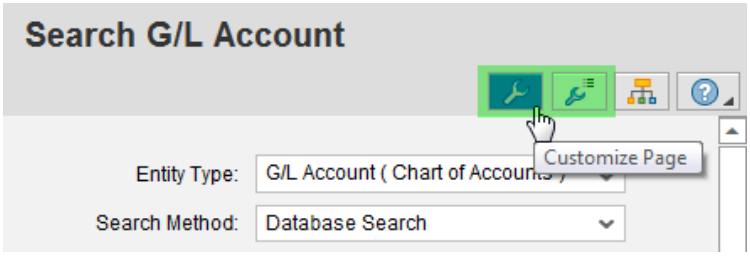
There are different ways to start the Web Dynpro component customizing configurator for user interfaces:

1. Favorite in the SAP Menu
    a. Logon to your MDG Hub system.
    b. Create a new entry in you SAP Menu Favorite List.
    c. Choose *Web Dynpro Application*.
    d. Define the *Web Dynpro Application* as **CUSTOMIZE_COMPONENT**.
    e. Define any *Description*.



2. Configure the own user as FPM Configuration Expert
    a. Logon to your MDG Hub system.
    b. In the top-menu bar choose *System* □ *User Profile* □ *Own Data*.
    c. Switch to tab *Parameters*.
    d. Enter *Parameter ID* **FPM_CONFIG_EXPERT**and set its value to **X**.
    e. Save the changes.

If you now start any user interface that is based upon FPM, you'll notice two new buttons in the upper right area of the user interface. You can use both buttons to start the customizing configurator. The screenshot below uses the search for account as

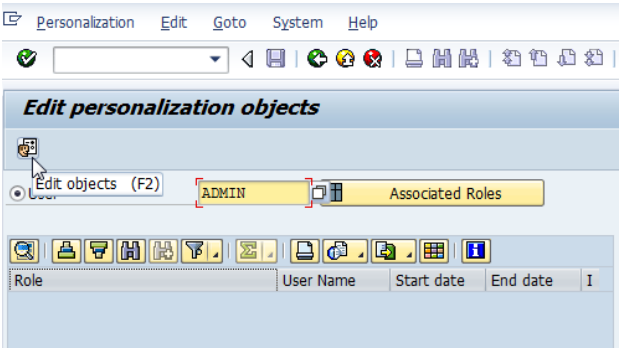


example.

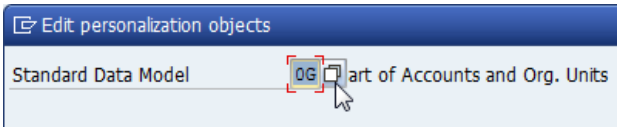
3.47.5 Q: Why is there a short dump when I try to customize a user interface?

Before you can create or modify user interface customizing, you must define the **Standard Data Model** in your user personalization. The model must be set according to the user interface that you want to customize. Ensure that the value is set to **0G**for MDG-F related user interfaces:

1. Start transaction **SPERS_MAINT**.
2. Enter your user name and choose *Edit objects (F2)* above your user name.



3. Use the search with search term *MDM* to locate the entry *SAP Master Data Governance – **R_FMDM_MODEL***in the table of personalization objects.
4. Double-click on the entry.
5. In the pop-up *Edit personalization objects* set the *Standard Data Model* to *0G*.



6. Save your changes.

3.47.6 Q: What does the error message “Feeder class does not provide a valid field catalog” mean?

This error message might be displayed if you try to open the web dynpro component configuration for any UIBB. The root cause is the same as for question “Why is there a short dump when I try to customize a user interface?”. The SAP New Weaver Basis release 7.40 displays the error message instead of causing a short dump. To resolve the issue you need to define the **Standard Data Model** as **0G** as described above.

3.47.7 Q: Why is there still a dump / error message although I have defined the Standard Data Model as 0G?

Technically the assignment of the standard data model to your user is stored in data base table **USMD2000**. You could either check by yourself or ask your system administrator to check, if there is a valid entry for your user in the table.

In systems that have been upgraded from an older MDG-F release to MDG-F 7.0 (or newer), it might be possible that:

a) Your user is still assigned to the old MDG-F roles.
b) The old MDG-F roles define a standard data model via customizing table **USMD2001**.

If this is the case, the standard data model being displayed by transaction **SPERS_MAINT** is determined according to your roles. This does not work for the user interface configuration of MDG-F 7.0. The invalid customizing settings (especially, the role assignment for your user to outdated MDG-F roles) have to be deleted. Afterwards you can define the standard data model.

3.47.8 Q: Why does the context based adaptation (CBA) not change the layout of the overview page (OVP)?

You can only use CBAs to change the layout of an OVP during the startup of the application. It is not possible to change the OVP (for example, the sequence of UIBBs) using a CBA during UI round-trips. CBAs can only change the layout of single UIBBs for each round-trip.

3.47.9 Q: Why are recent changes applied to the user interface not visible for other users?

The actual UI that is being displayed to a user in the web browser is determined from various components of the UI configuration:

- Personalization
- Enhancements
- Context Based Adaptations
- Base Configuration

The general rule is that the personalization is the strongest component. This is best explained with an example:

The base configuration defines the overview page as a list of UIBBs. Since a user does not want to scroll, he or she creates a personalization of the page introducing a stacking of the UIBBs in tab-strips. A UI designer decides to create a context-based adaptation that sets a single UIBB to “hidden and excluded from event loop”. All users not having a personalization no longer see this UIBB anymore. The user with the personalization set is unaffected by this change. This is because the UIBBs that are hidden and excluded from event loop still belong to the OVP. They can be added to the OVP using personalization. Since the user has created a personalization that shows the UIBB (the personalization was created before the CBA), the UIBB is still visible. To exclude the UIBB, you must either reset personalization or delete the UIBB in the CBA.

System administrators might use the ABAP Web Dynpro application **WD_ANALYZE_CONFIG_USER** to check and/or reset user personalization centrally.

3.47.10 Q: Create a new User Interface for custom Entity with Storage and Use Type 1?

If you have enhanced the standard data model 0G with a custom entity type with SUT 1, you need to create a new user interface, too. Follow the steps mentioned in the SAP Help for Creating User Interfaces for Single Object Processing. Of course you can use any of the existing MDG-F user interfaces as template.

3.48 Data Maintenance

3.48.1 Q: What does the error message “Relation <ABC> is not supported by Object <XYZ> mean?”

This error message is thrown by the genIL layer if the user interfaces requests a relation that is unknown in genIL.

This could be caused by an invalid Wire Configuration between UIBBs of the Overview Page. You should first check if both the genIL model and the OVP wiring are consistent.

In productive customer systems it could be the case that the shared memory component used by genIL is outdated or corrupt (genIL uses shared memory only in productive systems but not in development or test system). You can use transaction **SHMM** to invalidate the shared memory area **CL_CRM_GENIL_MODEL_SHM_AREA**. Afterwards simply restart the user interface.

3.48.2 Q: Why does the Undo button not work for key fields on the Overview Page?

All pre-defined MDG-F entities with **SUT1** do not allow key changes. This rule applies to choosing the ***Undo*** button as well. After you enter a valid value for a key of the main entity, you cannot undo this action. You can also not undo any attribute change you have done at the same time as the key change.

The MDG-F 7.0 Feature Pack offers changing IDs. The how to guide Enable changeable IDs in MDG-F explains the new functionality in more details.

3.48.3 Q: How can I search for an Edition instead of a Valid On date?

All database searches of the MDG-F user interfaces support both searching for specific validity dates or editions. The pre-defined user interface configurations use only the validity dates. If you want to switch to the edition instead, you need to change the **application configuration** of the related user interface.

1. Copy a pre-defined application configuration (for example **MDGF_0G_OVP_CCTR**for Cost Centers) to your custom namespace.
2. Edit the application parameter ***USMD_SEARCH_EDITION_MODE***and set its value to X. Note that you might have to adapt existing links (for example in the work centers) accordingly.

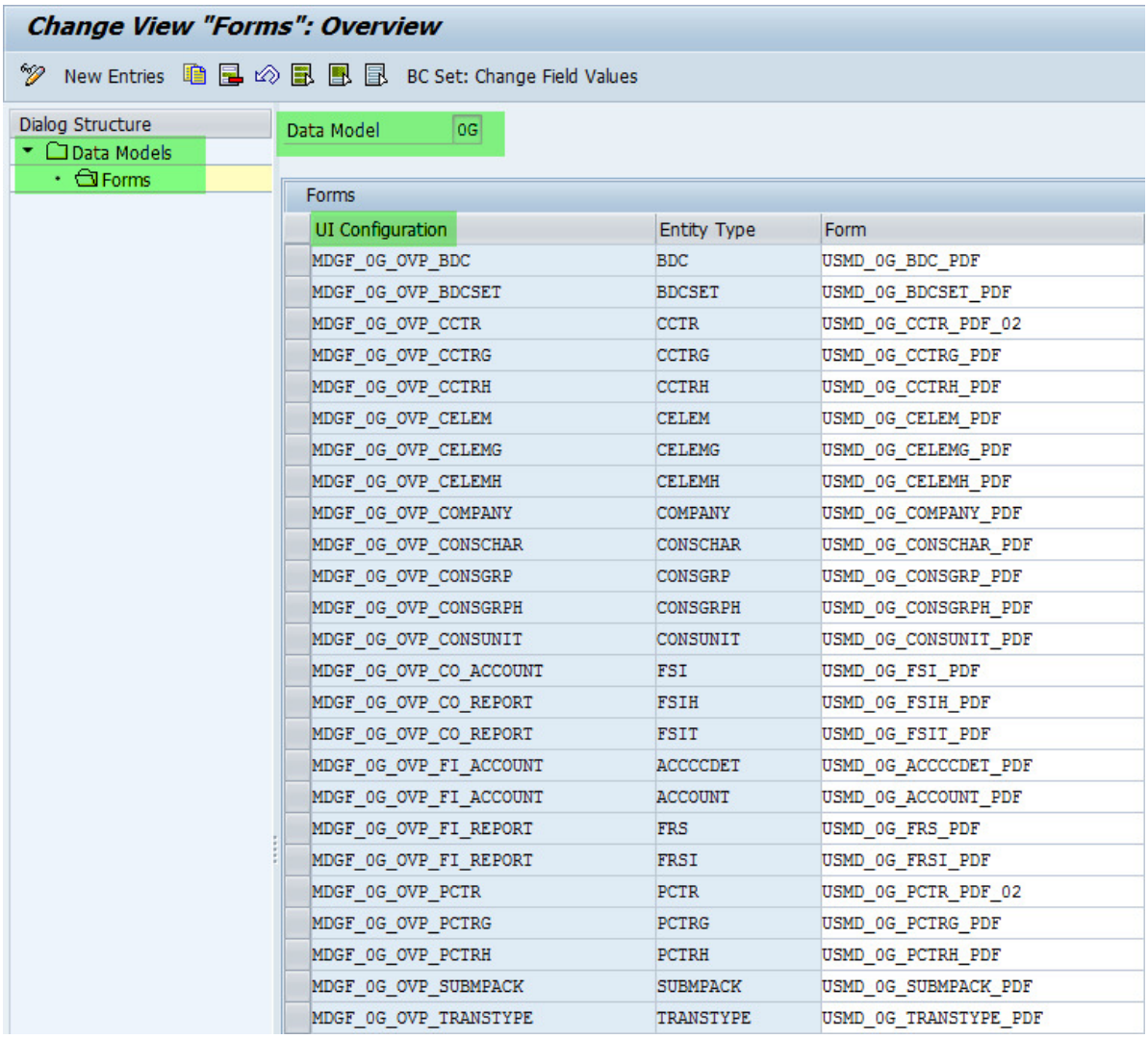
3.48.4 Q: Why does the Search not find my Object that I have saved as Draft?

The minimal requirements for saving an object as a draft are the object keys. Once the keys are defined, saving is possible. Nevertheless, MDG does not create an inactive record in that case but stores the object only within the object list of a change request. The object list is not accessed by the search. The search can only display active and inactive records. You have to access a draft that contains only the object key via the My Change Request application. Alternatively ensure to maintain at least a single object attribute besides the object keys before saving the object as draft.

3.48.5 Q: Why does the Print Preview not work in my copied Application Configuration?

The print forms that are used for the print preview functionality are bound to the application configuration. If you have copied the application configuration to a custom version, you need to create new entries in the related customizing table:

1. Start transaction *MDGIMG* and navigate to *General Settings* □ *UI Modeling* □ *Assign Print Forms for Single Processing*.
2. Mark data model *0G* and double click on *Forms*.
3. Add entries using the name of your custom application configuration. The screenshot below shows the default values pre-defined by SAP.



UI Configuration	Entity Type	Form
MDGF_0G_OVP_BDC	BDC	USMD_0G_BDC_PDF
MDGF_0G_OVP_BDCSET	BDCSET	USMD_0G_BDCSET_PDF
MDGF_0G_OVP_CCTR	CCTR	USMD_0G_CCTR_PDF_02
MDGF_0G_OVP_CCTRG	CCTRG	USMD_0G_CCTRG_PDF
MDGF_0G_OVP_CCTRH	CCTRH	USMD_0G_CCTRH_PDF
MDGF_0G_OVP_CELEM	CELEM	USMD_0G_CELEM_PDF
MDGF_0G_OVP_CELEMG	CELEMG	USMD_0G_CELEMG_PDF
MDGF_0G_OVP_CELEMH	CELEMH	USMD_0G_CELEMH_PDF
MDGF_0G_OVP_COMPANY	COMPANY	USMD_0G_COMPANY_PDF
MDGF_0G_OVP_CONSCHAR	CONSCHAR	USMD_0G_CONSCHAR_PDF
MDGF_0G_OVP_CONSGRP	CONSGRP	USMD_0G_CONSGRP_PDF
MDGF_0G_OVP_CONSGRPH	CONSGRPH	USMD_0G_CONSGRPH_PDF
MDGF_0G_OVP_CONSUNIT	CONSUNIT	USMD_0G_CONSUNIT_PDF
MDGF_0G_OVP_CO_ACCOUNT	FSI	USMD_0G_FSI_PDF
MDGF_0G_OVP_CO_REPORT	FSIH	USMD_0G_FSIH_PDF
MDGF_0G_OVP_CO_REPORT	FSIT	USMD_0G_FSIT_PDF
MDGF_0G_OVP_FI_ACCOUNT	ACCCCDET	USMD_0G_ACCCCDET_PDF
MDGF_0G_OVP_FI_ACCOUNT	ACCOUNT	USMD_0G_ACCOUNT_PDF
MDGF_0G_OVP_FI_REPORT	FRS	USMD_0G_FRS_PDF
MDGF_0G_OVP_FI_REPORT	FRSI	USMD_0G_FRSI_PDF
MDGF_0G_OVP_PCTR	PCTR	USMD_0G_PCTR_PDF_02
MDGF_0G_OVP_PCTRG	PCTRG	USMD_0G_PCTRG_PDF
MDGF_0G_OVP_PCTRH	PCTRH	USMD_0G_PCTRH_PDF
MDGF_0G_OVP_SUBMPACK	SUBMPACK	USMD_0G_SUBMPACK_PDF
MDGF_0G_OVP_TRANSTYPE	TRANSTYPE	USMD_0G_TRANSTYPE_PDF

3.49 Data Replication

3.49.1 Q: How does the MDG Hub handle key mapping for MDG-F entities?

Key mapping is not written automatically, for example it is not written during data import or data replication. If you require key mapping in your replication scenarios, you have to define the key mapping for each object manually. The key mapping maintenance can be accessed via the common user interface using its ***Show*** button.

3.49.2 Q: Why are attachments not replicated to ERP?

Most of the MDG-F entities support attachments. This enables adding documents and/or links to the master data. The attachments are a MDG specific functionality. They are not related to the SAP ECC document features and thus not part of data replication. Furthermore the IDOC and service interfaces of the financial objects do not support attachments.

3.49.3 Q: Why don't ERP transactions show a change history for master data changed in MDG?

When object changes in MDG are replicated to ERP via IDOCs or services the ERP inbound works with a simple and efficient logic: all existing data are being deleted before the changed data coming from MDG are inserted. Unfortunately, no change documents are created for the deletion and the insert.

For this reason, we recommend using the change tracking in MDG when you run central master data maintenance.

3.49.4 Q: Replicate MDG-F entities into the same client of my system?

MDG-F entities are completely stored within the generated tables of MDG. If you want to use the same system and client for productive usage of the data within SAP ERP, too, you need to replicate the MDG-F data into the same system and client. How-to guide "ALE Replication from MDG Hub to ERP using the Same Client in MDG-F" explains the required configuration.

3.49.5 Q: Why are some of my Account IDocs not processed immediately?

The ALE IDoc based replication might show some IDocs with *status* **64**and the following messages in the target system:

- **IDoc: <IDoc Number> Status: IDoc ready to be transferred to application**
- **Immediate Processing was Canceled**

This might happen if the replication contained both the creation and change of accounts or accounts in company codes at the same time. If IDocs are processed in parallel by the receiving system, locking issues may occur.

We recommend using transaction **WE02**in the target system to check the status of the IDocs. You can also use transaction **BD87**to post-process the IDoc.

3.50 SAP Notes

- 1637249 specifying required information for OSS support
- 2105467 specifying required information for Performance Issues

3.51 ALE Replication using the same Client in MDG-F

For the replication of master data created or changed in MDG into the client systems (ERP) SAP recommends the usage of enterprise services or ALE IDocs. The standard replication scenarios require that MDG hub and MDG client are two different systems, or at least two different physical clients within the same system.

For MDG-F, however, customers might want to run the MDG hub in the same physical client as the ERP client due to the nature of the MDG-F data model. Since all MDG-F entity types use the flexible mode their data is stored only in the MDG generated tables and not within ERP tables.

This document explains how-to setup an ALE scenario for data replication where MDG hub and ERP client share the same physical client of the same system.

We recommend that you study the following how-to guide before working with the current guide:

Extensibility Options for SAP Master Data Governance for Financial Data Overview about MDG-F

Strong ALE configuration expertise is required, too.

3.52 Steps for the ALE Scenario Configuration

The guide uses the system **E8B** and its client **001** as sample data. When you configure this scenario for your landscape, ensure you replace the system ID and the client ID with your own system data.

3.52.1 Define Logical Systems

Start transaction **SALE**. In the tree menu navigate to **Basic Settings -> Logical Systems -> Define Logical System**. Define two logical systems:

- One for the MDG hub (in this example E8BCLNT001)
- One for the MDG client (in this example E8B_001)

3.52.2 Define a RFC Connection

Start transaction **SALE**. In the tree menu navigate to **Communication -> Create RFC Connections** (alternatively start transaction **SM59**).

Create a RFC connection using Connection Type 3 (ABAP Connection) into the same client.

3.52.3 Define an ALE tRFC Port

Start transaction **WE21**. Create a port for transactional RFC. The port uses the RFC destination created in the step before.

3.52.4 Define an ALE Distribution Model

Start transaction **SALE**. In the tree menu navigate to **Modelling and Implementing Business Processes -> Maintain Distribution Model and Distribute Views** (alternatively start transaction **BD64**).

Use the previously created logical systems as sender (MDG Hub) and receiver (MDG client). Add the required message types (e.g. COSMAS for cost centers) according to your replication scenario.

3.52.5 Define Partner Profiles

There are two options for the definition of partner profiles. If you are still in the *Change Distribution View* of the previous step, use the top menu bar and execute **Environment -> Generate Partner Profiles**. The screenshots below show needed parameters. Alternatively start transaction **WE20**.

Locate the MDG Client (in this example **E8B_001**) below the tree node **Partner Type LS (Logical System)** to maintain the settings for the IDoc receiver. Add all message types that you use in your replication scenario as **Outbound Parameters**.

Define the previously created ALE tRFC Port for each message. This enables the connection to thee own client as IDoc receiver.

Locate the MDG Hub (in this example **E8BCLNT001**) below the tree node **Partner Type LS (Logical System)** to maintain the settings for the IDoc sender. Add all message types that you use in your replication scenario as **Inbound Parameters**

3.52.6 Steps for the MDG Data Replication Framework (DRF) Configuration

The customizing transactions for the steps in this section are collected in the IMG of the data replication framework. Use transaction **DRFIMG** to display the DRF implementation guide.

3.52.7 Define Business Systems

Navigate to **Define Custom Settings for Data Replication -> Define Technical Settings -> Define Technical Settings for Business Systems**. Create a Business System for the IDoc Receiver (in this example **Logical System E8B_001**). Use the RFC Destination to the own client.

Assign the related Business Object Types for all MDG-F entity types you want to replicate (for example, 158 Cost Center).

Specify the communication channel for IDoc replication for each Business Object Type.

3.52.8 Define a Replication Model

Navigate to **Define Custom Settings for Data Replication -> Define Replication Models**. Assign your own client (in this example **E8B_001**) as the *Target Business System* for each IDoc *Replication Model* and *Outbound Implementation*.

Activate the replication model.

3.52.9 Confirmation IDocs (ALEAUD) for setting replication status in MDG

In standard ALE replication scenarios where sender and receiving system are at least located in different physical clients you can use IDocs of message type ALEAUD for setting the replication status in MDG DRF. These ALE audit IDocs are triggered by report RBDSTATE which periodically runs in a background job.

In this special scenario where you have only one physical client this would require an ALE replication model for IDOC type ALEAUD with sender system = receiver system. Unfortunately this is not allowed in transaction BD64.

As a workaround you can create a copy of program RBDSTATE which sends ALE audit documents within the same client without having the distribution model for ALEAUD defined in BD64. MDG specific enhancements are required in the copied report.

3.52.10 SAP Notes

- 1637249 specifying required information for OSS support
- 2105467 specifying required information for Performance Issues

3.52.11 Example

```
REPORT zmdgrbdstate MESSAGE-ID b1.
** Copy of program RBDSTATE - Use for MDG only!
** Required for sending ALEAUDIT if ALE distribution for MDG objects
** is setup within same physical client.
** MDG specific enhancements are marked with "MDG Enhancement
**
** TODO:
** Copy GUI Status "STANDARD" from program RBDSTATE to this report!
INCLUDE mbdconst.
INCLUDE bdcstaud.
CONSTANTS:
c_max_idocs TYPE i VALUE '500'.
*** Begin MDG Enhancement
```

```
TYPES:
tyt_idoc_list TYPE TABLE OF bdidocs,
tyt_idoc_contrl TYPE TABLE OF edidc,
tyt_idoc_data TYPE TABLE OF edidd,
tyt_idoc_status TYPE TABLE OF bdidocstat,
tyt_edi_status_records TYPE TABLE OF edi_ds40.
DATA:
gt_idoc_contrl TYPE tyt_idoc_contrl,
gt_idoc_data TYPE tyt_idoc_data,
gt_idoc_status TYPE tyt_idoc_status.
*** End MDG Enhancement
TABLES: edidc, sscrfields, bdaudstate.
DATA: gs_layout TYPE slis_layout_alv,
gt_fieldcat TYPE slis_t_fieldcat_alv,
gs_excluding TYPE slis_t_extab,
gs_excl_head TYPE slis_extab,
```

```
g_status_set TYPE slis_formname VALUE 'PF_STATUS_SET',
g_user_command TYPE slis_formname VALUE 'USER_COMMAND',
header TYPE lvc_title.
SELECT-OPTIONS:
s_sndsys FOR bdaudstate-rcv_system,
s_recsys FOR bdaudstate-snd_system. "MDG Enhancement
SELECTION-SCREEN SKIP 1.
SELECT-OPTIONS:
s_mestyp FOR bdaudstate-mess_type,
s_mescod FOR bdaudstate-mess_code,
s_mesfct FOR bdaudstate-mess_funct,
status FOR edidc-status.
SELECTION-SCREEN SKIP 1.
SELECT-OPTIONS: s_upddat FOR edidc-upddat NO-EXTENSION .
DATA:
left_date LIKE edidc-upddat,
left_time LIKE edidc-updtim,
right_date LIKE edidc-upddat,
right_time LIKE edidc-updtim,
t_idoc_control TYPE audit_idoc_control_tab,
t_idoc_control_all TYPE audit_idoc_control_tab,
s_idoc_control TYPE audit_idoc_control_t ,
nothing_to_do,
resulting_idocs LIKE bdidocs OCCURS 0 WITH HEADER LINE.
DATA: paket TYPE i,
t_idoc_control_all_max TYPE i,
t_idoc_control_all_anz TYPE i.
START-OF-SELECTION.
IF NOT sy-batch IS INITIAL AND s_upddat[] IS INITIAL.
PERFORM time_interval_get
CHANGING
left_date
```

```
left_time
right_date
right_time.
MESSAGE i147 WITH left_date left_time right_date right_time.
* Selektionszeitraum für die AUDIT Idocs: &1 &2 -> &3 &4 .
* comparisons are only based on numbers (date and time). This is
* portable to AS/400.
SELECT sndprn mestyp mescod mesfct credat cretim docnum status
FROM edidc INTO TABLE t_idoc_control_all
WHERE ( upddat = left_date AND updtim >= left_time "#EC PORTABLE
OR upddat > left_date ) "#EC PORTABLE
AND ( upddat < right_date "#EC PORTABLE
OR upddat = right_date AND updtim <= right_time ) "#EC PORTABLE
AND sndprt = c_prt_logical_system
AND sndprn IN s_sndsys
AND rcvprn IN s_recsys "MDG Enhancement
AND mestyp IN s_mestyp
AND mescod IN s_mescod
AND mesfct IN s_mesfct
AND status IN status
AND status <> c_status_in_archive_reload
AND status <> c_status_in_archived
AND status <> c_status_in_orig_of_edited
ORDER BY sndprn mestyp mesfct mescod credat cretim.
IF sy-subrc <> 0.
nothing_to_do = 'X'.
ELSE.
DESCRIBE TABLE t_idoc_control_all LINES t_idoc_control_all_max.
LOOP AT t_idoc_control_all INTO s_idoc_control.
APPEND s_idoc_control TO t_idoc_control.
t_idoc_control_all_anz = t_idoc_control_all_anz + 1.
paket = paket + 1.
```

```
IF paket = c_max_idocs OR
t_idoc_control_all_anz = t_idoc_control_all_max.
PERFORM idocs_create
TABLES
resulting_idocs
USING
t_idoc_control.
CLEAR paket.
REFRESH t_idoc_control.
ENDIF.
ENDLOOP.
ENDIF.
ELSE.
MESSAGE s147 WITH
s_upddat-low '00:00:00' s_upddat-high '24:00:00' .
* Selektionszeitraum für die AUDIT Idocs: &1 &2 -> &3 &4 .
SELECT sndprn mestyp mescod mesfct credat cretim docnum status
FROM edidc INTO TABLE t_idoc_control_all
WHERE upddat IN s_upddat
AND sndprt = c_prt_logical_system
AND sndprn IN s_sndsys
AND rcvprn IN s_recsys "MDG Enhancement
AND mestyp IN s_mestyp
AND mescod IN s_mescod
AND mesfct IN s_mesfct
AND status IN status
AND status <> c_status_in_archive_reload
AND status <> c_status_in_archived
AND status <> c_status_in_orig_of_edited
ORDER BY sndprn mestyp mesfct mescod credat cretim.
IF sy-subrc <> 0.
```

```
nothing_to_do = 'X'.
ELSE.
DESCRIBE TABLE t_idoc_control_all LINES t_idoc_control_all_max.
LOOP AT t_idoc_control_all INTO s_idoc_control.
APPEND s_idoc_control TO t_idoc_control.
t_idoc_control_all_anz = t_idoc_control_all_anz + 1.
paket = paket + 1.
IF paket = c_max_idocs OR
t_idoc_control_all_anz = t_idoc_control_all_max.
PERFORM idocs_create
TABLES
resulting_idocs
USING
t_idoc_control.
CLEAR paket.
REFRESH t_idoc_control.
ENDIF.
ENDLOOP.
ENDIF.
ENDIF.
IF NOT nothing_to_do IS INITIAL.
MESSAGE i139.
ELSEIF NOT resulting_idocs[] IS INITIAL.
*** BEGIN MDG Enhancement
" Select IDOC data from database
PERFORM read_idocs
USING resulting_idocs[]
CHANGING gt_idoc_contrl
gt_idoc_data.
```

```
" Call DRF inboung for ALEAUD idocs
PERFORM drf_idoc_input_aleaud_process
USING gt_idoc_contrl
gt_idoc_data
CHANGING gt_idoc_status.
" Set IDOC status
PERFORM set_idoc_status
USING gt_idoc_status.
*** End MDG Enhancement
CALL FUNCTION 'DB_COMMIT'.
CALL FUNCTION 'DEQUEUE_ALL'.
COMMIT WORK.
PERFORM output_list.
ELSE.
MESSAGE i158.
* Für die selektierten Daten gibt es keine Interessenten
ENDIF.
AT SELECTION-SCREEN ON s_upddat.
IF sy-batch IS INITIAL AND
( sscrfields-ucomm = 'ONLI' OR sscrfields-ucomm = 'PRIN' ).
IF s_upddat[] IS INITIAL.
MESSAGE e142.
* Es muß ein Änderungszeitraum angegeben werden.
ENDIF.
ENDIF.
*&---------------------------------------------------------------------*
*& Form TIME_INTERVAL_GET
*&---------------------------------------------------------------------*
* text
*----------------------------------------------------------------------*
* <--P_LEFT_DATE text *
```

```
* <--P_LEFT_TIME text *
* <--P_RIGHT_DATE text *
* <--P_RIGHT_TIME text *
*---------------------------------------------------------------------*
FORM time_interval_get
CHANGING
left_date LIKE edidc-upddat
left_time LIKE edidc-updtim
right_date LIKE edidc-upddat
right_time LIKE edidc-updtim.
DATA:
job_name LIKE tbtcjob-jobname,
job_count LIKE tbtcjob-jobcount,
job_head LIKE tbtcjob,
cmp_time LIKE edidc-updtim.
CALL FUNCTION 'GET_JOB_RUNTIME_INFO'
IMPORTING
jobcount = job_count
jobname = job_name.
CALL FUNCTION 'BP_JOB_READ'
EXPORTING
job_read_jobcount = job_count
job_read_jobname = job_name
job_read_opcode = 19
IMPORTING
job_read_jobhead = job_head.
left_time = job_head-sdltime - 300. " 300 seconds = 5 minutes
cmp_time = -300. "23:55
IF left_time < cmp_time.
left_date = job_head-sdldate.
ELSE.
* date has to changed
left_date = job_head-sdldate - 1.
ENDIF
```

```
cmp_time = -240. "23:56
IF right_time < cmp_time.
right_date = job_head-strtdate.
ELSE.
* date has to be changed
right_date = job_head-strtdate - 1.
ENDIF.
ENDFORM. " TIME_INTERVAL_GET
*&---------------------------------------------------------------------*
*& Form IDOCS_CREATE
*&---------------------------------------------------------------------*
* text
*----------------------------------------------------------------------*
* --> p1 text
* <-- p2 text
*----------------------------------------------------------------------*
FORM idocs_create
TABLES
resulting_idocs STRUCTURE bdidocs
USING
idoc_controls TYPE audit_idoc_control_tab.
CONSTANTS:
c_filter_mestyp LIKE tbd10-objtype VALUE 'MESTYP'.
DATA: idoc_control TYPE audit_idoc_control_t,
receiver_input LIKE bdi_logsys OCCURS 0 WITH HEADER LINE,
receiver_output LIKE bdi_logsys OCCURS 0 WITH HEADER LINE,
filter_objects LIKE bdi_fltval OCCURS 0 WITH HEADER LINE,
to_send TYPE c,
idoc_info TYPE audit_idoc,
control_package TYPE audit_idoc_tab.
LOOP AT idoc_controls INTO idoc_control.
AT NEW mestyp.
REFRESH receiver_input.
```

```
receiver_input-logsys = idoc_control-sndprn.
APPEND receiver_input.
REFRESH filter_objects.
* append entry for filter object MESTYP
filter_objects-objtype = c_filter_mestyp.
filter_objects-objvalue = idoc_control-mestyp.
APPEND filter_objects.
CALL FUNCTION 'ALE_MESTYPE_GET_RECEIVER'
EXPORTING
message_type = c_mestyp_aleaud
TABLES
receiver_input = receiver_input
receivers = receiver_output
filterobject_values = filter_objects.
IF receiver_output[] IS INITIAL. "MDG Enhancement
APPEND LINES OF receiver_input TO receiver_output. "MDG Enhancement
ENDIF. "MDG Enhancement
IF NOT receiver_output[] IS INITIAL.
to_send = 'X'.
ELSE.
CLEAR to_send.
ENDIF.
ENDAT.
IF NOT to_send IS INITIAL.
MOVE-CORRESPONDING idoc_control TO idoc_info.
APPEND idoc_info TO control_package.
ENDIF.
AT END OF sndprn.
IF NOT control_package[] IS INITIAL.
CALL FUNCTION 'AUDIT_IDOC_CREATE'
EXPORTING
rcv_system = idoc_control-sndprn
```

```
IMPORTING
idoc_number = resulting_idocs-docnum
CHANGING
idoc_info_records = control_package.
* exceptions
* others = 1.
REFRESH control_package.
IF NOT resulting_idocs IS INITIAL.
APPEND resulting_idocs.
ENDIF.
ENDIF.
ENDAT.
ENDLOOP.
ENDFORM. " IDOCS_CREATE
*&---------------------------------------------------------------------*
*& Form output_list
*&---------------------------------------------------------------------*
* text
*----------------------------------------------------------------------*
* -->P_RESULTING_IDOCS text
*----------------------------------------------------------------------*
FORM output_list.
header = text-001.
PERFORM fieldcat_init USING gt_fieldcat[].
PERFORM layout_init USING gs_layout.
PERFORM icon_excluding USING gs_excluding.
CALL FUNCTION 'REUSE_ALV_GRID_DISPLAY'
EXPORTING
i_callback_program = sy-repid
i_callback_pf_status_set = g_status_set
i_callback_user_command = g_user_command
i_grid_title = header
is_layout = gs_layout
it_fieldcat = gt_fieldcat[]
```

```
it_excluding = gs_excluding[]
TABLES
t_outtab = resulting_idocs.
ENDFORM. " output_list
*&---------------------------------------------------------------------*
*& Form fieldcat_init
*&---------------------------------------------------------------------*
* text
*----------------------------------------------------------------------*
* -->P_GT_FIELDCAT[] text
*----------------------------------------------------------------------*
FORM fieldcat_init USING rt_fieldcat
TYPE slis_t_fieldcat_alv.
DATA: ls_fieldcat TYPE slis_fieldcat_alv.
CLEAR ls_fieldcat.
ls_fieldcat-fieldname = 'DOCNUM'.
ls_fieldcat-seltext_l = text-200.
ls_fieldcat-outputlen = '30'.
APPEND ls_fieldcat TO rt_fieldcat.
ENDFORM. " fieldcat_init
*&---------------------------------------------------------------------*
*& Form layout_init
*&---------------------------------------------------------------------*
* text
*----------------------------------------------------------------------*
* -->P_GS_LAYOUT text
*----------------------------------------------------------------------*
FORM layout_init USING rs_layout TYPE slis_layout_alv.
*doubleclick
rs_layout-f2code = 'IDOC'.
rs_layout-colwidth_optimize = 'X'.
```

```
ENDFORM. " LAYOUT_INIT
*&---------------------------------------------------------------------*
*& Form icon_excluding
*&---------------------------------------------------------------------*
* text
*----------------------------------------------------------------------*
* -->P_GS_EXCLUDING text
*----------------------------------------------------------------------*
FORM icon_excluding USING p_gs_excluding TYPE slis_t_extab.
REFRESH p_gs_excluding[].
gs_excl_head-fcode = '&VEXCEL'. "Excel
APPEND gs_excl_head-fcode TO p_gs_excluding.
gs_excl_head-fcode = '&AQW'. "word
APPEND gs_excl_head-fcode TO p_gs_excluding.
gs_excl_head-fcode = '&GRAPH'.
APPEND gs_excl_head-fcode TO p_gs_excluding.
gs_excl_head-fcode = '&XXL'.
APPEND gs_excl_head-fcode TO p_gs_excluding.
gs_excl_head-fcode = '&CRBATCH'.
APPEND gs_excl_head-fcode TO p_gs_excluding.
gs_excl_head-fcode = '&CRTEMPL'.
APPEND gs_excl_head-fcode TO p_gs_excluding.
gs_excl_head-fcode = '&XINT'.
APPEND gs_excl_head-fcode TO p_gs_excluding.
gs_excl_head-fcode = '&URL'.
APPEND gs_excl_head-fcode TO p_gs_excluding.
gs_excl_head-fcode = '&CRDESIG'.
APPEND gs_excl_head-fcode TO p_gs_excluding.
```

```
gs_excl_head-fcode = '&VLOTUS'.
APPEND gs_excl_head-fcode TO p_gs_excluding.
gs_excl_head-fcode = '&VCRYSTAL'.
APPEND gs_excl_head-fcode TO p_gs_excluding.
gs_excl_head-fcode = '&OL0'.
APPEND gs_excl_head-fcode TO p_gs_excluding.
gs_excl_head-fcode = '&XPA'.
APPEND gs_excl_head-fcode TO p_gs_excluding.
gs_excl_head-fcode = '&OMP'.
APPEND gs_excl_head-fcode TO p_gs_excluding.
gs_excl_head-fcode = '&ILT'.
APPEND gs_excl_head-fcode TO p_gs_excluding.
ENDFORM. " ICON_EXCLUDING
*&---------------------------------------------------------------------*
*& Form PF_STATUS_SET
*&---------------------------------------------------------------------*
* text
*----------------------------------------------------------------------*
* --> p1 text
* <-- p2 text
*----------------------------------------------------------------------*
FORM pf_status_set USING rt_extab TYPE slis_t_extab.
DATA: l_status LIKE sy-pfkey VALUE 'STANDARD'.
* EXCLUDING FCODES GIVEN BY ABAP LISTVIEWER *
SET PF-STATUS l_status EXCLUDING rt_extab.
ENDFORM. " PF_STATUS_SET
*&---------------------------------------------------------------------*
```

```
*& Form USER_COMMAND
*&---------------------------------------------------------------------*
* text
*----------------------------------------------------------------------*
* --> p1 text
* <-- p2 text
*----------------------------------------------------------------------*
FORM user_command USING rf_ucomm LIKE sy-ucomm
rs_selfield TYPE slis_selfield.
CASE rf_ucomm.
WHEN 'IDOC'.
READ TABLE resulting_idocs INDEX rs_selfield-tabindex.
IF sy-subrc = 0.
SUBMIT idoc_tree_control WITH docnum = resulting_idocs-docnum
AND RETURN.
ELSE.
MESSAGE s010.
* Bitte Cursor richtig positionieren
ENDIF.
CLEAR rf_ucomm.
WHEN OTHERS.
ENDCASE.
ENDFORM. " USER_COMMAND
*&---------------------------------------------------------------------*
*& Form read_idocs
*&---------------------------------------------------------------------*
* text
*----------------------------------------------------------------------*
* -->IT_IDOC_LIST text
* -->CT_IDOC_CONTRL text
* -->CT_IDOC_DATA text
*----------------------------------------------------------------------*
FORM read_idocs
USING it_idoc_list TYPE tyt_idoc_list
CHANGING et_idoc_contrl TYPE tyt_idoc_contrl
```

```
et_idoc_data TYPE tyt_idoc_data.
DATA:
lv_docnum TYPE edidc-docnum,
ls_idoc_contrl TYPE edidc,
lt_idoc_data TYPE tyt_idoc_data.
CLEAR et_idoc_contrl.
CLEAR et_idoc_data.
LOOP AT it_idoc_list INTO lv_docnum.
CLEAR lt_idoc_data.
CALL FUNCTION 'EDI_DOCUMENT_OPEN_FOR_READ'
EXPORTING
document_number = lv_docnum
IMPORTING
idoc_control = ls_idoc_contrl
EXCEPTIONS
OTHERS = 4.
IF sy-subrc = 0.
APPEND ls_idoc_contrl TO et_idoc_contrl.
CALL FUNCTION 'EDI_SEGMENTS_GET_ALL'
EXPORTING
document_number = lv_docnum
TABLES
idoc_containers = lt_idoc_data
EXCEPTIONS
document_number_invalid = 1
end_of_document = 2
OTHERS = 3.
IF sy-subrc <> 0.
MESSAGE ID sy-msgid TYPE sy-msgty NUMBER sy-msgno
WITH sy-msgv1 sy-msgv2 sy-msgv3 sy-msgv4.
ELSE.
```

```
APPEND LINES OF lt_idoc_data TO et_idoc_data.
ENDIF.
ENDIF.
CALL FUNCTION 'EDI_DOCUMENT_CLOSE_READ'
EXPORTING
document_number = lv_docnum.
ENDLOOP.
ENDFORM. "read_idocs
*&---------------------------------------------------------------------*
*& Form set_idoc_status
*&---------------------------------------------------------------------*
* text
*----------------------------------------------------------------------*
* -->IT_IDOC_STATUS text
*----------------------------------------------------------------------*
FORM set_idoc_status
USING it_idoc_status TYPE tyt_idoc_status.
DATA:
ls_idoc_status TYPE bdidocstat,
lt_idoc_data TYPE tyt_idoc_data,
ls_edi_status_record TYPE LINE OF tyt_edi_status_records,
lt_edi_status_records TYPE tyt_edi_status_records.
LOOP AT it_idoc_status INTO ls_idoc_status.
CALL FUNCTION 'EDI_DOCUMENT_OPEN_FOR_EDIT'
EXPORTING
document_number = ls_idoc_status-docnum
* ALREADY_OPEN = 'N'
* IMPORTING
* IDOC_CONTROL =
TABLES
idoc_data = lt_idoc_data
EXCEPTIONS
document_foreign_lock = 1
```

```
document_not_exist = 2
document_not_open = 3
status_is_unable_for_changing = 4
OTHERS = 5
.
IF sy-subrc <> 0.
* Implement suitable error handling here
ENDIF.
CLEAR ls_edi_status_record.
CLEAR lt_edi_status_records.
MOVE-CORRESPONDING ls_idoc_status TO ls_edi_status_record.
ls_edi_status_record-mandt = sy-mandt.
ls_edi_status_record-logdat = sy-datum.
ls_edi_status_record-logtim = sy-uzeit.
ls_edi_status_record-uname = sy-uname.
ls_edi_status_record-status = ls_idoc_status-status.
ls_edi_status_record-repid = sy-repid.
APPEND ls_edi_status_record TO lt_edi_status_records.
CALL FUNCTION 'EDI_DOCUMENT_CLOSE_EDIT'
EXPORTING
document_number = ls_idoc_status-docnum
* DO_COMMIT = 'X'
* DO_UPDATE = 'X'
* WRITE_ALL_STATUS = 'X'
* STATUS_75 = ' '
TABLES
status_records = lt_edi_status_records
EXCEPTIONS
idoc_not_open = 1
db_error = 2
OTHERS = 3.
IF sy-subrc <> 0.
* Implement suitable error handling here
ENDIF.
ENDLOOP.
```

```
ENDFORM. "set_idoc_status
*&---------------------------------------------------------------------*
*& Form drf_idoc_input_aleaud_process
*&---------------------------------------------------------------------*
* text
*----------------------------------------------------------------------*
* -->IT_IDOC_CONTRL text
* -->IT_IDOC_DATA text
* -->ET_IDOC_STATUS text
*----------------------------------------------------------------------*
FORM drf_idoc_input_aleaud_process
USING it_idoc_contrl TYPE tyt_idoc_contrl
it_idoc_data TYPE tyt_idoc_data
CHANGING et_idoc_status TYPE tyt_idoc_status.
DATA:
lt_idoc_status TYPE TABLE OF bdidocstat,
lt_return_variables TYPE TABLE OF bdwfretvar,
lt_serialization_info TYPE TABLE OF bdi_ser.
CLEAR et_idoc_status.
" Call DRF inboung for ALEAUD idocs
CALL FUNCTION 'DRF_IDOC_INPUT_ALEAUD'
EXPORTING
input_method = space
mass_processing = space
TABLES
idoc_contrl = it_idoc_contrl
idoc_data = it_idoc_data
idoc_status = et_idoc_status
return_variables = lt_return_variables
serialization_info = lt_serialization_info
EXCEPTIONS
```

```
wrong_function_called = 1
OTHERS = 2.
ASSERT sy-subrc = 0.
ENDFORM. "drf_idoc_input_aleaud_process
```

3.53 New function - Master Data Governancefor FI Contract Account

CONTRACT ACCOUNTS RECEIVABLE AND PAYABLE (FI-CA)

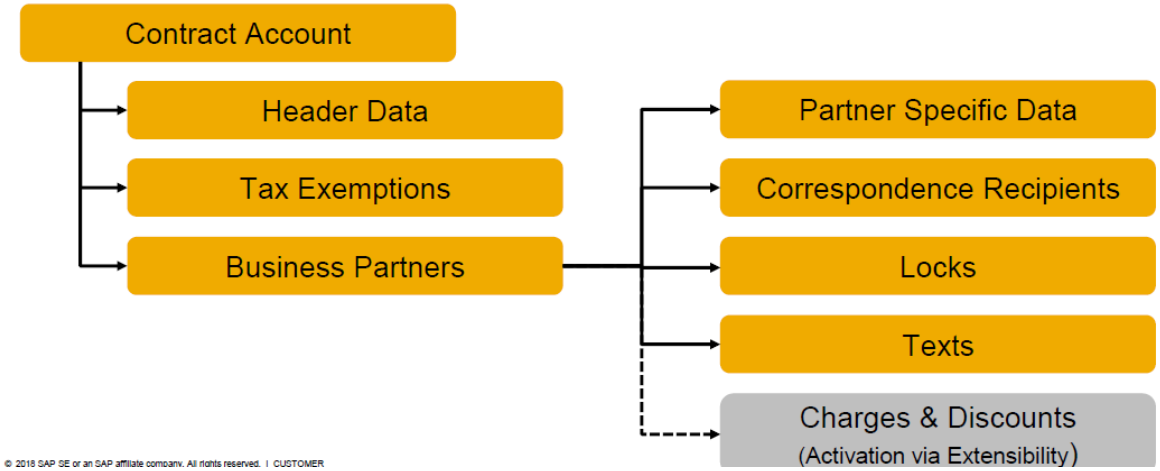
- Sub ledger accounting for processing large document volumes
- Realizes the typical accounts receivable functions
- Originally used in Service Industries like Insurance, Utilities, Telecom, Public Sector, Media etc.; more and more used for all kind of mass business
- Based on SAP Business Partner (ERP-Customer / ERP-Vendor Master are not used)
- Each business partner posting is assigned to one Business Partner and to one Contract Account

Contract Accounts

- Define, for each business partner, the procedures that apply when posting and processing line items
- Control parameters for billing, payment, dunning, correspondence, etc.
- Independent entity
  - One Contract Account can include assignments to multiple Business Partners
  - One Business Partner can be assigned to multiple Contract Accounts

MDG for FI Contract Accounts available in MDG on S/4HANA 1709 SPS1 and MDG 9.1

- Contract Account data model consists of generic core data as well as industry-specific data
- MDG supports generic core data out-of-the-box as well as extensibility options
- In MDG Contract Account data is part of Data Model BP for Business Partners



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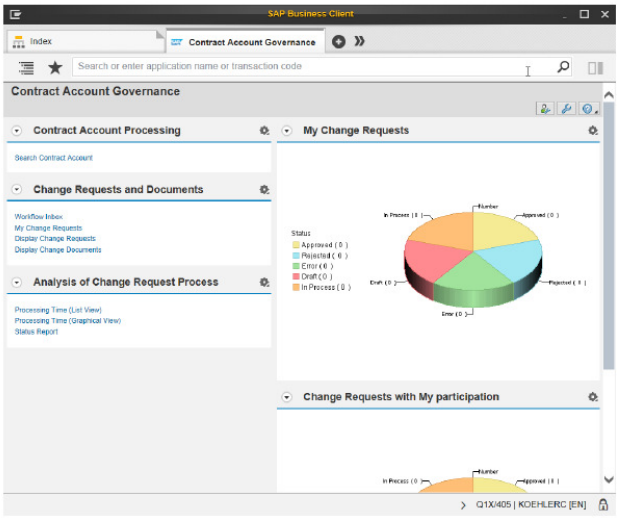
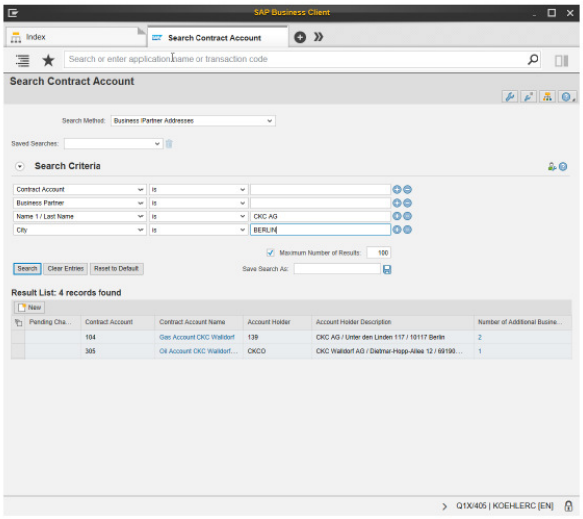
Launchpad

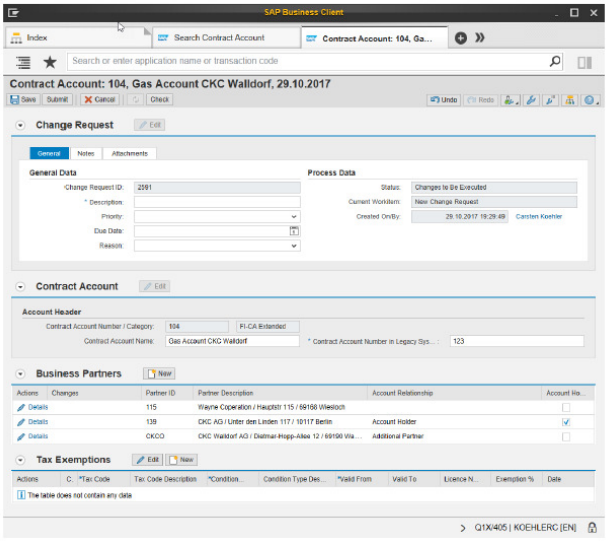


Search Screen

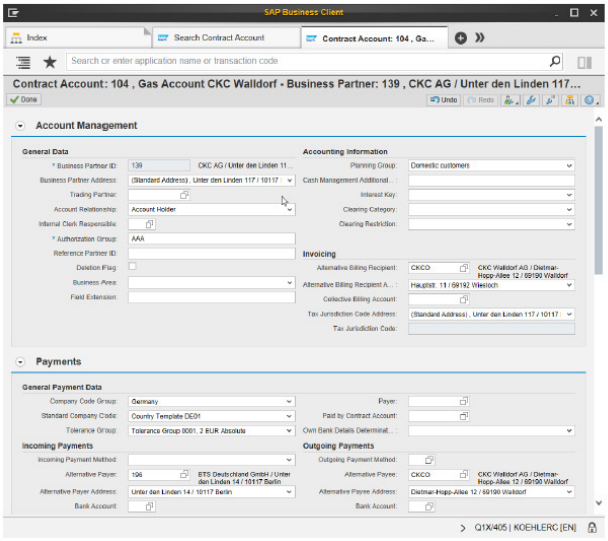
Search via Contract Account or BusPartner data



Header Data



Partner Specific Data



Activation

- Business Function for FI-CA to be activated before starting with MDG for FI Contract Account

Configuration

- Using configuration steps in IMG under Financial Accounting (New) → Contract Accounts Receivable and Payable → Basic Functions → Contract Accounts (Transaction FICAIMG)

Validations

- Standard checks implemented in SAP-GUI UI
  - Number Ranges and Contract Account Categories
  - Contract Account / Business Partner Relationships etc.
- Custom Validations via BRF+
- Field Modification per Contract Account Category and per Activitiy
- Authority Checks for Contract Account Type, Company Code, Authorization Group etc.

Derivations via BRF+

3.53.1 Prozess modelling – new function

Business Activities

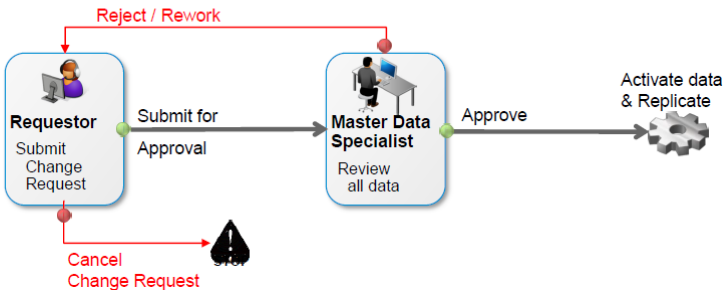
- FKK1 Create Contract Account
- FKK2 Process Contract Account
- FKK3 Display Contract Account

Change Request Types (pre-delivered, can be adjusted within project)

- FKK1P1 Create Contract Account
- FKK2P1 Change Contract Account

Workflow

- Two-step workflow for four-eyes-principle
- Implemented via Rules-based Workflow



Field Extensibility

- Pre-delivered MDG data model consists generic fields
- Industry-specific fields can be added to pre-delivered entity types via field extensibility
- Custom-specific fields can be added to pre-delivered entity types via field extensibility
- Field Extensibility Guide with step-by-step explanation available

Node Extensibility

- Only industry-specific entity type is Charges & Discounts
- UI / logic for Charges & Discounts is pre-delivered, but inactive
- Charges & Discounts can easily be activated following the Node Extensibility Guide for Charges & Discounts

3.54 Planned functionality

Data Replication via SOA-Services

- Currently replication via CRM-MW available only, cannot be used for replication from MDG Hub to SAP-ERP systems and Non-SAP systems
- Development of SOA replication services planned
  - Replication message
  - Confirmation message
- Data Load from ERP system to MD Hub via SOA-Services

Client Maintenance

- Enable Client Maintenance for Contract Accounts in MDG; incoming data can be
  - Activated directly or
  - Governed via a Change Request
- Client Maintenance of incoming Business Agreements from SAP-CRM (Hybris Billing)

Planned changes

- Maintain planned changes for Contract Accounts in MDG-UI

If in customer master data a payment condition is missing, if the delivery or the payment address has incorrect entries, or the sales tax identification number contains errors, not only orders are at stake but in worst case even the customer relationship endangered. If your colleagues can rely on trustworthy data records, they do not have to spend extra time on corrections and positive customer correspondence is ensured.

Information on business partners are used by various departments within companies. Thus, knowledge of contents on the different views of a data record is decentralized. The experts within departments such as sales or accounting are the best contributors to maintain high quality data.

With SAP MDG-C, SAP provides a Master Data tool ideal to combine with an SAP ERP system. The reuse of existing technologies and structures such as types, checks and configuration ensures a neat less integration. The user friendly front end and the individually adaptable process control provide the opportunity to create efficient workflows and gain data records of high quality. Automated rules for the derivation of values and validations, email notifications as well as service level agreements (SLAs) are further tools to facilitate maintenance processes. Ideally, SAP MDG is defined as the leading system for master data. The solution provides not only SAP standard integration technologies but additional functionality to harmonize master data.

For efficient supplier management, where procurement counts as a value driver for the entire company, a stable and reliable database is essential.

To date, companies have followed a strategically inadequate approach to the maintenance of supplier master data. The complex interrelationships of purchasing organizations, company codes and business partners make it difficult to efficiently maintain master data. This is particularly the case if the maintenance processes and the system landscape have not been centralized. In addition to the purchasing department, the accounting department also requires access to supplier master data in order to maintain accounting views, as well as bank data and control data of business partners. Supplier relationships can become strained if, for instance, invoices cannot be settled due to insufficient master data in suppliers' bank data.

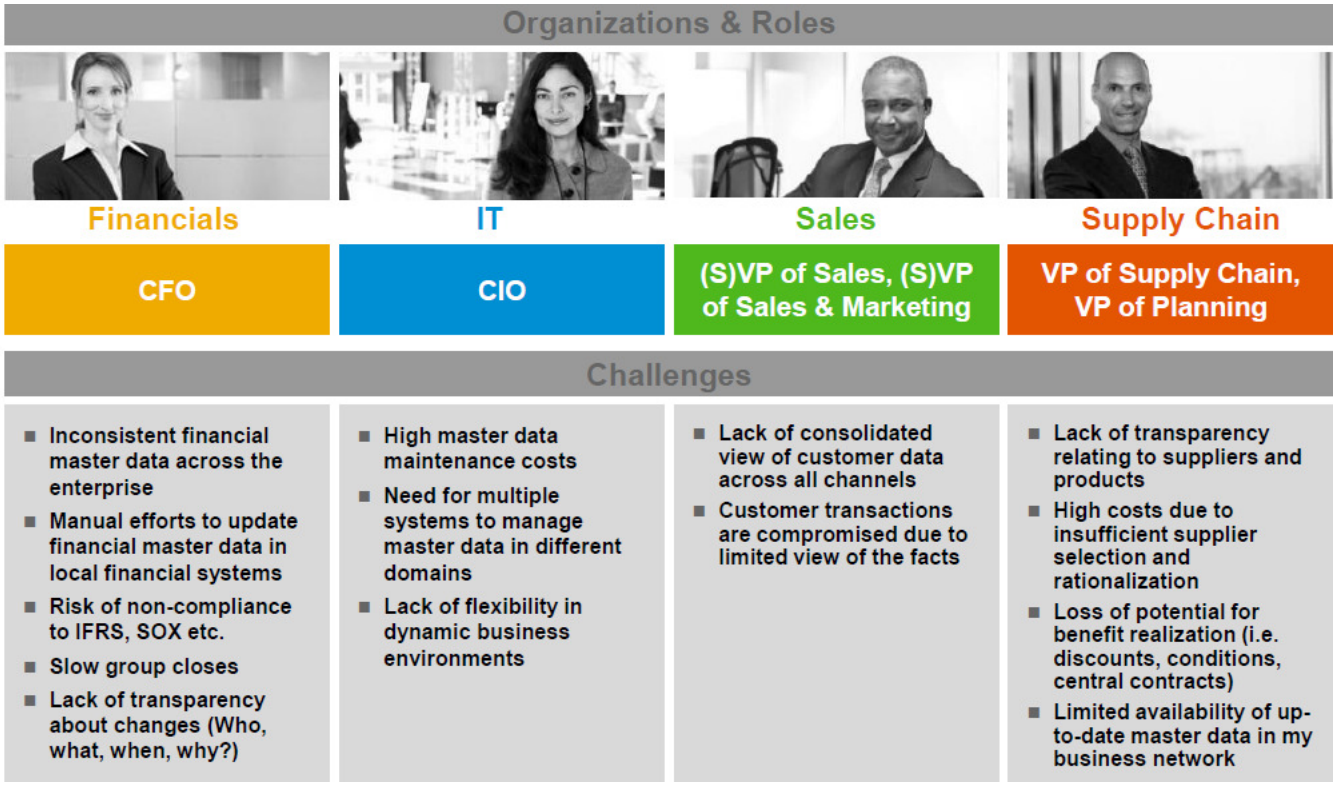
SAP MDG-S offers you a solution that is not only compatible with your SAP ERP, but which also reuses any data structures, checks and validations already available there. **The integration of MDG-S into your system landscape, on the basis of SAP ERP**, thus presents an effective way to harmonize your supplier master data by means of a central control facility. The rule-based workflows in MDG-S facilitate the involvement of all the parties charged with the task of maintaining supplier master data. The simple

integration of address data validations or the incorporation of external services, such as the comparison VAT registration numbers or Dun & Bradstreet data, improve the data quality, facilitate correct evaluations and provide the basis for the observance of legal regulations.

The MDG Consolidation can be used for the migration of your existent master data. The data from different source systems can be consolidated to “golden records” supported by specific processes. This ensures data correctness and the creation of duplicates can be avoided.

4.1 Overview

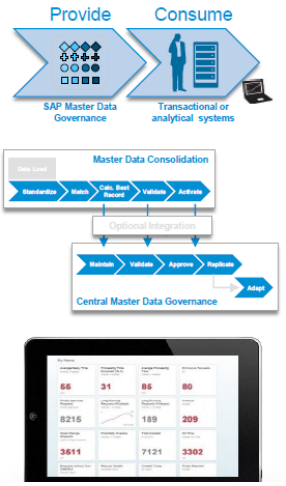
4.1.1 Master data quality impacts all Lines of Business



4.1.2 Centrally govern and consolidate master data for your enterprise

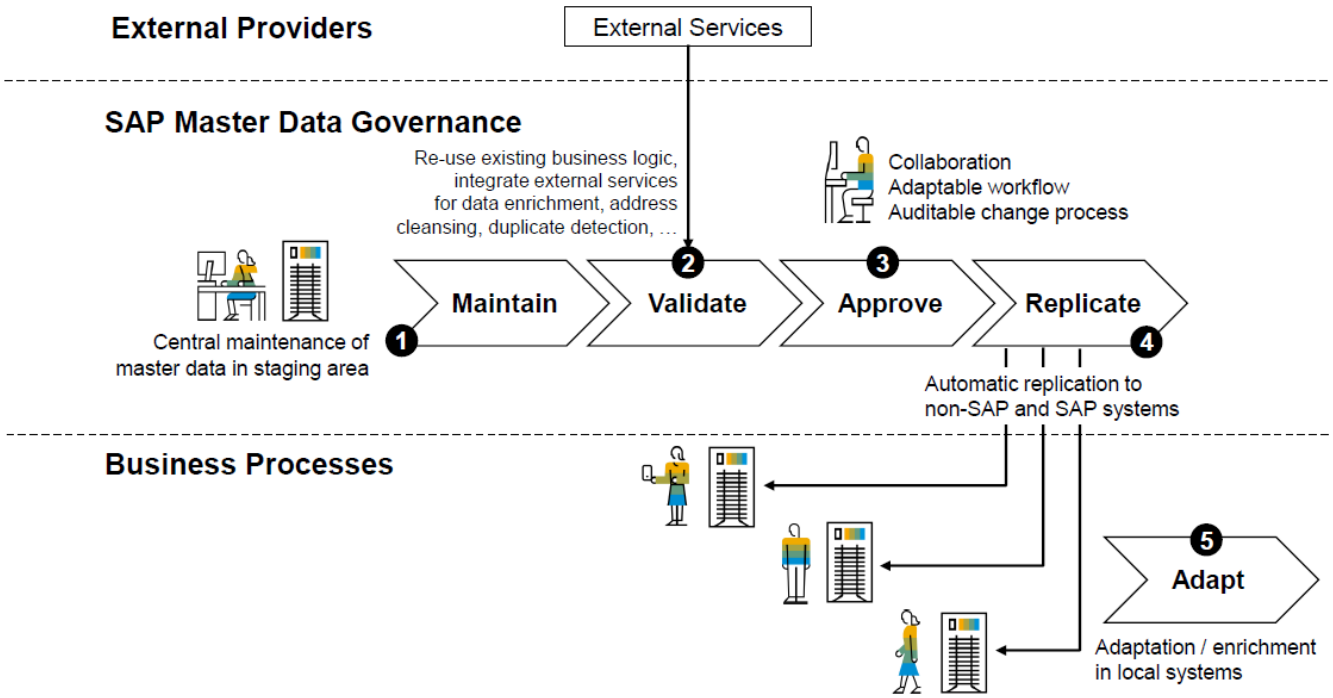
SAP MDG on S/4HANA is a state-of-the-art master data mgmt solution, providing out-of-the-box, domain-specific master data governance to centrally create, change and distribute or to consolidate master data across your complete enterprise system landscape

Governance	Enables governance, compliance and transparency of master data during creation and change through integrated staging, approval and central audit trail
Consistency	Delivers consistent definition, authorization and replication of key master data entities. Eliminates error prone manual maintenance processes for master data in multiple systems
Consolidation	Consolidate master data in any enterprise system landscape, create best records and key mapping between duplicates, and optionally combine consolidation with central governance for sustained master data quality
Integration	Provides native integration in SAP solutions, incl. re-use of data model, business logic, and configuration for validation while offering openness to integrate 3rd party services
Flexibility	Open to extend the standard models and to create governance for your own master data and flexibly for non-SAP environments
Data Quality	Measure Process quality using SAP Smart Business and integrate with SAP Data Services & SAP Information Steward for quality, enrichment, and data remediation

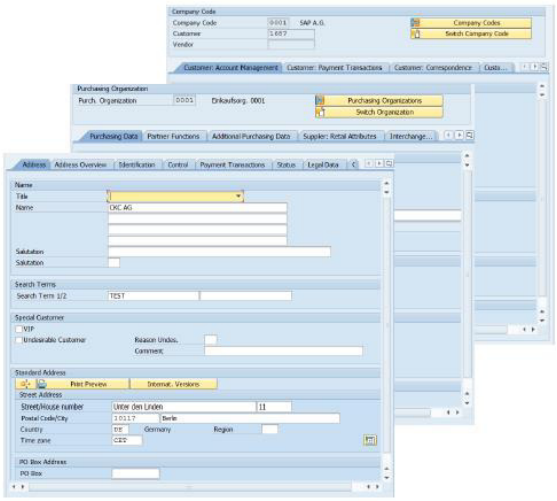


4.2 SAP Master Data Governance , Central Governance

4.2.1 Process Flow

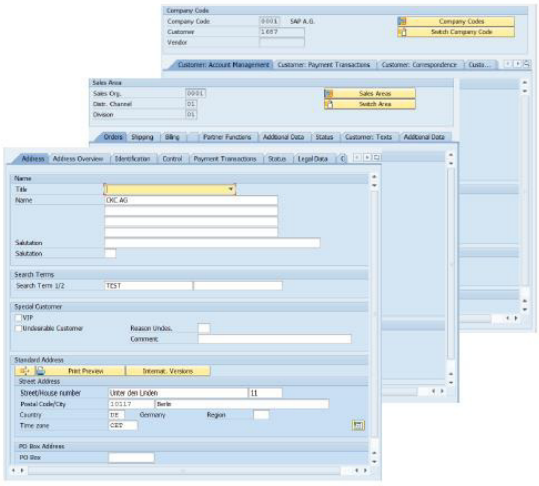


4.2.2 Master Data – A multitude of processesand facets> Supplier Master Data



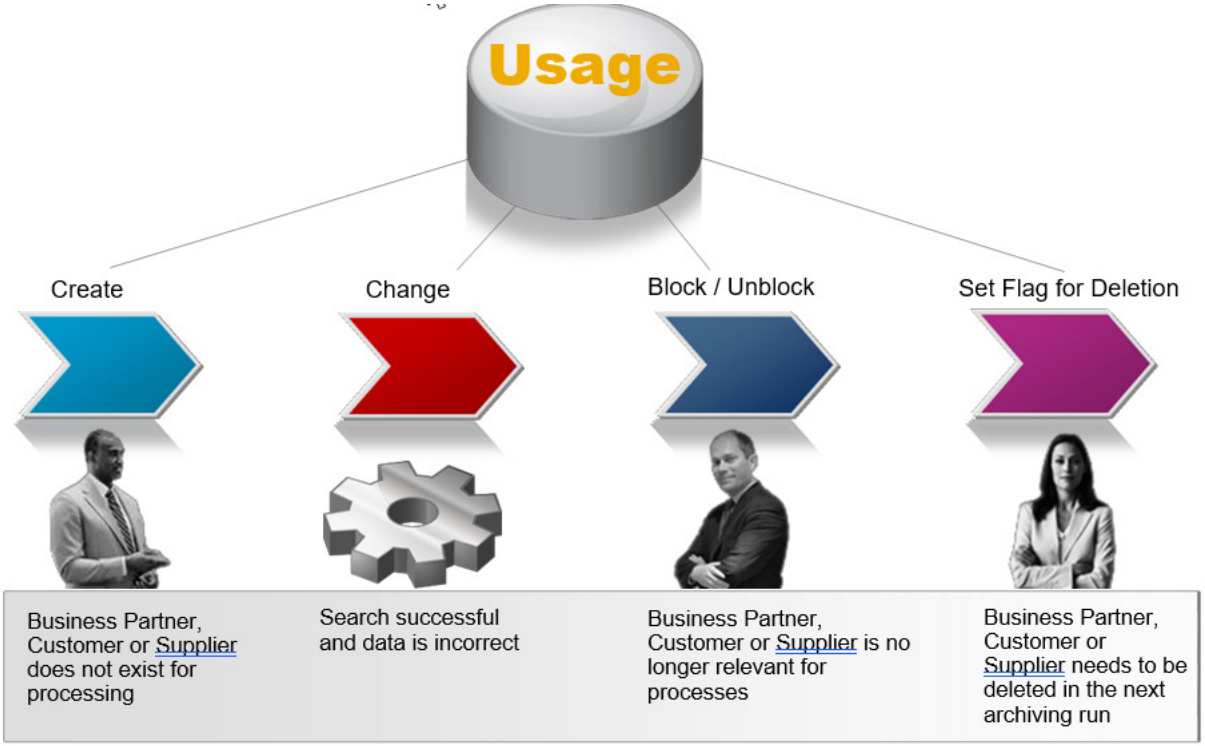
Is there one single person in your organization who knows all this?

4.2.3 Master Data – A multitude of processesand facets > Customer Master Data

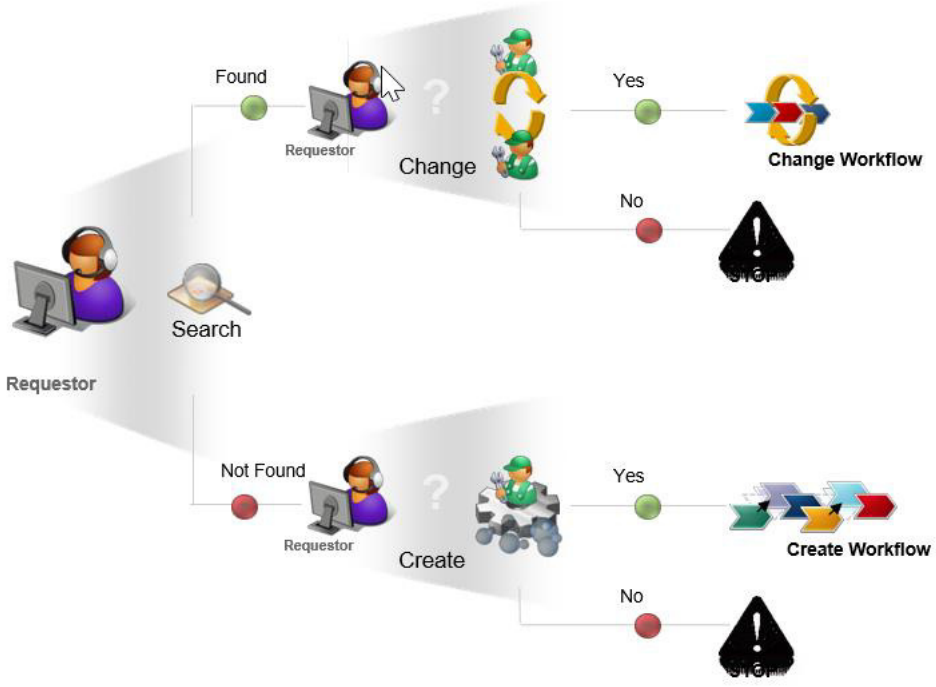


Is there one single person in your organization who knows all this?

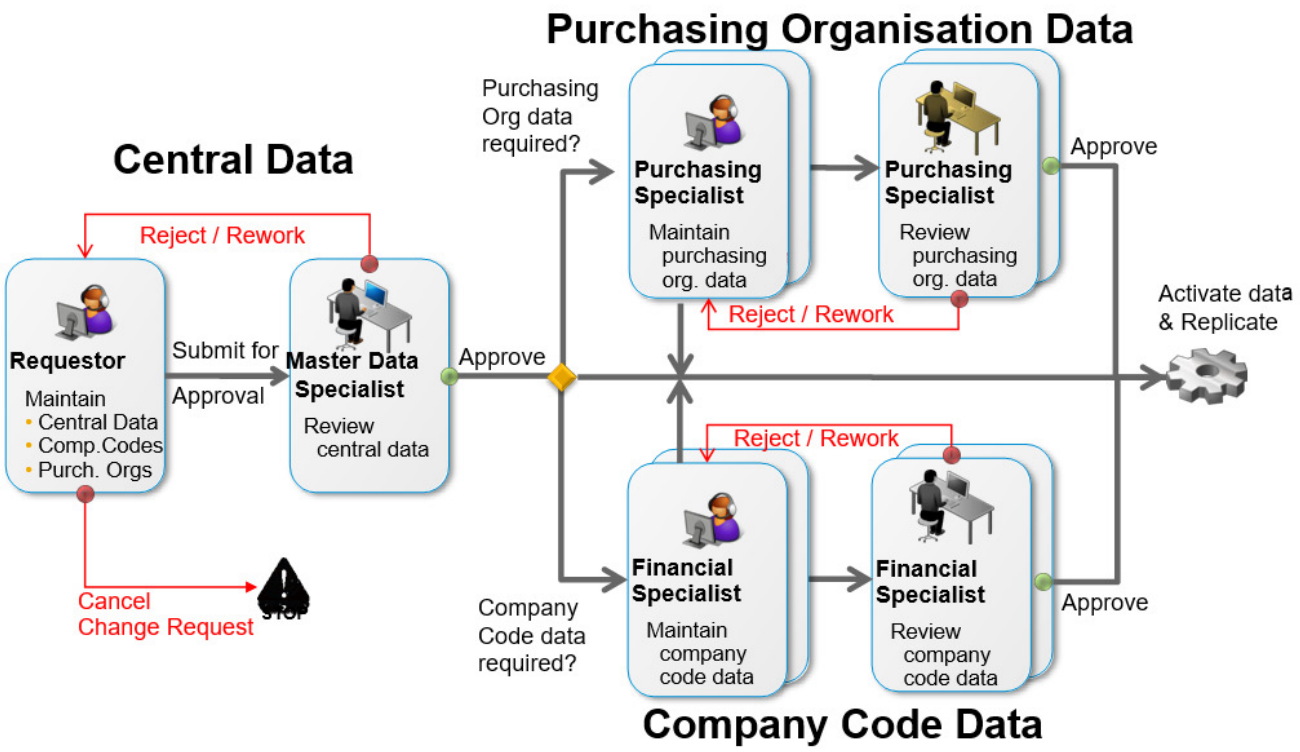
4.3 When to Use Which Process?



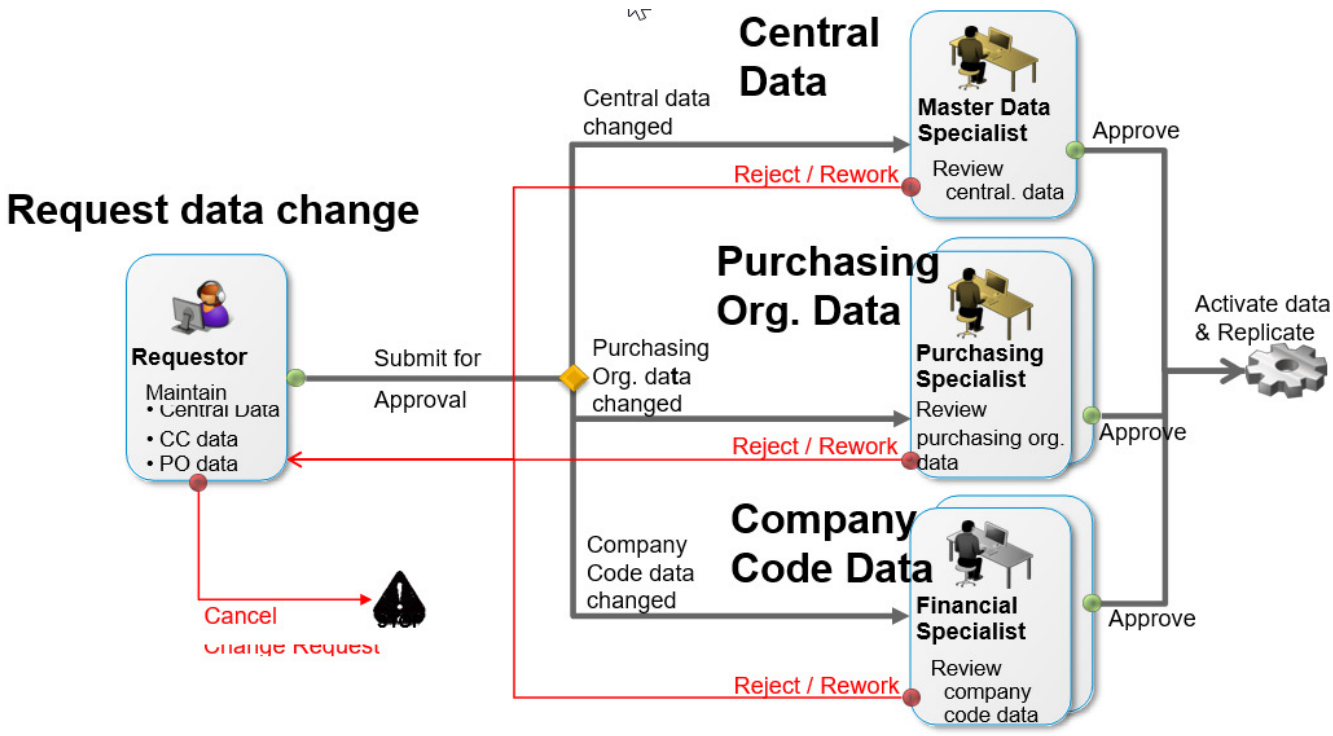
4.3.1 Navigation to Create / Change Process



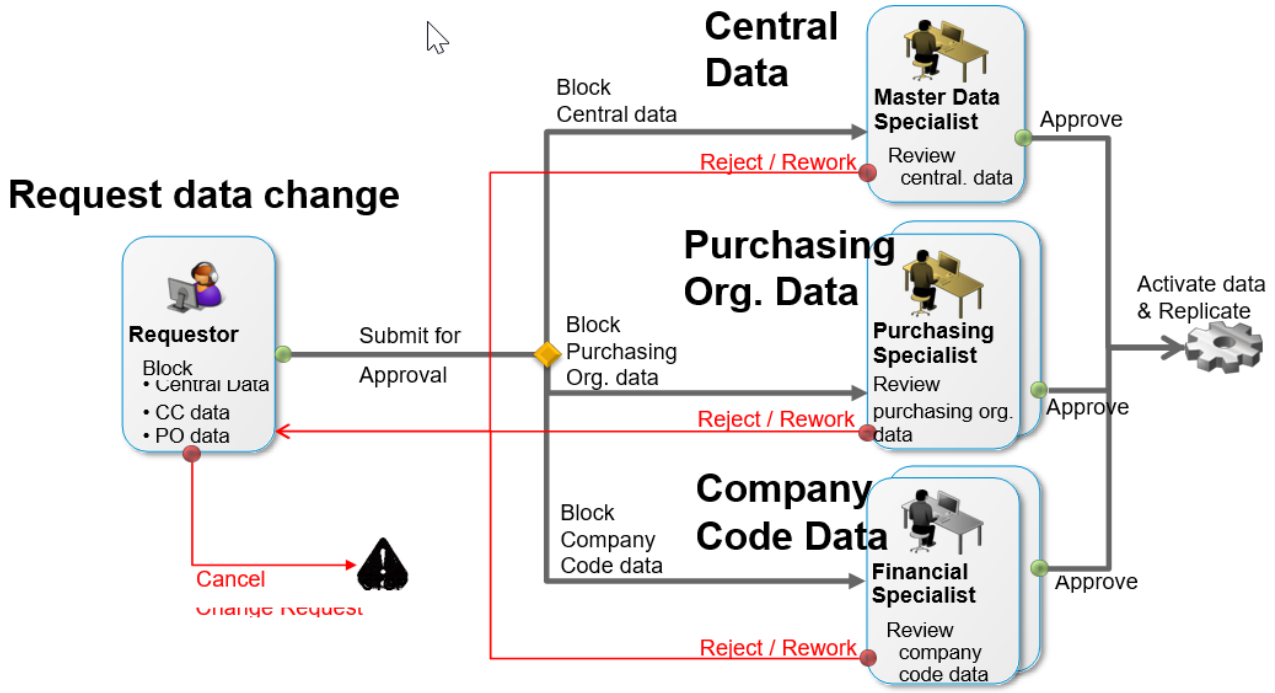
4.3.2 Workflow Template: Create Supplier (Distributed)



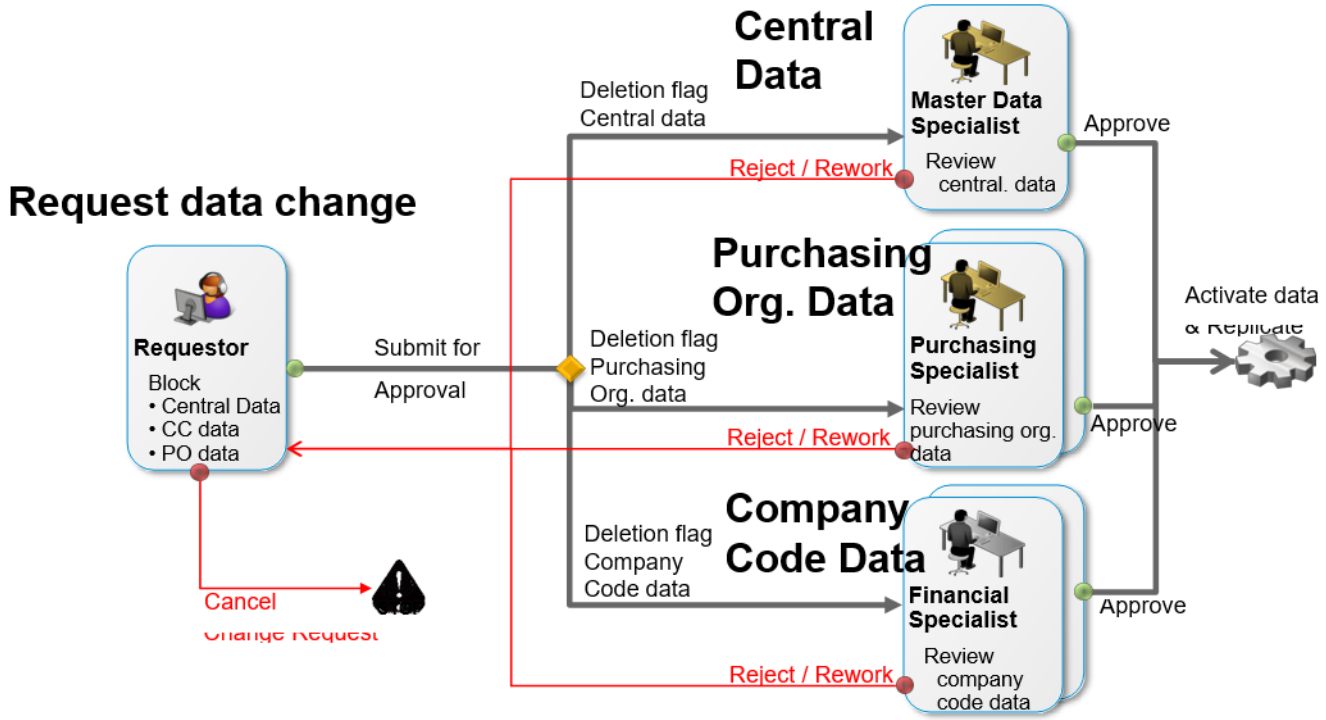
4.3.3 Workflow Template: Change Supplier (Distributed)



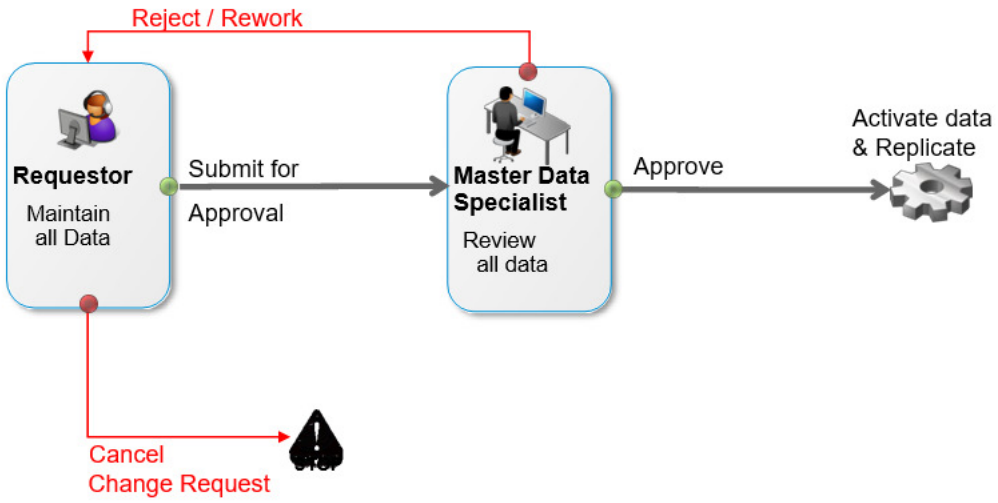
4.3.4 Workflow Template: Block / Unblock Supplier (Distributed)



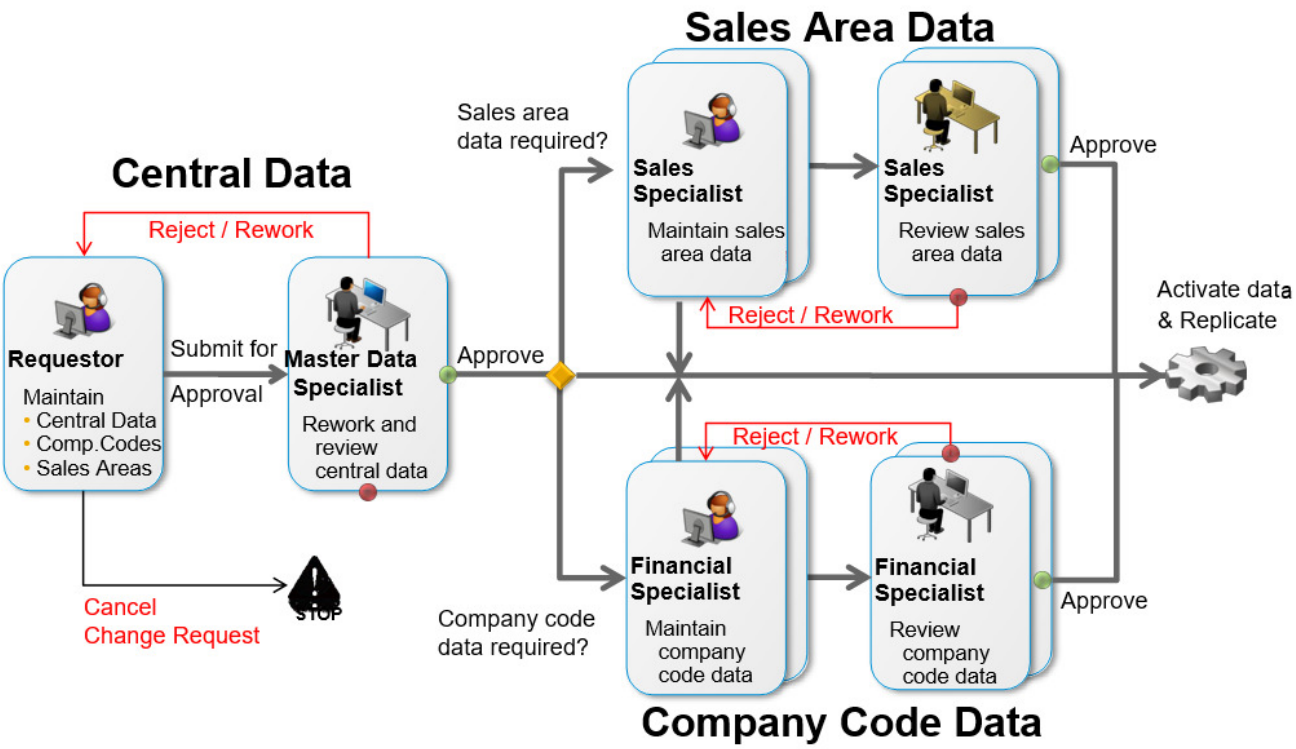
4.3.5 Workflow Template: Set Flag for Deletion Supplier (Distributed)



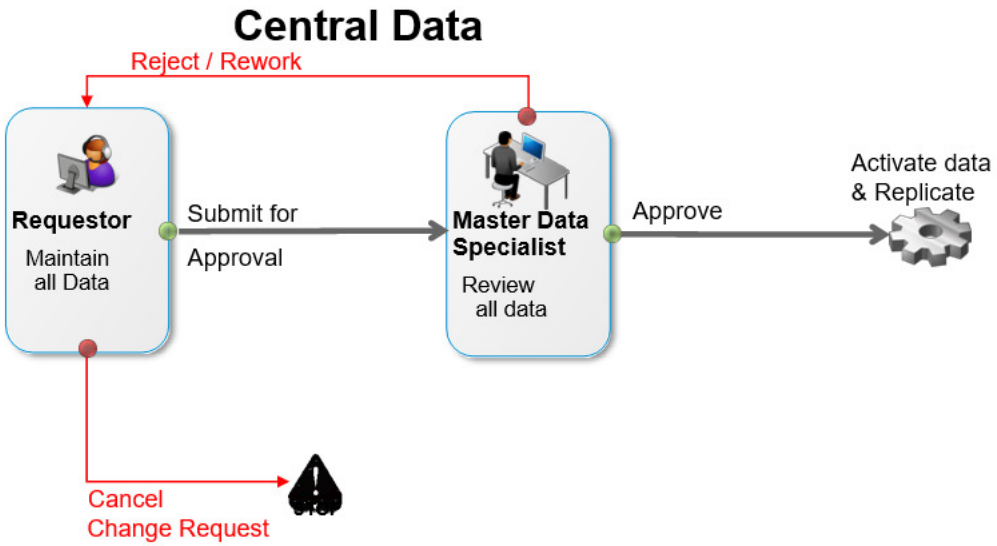
4.3.6 Workflow Template: Create / Change / Block / Set Flag for Deletion Supplier (Simple)



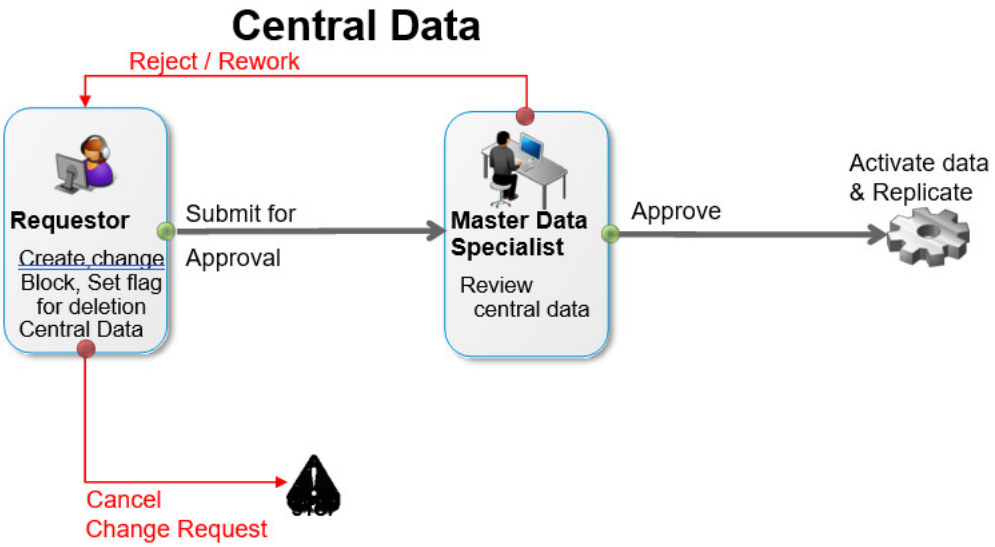
4.3.7 Workflow Template: Create Customer (Distributed)



4.3.8 Workflow Template: Create / Change / Block / Set Flag for Deletion Customer(Simple)



4.3.9 Create / Change / Block / Set Flag for Deletion Business Partner



4.3.10 Workflow Templates out-of-the-box

Title	Process	WF Technique
Simple workflow	Two step workflow • starting with a requester and • completed by an approver also being allowed to change / enrich the data.	SAP Business Workflow (Supplier + Customer) Rule-based Workflow (Business Partner)
Distributed workflow	Separate process steps for Central data, Financials data, and Sales / Purchasing data providing 4-eyes principle for each segment. Process for Financials data and Sales / Purchasing data run in parallel.	SAP Business Workflow
Fiori workflow	Three step workflow with • a requester maintaining a limited number of fields • a data steward changing / completing the data and • another data steward to finally approve the data.	Rule-based Workflow

4.3.11 Multiple Change Requests per Customer / Supplier at the Same Time

Creation of customer / supplier with initial Change Request

- One Change Request for create customer / supplier

Parallel Change Requests for subsequent individual processing

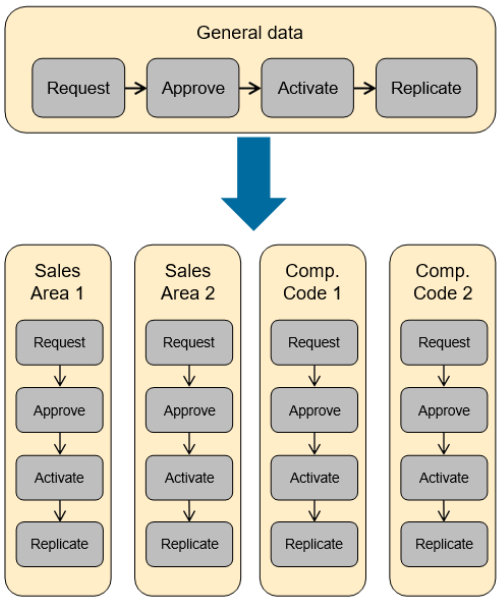
- Parallel Change Requests for extending customer per Sales Area and / or per Company Code
- Parallel Change Request for extending supplier per Purchasing Organization and / or per Company Code

Activation for general data and per organizational unit directly after approval

- Workflow creation for each selected organizational unit, activation after approval

Lock on entity level (interlocking)

- Instead of lock on object level only
- Change request type determines the maintainable fields

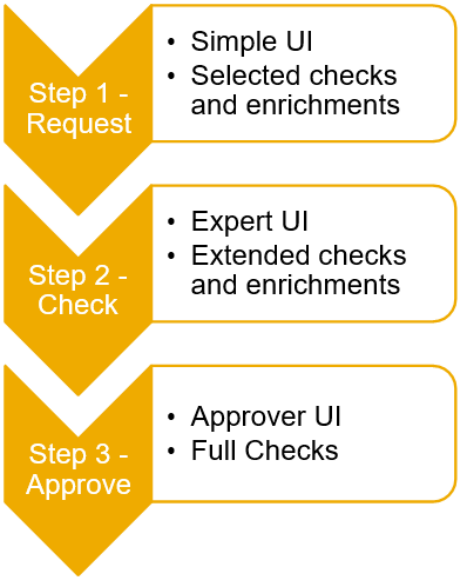


4.3.12 Workflow-Step Dependent Configuration Enables Flexible UI / Process Design

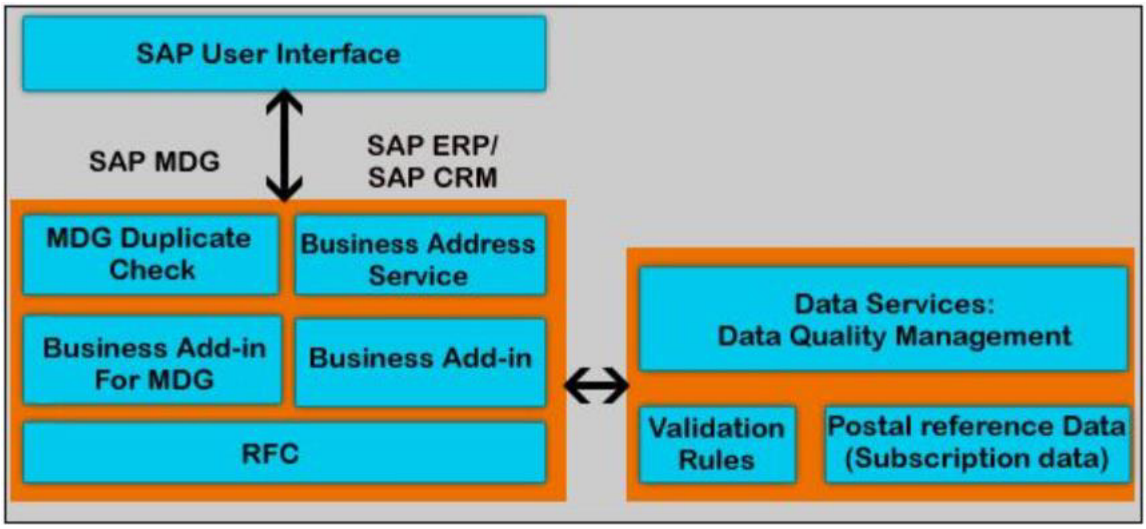
UI Configuration
Provide different User Interface per WF-step based on task / knowhow of the user
Example: Lean Request UI

Validations
Enable or disable checks / enrichments
Example: Switch off required field check and address validation in Lean Request UI

Field properties
Set some data to read-only
Example: Processor for company code 0001 can only maintain these fields



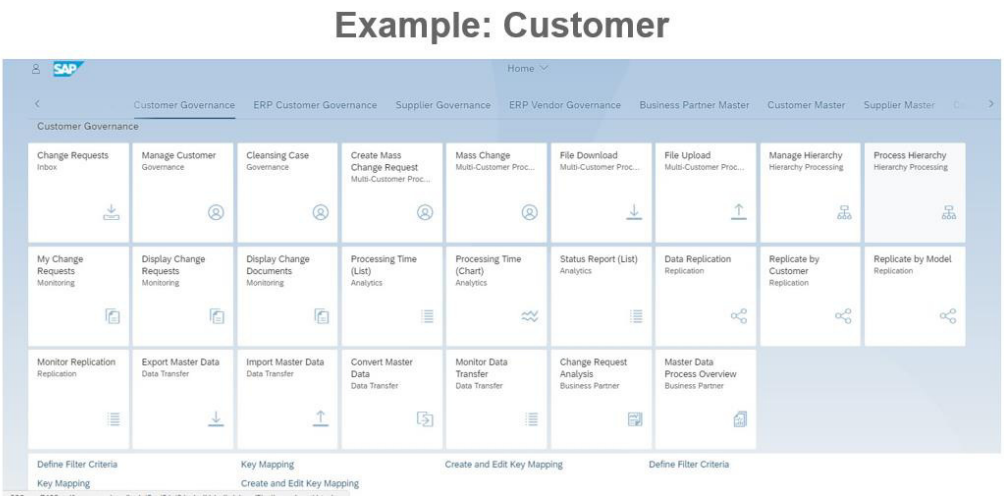
4.4 User Interface



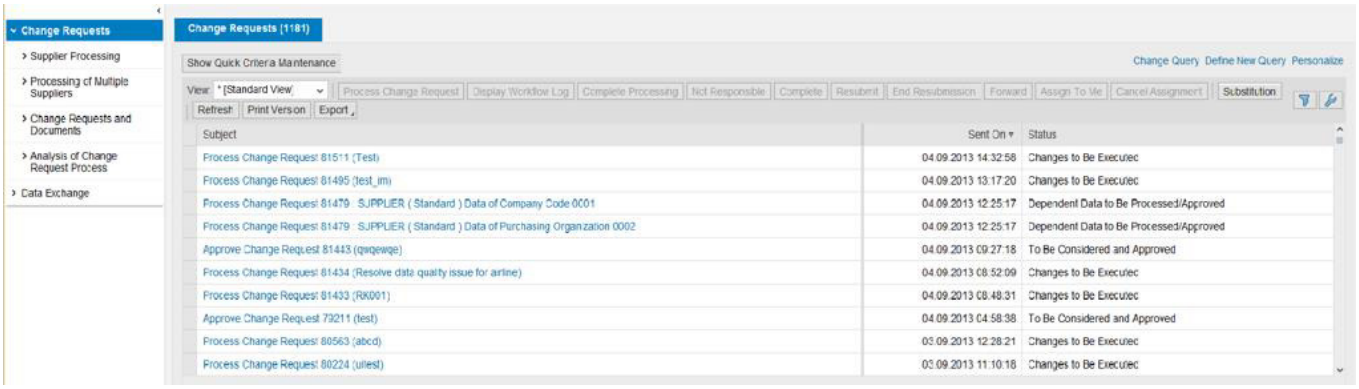
4.4.1 Landing Page for MDG on S/4HANA

All relevant tasks reachable with one click

Easy to configure and personalize



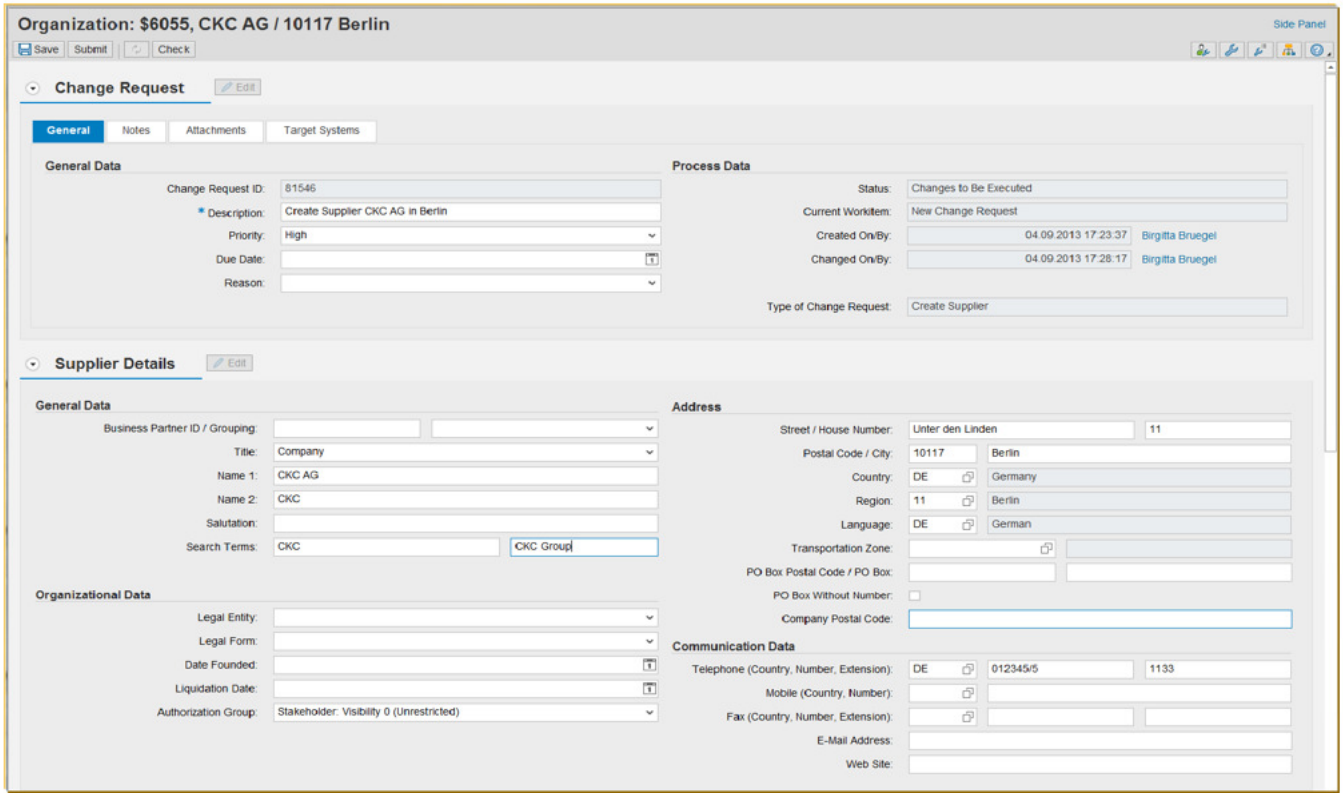
4.4.2 Workflow Inbox



4.4.3 UI Configurations: Overview out-of-the-box configurations

Title	Process	Configuration
BP-UI	UI consisting of all data (central data as well as organizational data) of a BP, customer and / or supplier. Separate pages for BP data, central customer / supplier data as well as organizational data.	WD-App: BS_OVP_BP BS_OVP_BP BS_OVP_CU BS_OVP_SP BS_OVP_BP_ALL
ERP Customer / ERP Vendor like UI	UI consisting of all data (central data as well as organizational data) of a customer or a supplier. Look & feel very similar to ERP Customer / ERP Vendor UI in ERP standard..	WD-App: BS_OVP_BP BS_OVP_CU_CL BS_OVP_SP_VL
Specialized UIs for org. data	UI for Sales, Purchasing, Financial specialists consisting of organizational data for one organizational unit.	WD-App: BS_OVP_BP BS_OVP_CU_ORG BS_OVP_SP_ORG
Fiori UIs	Lean UIs to request new customer, supplier or Business Partner; intended for business users with no / limited MDG knowledge UIs to approve new or changed customer / supplier; intended for experts approving MD changes Implemented as FIORI application running on all devices.	

4.4.4 UI Configurations: BP Maintenance UI



4.4.5 UI Configurations: ERP Customer-like UI

Goal: Support data maintenance in a way being very near to ERP Customer Master

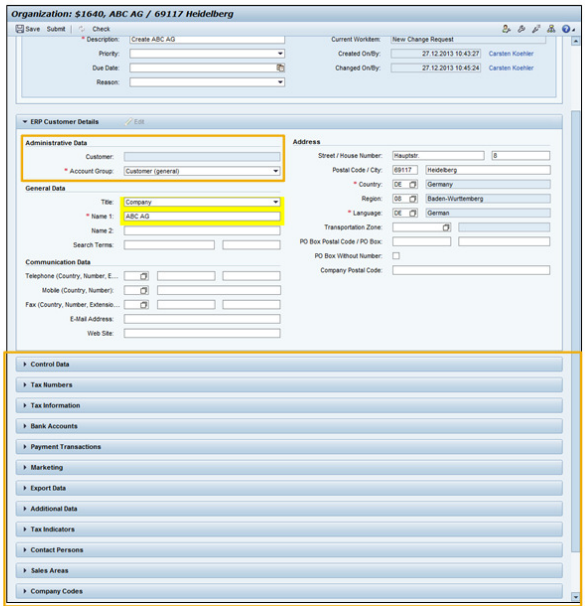
Every new BP automatically becomes an ERP Customer

Display ERP Customer number and select account group instead of BP number and BP grouping

Identifying data (name, address, bank accounts, etc.) and general customer data (Control data, Company codes, Sales areas, etc.) on one screen

BP role derived in the background

Maintain contact persons similar to ERP Customer Master



4.4.6 UI Configurations: ERP Vendor-like UI

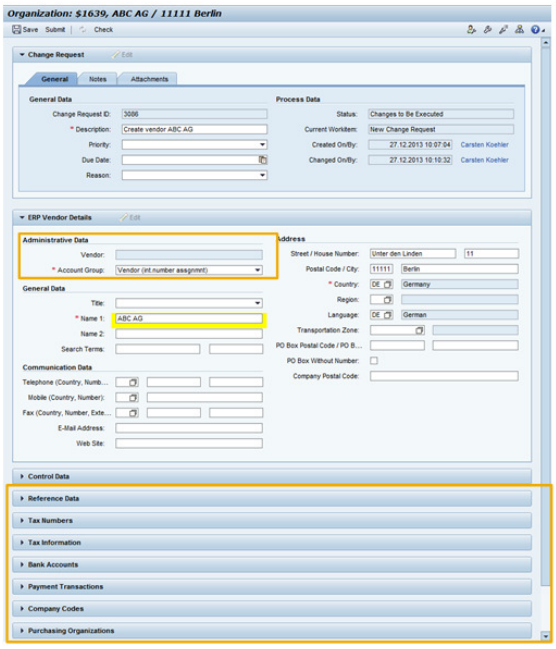
Goal: Support data maintenance in a way being very similar to ERP Vendor Master

Every new BP automatically becomes an ERP Vendor

Display ERP Vendor number and select account group instead of BP number and BP grouping

Identifying data (name, address, bank accounts, etc.) and general vendor data (Control data, Company codes, Purchasing organizations, etc.) on one screen

BP role derived in the background

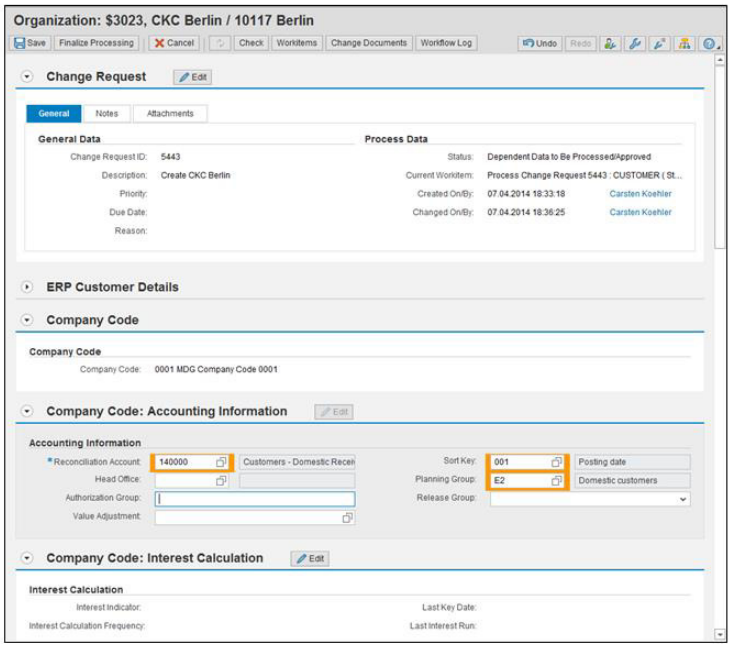


4.4.7 UI Configurations: Specialized UI, Maintain / approve Company Code data

UI contains data of the relevant company code only

Direct navigation from inbox to the UI

Finalize processing of the task on the same page

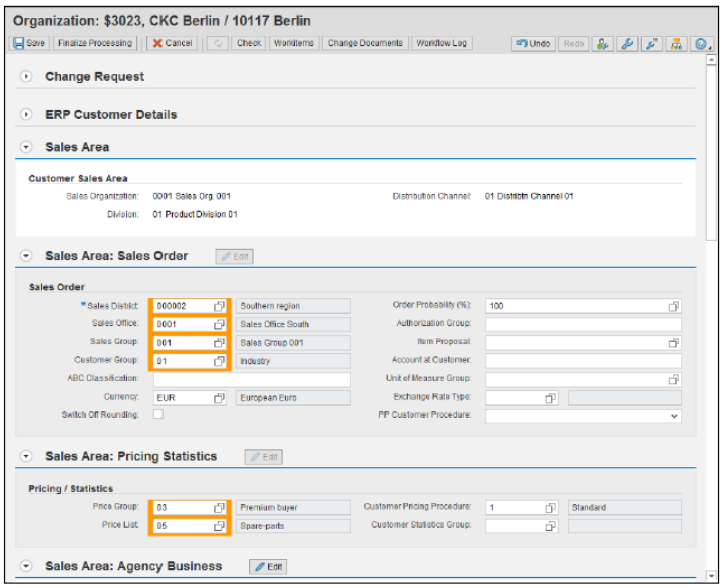


4.4.8 UI Configurations: Specialized UI, Maintain / approve Sales Areadata

UI contains data of the relevant sales area only

Direct navigation from inbox to the UI

Finalize processing of the task on the same page

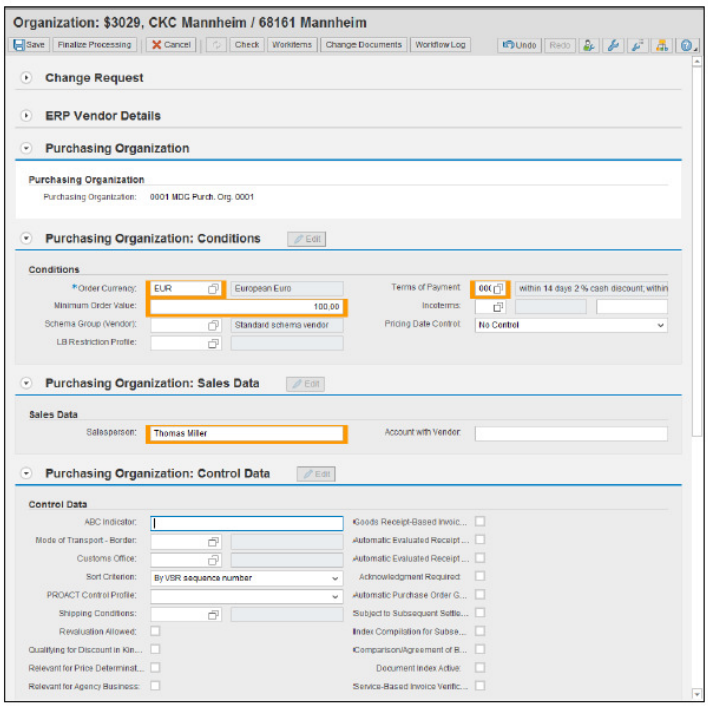


4.4.9 UI Configurations: Specialized UI, Maintain / approve Purchasing Organization data

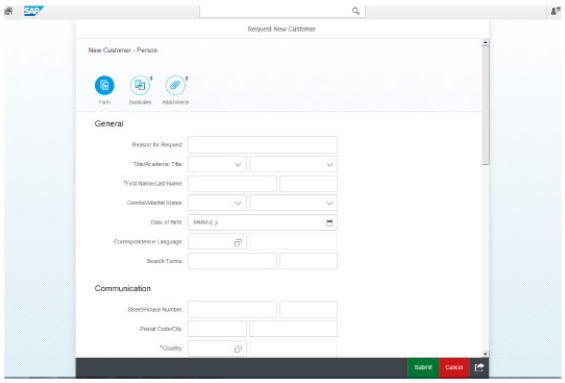
UI contains data of the relevant purchasing organization only

Direct navigation from inbox to the UI

Finalize processing of the task on the same page



4.4.10 UI Configurations: Fiori UI, Lean Request for new Customer[15]



Facts & Figures

Help Portal:	Click here for help documentation
Role:	A general business user
Target Segment:	All employees even having no MDG or SAP knowledge
Functional and Technical Prerequisites:	MDG 7.0 SP02
Device and Browser Support:	Desktop, iPad, Smartphone, IE9, Chrome (check configuration guide for more details of support matrix)

Business/Process Context

- MDG Fiori lean request for Customer is meant for a business user who does not have much expertise in MDG master data handling but still would like to request a new MDG Customer
- Such a business user even being on the move may require a new Customer for his product. Rather than sending an email or phone call to Master Data Expert, he/she can raise a request to create new customer by using Lean MDG Fiori application for customer

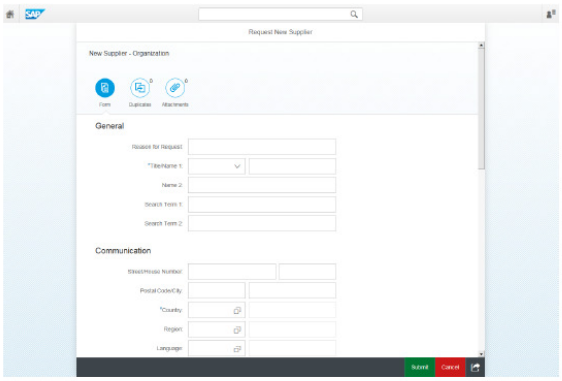
Typical Business Challenges

- Manual efforts (emails, phone calls) to create MDG customer when request comes from users either having no access to MDG or have no MDG / SAP knowledge
- Dependency on few employees having MDG knowledge to trigger MDG master data creation
- High TCO and TCD for improving the user experience for commonly used business functions of SAP MDG software

Features and Capabilities

- An easy, simple and user friendly UI for a user having no master data expertise
- Provides a simple, easy to use and coherent experience across devices – desktop, tablet, or smartphone
- Leverages responsive design, the concept of making apps sensitive to the form factor of devices with single user interface
- Automatic duplicate check when creating a new customer
- Attachment functionality to attach a document with request
- Supports creation of Organization or Person for a customer
- Supports US and Europe address form layouts for customer

4.4.11 UI Configurations: Fiori UI, Lean Request for new Supplier



Facts & Figures

Help Portal:	Click here for help documentation
Role:	A general business user
Target Segment:	All employees even having no MDG or SAP knowledge
Functional and Technical Prerequisites:	MDG 7.0 SP02
Device and Browser Support:	Desktop, iPad, Smartphone, IE9, Chrome (check configuration guide for more details of support matrix)

Business/Process Context

- MDG Fiori lean request for Supplier is meant for a business user who does not have much expertise in MDG master data handling but still would like to request a new MDG Supplier
- Such a business user even being on the move may require a new supplier for his running project. Rather than sending an email or phone call to Master Data Expert, he/she can raise a request to create new supplier by using Lean MDG Fiori application for supplier

Typical Business Challenges

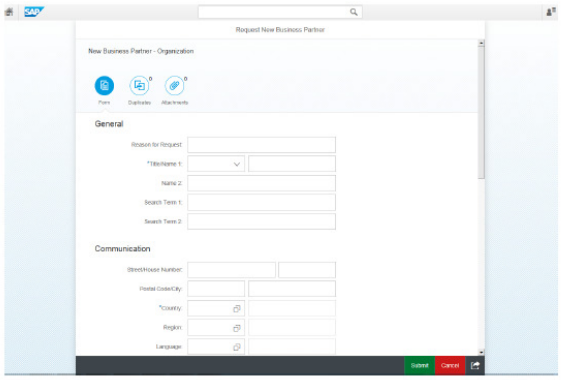
- Manual efforts (emails, phone calls) to create MDG supplier when request comes from users either having no access to MDG or have no MDG / SAP knowledge
- Dependency on few employees having MDG knowledge to trigger MDG master data creation
- High TCO and TCD for improving the user experience for commonly used business functions of SAP MDG software

Features and Capabilities

- An easy, simple and user friendly UI for a user having no master data expertise
- Provides a simple, easy to use and coherent experience across devices – desktop, tablet, or smartphone
- Leverages responsive design, the concept of making apps sensitive to the form factor of devices with single user interface
- Automatic duplicate check when creating a new supplier
- Attachment functionality to attach a document with request
- Supports creation of Organization or Person for a supplier
- Supports US and Europe address form layouts for supplier

[15] Quelle : SAP SE

4.4.12 UI Configurations: Fiori UI, Lean Request for new Business Partner



Facts & Figures

Help Portal:	Click here for help documentation
Role:	A general business user
Target Segment:	All employees even having no MDG or SAP knowledge
Functional and Technical Prerequisites:	MDG 7.0 SP02
Device and Browser Support:	Desktop, iPad, Smartphone, IE9, Chrome (check configuration guide for more details of support matrix)

Business/Process Context

- MDG Fiori lean request for Business Partner is meant for a business user who does not have much expertise in MDG master data handling but still would like to request a new MDG business partner
- Such a business user even being on the move may require a new business partner for his project. Rather than sending an email or phone call to Master Data Expert, he/she can raise a request to create new business partner by using Lean MDG Fiori application for business partner

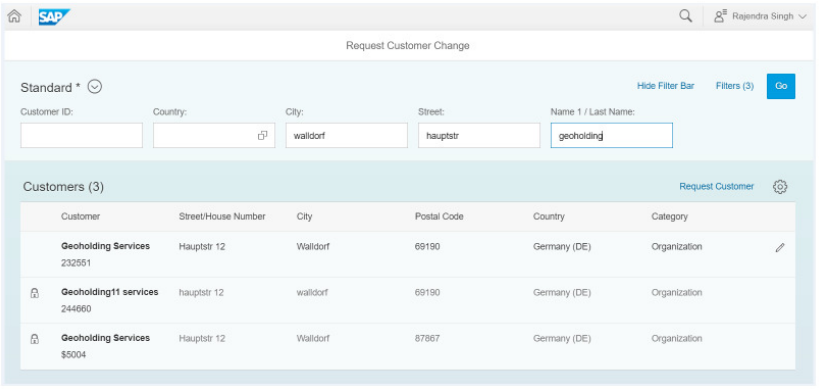
Typical Business Challenges

- Manual efforts (emails, phone calls) to create MDG business partner when request comes from users either having no access to MDG or have no MDG / SAP knowledge
- Dependency on few employees having MDG knowledge to trigger MDG master data creation
- High TCO and TCD for improving the user experience for commonly used business functions of SAP MDG software

Features and Capabilities

- An easy, simple and user friendly UI for a user having no master data expertise
- Provides a simple, easy to use and coherent experience across devices – desktop, tablet, or smartphone
- Leverages responsive design, the concept of making apps sensitive to the form factor of devices with single user interface
- Automatic duplicate check when creating a new business partner
- Attachment functionality to attach a document with request
- Supports Organization or Person for a business partner creation
- Supports US and Europe address layouts for business partner

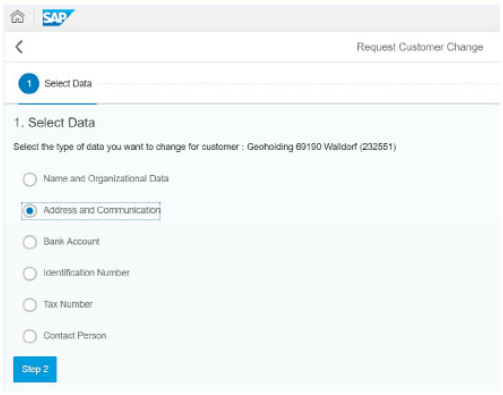
4.4.13 UI Configurations: Fiori UI, Lean Request for change of Customer

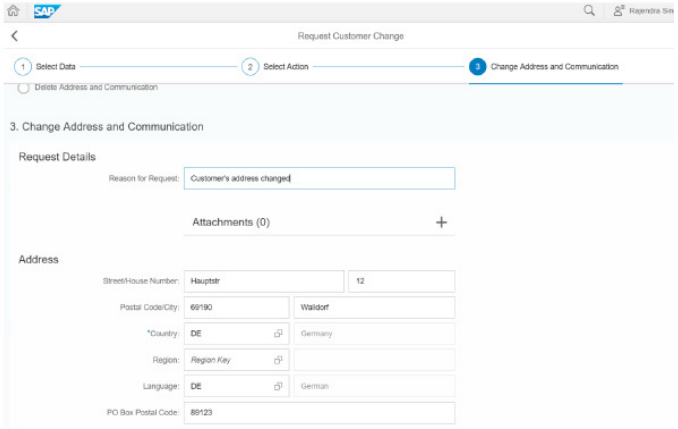


Search UI

- Uses standard search defined in MDG Configuration
- Configure search fields visible on main screen
- More search options via Filters link
- Save predefined filters to reload next time
- Navigate to FIORI app to request a new customer

4.4.14 UI Configurations: Fiori UI, Lean Request for change of Customer (ChangeData)



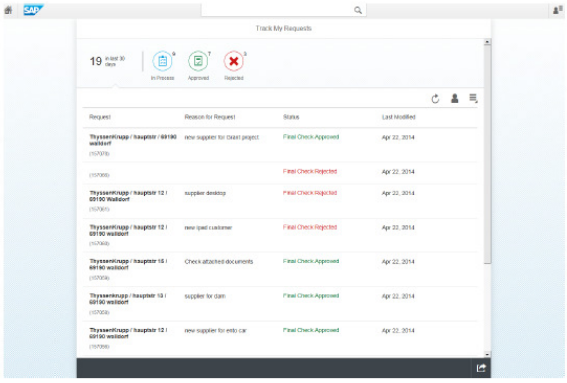


Same data available like in FIORI Lean Request for new Customer

Guided step-by-step UI to support unexperienced user

- **Select Data** to be changed (e.g. Name & Organization Data, Address, Bank Account, etc.)
- **Select Action** to be executed on selected data (e.g. Change selected address)
- **Change data:** Possibility to finalize change or add more changes

4.4.15 UI Configurations: Fiori UI, Track My Request



Facts & Figures

Help Portal:	Click here for help documentation
Role:	A general business user
Target Segment:	All employees even having no MDG or SAP knowledge
Functional and Technical Prerequisites:	MDG 7.0 SP02
Device and Browser Support:	Desktop, iPad, Smartphone, IE9, Chrome (check configuration guide for more details of support matrix)

Business/Process Context

- MDG Fiori application 'Track My Requests' is meant for a business user who does not have much expertise in MDG master data handling
- Such a business user after raising the request for a new MDG master data would like to see the status of his/her request if the request is approved or not

Typical Business Challenges

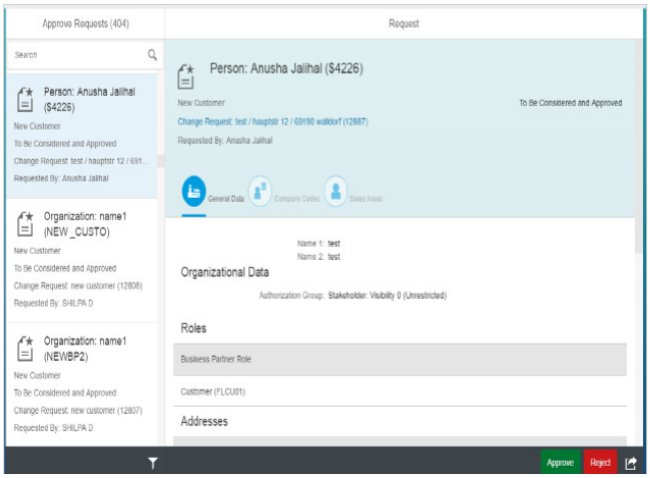
- Manual efforts (emails, phone calls) to contact and check the status of raised request with MDG expert
- Dependency on few employees having MDG knowledge to convey the status of MDG master data requests
- High TCO and TCD for improving the user experience for commonly used business functions of SAP MDG software

Features and Capabilities

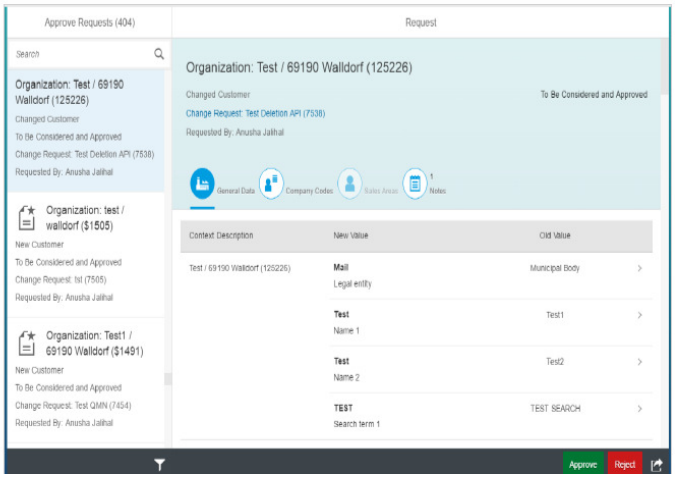
- An easy, simple and user friendly UI for a user having no master data expertise
- Provides a simple, easy to use and coherent experience across devices – desktop, tablet, or smartphone
- Leverages responsive design, the concept of making apps sensitive to the form factor of devices with single user interface
- Filter and sort functionality for selected requests
- Configurable duration for which user wants to see the raised requests

4.4.16 UI Configurations: Fiori UI, Approve Customer

Approve New Customer

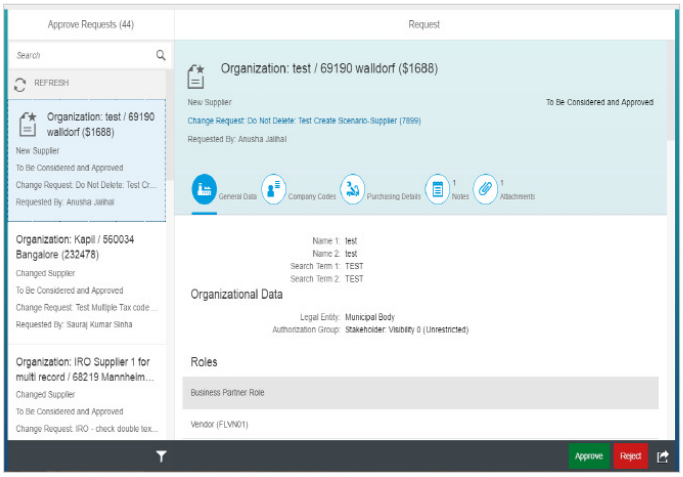


Approve Changed Customer

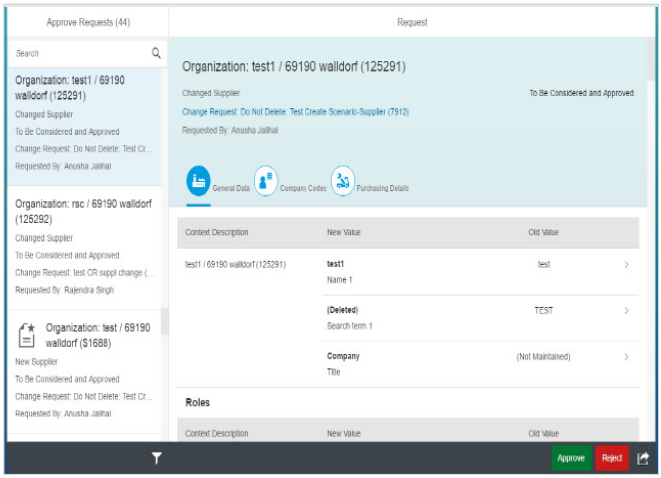


4.4.17 UI Configurations: Fiori UI, Approve Supplier

Approve New Supplier



Approve Changed Supplier



4.4.18 Field changes

Highlight Saved Changes: Supports processor to decide about next action

Highlight Unsaved Changes: Provides information what can be “un-done”.

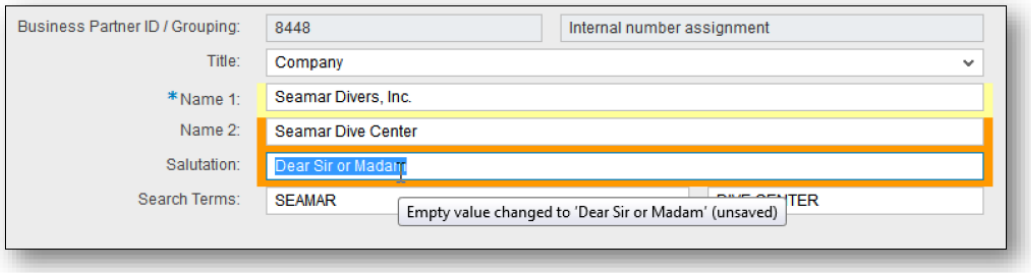
Saved changes and unsaved changes in two different colors.

Highlighting of Table Rows

Highlighted table rows refer to changes that you can only see when navigating from a table row to the details of a dependent entity.

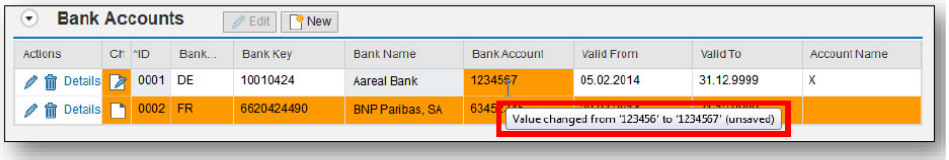
Tooltip Information

Field tooltip shows previous value.

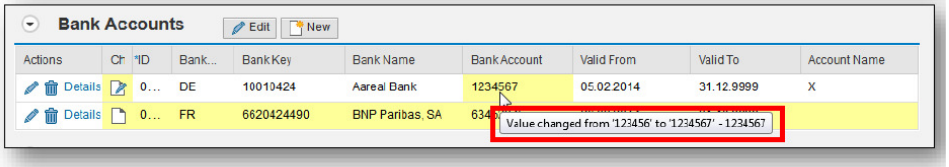


4.4.19 Table changes, Create / Update

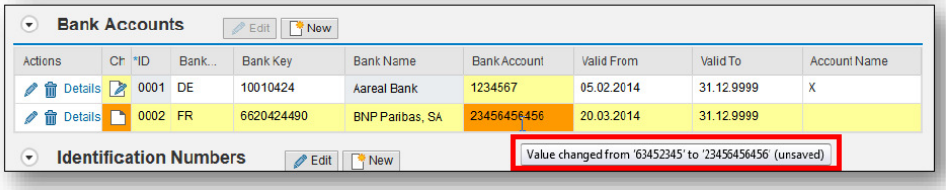
Unsaved changes
1st account changed
2nd account created



Saved changes
1st account changed
2nd account created



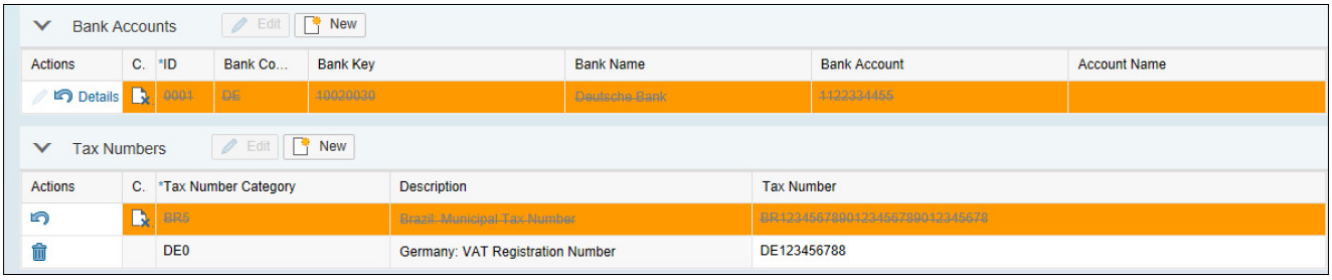
Changes after save
1st account changed and saved
2nd account created, saved, and changed again



4.4.20 Highlight Changes: Table Changes, Delete

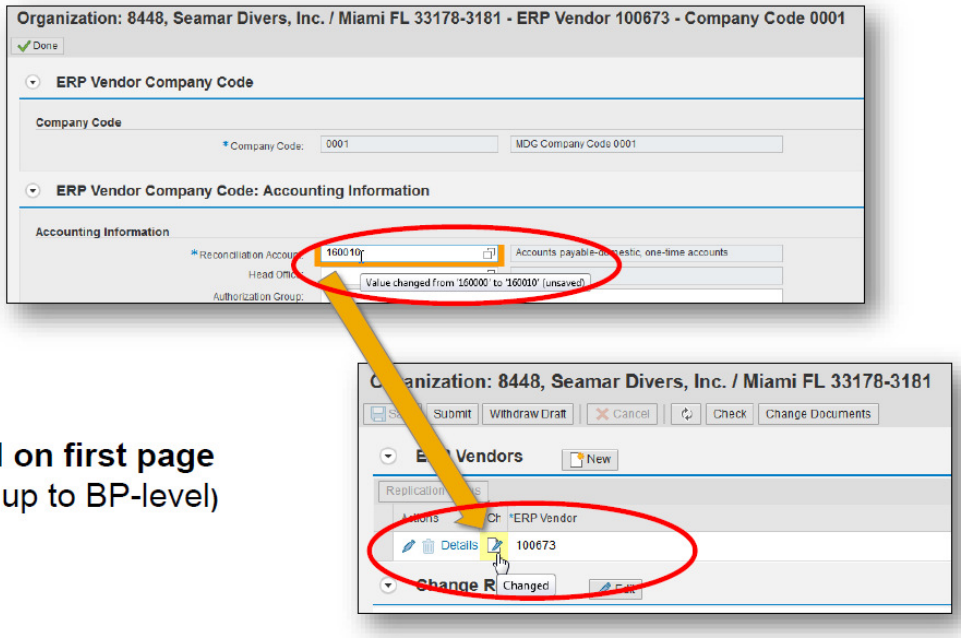
It enables you to highlight deleted values, including deleted rows that are shown in lists, and to restore the latest active version of the deleted record.

The system highlights the relevant rows and displays deleted values as strikethrough, supporting you in quickly identifying items marked for deletion when checking change requests, and offering a new, easy way to restore data.



4.4.21 Changes on Detail Pages

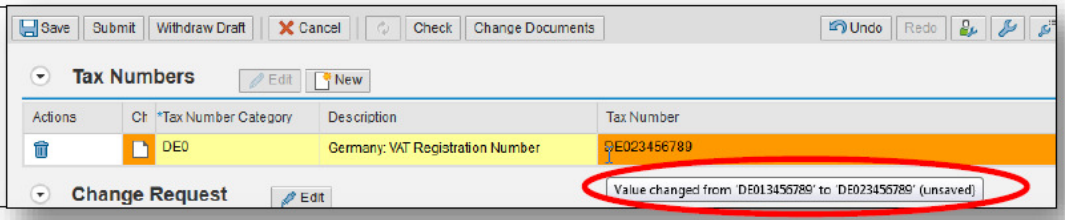
Change in vendor's company code data…



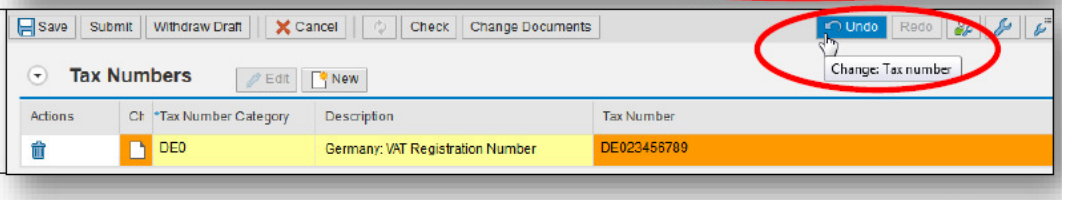
Highlighted on first page
Propagated up to BP-level)

4.4.22 Undo Changes

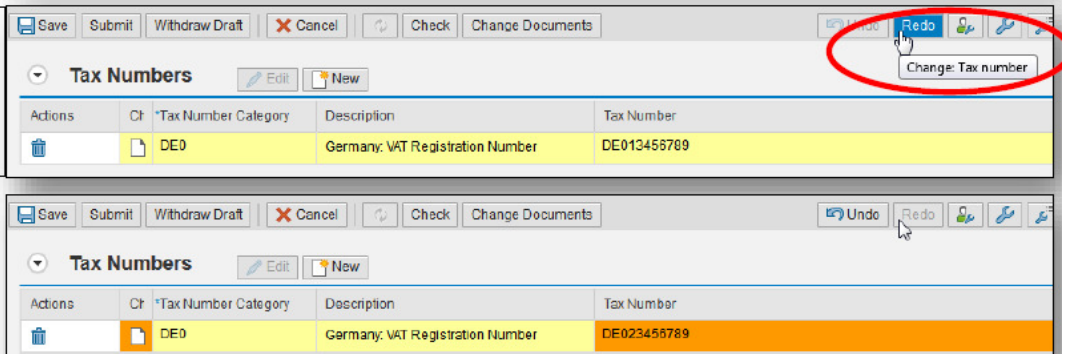
All actions between two roundtrips are collected in one step that can be ***undone***.



You can undo steps (since last save) using ***Undo***.



After having used *Undo*, use ***Redo*** to recover actions.



4.4.23 Create with reference

User Interaction

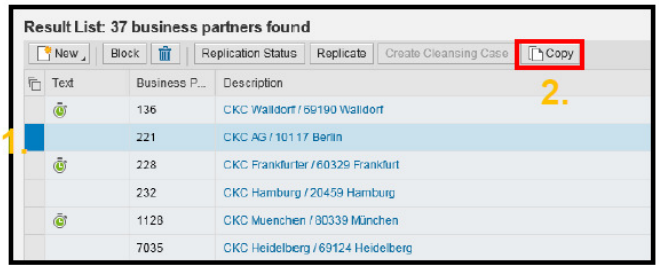
1. Mark business partner to be copied in search result list.
2. Select Copy button
3. UI to create customer /supplier appears, fields to be copied are taken over from source business partner

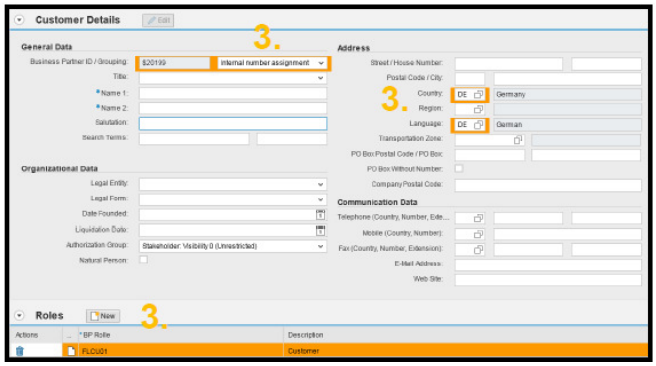
Copy Pattern

Two patterns delivered by SAP:

- Standard: Copy non-individual data only
- Complete: Copy all except fields causing business error, e.g. do not copy 'Date of last dunning notice'

More information see note 2020896.

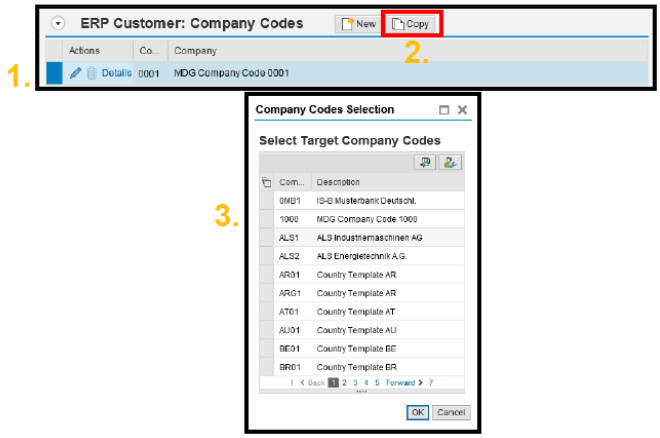




4.4.24 Copy segments

Copy organizational data

1. Mark already available organization
2. Select ‘Copy’ button
3. Select target organization(s) not yet being maintained.
4. Data is copied and can be reworked



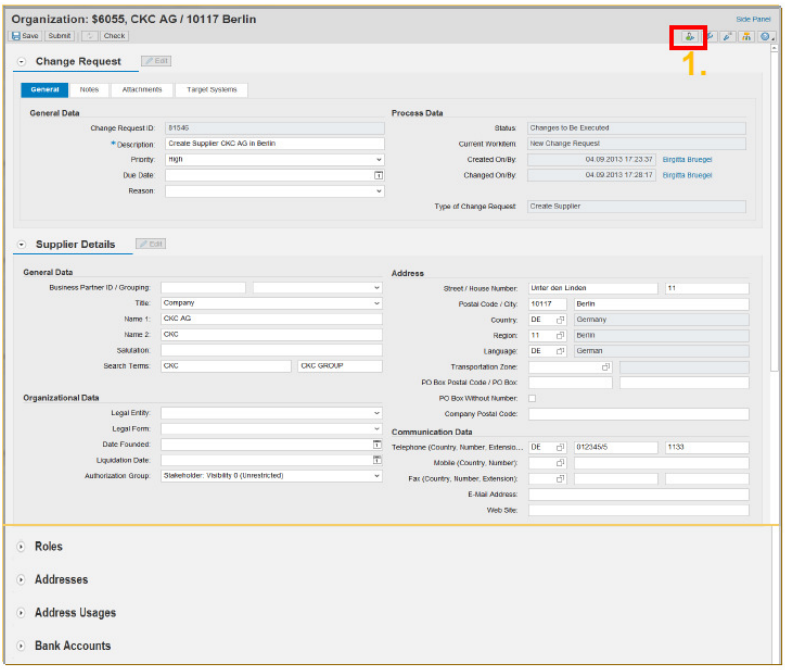
Copy org. data: Entities

- Customer: Company Code data
- Customer: Sales Area data
- Vendor: Company Code data
- Vendor: Purchasing Org. data

Copy assigned Customer / Vendor: User interaction

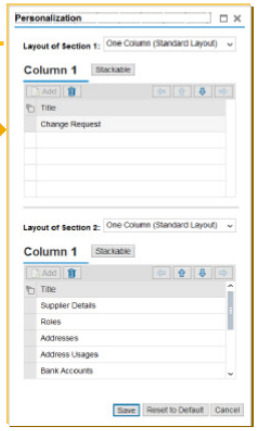
1. Mark assigned Customer / Vendor in UIBB.
2. Choose ‘Copy’ button
3. Customer / Vendor is copied and can be reworked

4.4.25 Flexible UI via Configuration / Personalization: Call personalization



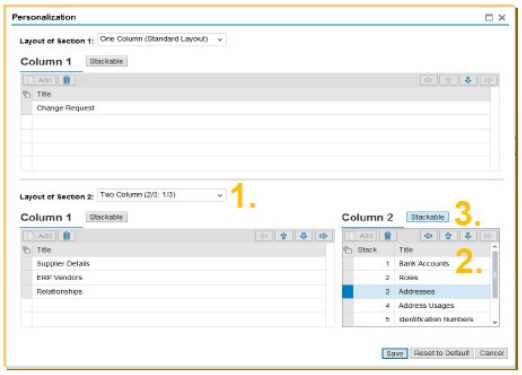
UI easy to configure and personalize

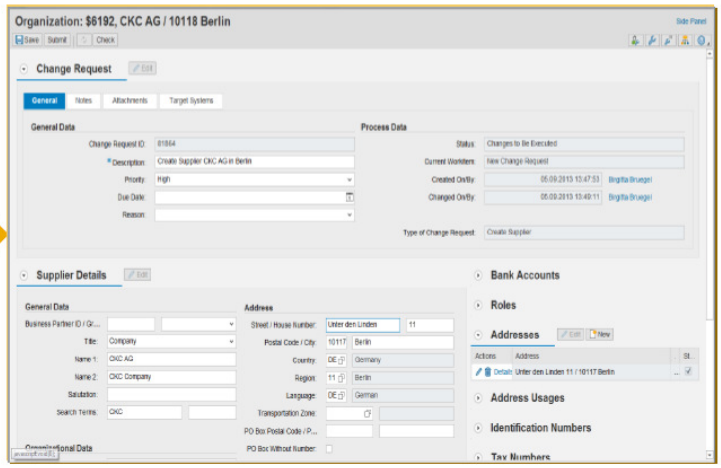
1. Press ‘Personalize’ button
2. Personalization popup appears



4.4.26 Flexible UI via Configuration / Personalization: Change layout

1. Multi column (up to 3) design
2. Change sequence
3. Stacked UIBBs (Tabs)
4. Expand / compress / suppress blocks

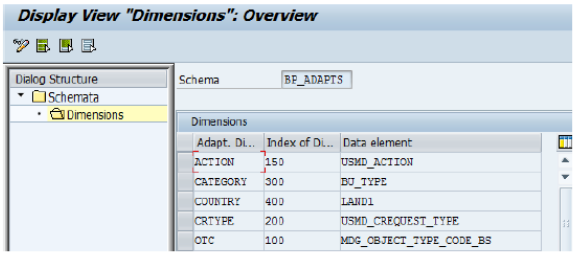




4.4.27 Flexible UI via Context Based Adaptations (CBA)

Integration of Context Based Adaptations:

- Pre-delivered CBA dimensions:



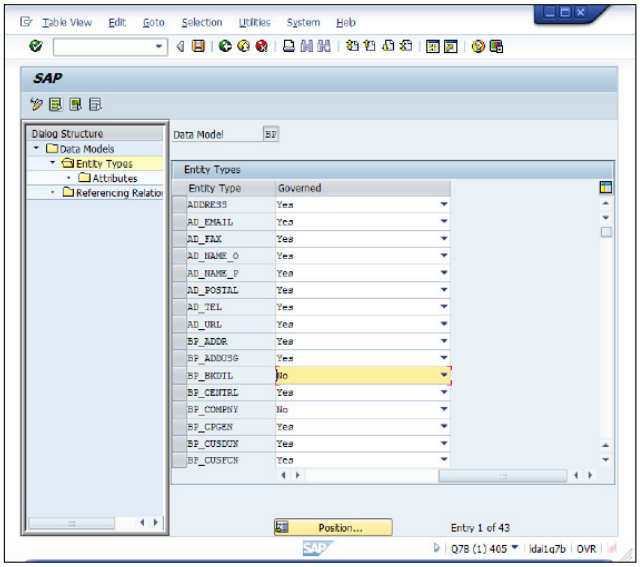
- Examples of pre-delivered dimension usages:
  - 'Category' used for BP category specific attributes like name and e.g. birth date / marital status for persons
  - 'Country' is used for country specific address layout

- CBA schema can be reused and enhanced with custom-defined dimensions
- Delivered CBA schema can be substituted by custom-defined schema
- Customizing: Configure Properties of Change Request Step, if the CBA is not used in the UI Configuration

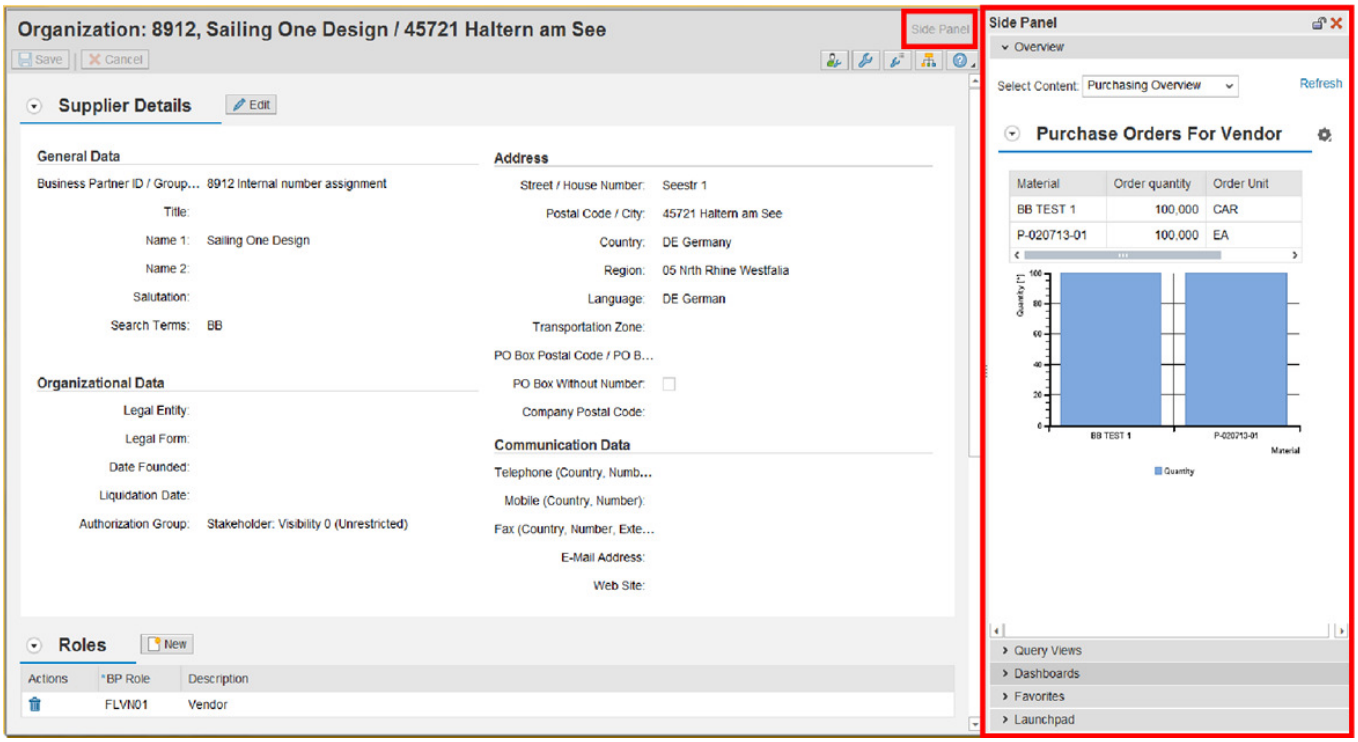
4.4.28 Governance Scope - Deactivate Entity Types / Fields

Governance Scope

- Only parts of the data model shall be put under governance
- Entity types / fields can be excluded from governance
- Excluded entity types / fields
  - can not be changed within a Change Request
  - are displayed only within the UI; can be suppressed by changing the UI configuration
  - can still be loaded to Active Area but not via CR / Staging
  - Can easily be re-added to the governance scope afterwards

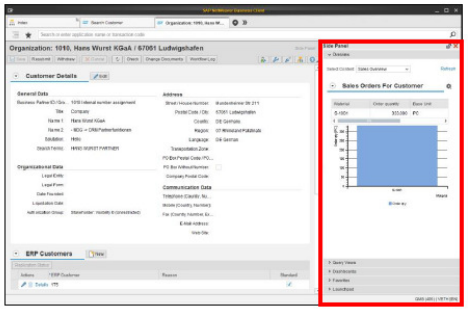


4.4.29 Integrate Additional Information via BCV Side Panel: Open Side Panel

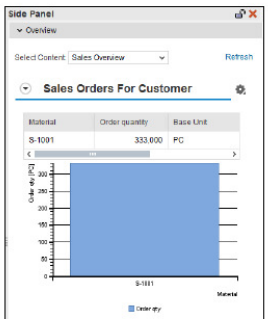


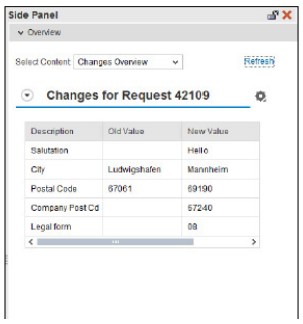
4.4.30 Integrate Additional Information via BCV Side Panel: Content for MDGCustomer

- Reuse of generic component Business Context Viewer (BCV)
- Customers can easily add content to the Side Panel



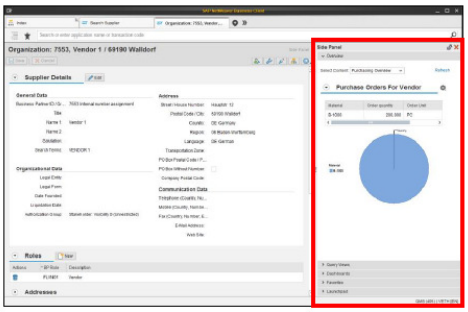
- ***Sales Overview***: Sales orders created for the current customer number.
- ***Changes Overview***: Changes raised by the current MDG change request.



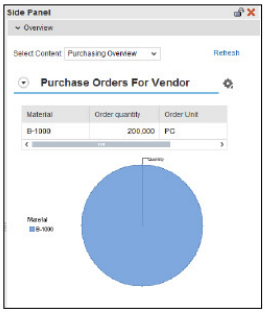


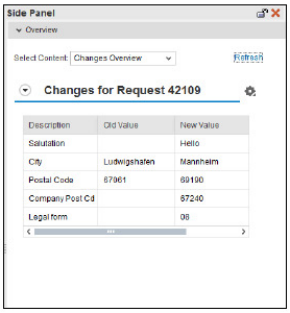
4.4.31 Integrate Additional Information via BCV Side Panel: Content for MDGSupplier

- Reuse of generic component Business Context Viewer (BCV)
- Customers can easily add content to the Side Panel



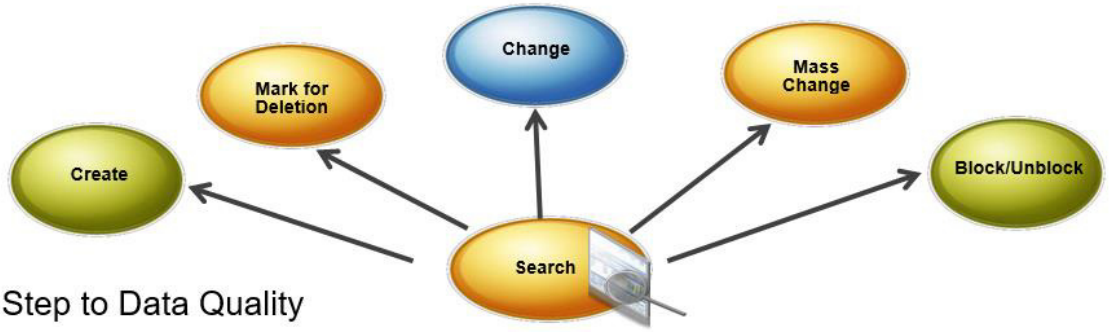
- ***Purchasing Overview***: Purchase orders created for the current vendor number.
- ***Changes Overview***: Changes raised by the current MDG change request.





4.5 Data Quality

4.5.1 Search Overview



Search – First Step to Data Quality

Search is the initial action, if a user wants to “do something” with the data.

Several Flavors Supported via Search Provider Concept

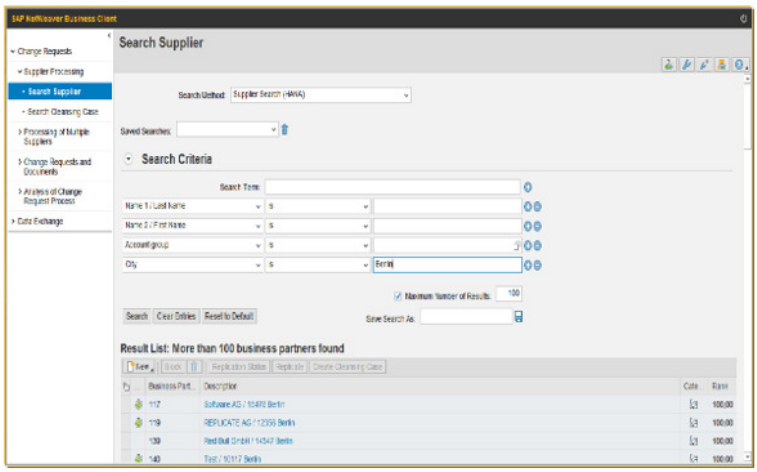
Standard providers:

- DB-based search
- Enterprise Search
- Fuzzy search via BAS-DES (e.g. SAP Data Quality Management)
- Hana Search

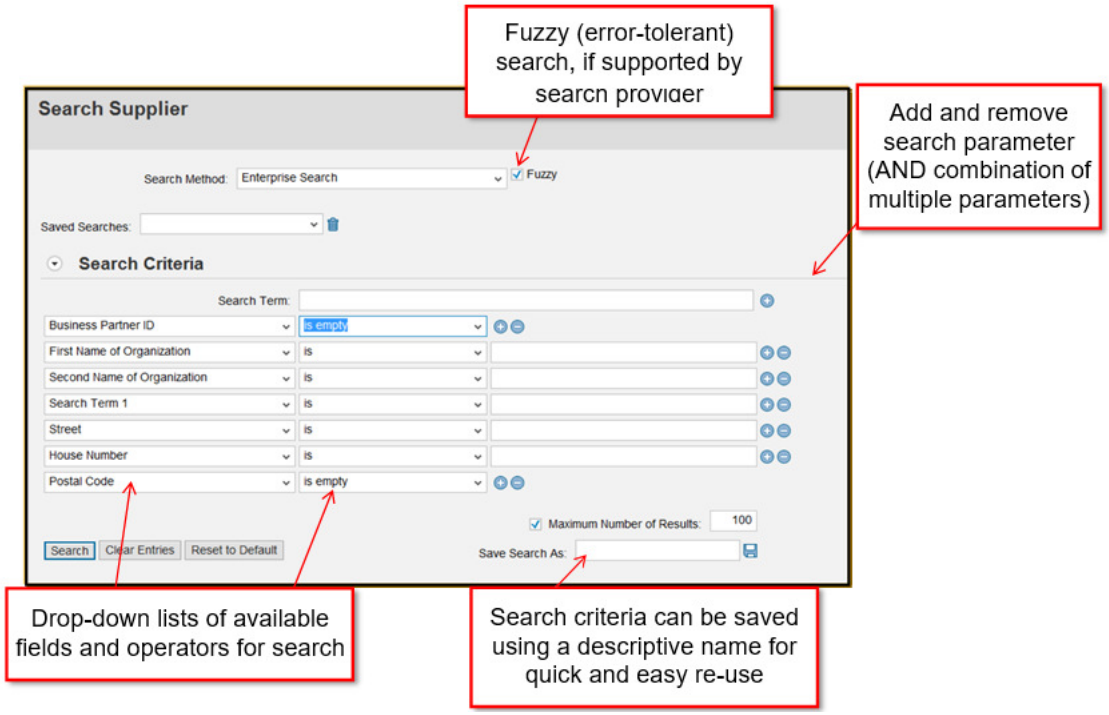
4.5.2 Search Capabilities

Consistent search UI containing all search capabilities

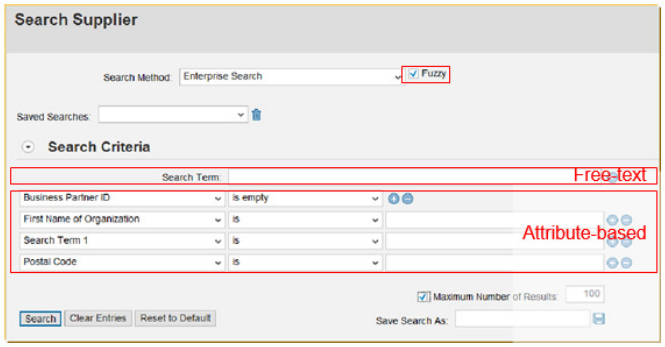
- Search methods easy to configure, more search methods can be added
- Search for hits within active and inactive data
- Result can be downloaded to Excel
- Result list can be personalized



4.5.3 Search Options



4.5.4 Enterprise Search



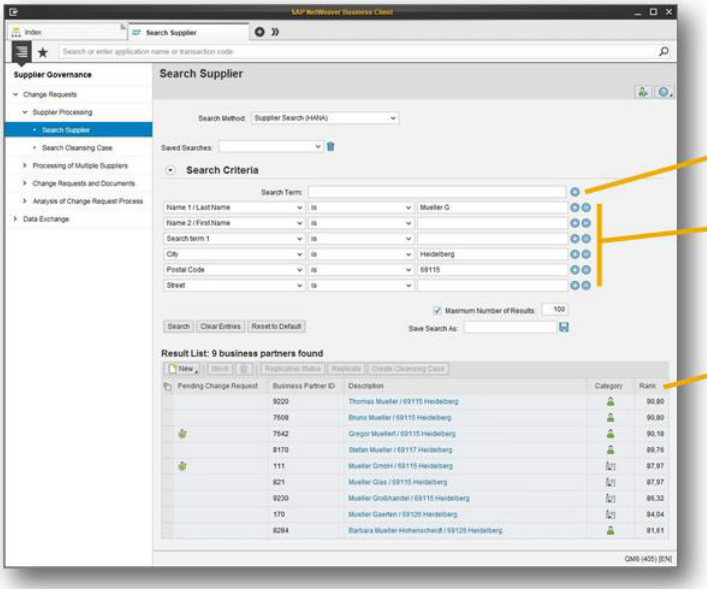
Advantages of using Enterprise Search (compared with database search)

- Free-text search
- Fuzzy (error-tolerant) search

Using Enterprise Search with MDG provides

- Free-text search across many attributes
- Attribute-based search in specific fields only
- Search active data only to limit search to active area
- Fuzzy search checkbox to enable error-tolerant search mode

4.5.5 HANA Fuzzy Search capabilities



Search terms and attribute-based search

- Using a single term "free-style search" against all attributes
- Search per attribute with dedicated thresholds

Calculated similarity rank

- For each positive hit, HANA Fuzzy Search calculates a similarity score weighted across all attributes

4.5.6 Duplicate Check, Validations, Address Enrichment

Prevent creation of duplicates for increased effectiveness and efficiency

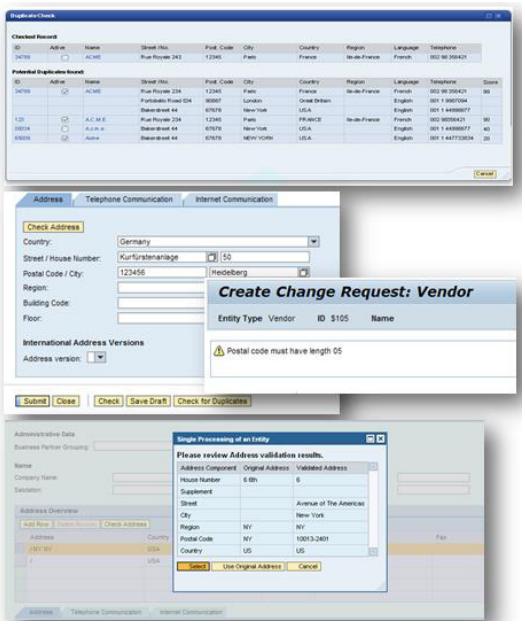
- Checked early and embedded in the process
- High detection quality of matching using Enterprise Search or HANA based search

Validations

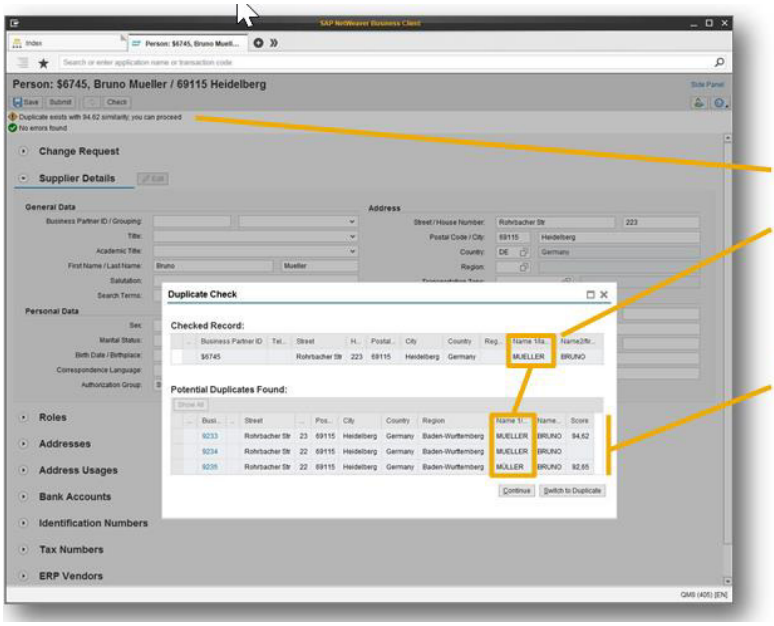
- Re-use of existing validation logic in S/4HANA
- Custom validations can be modeled and programmed (e.g. code lists, simple checks, or modeled rules via BRF+)

Address Enrichment

- Simple check and selection lists
- Integration with content provided by SAP Data Services
- Automatically adding Tax Jurisdiction Code re-using existing interfaces / providers



4.5.7 Duplicate detection via HANA Fuzzy Search



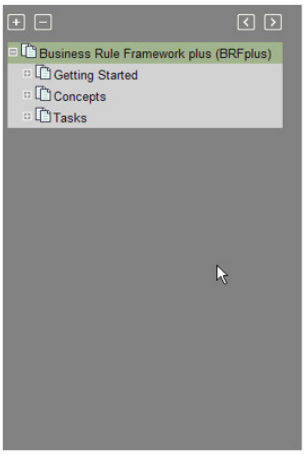
Prevent users from creating redundant master data

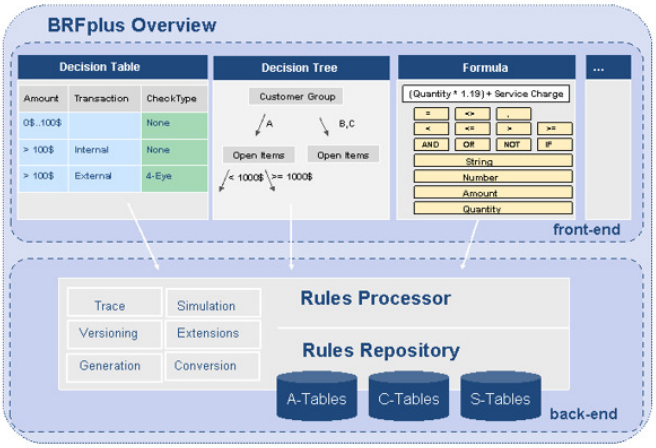
- Find duplicates early in the process
- High detection quality using SAP HANA fuzzy search and matching

Help users to decide if duplicates exist

- Similarity score used to identify and sort potential duplicates

4.5.8 Validations via BRFplus





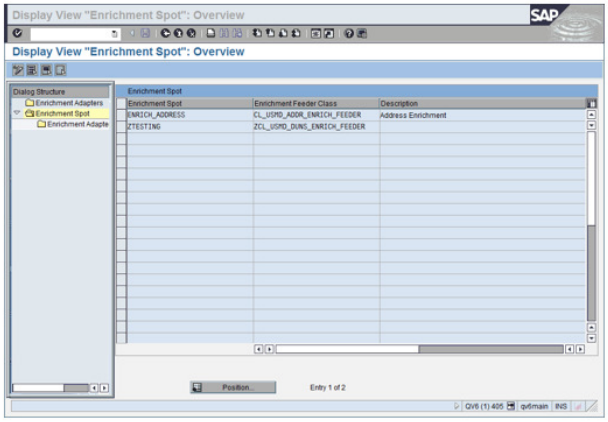
For general information on how to work with BRFplus, see

- SAP Help Portal
  - ☐ Search for “Business Rule Framework plus”
- SAP Documentation

4.5.9 Framework for Data Enrichment Spots

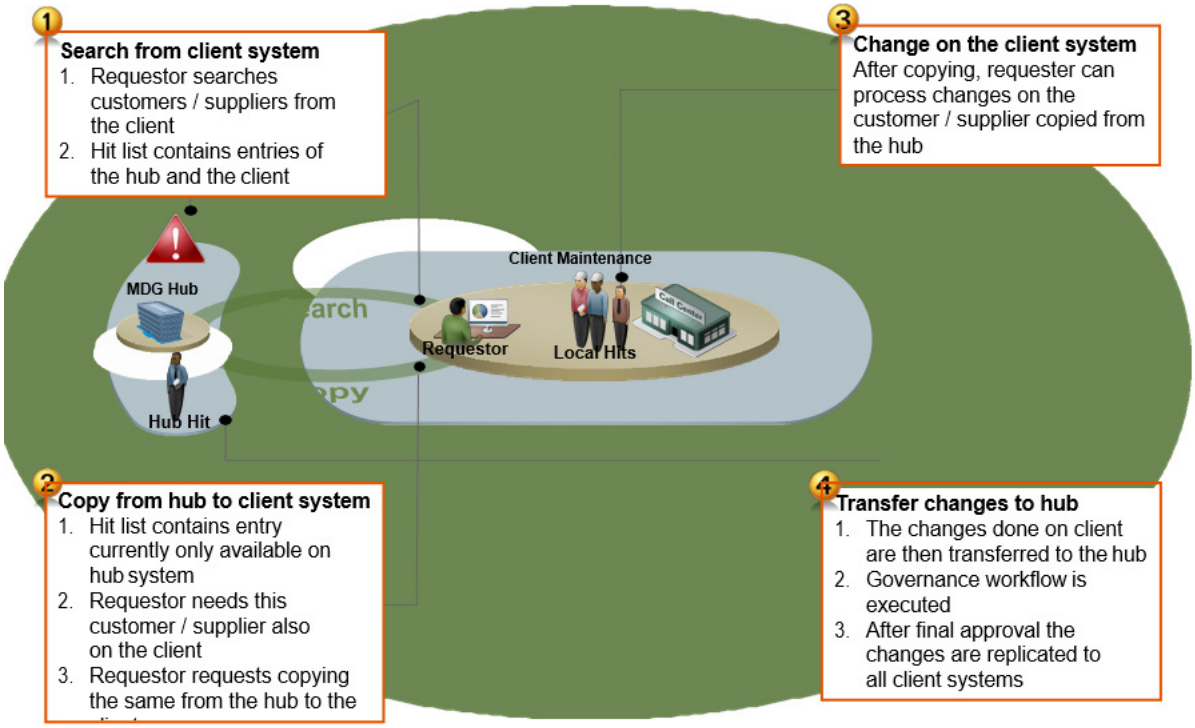
Data Enrichment Spots

- Flexible framework to define enrichment spots
- Is used by SAP for e.g. Address Validation / Enrichment and will be used for further spots in future
- Can be used by customers to define further enhancement spots (e.g. D&B services)



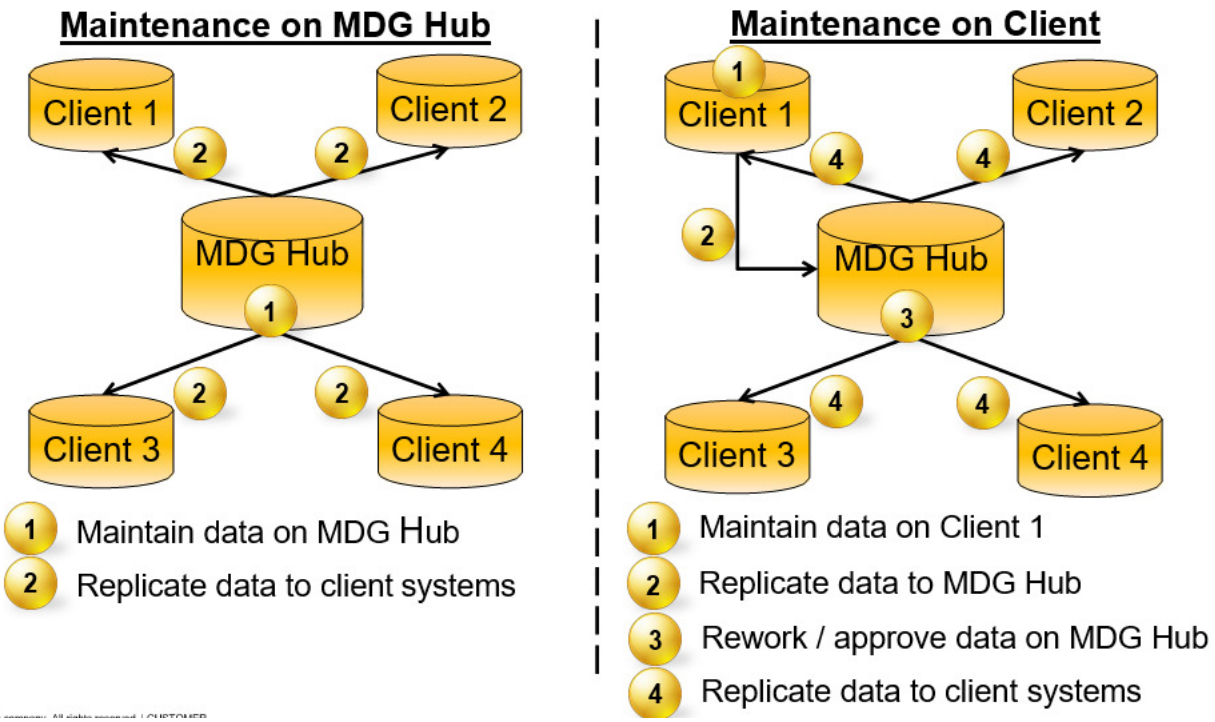
4.6 Client Maintenance

4.6.1 Process Steps



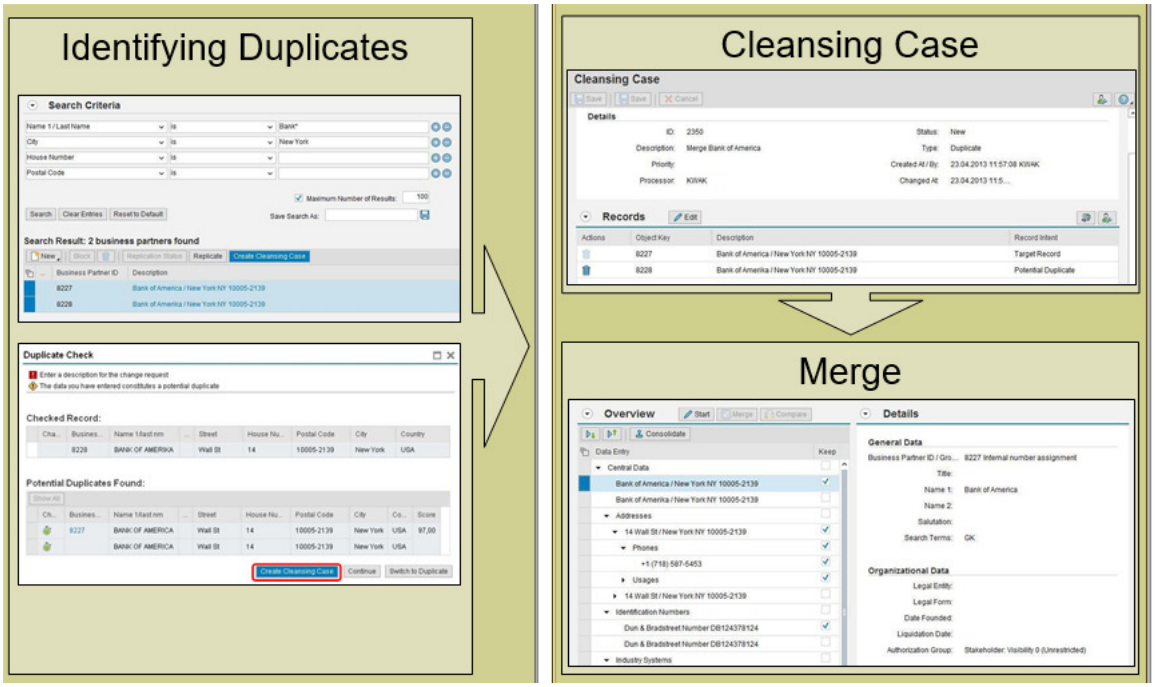
4.6.2 Client Maintenance ERP Customer: Search on Hub & Copy From Hub

4.6.3 Maintenance on Hub & on Client can be Used in Unison



4.7 Cleansing

4.7.1 Overview



4.7.2 Identifying Duplicates

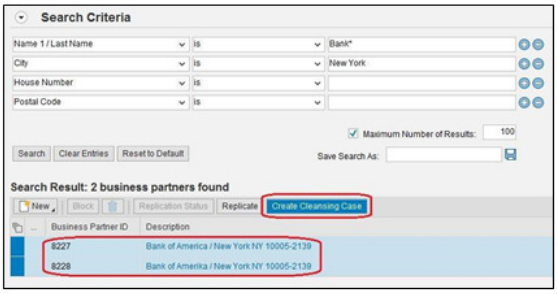
Before creating a cleansing case and do the merging the data potential duplicates have to be identified.

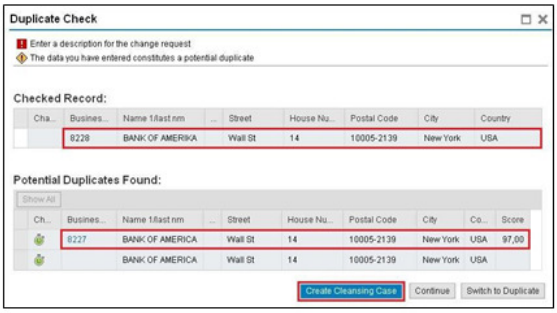
Within MDG duplicates can be identified during

- Search (within search result list)
- Single object maintenance (duplicate check)

Outside MDG

- Hand over cleansing case from client system to MDG Hub
- Full data base scan (using external tools)
- Identified potential duplicates will be stored within a cleansing case for further processing

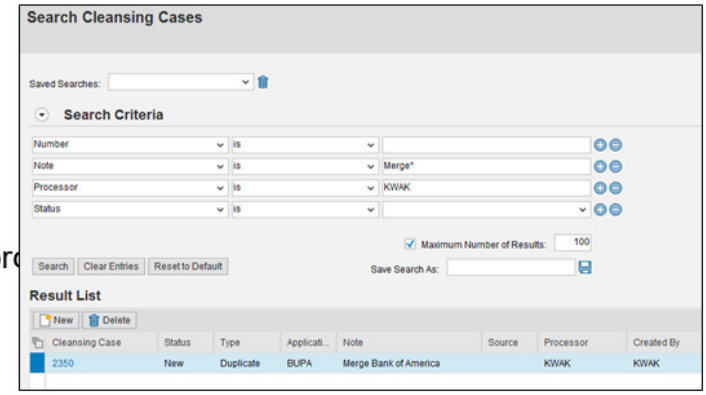


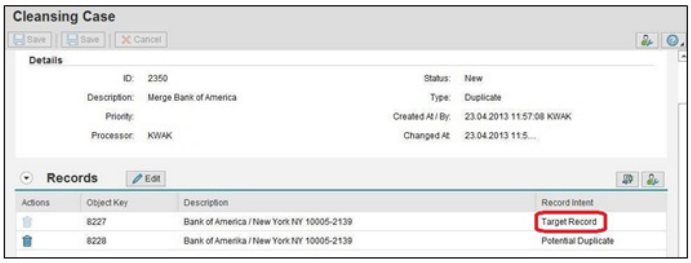


4.7.3 Cleansing Case

Process steps

- Create a new cleansing case, search and and assign potential duplicates
- Search cleansing case (can also be used as worklist)
- Check details of potential duplicate record
- Identify one „TargetRecord“
- Identify one or more „Potential Duplicates“ as well as „No Duplicates“
- Start Cleansing and create a Change Request





4.7.4 Merge

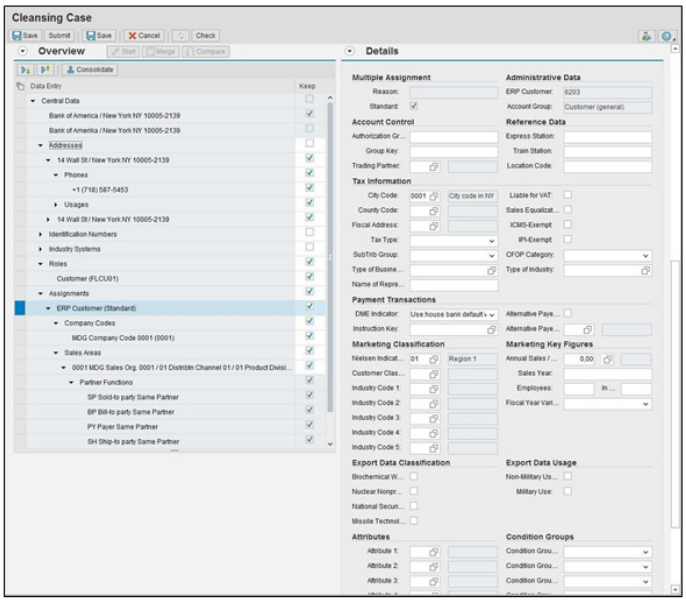
During the merge process all relevant data from „Potential Duplicate“ records can be taken over to the „Target Record“.

This can be done using the keep flag as well as using drag and drop functionalities.

Hereby it is possible to copy the data including all dependent data (e.g. address including its communication data).

In parallel it is also possible to edit data of „Target Record“ directly.

As result of the merge process a workflow process will be triggered



4.8 Process of Multiple Objects

Multi-Record Processing

- Well-defined governance process with multiple workflow steps for a limited number of Business Partners
- Typically a collaboration of multiple processors
- Governance processes exist for single and multiple objects
- Examples
  - Account manager wants to compare and harmonize sales data for all Customers belonging to the account
  - Strategic buyer wants to update purchasing conditions of all relevant Suppliers based on newly signed contract

Mass Processing

- Interactive master data change process for huge number of Business Partners
- Typically ad hoc / on-request / project-based processing by one data specialist
- Examples
  - Change dunning parameters for all C-Customers of a country based on the newly defined process
  - After reorganization in purchasing, Suppliers are assigned to newly responsible purchasing groups

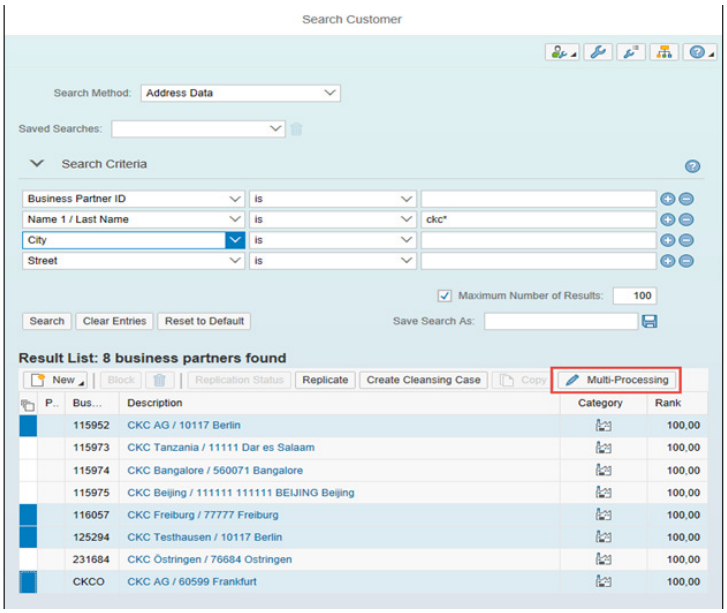
Process to change multiple Business Partners within a spreadsheet-like UI

Start from search result list

- Select multiple Business Partners
- Choose 'Multi-Processing' button

Pre-defined Change Request types

- Multi-Processing for Customer Sales
- Multi-Processing for Customer Financials
- Multi-Processing for Vendor Purchasing
- Multi-Processing for Vendor Financials



Functions for rows

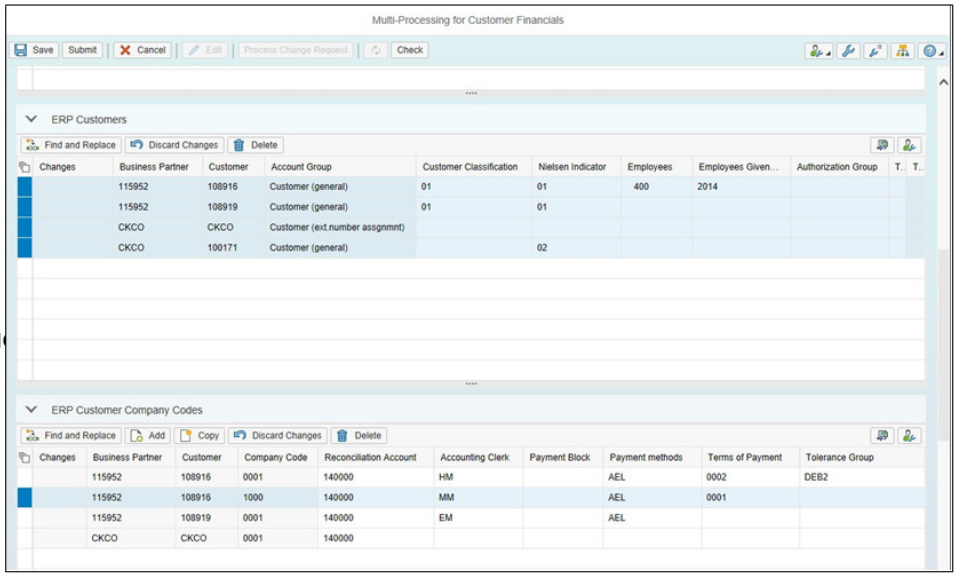
- Add new rows
- Copy rows
- Delete rows

Functions for field values

- Edit single fields
- Find and Replace
- Select rows with same valu
- Copy value to selected rows

Visualize / handle changes

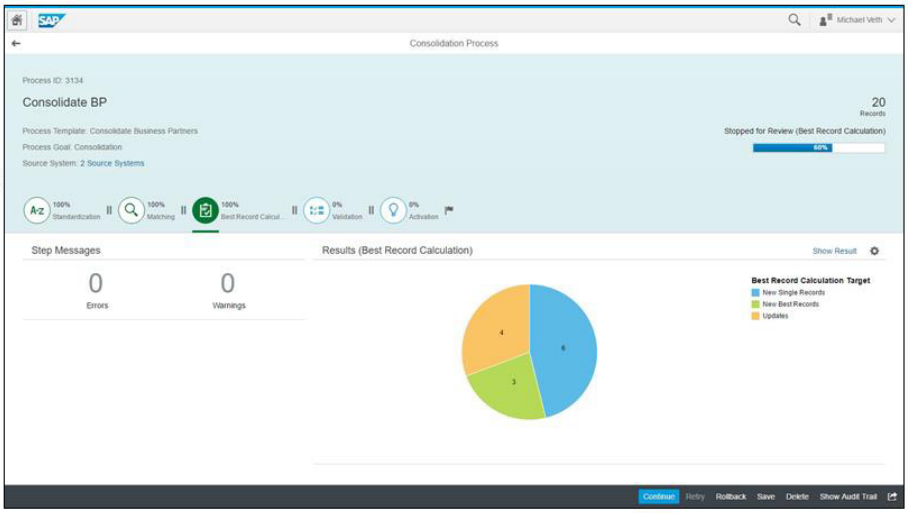
- Highlight changes
- Discard changes



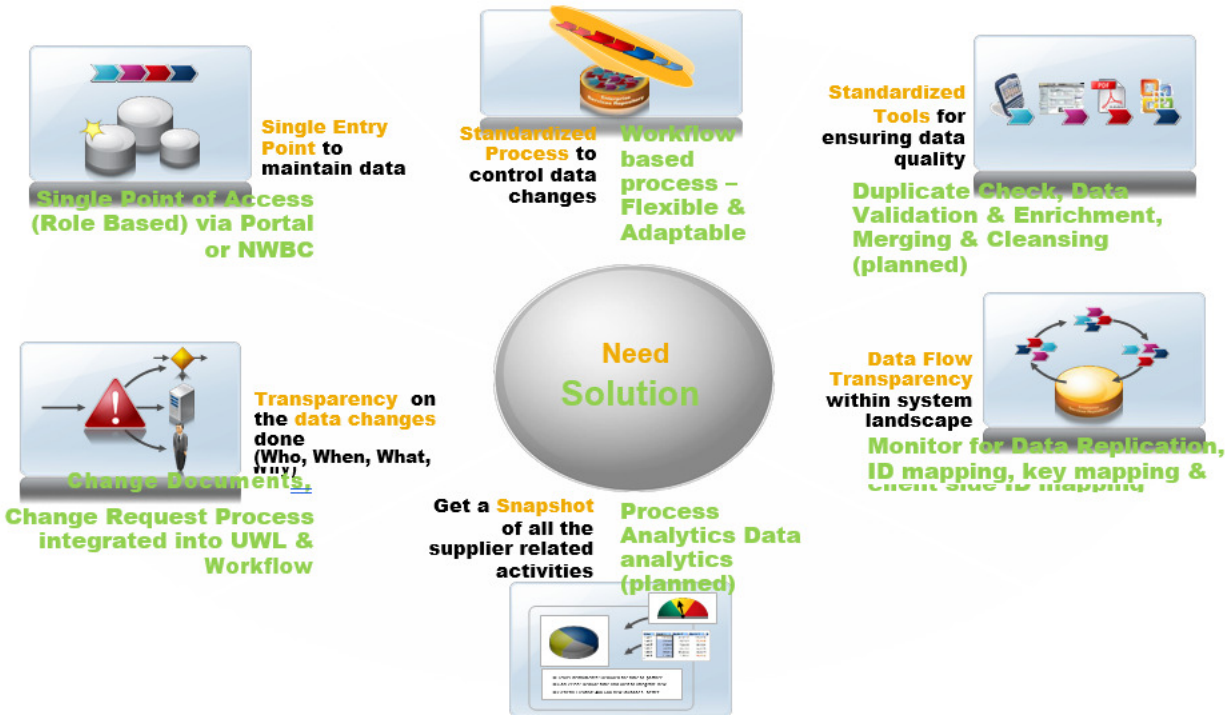
Process steps

- Enables to enrich and standardize data.
- Identify and merge duplicate records.
- Validate records and activate on Database.
- Replicate records to client systems.
- Process erroneous records with Central Governance.

For More information, see separate presentation on Consolidation



4.9 Benefits for MDG-S and MDG-C Solution[16]



Ensure Master Data Governance

Solution Strategy

- Deliver process-centric solution
- Workflow closely integrated to a framework for defining business rules
- Full Integration to SAP Systems and integration services for non-SAP
- Deliver roles on a web based UI
- Deliver a Data Model proven by several applications & industries
- Reuse of existing processing logic
- Out-of-the box data replication mechanisms

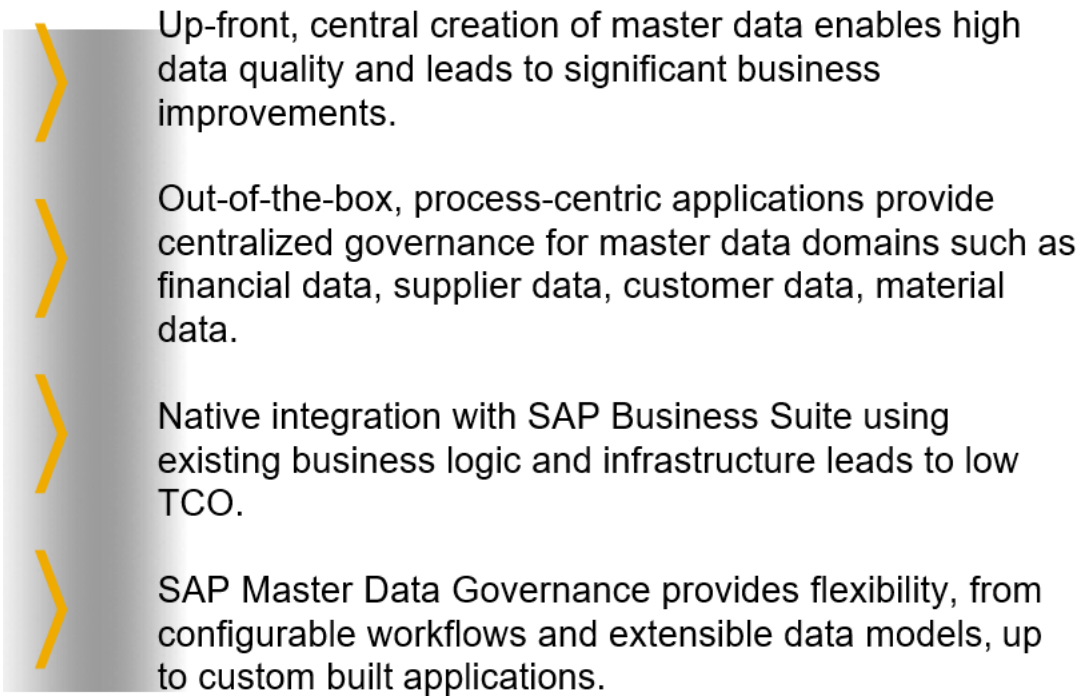
Basic Principles

- User-centricity : Central work center for the maintenance of materials
- Governance: Change requests with built-in approval process
- Workflow: standardized but flexible processes incl. rules management and tracking
- Architecture: Staging area to maintain inactive versions of material data

Common Services

- Search & Display
- Create, Change, Mass Change, Mark for Deletion
- Work lists (My Change Request, Display Change Requests, Change Documents)
- Duplicate Check
- Key- and Value-Mapping
- Data Replication
- File Upload / Initial Load
- Extensibility of Data Model & User Interface
- Deployment on top of operational ERP or as standalone hub

[16] SAP SE or an SAP affiliate company



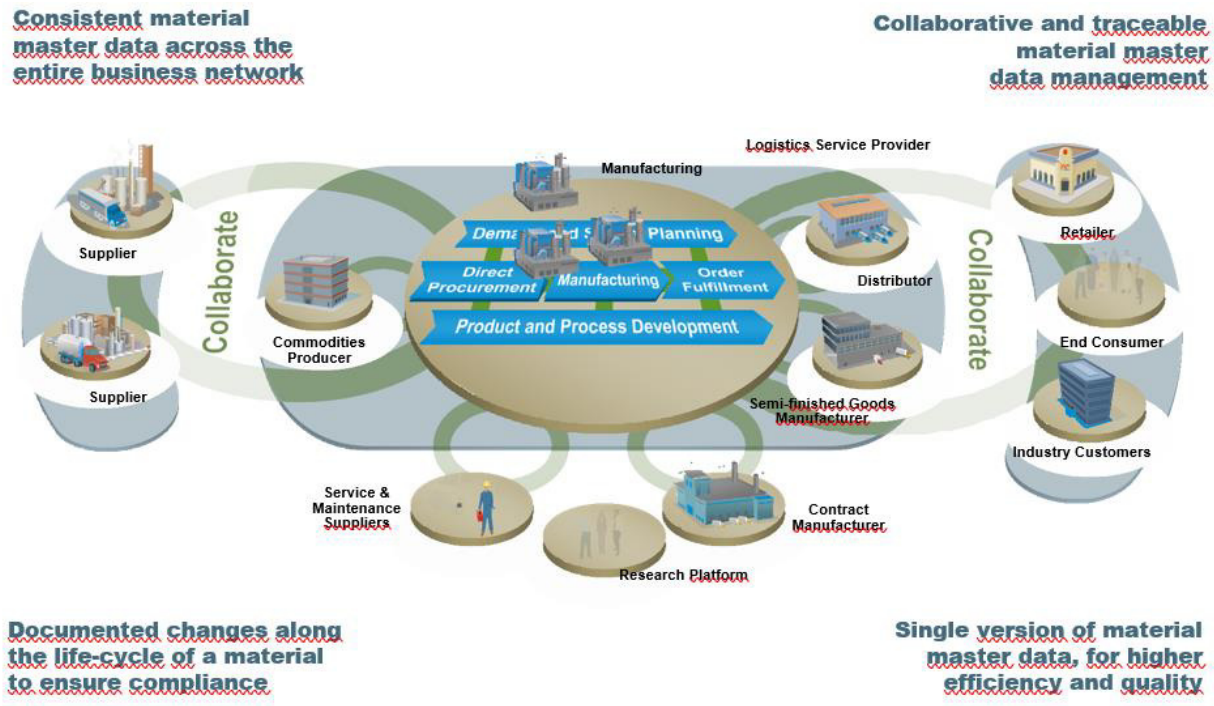
5 SAP MDG for Materials (MDG-M)

Maintenance processes for material master data are highly complex. From general information relating to plant, storage location and batch-specific data, through to price control and consumption data, countless data fields must be created and maintained for material master data. Access to these master data is required not only by purchasing and sales, but also, for example, by the accounting, warehouse management, QM and materials planning departments.

On account of inadequate and error-prone handling, many companies have moved away from manual workflow steps between the parties involved towards an IT-assisted, automated record entry and maintenance system. However, in the area of material master data a highly heterogeneous system landscape slows down business processes and makes it prone to errors. Consequently, a targeted, comprehensive and transparent master data solution is essential in order to permanently stabilize and speed up your processes based on the materials master data.

SAP MDG-M offers such a facility for permitting the comprehensive, standardized, efficient control of master data in the materials area. The central management of a whole array of different materials is possible; the maintenance processes are based on an automated contribution and approval process.

Above all, SAP MDG-M stands out by the fact that it is compatible and integrated into your SAP ERP and can reuse all the existing checks, validations and data structures.



5.1 Master Data Governance for material data

Master Data Governance for material data offers a central hub-based maintenance of material master data.

To address the key business issues related to such a governance solution for master data, MDG for material follows some basic principles that are shared across the domains of MDG

- Maintain once, use many
- One standardized process to be used by everybody
- One standardized user interface for everybody
- One standardized set of business logic

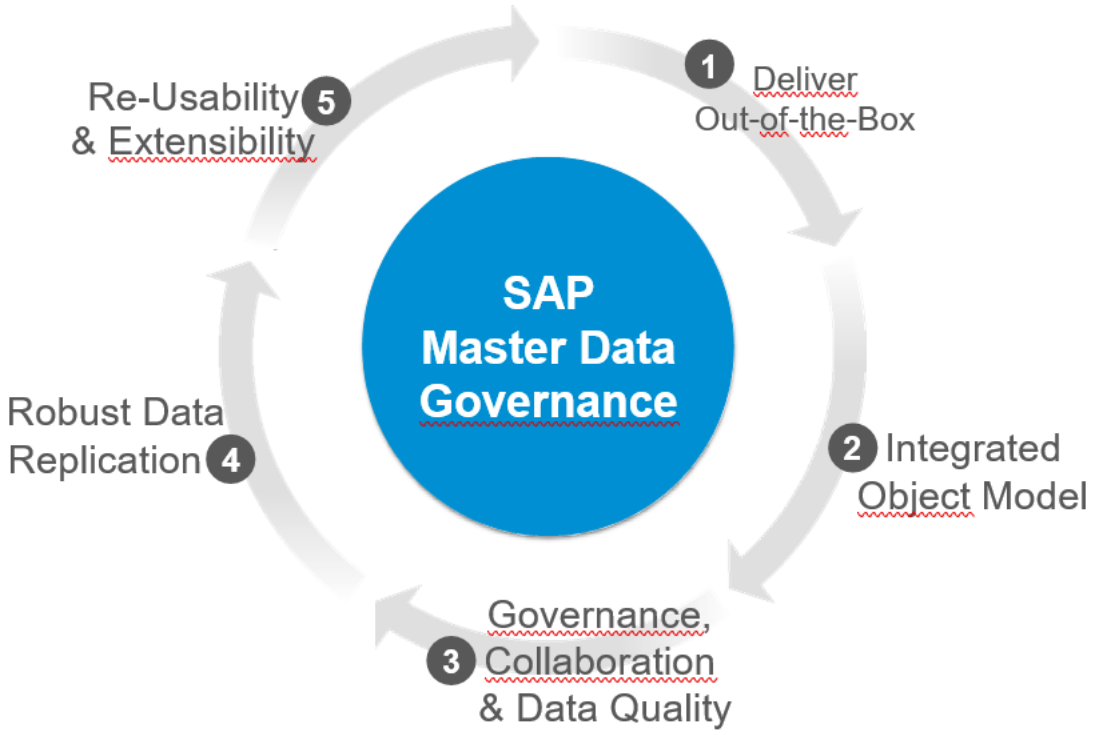
In order to fulfill these principles, SAP has built a Master Data Governance solution that offers

- Central based maintenance of master data
- Workflow-driven processes that are flexible while ensuring governance and standardization
- Harmonized user interface, based upon Enterprise Portal, Business Client or FIORI Launchpad
- Data model and business logic follow the best practices of SAP and reuse

existing data structures, business validations and various services

5.2 Capabilities

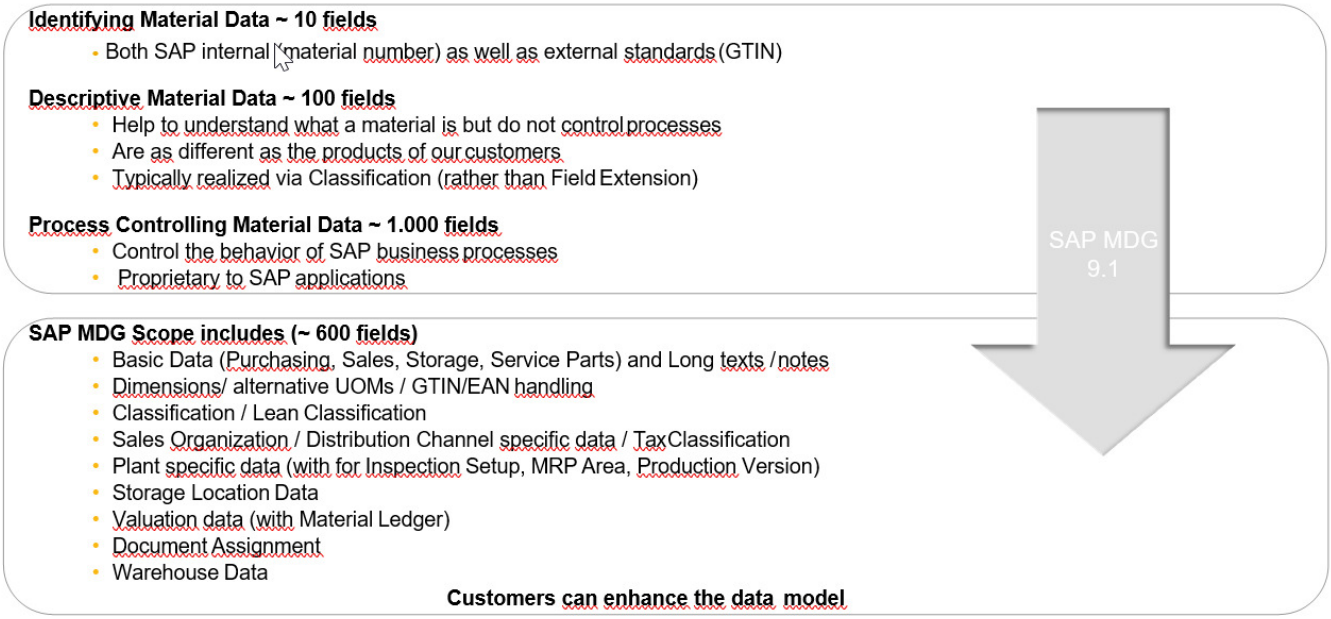
5.2.1 Establish Best Quality Master Data to Optimize Business Processes



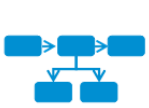
5.2.2 Deliver Out-of-the-Box

Low TCO	Full Integration to SAP systems as a client Offer integration services for non-SAP client systems
Flexibility	Workflow flexibility closely integrated to a framework for defining business rules
Ease of Consumption	Deliver a process-centric solution, with roles and a web-based user interface Deliver predefined data quality services
Best Practice	Deliver a data model proven by several applications & industries
Process Transparency	Reuse of existing processing logic
Integration	Out-of-the box data replication mechanisms, using stable mechanisms

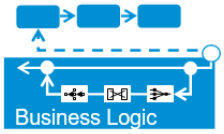
5.2.3 Integration Model



5.2.4 Governance, Collaboration



Linear or distributed and workflow based



Possibility to extend the process by adding business logic and thereby bringing in data quality

Has roles and responsibilities and also task authority (e.g., create, change, approve)

Data in process stored in a separated repository, will be transferred to operational database after final approval

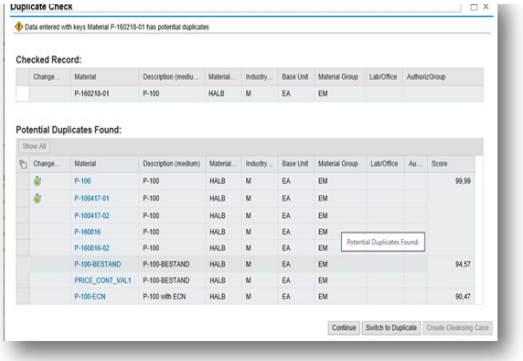
Can be adapted and tailored based on customer needs

Flexible enough to respect the distributed responsibilities existing in a company across various business units

5.2.5 Data Quality

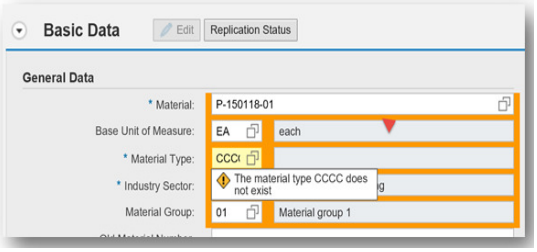
Prevent creation of duplicates for increased effectiveness and efficiency

- Checked early and embedded in the process
- High detection quality of matching using SAP HANA or Enterprise Search



Validations

- Re-use of existing validation logic
- Custom validations can be modeled and programmed (For example, code lists, simple checks, or rules using BRFplus)



Showing what was changed

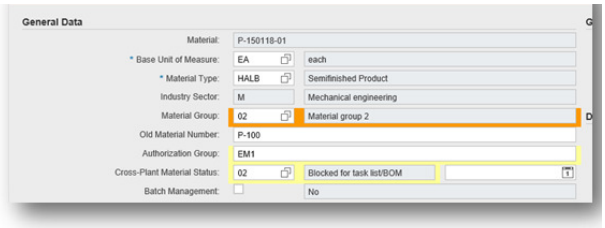
- Changes by earlier processors and own (unsaved) changes in two colors
- Previous, changed, and last saved value per attribute
- Available for all fields, including table cells, rows, navigation elements, etc.

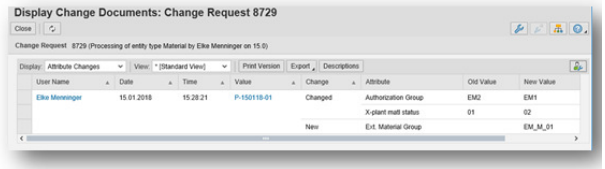
Helps people to do their job

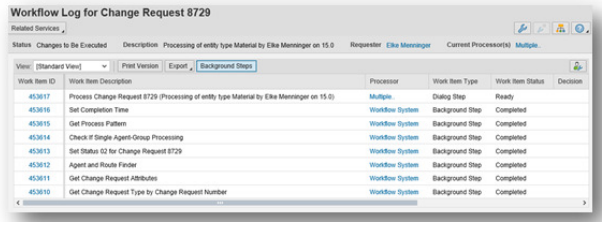
- Supports the processor to decide on their next action
- Supports approvers to quickly spot what they will approve
- Supports auditors to easily see what was changed

Transparency on what has happened earlier

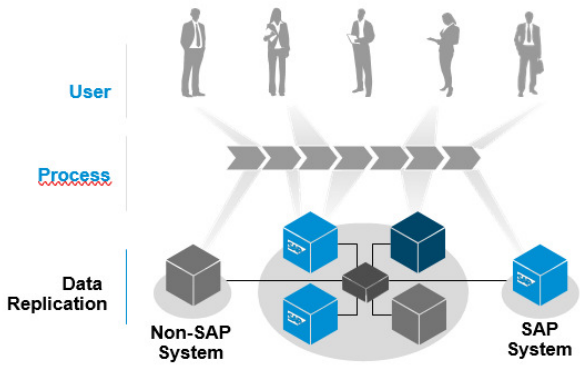
- Change documents provide information who changed which attribute from what value to what other value by when
- Workflow logs tell who was involved in the change request process and who approved which data change







5.2.6 Replication



Replication

- Direct, pooled or manual replication to SAP and non-SAP systems
- Peer-to-peer or mediated

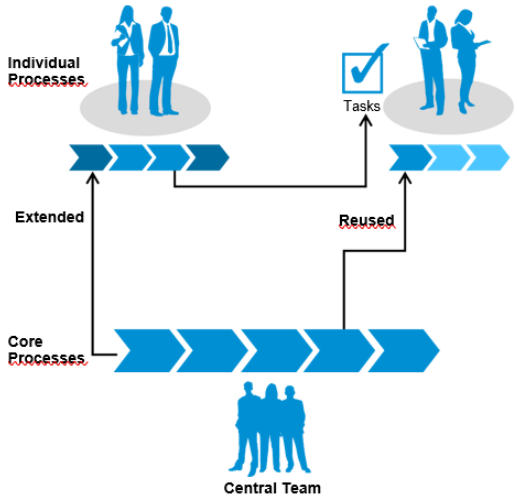
Replication Techniques

- DRF controlling replication mechanisms like
  - ALE for SAP S/HANA, SAP ECC, R/3 and non SAP systems
  - SOA service (only in S/4) for SAP and non SAP systems
- CIF for SAP SCM and MW for SAP CRM/SRM

Replication Features

- Flexibility to replicate selected data only to a specific client system by defining filters (controlled by DRF)
- Support of different object IDs or customizing keys between hub and client systems
  (via mapping in MDG or PI)
- Monitoring and error handling

5.2.7 Reusability & Extensibility



Reuse standard content

- Reuse existing data structures and functionality of SAP Business Suite

Enhance functionality

- Be able to enhance MDG functionality based on predefined content (e.g. plant or sales data) or customer-specific extensions

Extend data structures

- Provide easy-to-use extensibility capabilities for enhancing nodes and fields
  - data models
  - user interfaces
  - processes
  - data replication

5.3 Governance Principle

5.3.1 Central Maintenance rules

User-Centricity

- Central Maintenance of Materials
- Master data governance bundles together all activities for material master data maintenance.
- Central deployment and authorization ensures governance
- Local adaptation and personalization possible

Governance

- Change Requests With Built-In Approval Process
- Integrated process starts with a request to create or change a material, continues with processing and commenting by different expert users and stakeholders, and finishes with approval or rejection by authorized experts
- All processing steps are supported by workflow

Workflow

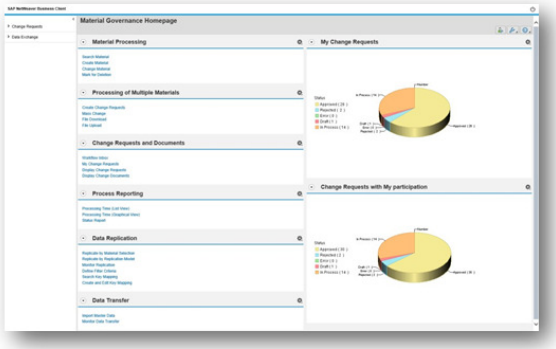
- Standardized but Flexible Processes Incl. Rules Management and Tracking
- The standard SAP Business Workflow is enhanced with a rule-based engine; therefore changes in responsibilities and in processes can be reflected immediately and with ease.

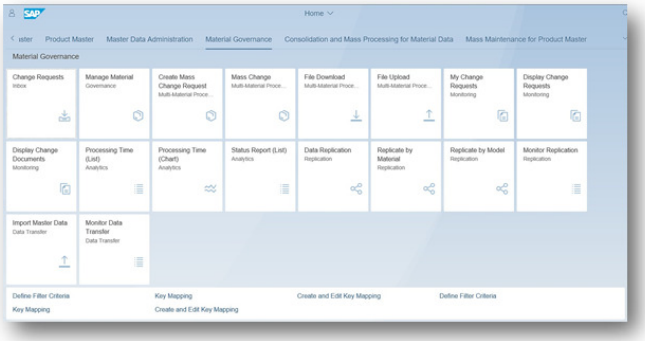
Architecture

- Staging Area to maintain inactive data of Material
- Active area – Holds the active data; Staging – Holds the data during the change request processing before approval

5.3.2 Central Maintenance of Materials

All activities around the creation and maintenance of materials are together. Companies can establish it as the standard hub for creating and changing materials.
Customers can use the SAP Enterprise Portal, SAP Business Client or Fiori Launchpad (only S/4 HANA)





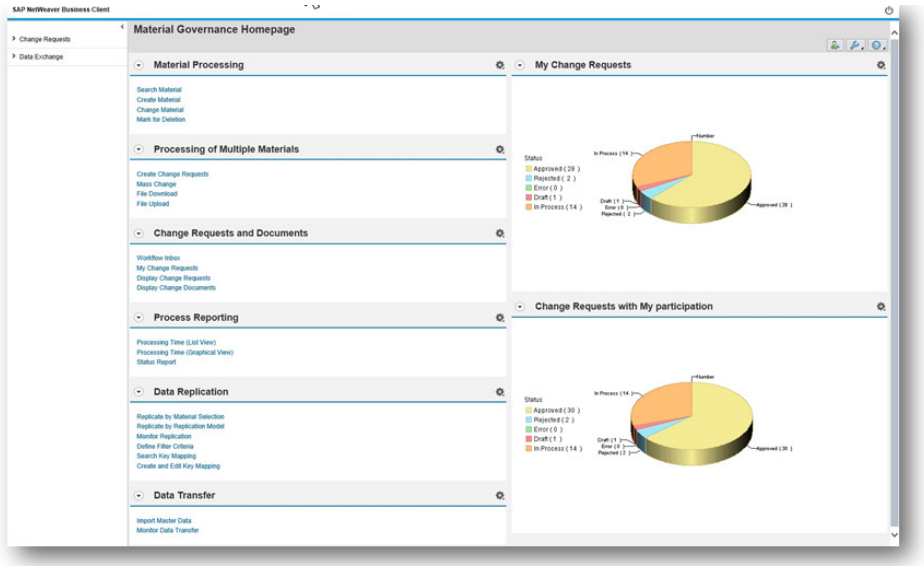
Customer business challenges

- Users maintained master data in different systems
- Different transactions were necessary to execute the maintenance tasks (change, create, delete)
- Planned changes were not transparent to other systems

Business value

- Material master data governance bundles together all activities for material master data maintenance.
- Central deployment and authorization ensures governance
- Local adaptation and personalization possible

5.3.3 Material Governance Homepage



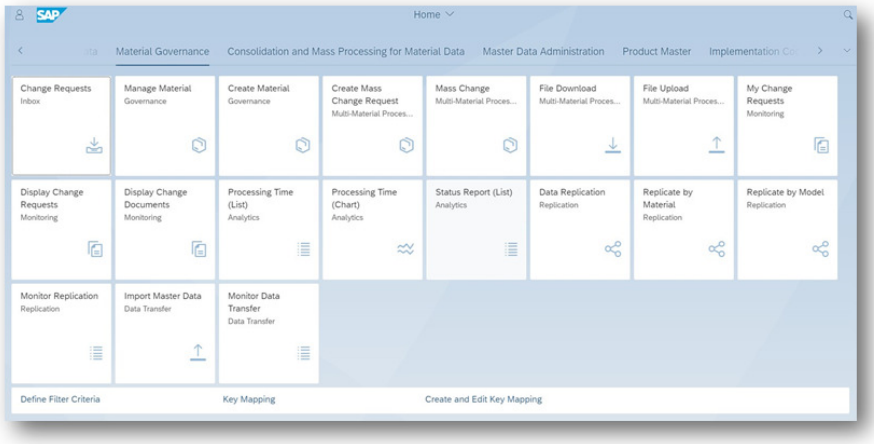
All relevant tasks reachable with one click

Easy to configure and personalize

PFCG Role SAP_MDGM_MENU_07

Start a Change Request with for example ‘Search Material’ or ‘Create Material’

5.3.4 Fiori Launchpad in S/4



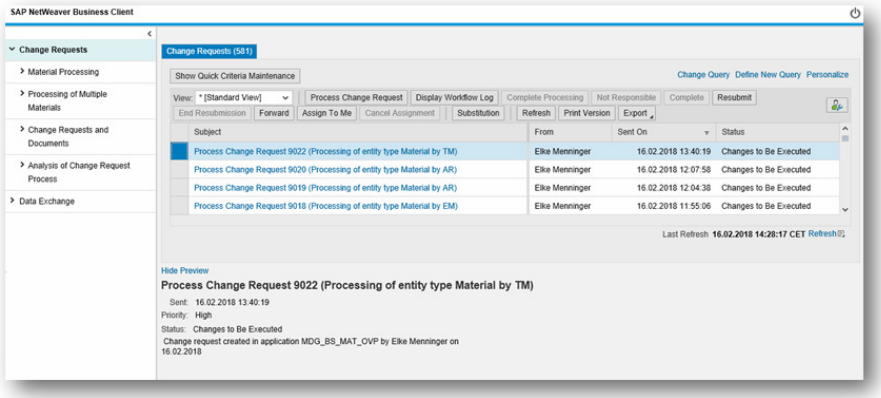
All relevant tasks reachable with one click

Easy to configure and personalize

Catalog SAP_MDG_BC_MATERIAL_DATA

Start a Change Request with for example ‘Manage Material’ or ‘Create Material’

5.3.5 Workflow Inbox



Within the workflow inbox, Change Requests are displayed, where you are the processor.

You can personalize this list to your needs.

Multiple options are available, for example your can open the Change Request or you can forward it.

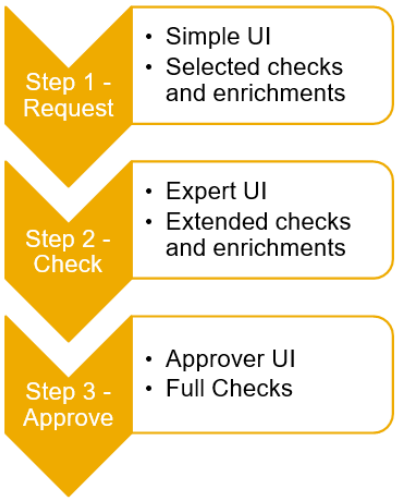
A preview of the Change Request is available.

5.3.6 Workflow-Step Dependent Configuration Enables Flexible UI / Process Design

UI Configuration
Provide different User Interfaces per WF-step based on the task or knowhow of the user

Validations
Enable or disable checks / enrichments
Example: Switch off required field check in Request UI

Field properties
Set some data to read-only
Example: Expert can only maintain dimensions and the basic data are read-only



5.3.7 Out-of-the-box configurations

User-Centric Principle
Out-of-the-box configurations

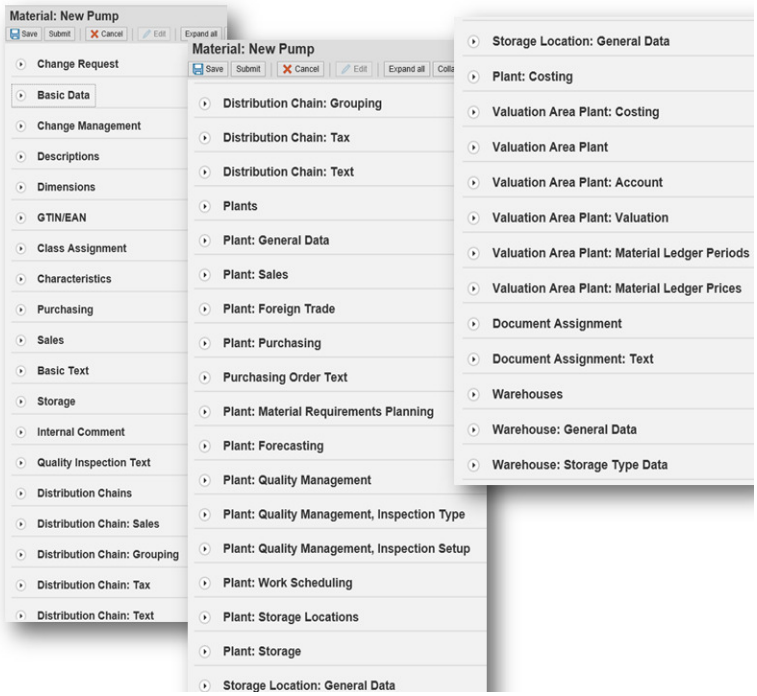
The delivered configurations can be used as template or can be adapted in different ways.

Application MDG_BS_MAT_OVP for Single Object Maintenance

FPM Application Configurations:
BS_MAT_INIT_07 (Initial Page)
BS_MAT_OVP_07 (Material create)
BS_MAT_OVP_07H (Material change)

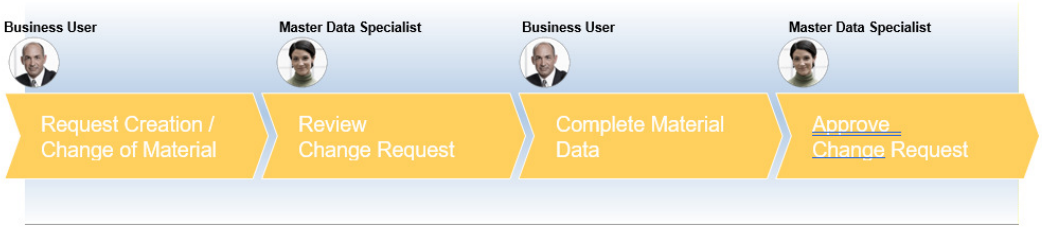
Application MDG_BS_MAT_SEARCH for Search

FPM Application Configurations:
MDG_BS_MAT_SEARCH_07
MDG_BS_MAT_SEARCH_04H



5.3.8 Change Requests With Built-In Approval Process

This function allows you to implement an integrated process of master data governance that starts with a request to create or change a material, continues with processing and commenting by different expert users and stakeholders, and finishes with approval or rejection by authorized experts.
All processing steps are supported by workflow.



Customer business challenges

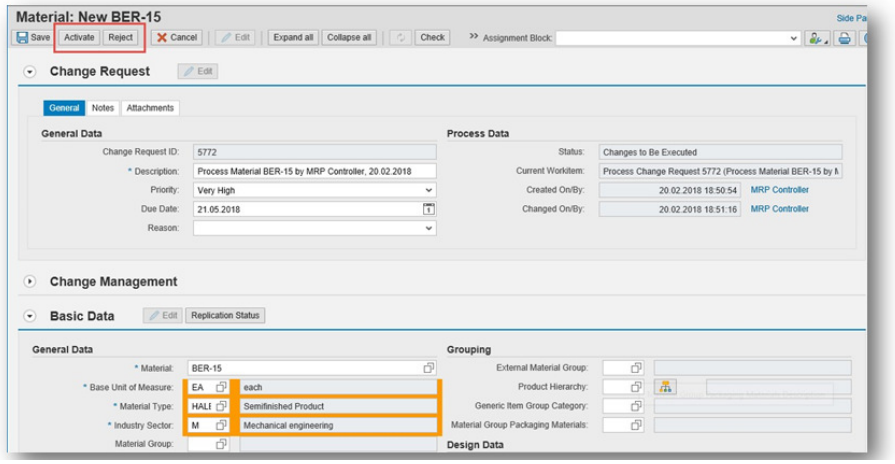
Requests and reasons for new or changed master data had to be communicated by phone or e-mail to master data specialists. They had to gather any further information in a time-consuming process from other experts and stakeholders. Follow-up processing by certain specialists had to be triggered manually. An approval process with a two man rule could not be assured.

Business value

The approval process for maintenance of master data enables:

- Fulfillment of compliance requirements
- Better data quality
- Faster and transparent decisions
- Relevant information in one place

5.3.9 Change Request Header



Notes and attachments can be assigned to the change request header.

Requester has to enter a description. Priority, due date and reason are optionally.

Last user can approve or reject the change request.

5.4 Workflow rules

5.4.1 Standardized but Flexible Processes Incl. Rules Management

The data maintenance activities are bundled via a change request that is linked to a workflow. The workflow can be linear or distributed. Besides automated steps, ad hoc steps, delegation etc. can be used.
The standard SAP Business Workflow is enhanced with a rule-based engine (Workflow template WS60800086); therefore changes in responsibilities and in processes can be reflected immediately and with ease. There is also a workflow template WS46000057without BRF+, which covers the same processes.

Customer business challenges

- Processes were unstructured and neither guided nor controlled
- High effort in data collection, change processing and issue resolution
- High TCO and low flexibility if hard-coded semi-automated processes are used

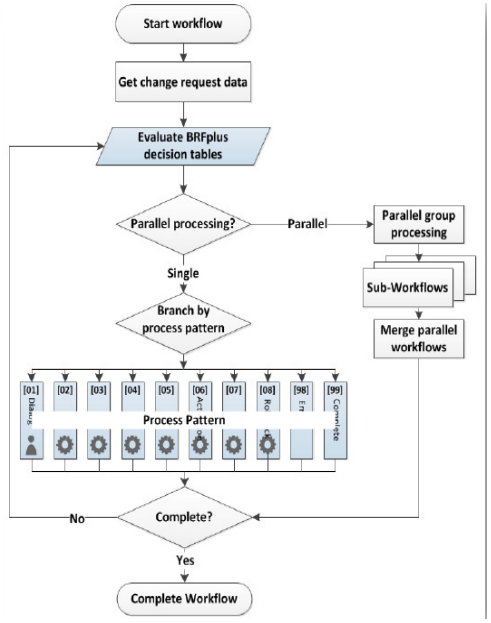
Business value

- Build-in guidance, monitoring and reporting using a standard workflow tool provides process and data governance
- Integrated tool for keeping data, process and issue resolution in one place
- Enhanced flexibility by integrated rules management

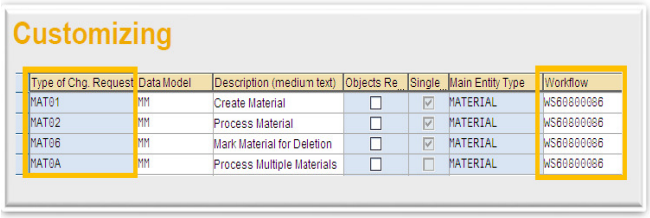
5.4.2 Rule-Based Workflow Template

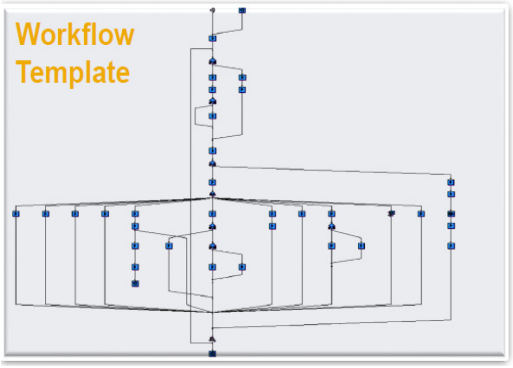
Rule-Based Workflow Template

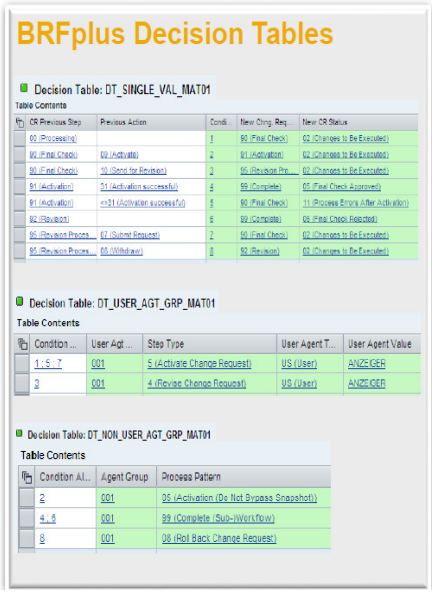
- The generic workflow template (WS60800086) is the foundation of the RBWF. The different processes are reflected by BRFplus decision tables, which are maintained in the customizing for each change request type.
- The combination of this workflow template with the BRFplus decision tables allows you to easily set up many different processes that can be flexibly changed through customizing.



5.4.3 Rule-Based Workflow Components



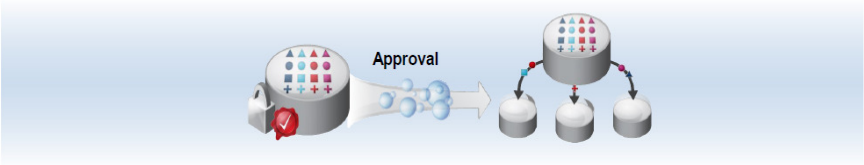




5.5 Architecture Rules

5.5.1 Staging Area to Maintain Inactive Versions of Material Data

As long as the change request for the creation or change of a material is in progress, the attributes of the material are stored in an inactive version that exists (in the case of a change) in parallel to the active version. When the change request is approved then the data is written as the active version to the master data tables and this new or updated record is then available for business processes.



Customer business challenges

- Several users from different departments are involved in the creation of a material item
- Material items were immediately available in the system as soon as the first user saved the entries
- Many incomplete material items and partial changes could exist
- Planned changes were usually not visible to the business users

Business value

- Incomplete material (changes) are stored in an inactive version so the material can only be used in business processes when it is approved (better process quality)
- Rejected change requests don't create a material number in SAP (better data quality)
- Planned changes are visible to the concerned user groups (better transparency)

5.5.2 Handling of Active and Inactive Data

To enable the master data governance process MDG separates

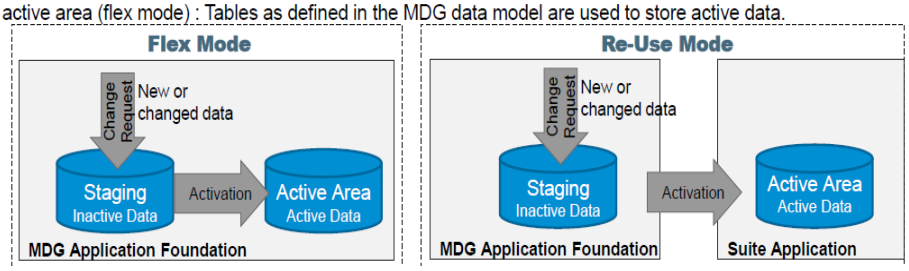
- Active data – Ready to be used by applications, can be distributed to client systems
- Inactive data – Not yet approved, currently part of a change request

Accordingly there are two separate storage locations

- Active area – Holds the active data
- Staging – Holds the data during the change request processing before approval

For optimal integration, MDG allows two modes

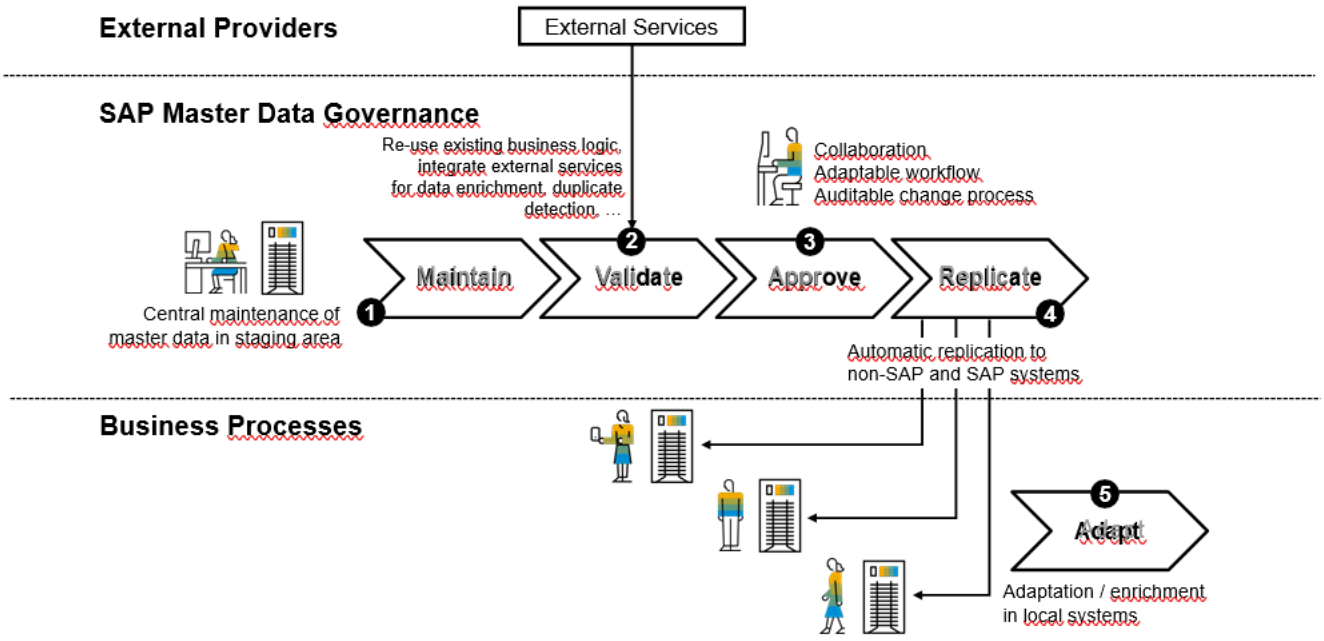
- Re-Use active area (re-use mode) : Existing structures of applications are used. For example, MDG for material use of the MARA table
- Generated active area (flex mode) : Tables as defined in the MDG data model are used to store active data.



5.6 Function in Detail

- Process Flow
- Material Processes: Search, Create Material, Duplicate Check, Display, Change Material, Side Panel, Copy, DMS and Classification Integration, Mark for Deletion
- SAP MDG Fiori Apps: Request new material and Approve new or changed material
- Processing multiple materials: Multiple Record Processing, SAP MDG Mass Processing and SAP MDG Consolidation
- MDG Framework Capabilities: Highlighting, Mass change, Parallel CR, Enrichment Spot, Governance Scope, Data Import, Key-and Value Mapping, Worklists, Data Replication, Process Analytics
- Master data remediation

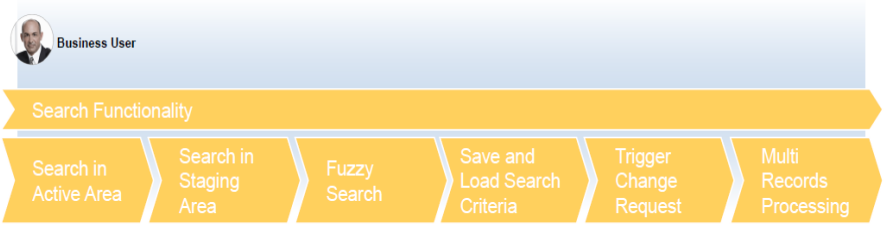
5.6.1 Process Flow



5.6.2 Material Processes

5.6.2.1 Search Functionality

- Users can search for materials that are stored in the active or in the staging area.
- Search criteria combine material attributes and classification.
- Search uses SAP HANA-based Search, Enterprise Search or alternative search providers.
- The user can start a change request for single or multi records processing directly from the results list.



Customer business challenges

- Search results only took operational or active data into account, not the inactive data in the staging area related to planned or future changes
- Business activities could not be directly triggered from a search result list

Business value

- Comprehensive search functionality simplifies daily work
- Integration into change request processing enhances productivity
- Including inactive data from staging area reduces parallel change processes

5.6.2.2 Search Options in SAP Master Data Governance

The following options are delivered:

You can use **Enterprise Search.**

- You have installed and connected Enterprise Search either in embedded or federated form.

You can use **SAP HANA-based search**.

- You can use it in a side-by-side approach or fully embedded if MDG itself runs on an SAP HANA database.
- Follow the Configuration Guide which exists on SAP Help.
- With SAP HANA-based search for MDG, the federated search is not supported. If you require search capabilities across systems and beyond the MDG system, Enterprise Search can still be used.

The following option is not delivered:

You can use **Database Search** in exceptional cases if the two delivered options are not feasible.

- Database Search is not released to be used in the context of master data governance for material for live systems, it is only intended to be used as a temporary workaround for POC/Test purposes.
- Please follow the extension guide Replace Enterprise Search by DB or alternative search provider that also lists the limited capabilities.

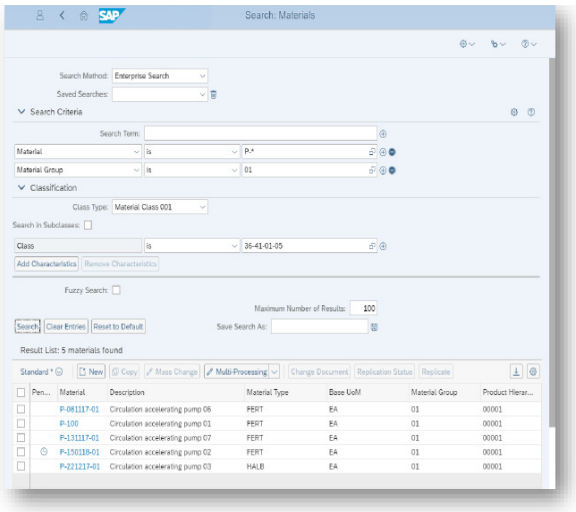
5.6.2.3 Capabilities of the Search Options of MDG for Material

Topic	Enterprise Search	SAP HANA-based search
Fuzzy Search	Yes	Yes
Support of Search Ranking (Score)	No (SAP Note 2284745)	Yes
Duplicate Check	Yes	Yes
Duplicate Check with Fuzziness	Yes, can't be switched off	Yes, configurable (defined by search view)
Long texts (multi languages)	Yes	Yes
Free-text	Yes	Yes
Classification	Yes	Yes
Time Dependent Classification	Yes	No (SAP Note 2284745)
Extensibility with fields	Yes (higher effort)	Yes (lower effort)
TCO	High (additional TREX hardware)	Low (embedded), High (side-by-side)
Multiple Search Views (UI)	No	Yes

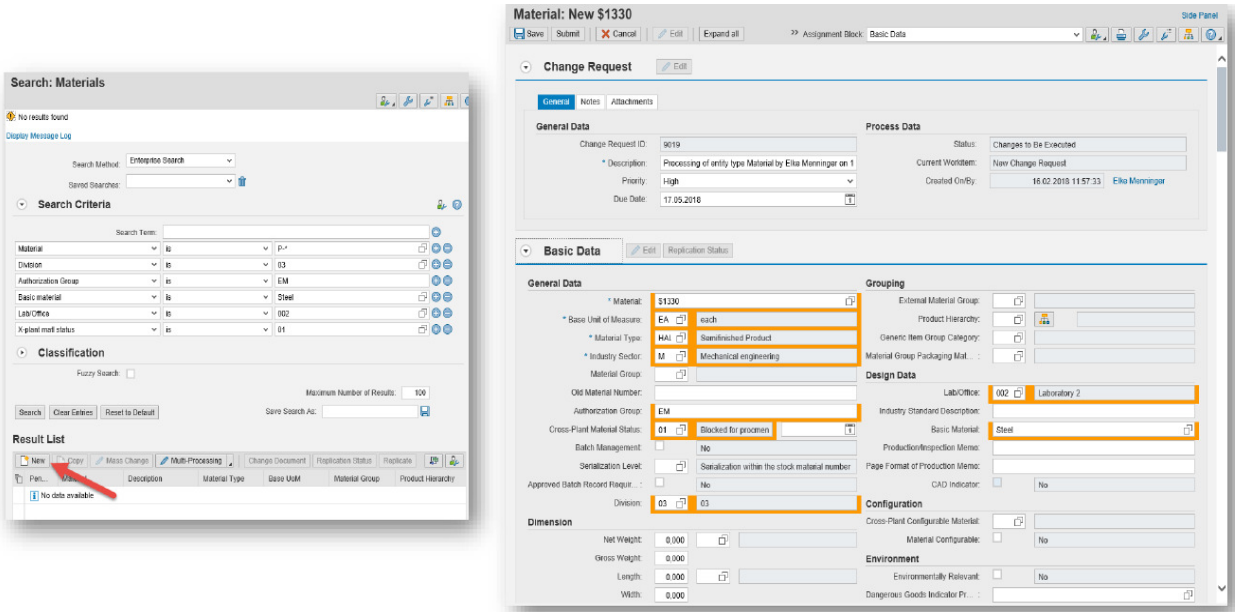
5.6.2.4 Search Functionality

Consistent search UI containing all search capabilities

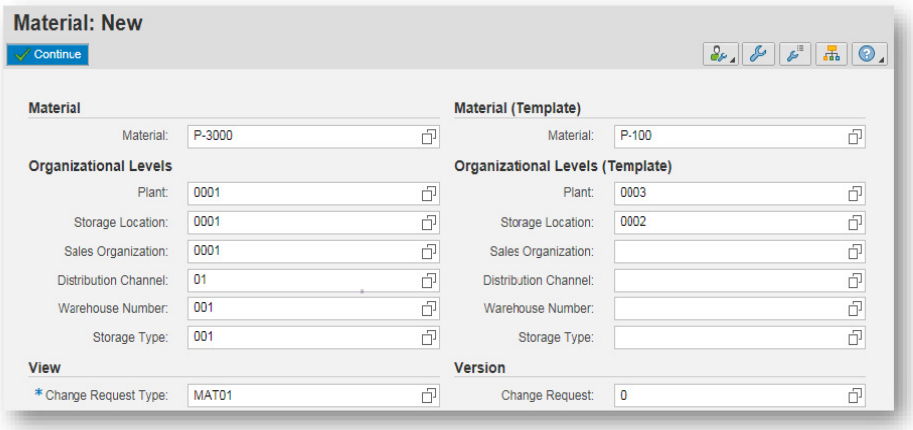
- Search methods easy to configure, more search methods can be added
- Search criteria combine material attributes and classification
- Drop-down lists of available fields and operators
- Add and remove search criteria (AND combination of multiple parameters)
- Search criteria can be saved using a descriptive name for quick and easy re-use
- Search for hits within active and inactive data
- Result can be downloaded to Excel
- Result list can be personalized



5.6.2.5 Search Functionality: Copy search criteria to Change Request

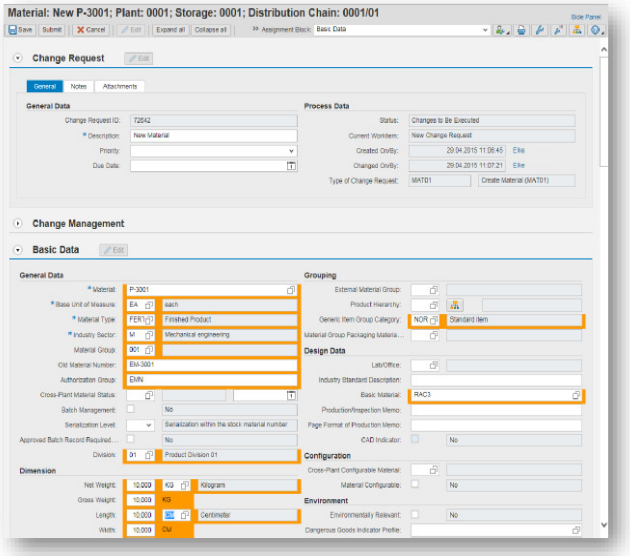


5.6.2.6 Create Material With Template, Initial Screen



Copy from a Template: Enter Material, and if necessary a Plant, Storage Location, Sales Organization and Distribution Channel, Warehouse Number and Storage Type
If you specify a Change Request as template than inactive data of the material is copied
Only specified organizational levels of template are copied. Otherwise no organizational level is copied.

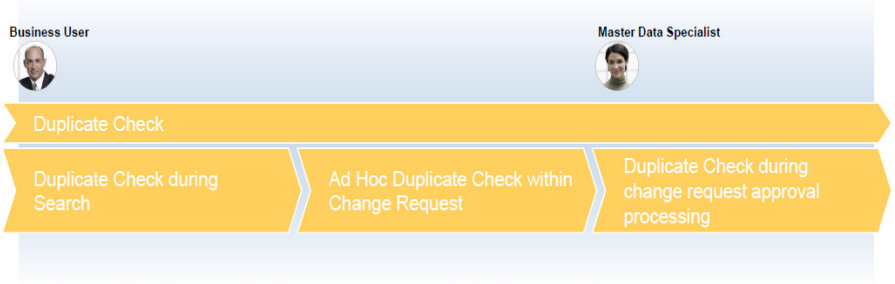
5.6.2.7 Create Material Overview Page



- Several actions are possible for processing a change request (Customizing)
- Workflow is controlled by the BRFplus decision tables when using the Rules based Workflow template or controlled by the Classic Workflow template
- Dedicated users or organizational units are derived for further processing
- Delivered template configuration: BS_MAT_OVP_07

5.6.2.8 Duplicate Check

A basic duplicate check is offered to avoid the creation of already existing materials. The information on potential duplicates is provided during when checking or submitting the change requests. It can be switched on or off per workflow steps.



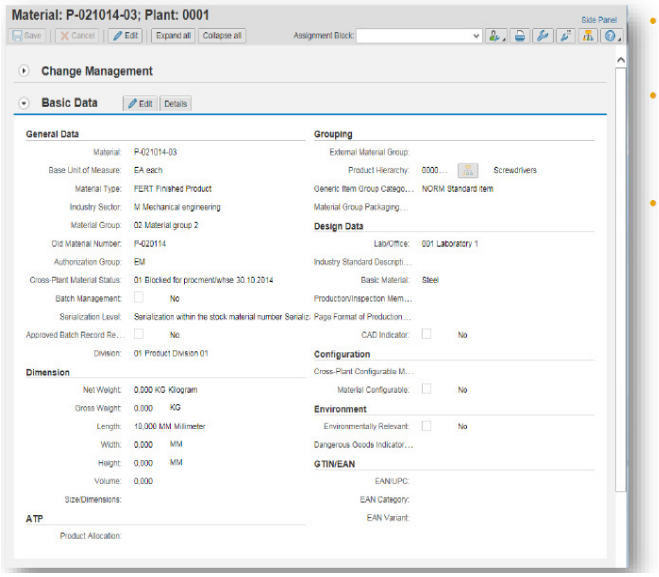
Customer business challenges

•Users often created a new material item even if the material master already existed in the system. This resulted not only in bad data quality, but also generated unnecessary cost since every master data record has its price tagged during its lifecycle.

Business value

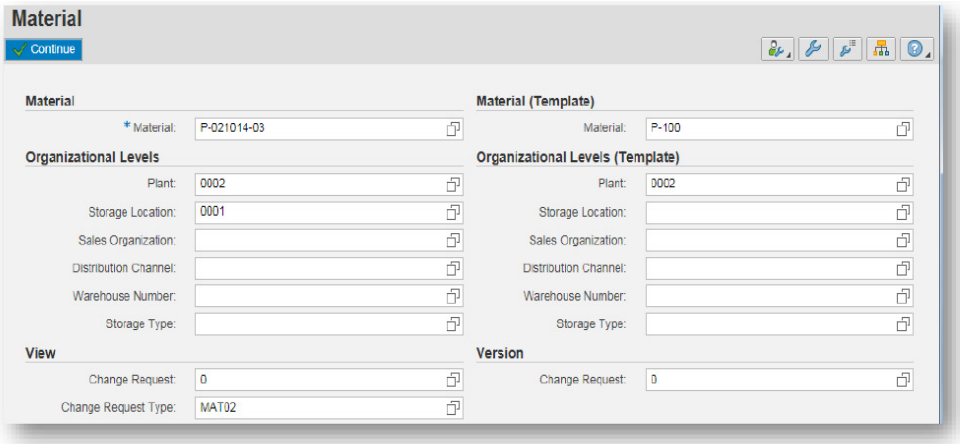
- The duplicate check improves the data quality as well as the decision and business process quality in the connected systems
- Avoiding duplicates as early as possible saves time and money in the master data process

5.6.2.9 Display Material



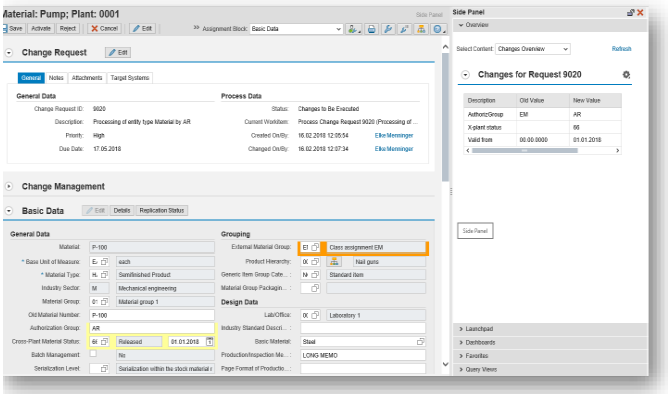
- Display material via WebDynpro Application Change Material
- Or display material via WebDynpro Application Search Material and click on the Material Number link
- Switch with the edit mode to create a new change request or to existing change request

5.6.2.10 Change Material With Template, Initial Screen



Copy from a Template: Enter Material, and if necessary a Plant, Storage Location, Sales Organization and Distribution Channel, Warehouse Number and Storage Type
If you specify a Change Request as template than inactive data of the material is copied
Only specified organizational levels of template are copied. Otherwise no organizational level is copied.

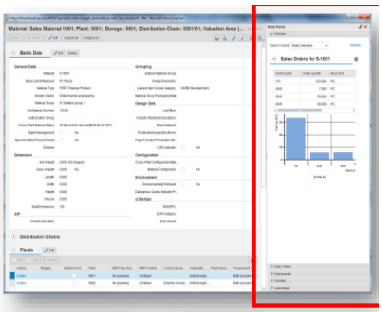
5.6.2.11 Change Material

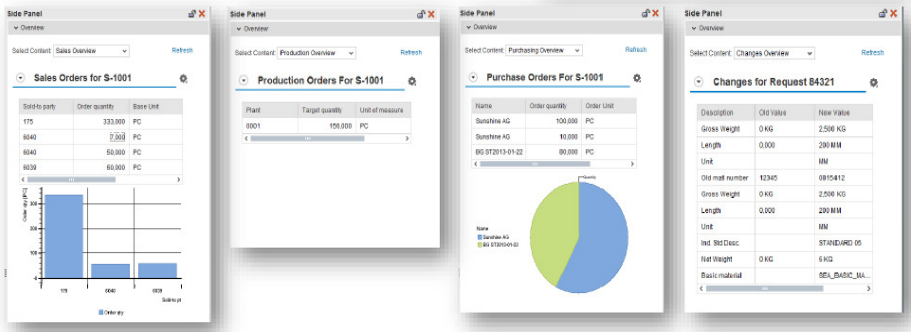


- Review Changes by showing the Change Documents, also in the Side Panel
- Several actions are possible for processing a CR (Customizing)
- Workflow is controlled by the BRFplus decision tables when using the Rules based Workflow template or controlled by the Classic Workflow template
- Dedicated users or organizational units are derived for further processing
- Delivered hierarchical template configuration: BS_MAT_OVP_07H

5.6.2.12 BCV Side Panel Content for MDG Material

- Sales Overview: Sales orders created for the current material master.
- Production Overview: Production orders created for the current material master.
- Purchasing Overview: Purchase orders created for the current material master.
- Changes Overview: Changes raised by the current MDG change request.





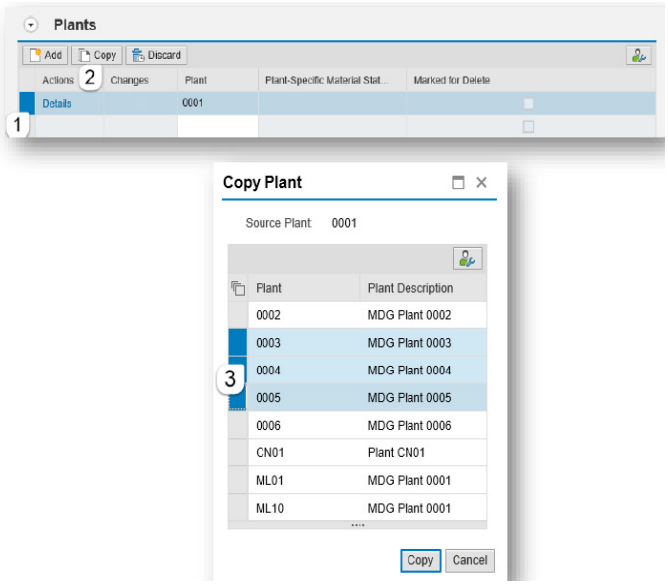
5.6.2.13 Copy organizational data

Copy organizational data

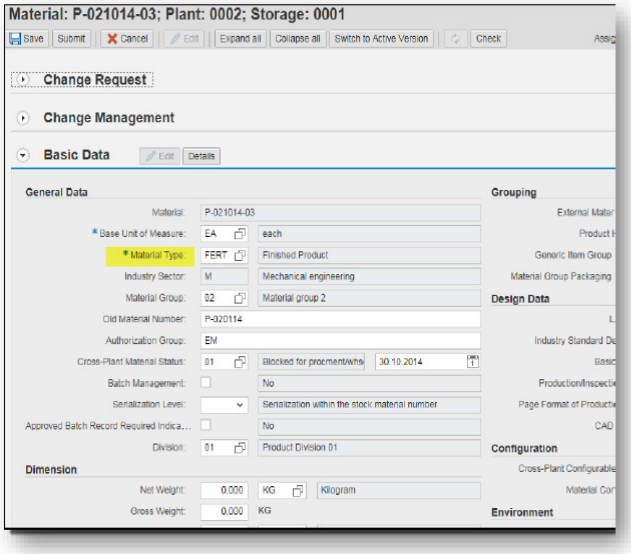
1. Mark already available organization
2. Select ‘Copy’ button
3. Select target organization(s) not yet being maintained.
4. Data is copied and can be reworked

Copy organizational data is available for

- Distribution Chains
- Plants
- Valuation Area Plant
- Warehouses
- Production Versions
- MRP Areas



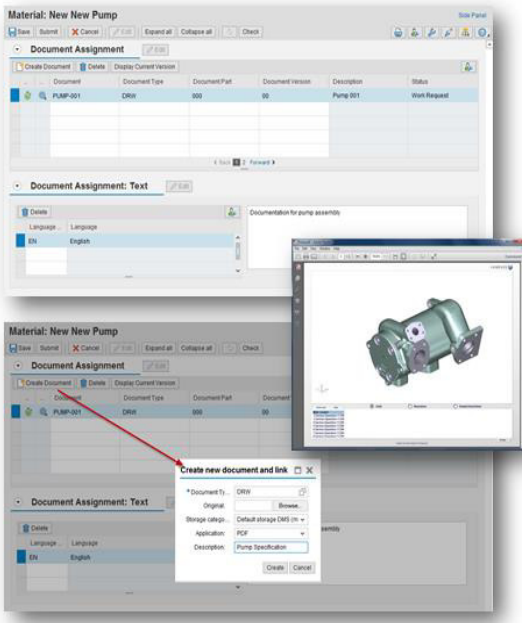
5.6.2.14 Change Material Type



- With the Material Change UI it is possible to change the Material Type
- Prerequisite: Authorization M_MATE_MTA

5.6.2.15 Document Management System (DMS) Integration

- Display, edit and delete links to document info records
- Maintain long text information for material link
- Easy creation of new documents (and material link)
- Display document info record dependent on user/system environment
  - Using PLM Web UI
  - Using SAP GUI HTML
  - Using SAP GUI (NWBC for desktop only)



5.6.2.16 Classification

Classification available since EhP5

Only for Material:

- MDG Single-Object Maintenance UI

UI re-used from PLM

Not mass-enabled

Restrictions (see SAP Note 2461516)

Predelivered CR-Types:
MAT01, MAT02, MAT06, MAT0A, MAT0B

Lean Classification since MDG 9.1

For Customer, Supplier and Material:

- MDG Single-Object Maintenance UI
- MDG Consolidation

Simplified UIs

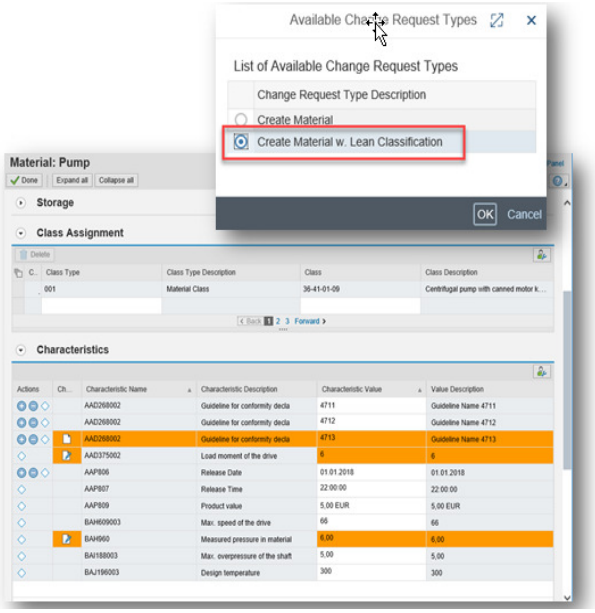
Mass-enabled API

Simplified backend features to reduce complexity

Predelivered CR-Types:
MATL01, MATL02, MATL0A, MATL0B, MATLCLF

Class assignment and characteristics valuation work like normal entities in data model

5.6.2.17 Lean Classification



Usage of lean classification in MDG-M can be configured using change request type

Assign Classes
List UIBB to assign multiple classes.

Enter Characteristic Values
Characteristics assigned to selected classes are visible in one row and can be filled with a value. You can use the ◇ icon to clear the value.

Multiple Values Characteristics
Multiple value characteristics are also visible in one row and can be recognized by '+' and '-' icons. You can use these icons to insert or delete rows.

5.6.2.18 Lean Classification Features and Restrictions

Supported Features

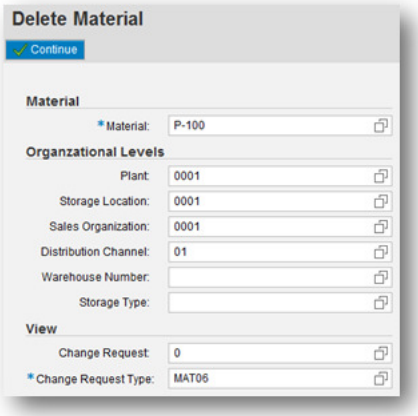
- Derivations using BAdI CROSS_ENTITY_BADI (incl. defaulting for class types and classes)
- Checks using BRFplus
- Authorizations (incl. display authorization)
- Change Documents
- Multiple class types and class assignments
- Characteristic values: Single or multiple values, additional values, intervals, mandatory checks (required entry)

Restrictions (not supported)

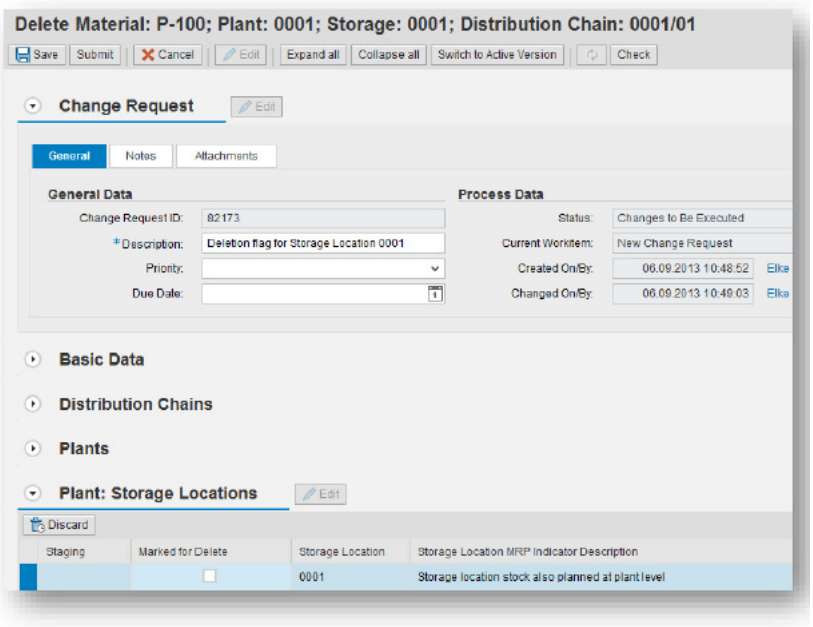
- Object dependencies (preconditions, actions, selection condition, procedures)
- Overwriting of characteristic values and overwriting of inherited values
- Engineering Change Number

You can find detailed information about included/excluded features in SAP Note 2479869.

5.6.2.19 Mark Material for Deletion



- Mark a single material for deletion
- Or set the flag on a org level
- A change request is triggered and data is copied into the staging area



- Workflow is triggered and processed the same way as for the other processes
- The deletion is limited to the material entity. Currently no remote checks about reuse of the material is done.

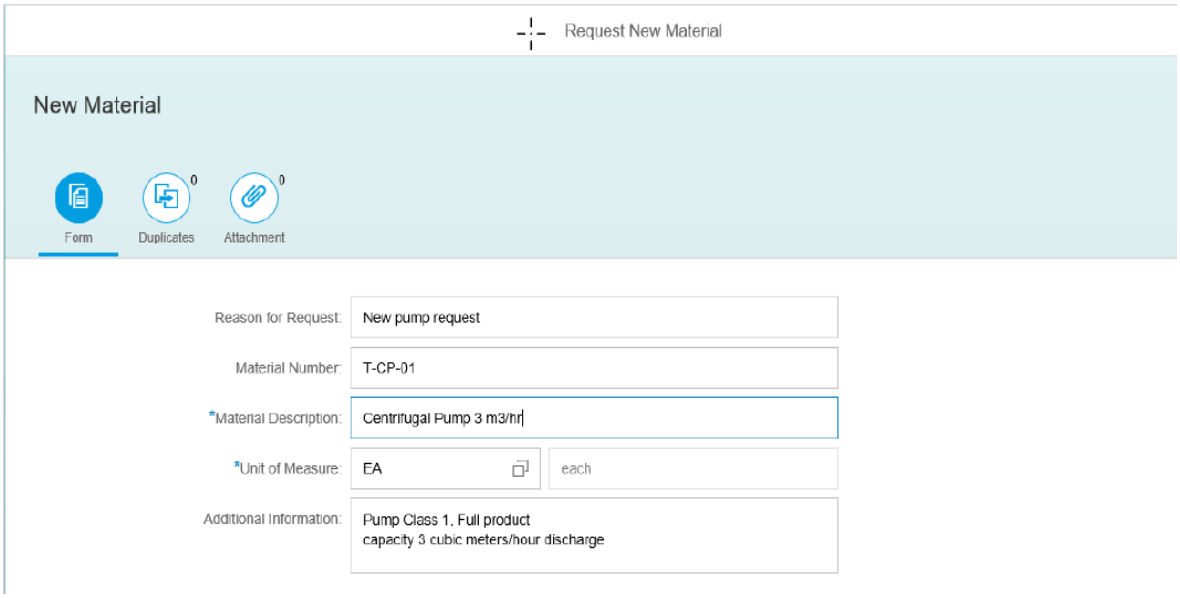
5.6.2.20 Apps Fiori MDG-M

The following SAP Fiori Applications are part of SAP MDG 9.0 and SAP S/4HANA 1610

MDG Fiori Applications		Backend	Frontend(UI)
MDG Fiori Request Applications	Lean Request for MDG Material	MDG7.0 SP07 Onwards OR MDG8.0 SP01 Onwards OR MDG 9.0 Onwards	MDG FIORI 1.0 (UIMDG001 100) SP06 Onwards
	Track status of My Requests		
MDG Fiori Approve Applications	Approve Material	MG8.0 SP02 Onwards OR MDG 9.0 Onwards	MDG FIORI 1.0 (UIMDG001 100) SP07 Onwards

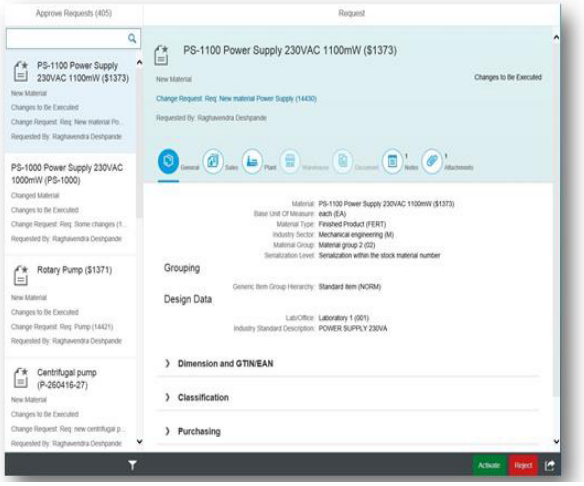
5.6.3 SAP MDG Fiori Apps: Request new material and Approve new or changed material

5.6.3.1 Request New Material -

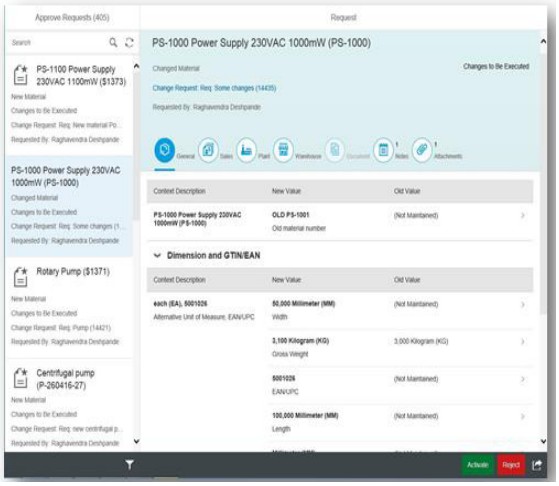


5.6.3.2 Approve Material

Approve New Material



Approve Changed Material



5.6.4 Multi-Record Processing

- Well-defined governance process with multiple workflow steps
- Typically, a collaboration of multiple processors
- Governance processes exist for single and multiple objects

  Examples:
  - New global contract leads to changes in multiple materials, supervisor checks and approves changes
  - MRP controller checks his materials and changes stock-relevant data, production supervisor needs to agree

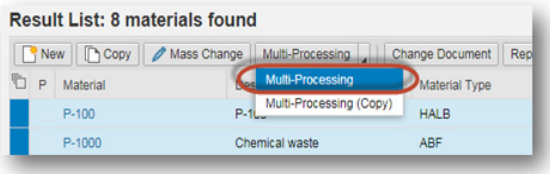
- Interactive master data change process
- Typically ad-hoc/on-request/project-based processing by one data specialist

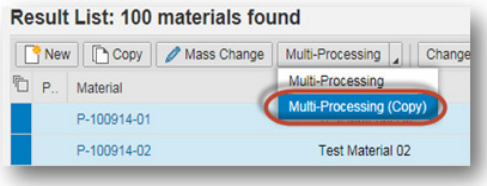
  Examples
  - Set distribution chain status of all materials of a closed sales organization
  - Exchange MRP controllers

    - Processing multiple materials: Multiple Record Processing, SAP MDG Mass Processing and SAP MDG Consolidation
    - MDG Framework Capabilities: Highlighting, Mass change, Parallel CR, Enrichment Spot, Governance Scope, Data Import, Key-and Value Mapping, Worklists, Data Replication, Process Analytics
    - Master data remediation

5.6.4.1 Multiple Record Processing

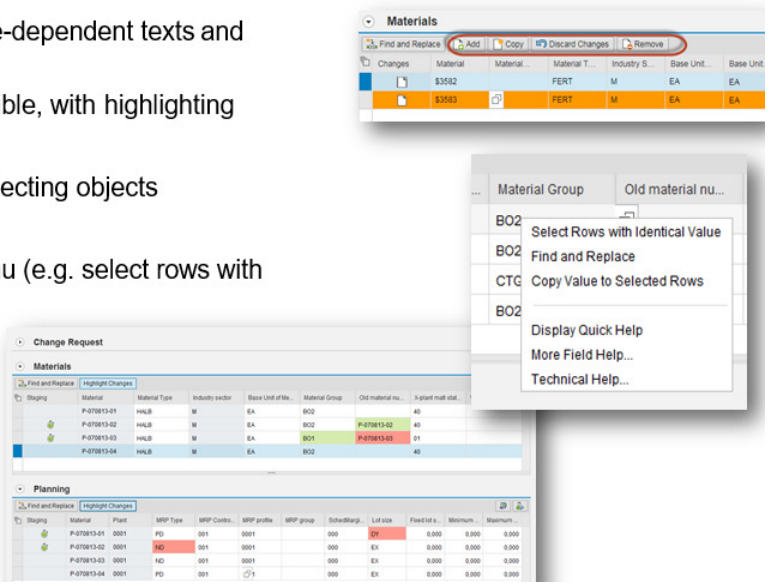
- Change selected materials when navigating from the search UI to the multiple record processing application
- Copy multiple materials with sub-objects when navigating from the search UI to the multiple record processing application
- Combined maintenance of different entity types
- List and Form UIBB's are supported
- Example UI configurations and customizing delivered by business configuration sets



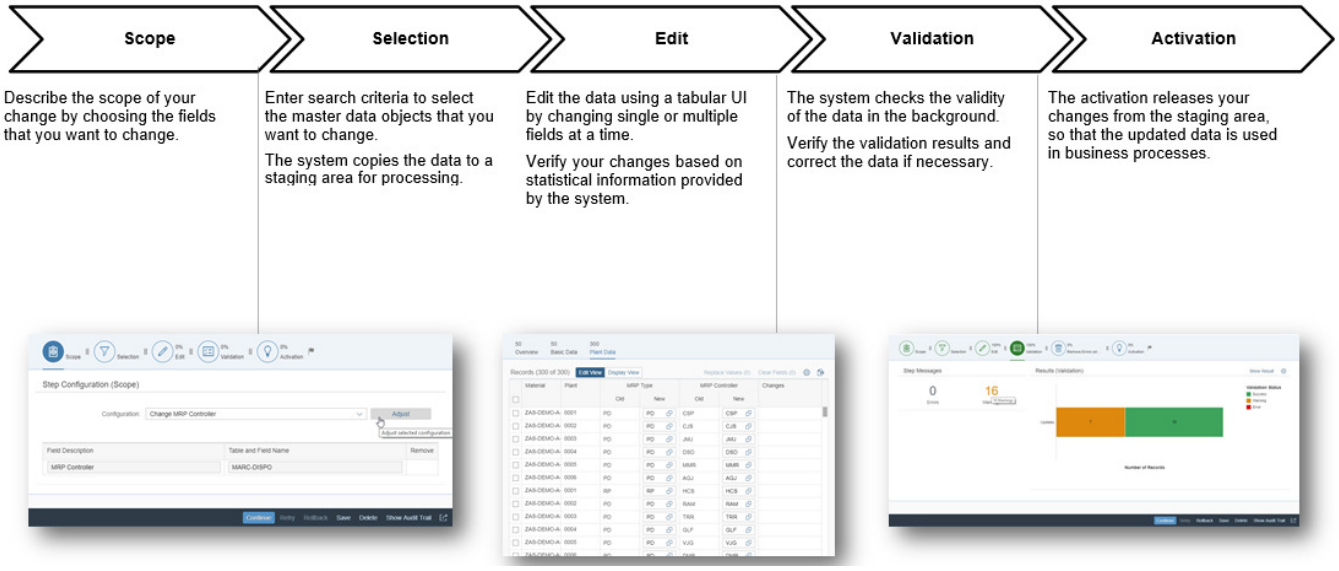


5.6.4.2 Multiple Record Processing Features

- Add, copy and change materials, language-dependent texts and organizational data in a change request
- Delete, remove and discard changes possible, with highlighting
- Copy with reference
- Create MuRP Change Request without selecting objects
- Add existing objects with New Button
- Several operators available in context menu (e.g. select rows with identical value)
- Highlighting changes with different colors
  - Saved changes
  - Current changes



5.6.4.3 Process flow



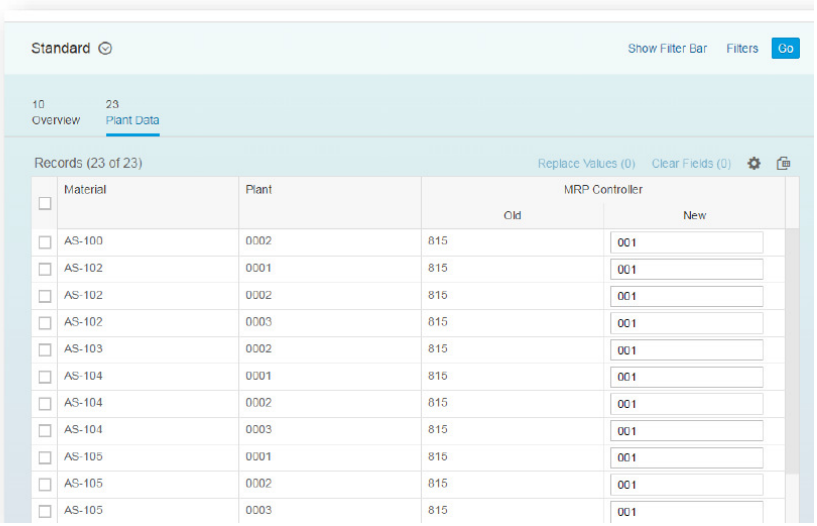
5.6.5 Mass Processing of materials

New process to execute mass changes for all attributes

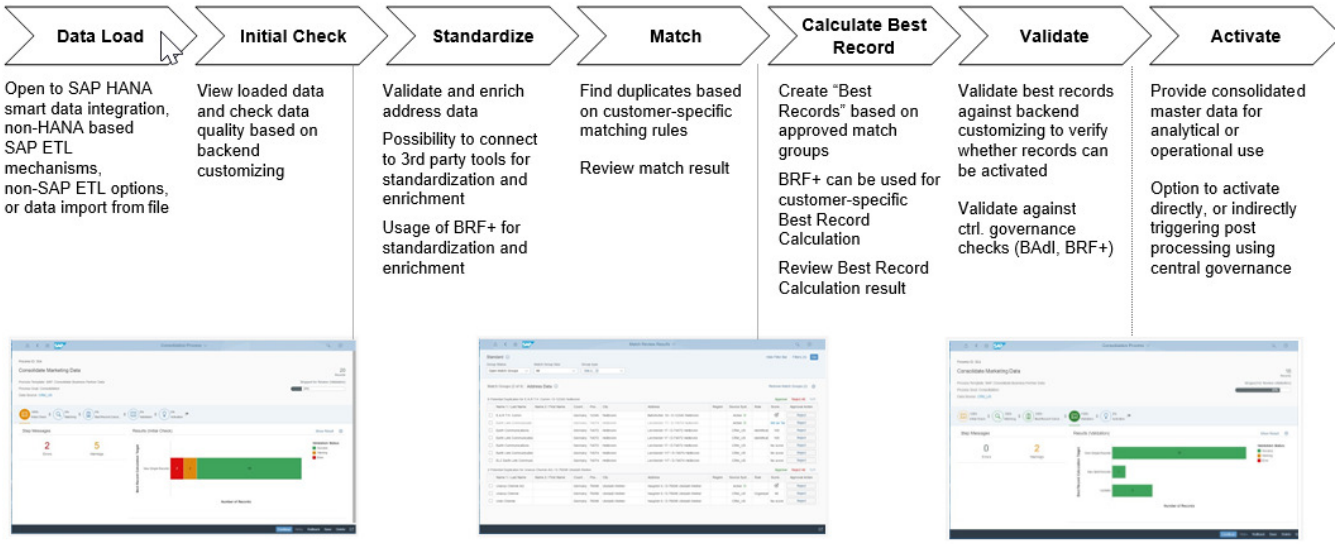
Process Steps

- Define fields to be changed
- Select materials to be changed
- Manually edit data to be changed
- Validate data against business rules
- Activate data changes

For more information, see separate Mass Processing presentation in the Learning Map.



5.6.6 Consolidation - Process flow

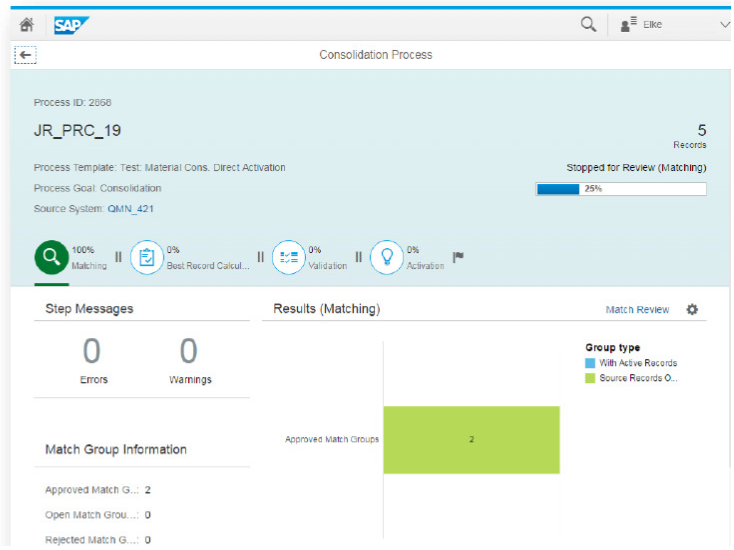


5.6.6.1 Consolidation of material

The new process SAP MDG, Consolidation enables you to enrich and standardize your data.

The solution helps you to identify and merge duplicate records to produce the best possible master data records. You can then validate and load your newly consolidated data into your target system. You can also integrate SAP MDG, consolidation with MDG, central governance.

For more information, see separate Consolidation presentation in the Learning Map.



5.6.7 MDG Framework Capabilities

5.6.7.1 Common Services of MDG for Material

Common services of change request processing

- Highlighting
- Mass change
- Support of parallel Change Request
- Enrichment Spot
- Governance Scope
- Key-and Value Mapping
- Import Master Data
- Worklists (My Change Request, Display Change Requests, Change Documents)
- Data Replication
- Process Analytics

5.6.7.2 Highlight Changes: Field changes

Highlight Saved Changes: Supports processor to decide about next action

Highlight Unsaved Changes: Provides information what can be “un-done”.

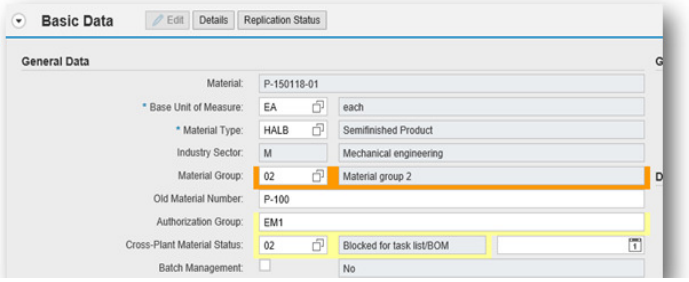
Saved changes and unsaved changes in two different colors.

Highlighting of Table Rows

Highlighted table rows refer to changes that you can only see when navigating from a table row to the details of a dependent entity.

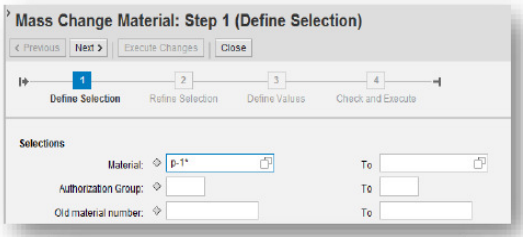
Tooltip Information

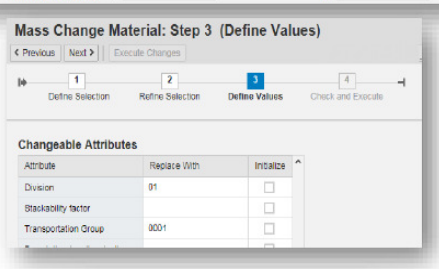
Field tooltip shows previous value.



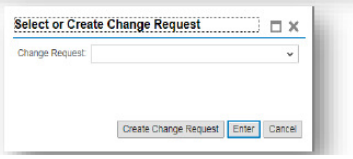
5.6.7.3 Mass Change

- Optionally, create a change request and submit it (typically w/o materials assigned, yet)
- Second, start the mass change and select your relevant materials by Define Selection (Step 1) and Refine Selection (Step 2)





- Step 3: Replace one or more fields with one dedicated value
- Step 4: You can review the results and then execute the change. If not yet done, you now need to assign an open mass change request that is assigned to you, or you can create a new change request



5.6.7.4 Support of Parallel Change Request (Multiple Change Requests per Material at the Same Time)

Creation of material with initial Change Request

- One Change Request for create material

Parallel Change Requests for subsequent individual processing

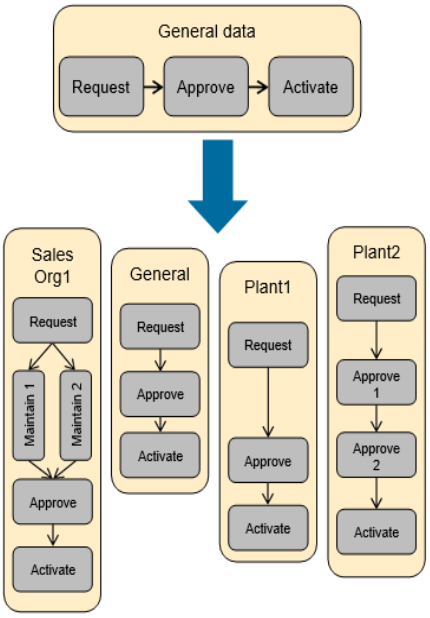
- Parallel Change Requests for extending material
- Extend material to different organizations or change existing data in parallel to extend the material

Activation for general data and per organizational unit directly after approval

- Workflow creation for each selected organizational unit, activation after approval

Lock on entity level (interlocking)

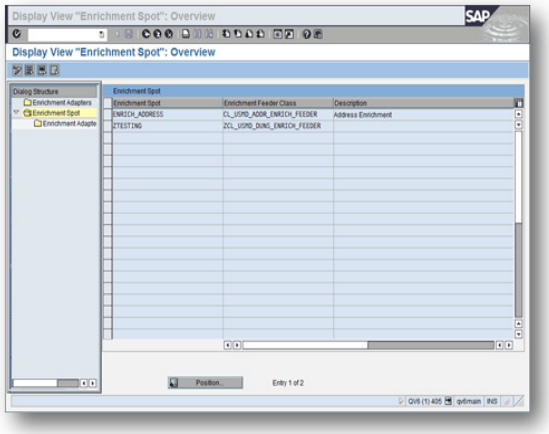
- Instead of lock on object level only
- Change request type determines the maintainable fields



5.6.7.5 Data Quality Framework for Data Enrichment Spots

Data Enrichment Spots

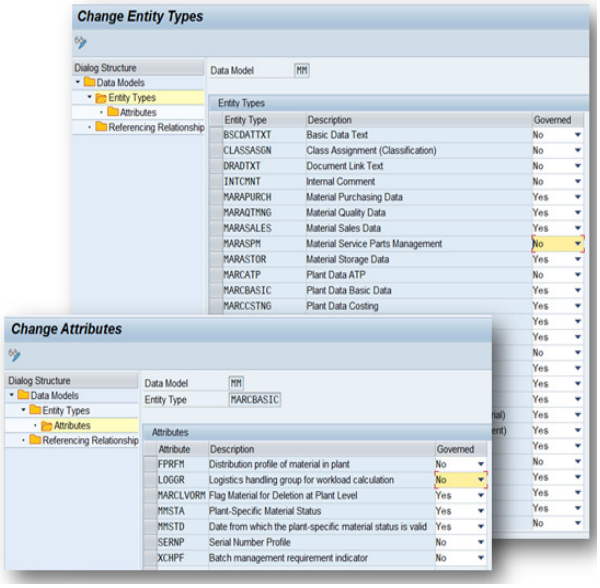
- Flexible framework to define enrichment spots
- Is used by SAP for example for Address Validation
- Can be used by customers to define further enhancement spots



5.6.7.6 Governance Scope - Deactivate Entity Types / Fields

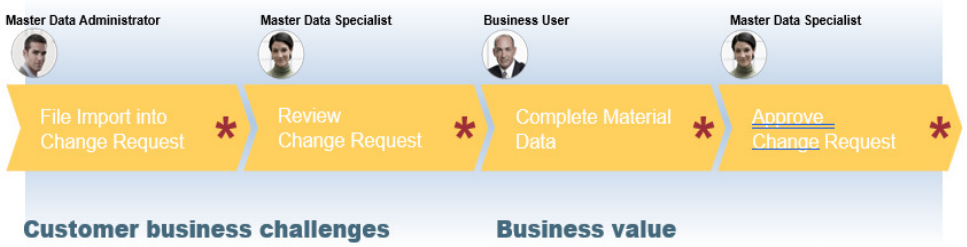
Governance Scope

- Only parts of the data model shall be put under governance
- Entity types / attributes can be excluded from governance
- Excluded entity types / attributes
  - can not be changed within a Change Request
  - are displayed only within the UI; can be suppressed by changing the UI configuration
  - can still be loaded to active area but not via CR / staging
  - Can easily be re-added to the governance scope afterwards



5.6.7.7 Import Master Data

Material master data, Key- and Value-Mapping can be uploaded to the MDG system from an XML file using the MDG Import Master Data transaction. The imported material master data can be adjusted or modified in a change request. Only when the change request is approved is data written to the SAP S/4HANA material master data (which is the active area).
If data quality is good, the data can also be written directly into the active area.



Customer business challenges

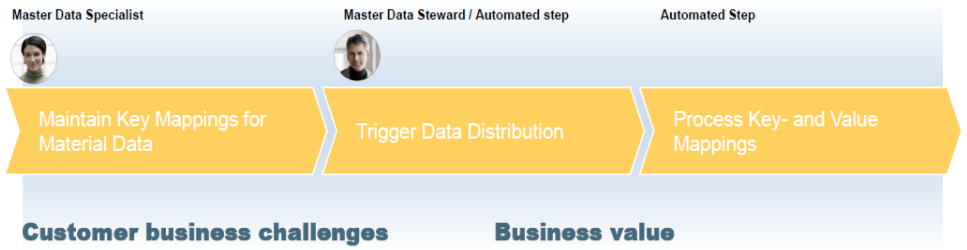
- No direct upload functionality was available within SAP ERP
- No governance process existed for the upload of mass data

Business value

- Upload to MDG system (Material, Key- and Value-Mapping)
- Built-in upload functionality for MDG increases the speed of updates and streamlines integration between systems
- Governance functionality avoids creation of duplicates and ensures completeness and accuracy of data

5.6.7.8 Key- and Value Mapping

This feature supports the replication of data into systems with non-harmonized customizing or with heterogeneous keys for material. For example, in cases where the same material has a different material number on the master data governance (MDG) hub than on a client (key mapping), or similar material groups have a different code in hub and clients (value mapping).



Customer business challenges

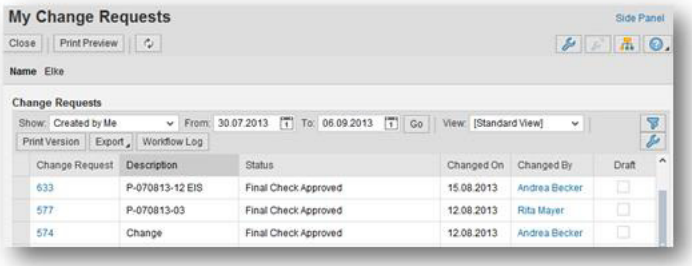
- Customers built their own solutions to map the values to distribute the data correctly to different systems

Business value

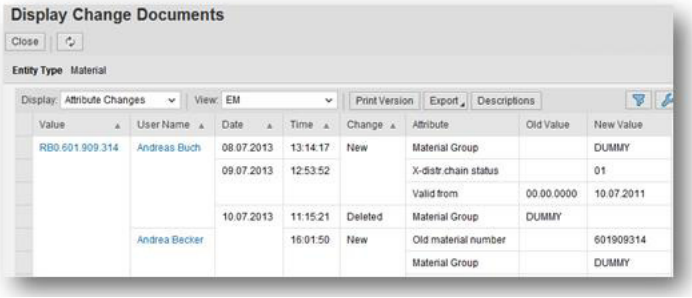
- Heterogeneous system landscapes can be supplied using the key- and value-mapping functionality offered with MDG

5.6.7.9 Worklists

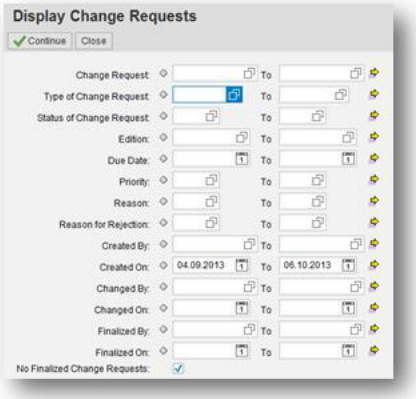
- My Change Requests



- Change Documents



- Display Change Requests



5.6.7.10 Data Replication for MDG Material, Outbound implementations

You can decide which outbound implementation you want to use in the replication models.

- DRF outbound implementation I_MAT

  The IDocs MATMAS (material) and CLFMAS (classification) are sent.

- DRF outbound implementation I_MAT_V2

  With the new outbound implementation, the IDocs DOLMAS (document assignment), ECMREV (change number and revision level) and MATQM (inspection lot) are sent in addition to MATMAS (material) and CLFMAS (classification).

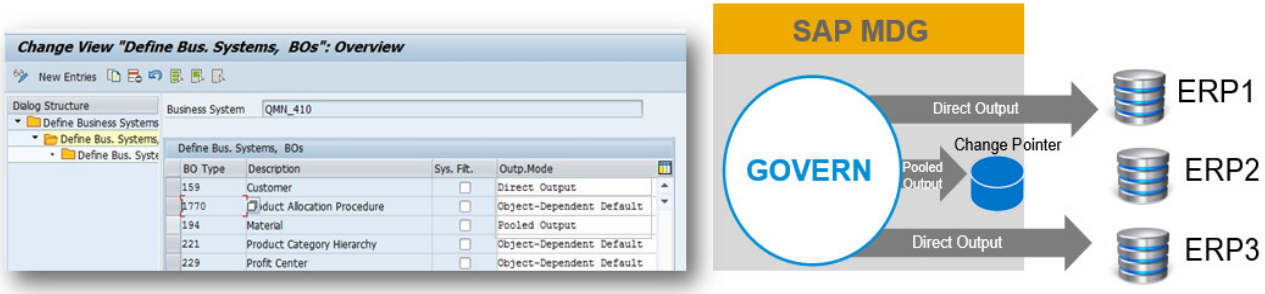
- DRF outbound implementation 194_3 (only S/4 HANA)

  The ProductMDMBulkReplicateRequest_Out SOAP service enables you to replicate product master records from the SAP S/4HANA to client systems.

5.6.7.11 Data Replication for MDG Material Enable Pooled or Direct Replication

You can decide if you want to enable direct replication or pooled replication for each combination of business object and target system. With direct replication the IDocs are sent with the activation of the change request, with pooled replication the IDocs are sent periodically using the DRF change pointer.

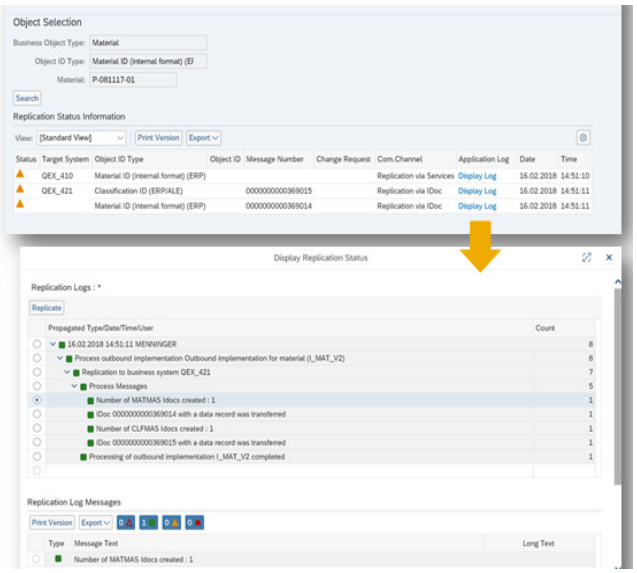
- **Direct Output:** Changes are directly transferred to a target system.
- **Pooled Output:** Changes are collected and transferred in a mass process at a later point in time.
  You can trigger the mass process for pooled output by starting or scheduling the report Execute Data Replication (RDRF_MESSAGE_OUT) for the relevant replication models.



5.6.7.12 DRF Replication Status Information

Replication Status Information

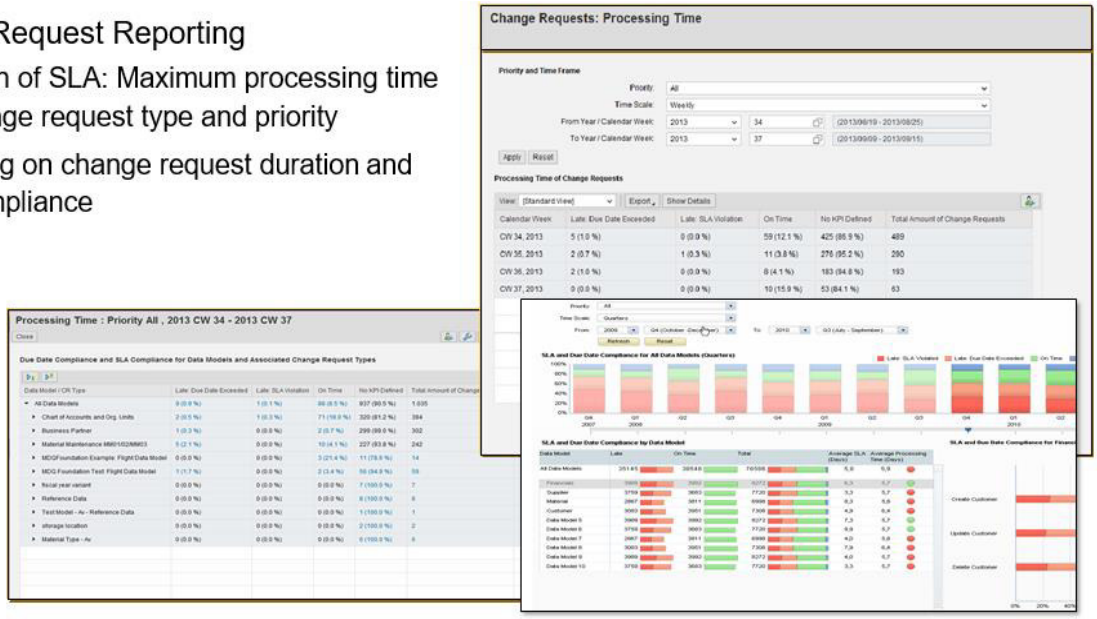
- Overview of replication status per material to all relevant target systems
- Status per target system
  - Green: OK
  - Yellow: OK, but answer from target system still pending
  - Red: Error occurred
- Detailed log per target systems available



5.6.7.13 Process Reporting

Change Request Reporting

- Definition of SLA: Maximum processing time per change request type and priority
- Reporting on change request duration and SLA compliance



5.6.8 Master data remediation (only S/4HANA)

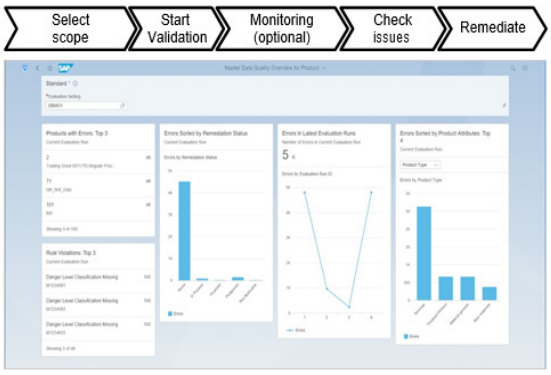
5.6.8.1 Trigger validation and remediation process for erroneous product master data

In a nutshell…

Data quality control for master data specialists seeking effective remediation of product data.

Business Value

- Enable data quality control for product data by validating data, revealing quality issues and starting the remediation process.



Process Related Capabilities:
Selection of records to be validated

- Validation results displayed in worklist
- Quality issues of product data displayed in worklist
- Filtering on specific object attributes, rule sets, groups and IDs
- Detailed description for every rule violation
- Navigation to object data
- Navigation to object maintenance applications incl. single or mass processing in MDG

5.6.9 Integration Scenarios

5.6.9.1 Integration with SAP Hybris Product Content Management (PCM) Integration with PLM

SAP MDG provides central governance of master data across the product, supplier,customer, financial and custom domains.

SAP Hybris PCM is a PIM (Product Information Management) product.

For many customers looking for a complete EIM solution in the product space, both solutions are needed.

MDG provides detailed governance, control and data quality over the creation and update of data used in transactional, analytical and reporting systems.

SAP Hybris PCM provides product catalogs and other PIM functionality.

A joint positioning of these products in a customers landscape will provide complementary benefit to our customers and a complete product data management solution.

MDG will govern and control core data in a landscape. For data needed in a product catalog, MDG will feed SAP Hybris PCM for further enrichment and for catalog management.

External data can (optionally) feed SAP Hybris PCM directly if it is not needed in transactional, analytical and reporting systems.

SAP's other EIM tools will support in an overall landscape perspective.

5.6.9.2 Decision Indicators for Product Data Management

Situation	Initial Guidance
Most typical scenario: Company runs Business Suite, has e.g. own production or goods logistics, maintains product data (partly) themselves, and needs PIM capabilities, e.g. catalogs	**MDG-M and Hybris PCM**
Special ERP focus scenario: Company runs Business Suite, has no need for rich product information handling, is not going into eCommerce	**MDG-M**
Special eCommerce focus scenario: Company resells goods from other suppliers, acquires product data from suppliers and uses catalogs or multi-channel eCommerce tools, without need to manually maintain product data, without running own warehouse logistics or production	**Hybris PCM**

5.6.9.3 Joint Value Proposition

SAP PLM allows customers to manage their product data along the entire product life cycle from idea through portfolio management, engineering, release processes and costing to handover to manufacturing and service.

SAP MDG provides central governance of master data across the material, supplier, customer, financial and custom domains.

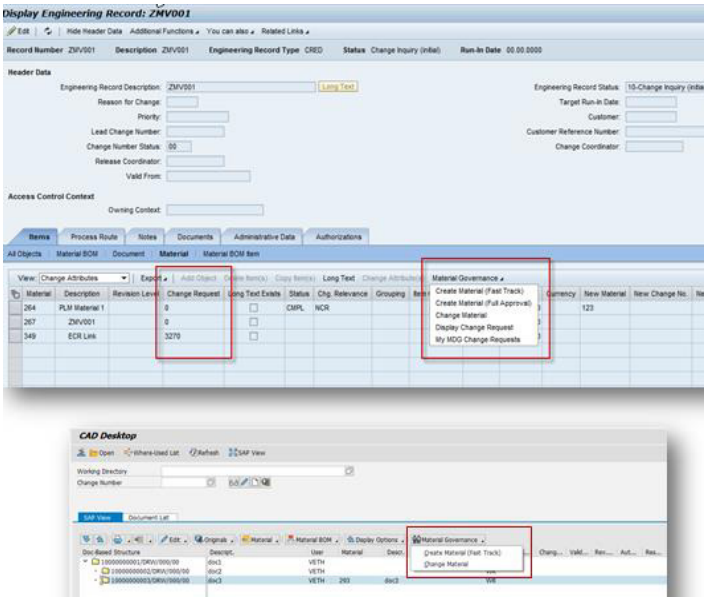
For many customers looking for a complete solution in the product space, both solutions are needed.

- SAP PLM orchestrates the processes of managing product data
- MDG provides detailed governance, control and data quality over the creation and update of material data used in transactional, analytical and reporting systems.

A joint positioning of these products in a customers landscape will provide complementary benefit to our customers and a complete product data management solution.

5.6.9.4 Process Integration Scenarios

- Enablement for PLM integration scenarios:
  - PLM Engineering Record
    - Create New Material
    - Change Existing Material
    - Check Status Change Request
  - CAD Desktop
    - Create New Material
    - Change Existing Material
- Available as SAP Consulting solution
- Concept reusable for integration into other objects



5.7 Benefit from the Solution MDG-M

The efficient way towards consistency and quality of your core information assets

SAP Master Data Governance (SAP MDG) provides end-to-end processes for the entire master data lifecycle. MDG helps empower business agility with SAP HANA-based analytics and consumer-grade user experience. Built on a strong foundation, it provides improved business efficiency and process flexibility, enabling simplification.

There are two versions of the product: **SAP MDG on SAP S/4HANA** and **SAP MDG** (which technically runs on SAP ERP 6.0). Out-of-the-box applications for central MDM

- SAP MDG for financial data (financial accounting / consolidation data, companies, profit & cost centers)
- SAP MDG for supplier data (corporate, company, and purchasing related data, partner functions)
- SAP MDG for customer data (corporate, company, and sales related data, partner functions)
- SAP MDG for material data (corporate, sales org, plant, storage, warehouse, valuation & costing data)
- SAP MDG for enterprise asset management data* (asset structures, work management, bills of material)
- SAP MDG for retail and fashion management* (single & generic article, listings, stores, DCs, procurement & sales)

Flexible framework for your own master data…

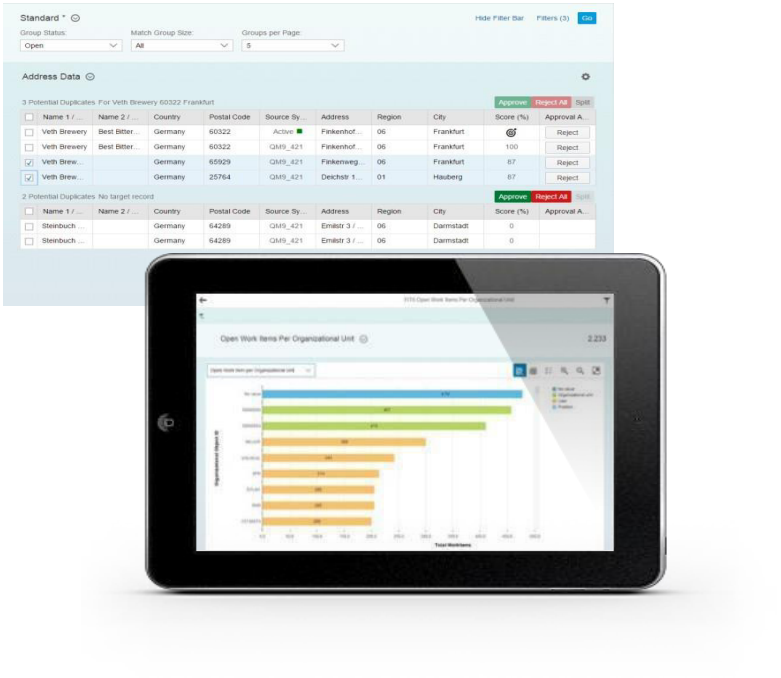
- …for enrichment of data models, processes, user interfaces, and to define governance for custom-objects

Master data consolidation

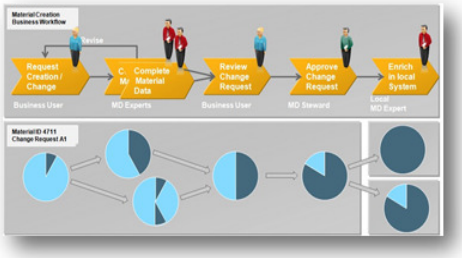
- Load, standardize, match, calculate best records (business partner, product master data or custom-objects)

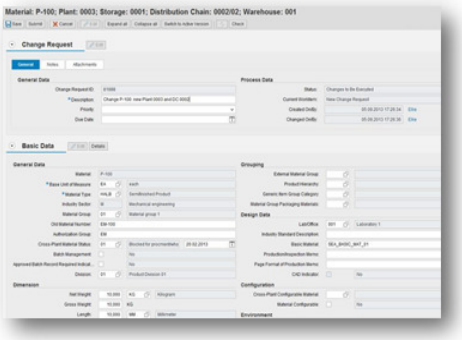
State-of-the-art SAP MDM solution

- Compelling user experience and dedicated SAP Fiori apps e.g. for master data request and approval
- Advanced data standardization, search, and duplicate detection (SAP HANA as an option)
- Governance of master data hierarchies as a basis for trusted business insight
- Interactive mass processing for ad hoc or project-based bulk changes by a data specialists
- Process quality trends or KPI reporting and data quality analytics with remediation
- Data replication to all SAP and non-SAP, cloud and on-premise systems in the landscape



5.7.1.1 Integrated, Centralized Product/Material Data for Process Excellence





Material Processing

- Workflow-driven creation and change process for Materials
- Change requests steer the maintenance & approval process of new or changed data
- Document changes for audits
- Work lists organize due changes
- Adaptable user interfaces and validations can be different per workflow step if needed

Distribution

- Changes are kept inactive until approval
- They can then be distributed to operational group or local systems via DRF using ALE

Scope of Master Data

- Search, Display, Change, Creation of single Material Master Data
- Mass and multi processing of changes
- File upload for initial load, system consolidation
- Broad out-of-the-box coverage of material attributes – incl. central and classification data as well as organizational unit data

6 Business Partner (BP)

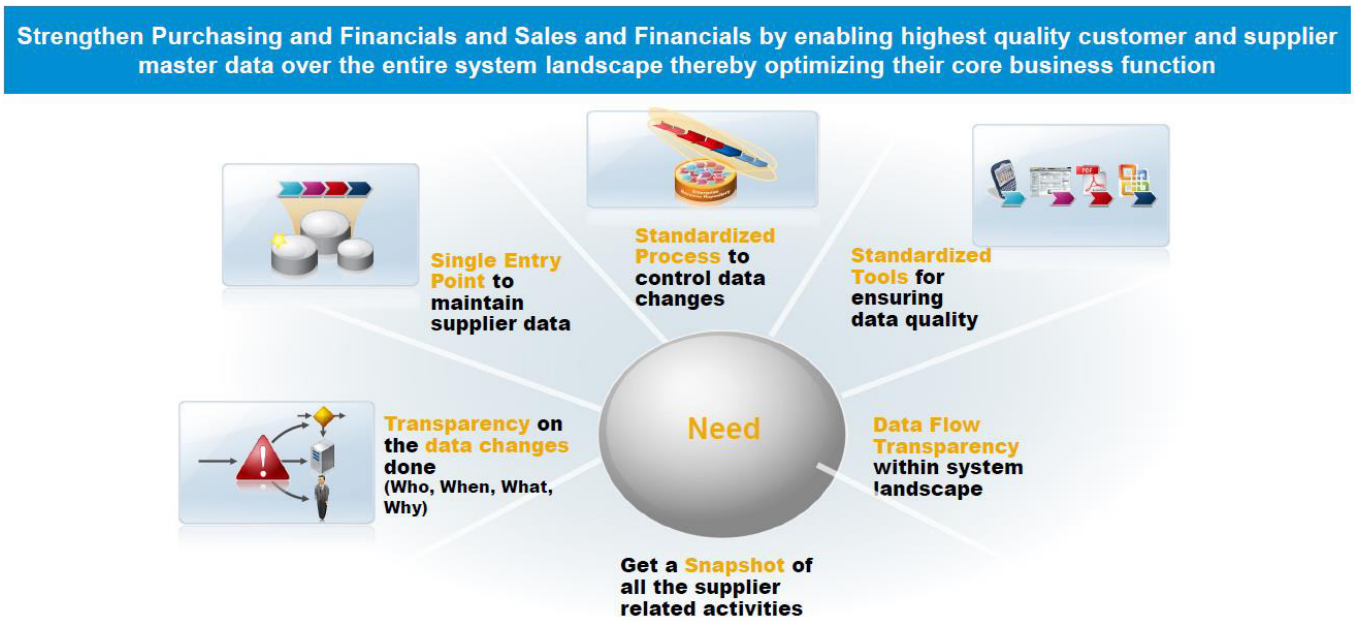
Organization (firm, branch office), person or a group of persons or organizations in which your company has a business interest.

You can create and manage your business partners centrally for different business transactions.

This is of prime interest if a business partner plays different roles for a company, such as sold-to party and ship-to party.

You can create a business partner in different business partner roles. During the course of the business relationship, the business partner can assume other business partner roles. You do not need to create the general data, which is independent of a business partner's function or of application-specific extensions, again in each case for this business partner. This prevents data being created and stored redundantly.

6.1 Goals for Central Governance of Business Partners



6.2 Structure

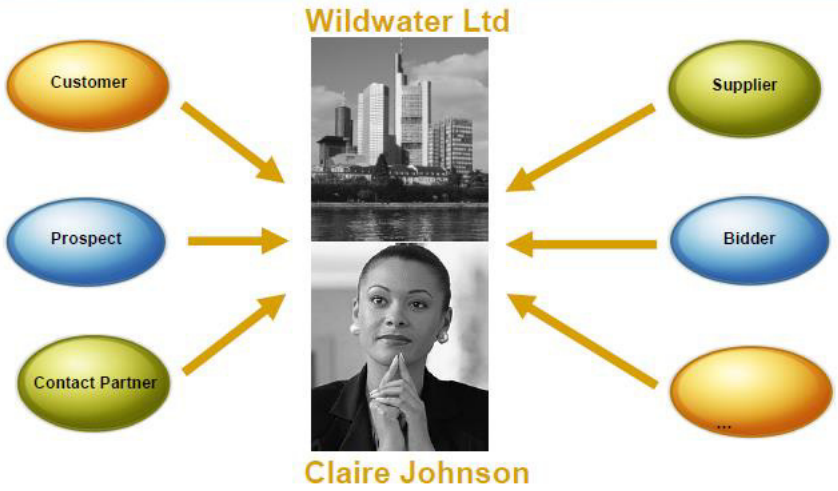
You can use the following basic elements, which form part of the business partner data:

Business Partner Concept

Concept	Remark
Business Partner Category	The data that is available for your business transactions depends on the business partner category .
Business Partner Role	A business partner can take on different business partner roles , which allow different views of the business partner data depending on the business function.
Business Partner Relationship	A relationship connects two business partners and is characterized by the business partner relationship category .
Business Partner Group Hierarchy	You can represent the structure of an organization using the business partner group hierarchy

6.3 Integrated Object Model: Business Partners

- Component for partner management.
- Arranges partner data in a cross-component general part and in any number of component-specific parts (role concept).
- Roles are defined as being meaningful groupings of attributes of a business partner.



The Entity type *Business Partner* is a generic application for business partner management (maintaining organizations, persons and groups of persons)

General Data

- Consists of core attributes and business attributes from various applications
- Customers/Partners can enhance the BP by adding their own attributes
- Time dependent with Technical validity and business validity

Role Concept

- Roles are views on the attributes and define the relevance by displaying the specific fields ONLY
- Classification according to business criteria
- Different attributes relevant for each BP Role
- **Examples:** Customer, Supplier, Prospect, Bidder, Payee, Employee

Relationships

- BPs can be connected using relationships
- Relationship categories depict types of business connection between BPs
- Categories can have attributes and the categories and the attributes can be enhanced by the customer

Usage

- Suite Ext.Applications (CRM, SRM)
- Industry solutions
- Suite Ext. Industries (Banking, Insurance, Utilities, Telecom, Media, …) *to meet generic and industry-specific requirements for business partner management*

6.4 Business Partner Tables

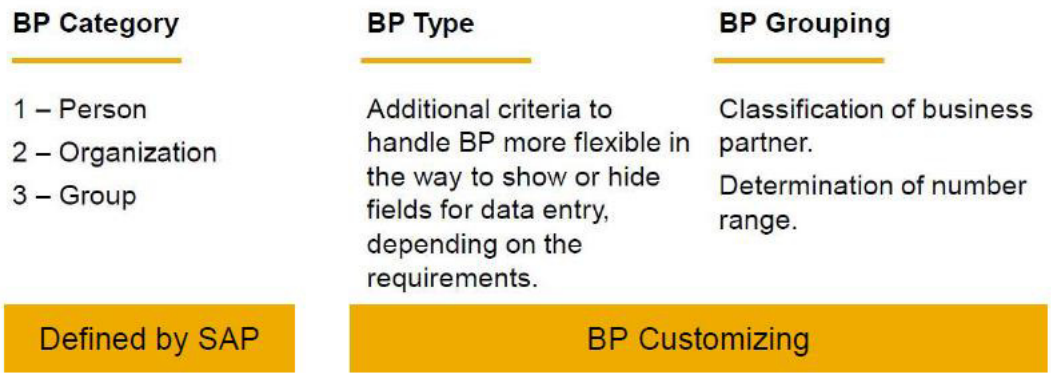
Business partner tables:

- BUT000: BP General data I
- BUT020: BP Addresses
- BUT021: BP Address usages
- BUT050: BP relationship/role definition
- BUT0ID: BP ID Numbers
- BUT0IS: BP industries sector
- BUT100: BP roles

Business address tables:

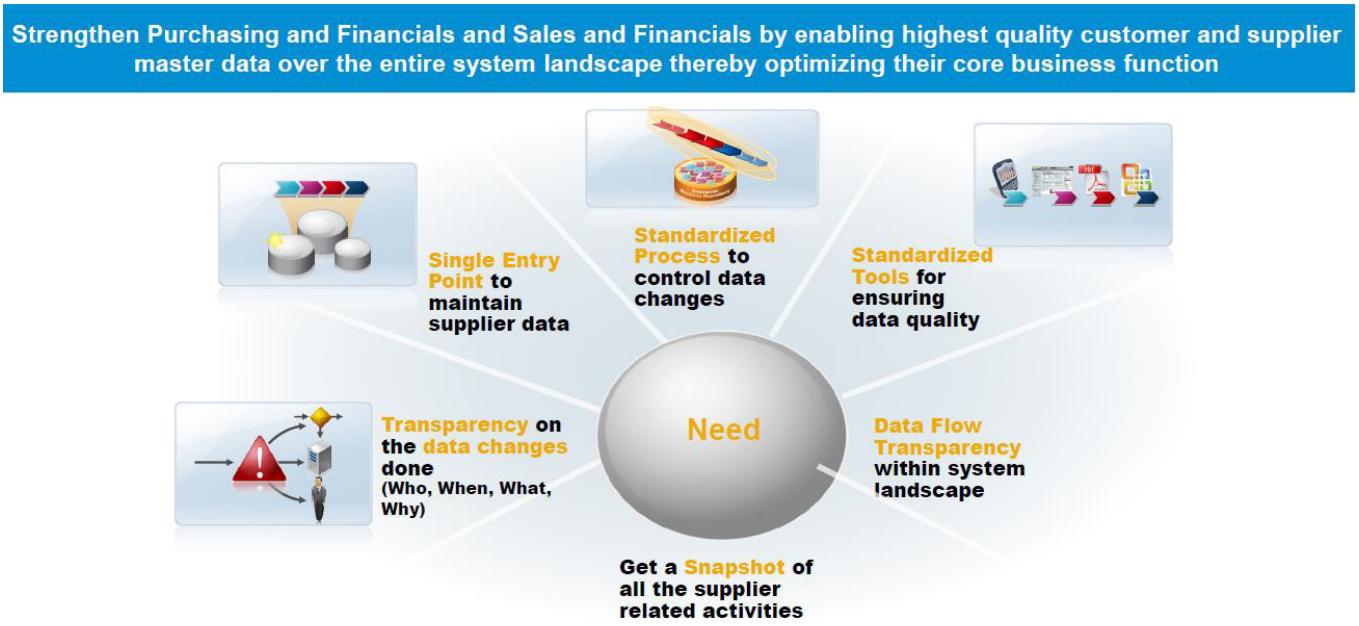
- ADRC: Addresses
- ADR2: Telephone Numbers
- ADR3: Fax Numbers
- ADR6: E-Mail Addresses
- ADRU: Table for Communication Usages
- ADRV: Address Where-Used List

6.5 BP Identification



6.6 Business Partner Capabilities

6.6.1 Integrated Object Model: SAP Business Partner Fundamentals



6.6.2 Integrated Object Model: Solutions for Customers, Supplier, and Business Partners

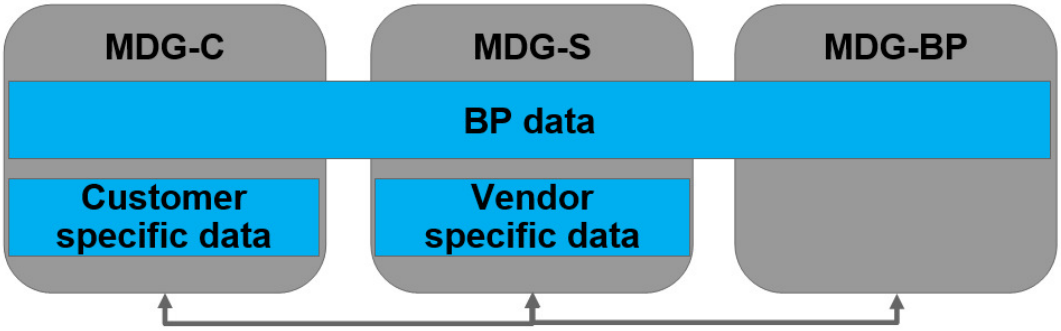
All solutions use the SAP-BP as central anchor for identifying data

- Business Partners being customer and supplier are represented by one BP only

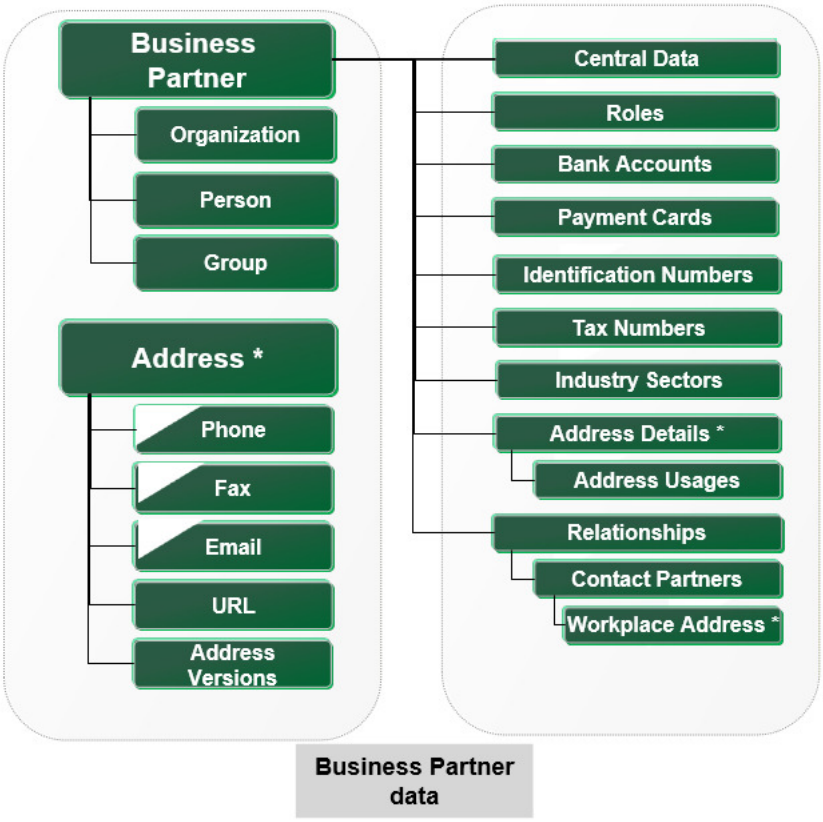
MDG-BP as basis for

- Customers limiting their governance scope to BP data only
- Solutions only being based on SAP-BP (e.g. Service Industries)

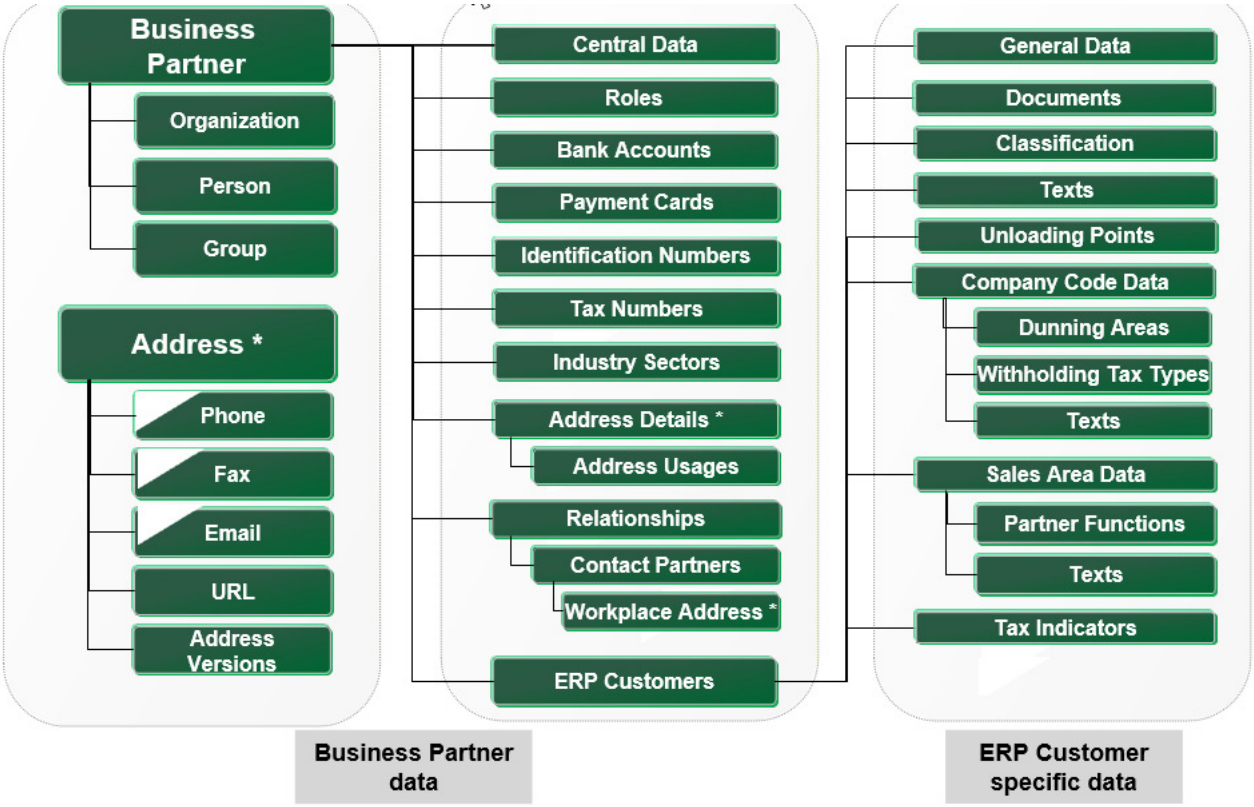
Transition from MDG-BP to MDG-S / MDG-C without disruption



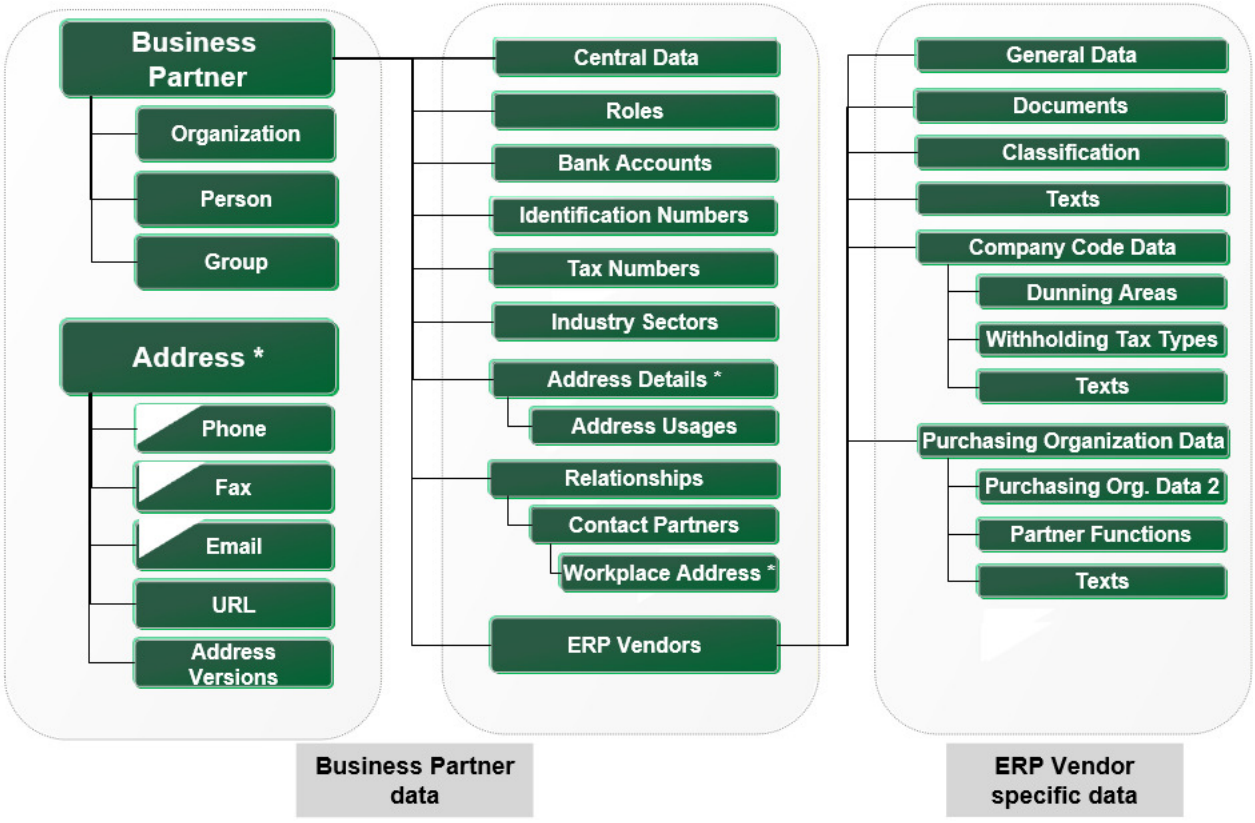
6.6.3 Integrated Object Model: Data Model MDG-BP



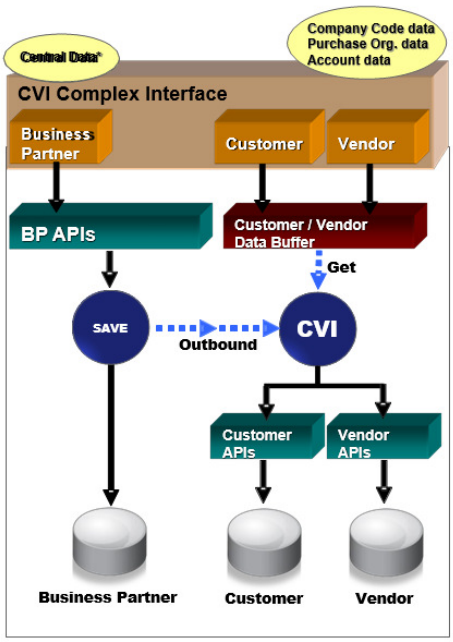
6.6.4 Integrated Object Model : MDG-C with BP



6.6.5 Integrated Object Model: Data Model MDG-S with BP



6.6.6 Integrated Object Model: Data Model Customer / Vendor Integration



Example: Vendor Creation

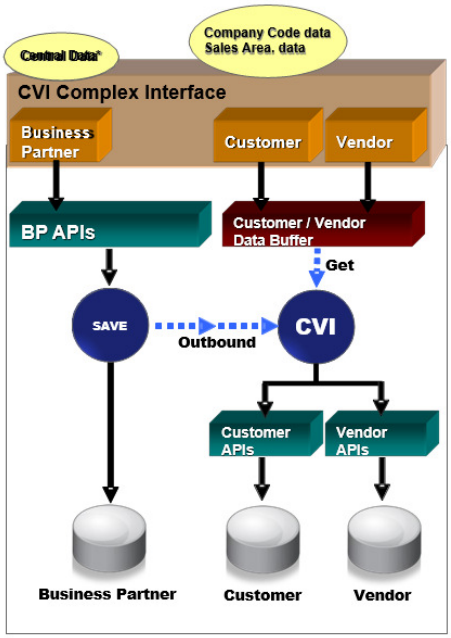
A **Business Partner** is always created when a customer or supplier is created.

The complex interface of the CVI (Customer-Vendor-Integration) contains **Business Partner** specific data as well as **Customer and Vendor** specific data.

Partially, the data of the Business Partner and Customer/Vendor are redundant (BUT000 against KNA1 & LFA1 data). *For instance 'Name and Address specific attributes' are available in both persistencies*.

Customer or Vendor specific data is routed through the Customer/Vendor specific interface and mixed up with the Business Partner central data.

On commit, the **Business Partner and corresponding Customer and/or Vendor** is maintained / created.



Example: Customer Creation

A **Business Partner** is always created when a customer or supplier is created.

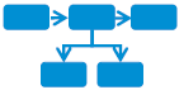
The complex interface of the CVI (Customer-Vendor-Integration) contains **Business Partner** specific data as well as **Customer and Vendor** specific data.

Partially, the data of the Business Partner and Customer/Vendor are redundant (BUT000 against KNA1 & LFA1 data). *For instance 'Name and Address specific attributes' are available in both persistencies.*

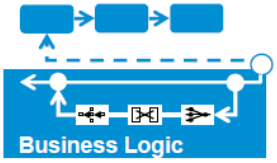
Customer or Vendor specific data is routed through the Customer/Vendor specific interface and mixed up with the Business Partner central data.

On commit, the **Business Partner and corresponding Customer and/or Vendor** is maintained / created.

6.6.7 Governance, Collaboration



Linear or distributed and workflow based



Possibility to extend the process by adding business logic and thereby bringing in data quality

Has roles and responsibilities and also task authority
(e.g., create, change, approve)

Data in process stored in a separated repository, will be transferred to operational database after final approval

Can be adapted and tailored based on customer needs

Flexible enough to respect the distributed responsibilities existing in a company across various business units

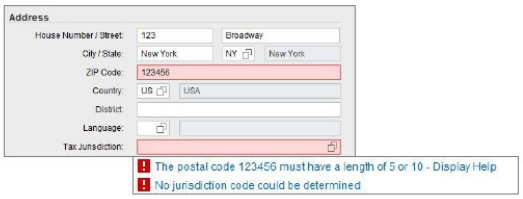
6.6.8 Data Quality

Prevent creation of duplicates for increased effectiveness and efficiency

- Checked early and embedded in the process
- High detection quality of matching using Enterprise Search or SAP Data Services

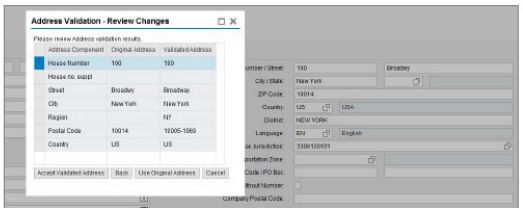
Validations

- Re-use of existing validation logic in S/4HANA
- Custom validations can be modeled and programmed (e.g. code lists, simple checks, or modeled rules via SAP BRF+)

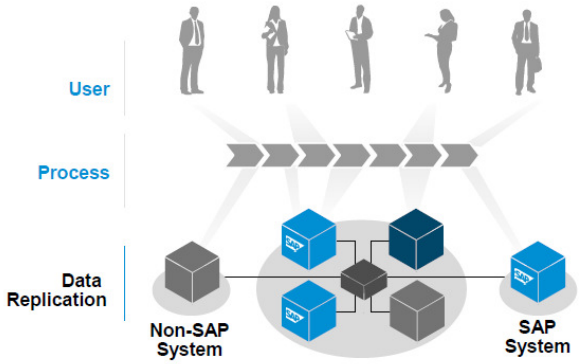


Address Enrichment

- Simple check and selection lists
- Integration with content provided by SAP Data Services
- Automatically adding Tax Jurisdiction Code re-using existing interfaces / providers



6.6.9 Data Replication



Replication

- Automatic or manual replication to SAP and non-SAP systems
- Peer-to-peer or mediated

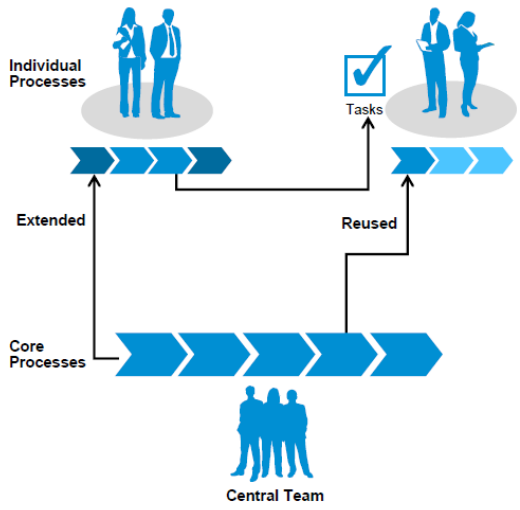
Replication Techniques

- DRF controlling replication mechanisms like
  - ALE for S/4HANA OP, SAP ECC, R/3 and non SAP systems
  - SOA service for SAP and non SAP systems
  - RFC and File for SAP and non SAP systems
- CIF for SAP SCM and MW for SAP CRM/SRM

Replication Features

- Flexibility to replicate selected data only to a specific client system by defining filters (controlled by DRF)
- Support of different object IDs or customizing keys between hub and client systems (via mapping in MDG or PI)
- Monitoring and error handling

6.6.10 Reusability & Extensibility



Reuse standard content

- Reuse existing data structures and functionality of SAP S/4HANA

Enhance functionality

- Be able to enhance MDG functionality based on predefined content (e.g. plant or sales data) or customer-specific extensions

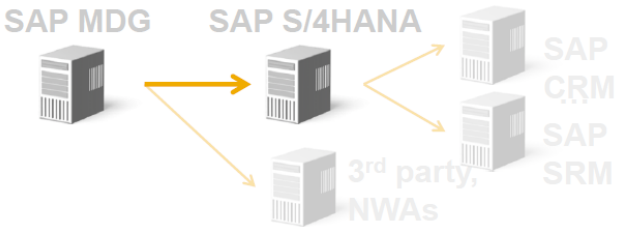
Extend data structures

- Provide easy-to-use extensibility capabilities for enhancing nodes and fields
  - data models
  - user interfaces
  - processes
  - data replication

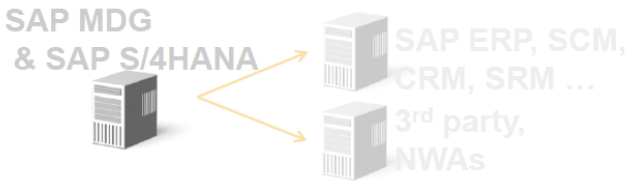
6.6.11 Deployment Options: Consciously Decide for a Central hub or Co-deployment

Deploy SAP MDG…

- … as standalone system / master data hub



- … on top of an operational S/4HANA / ECC



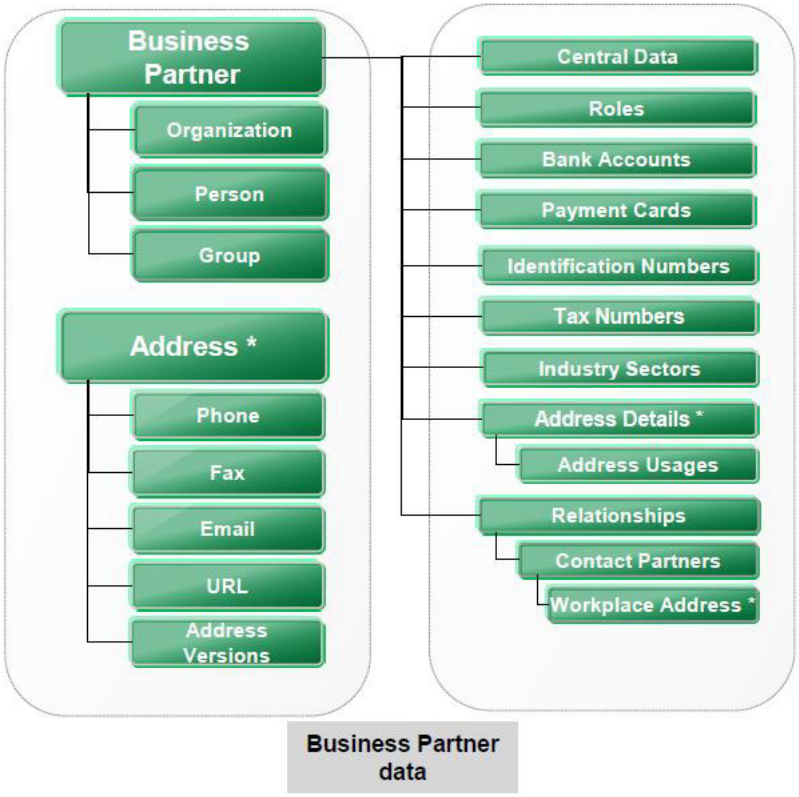
- **Customers typically choose to…**

… create a dedicated master data hub and distribute to all operative systems from there, or

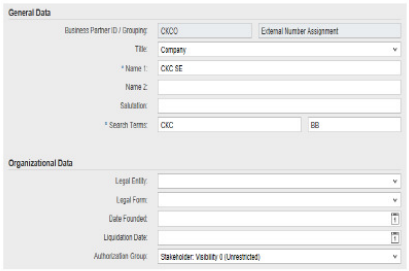
… co-deploy*) with their single instance SAP S/4HANA / ECC (or pick one of their SAP S/4HANA's / ECCs as the master for a particular domain) and create and distribute master data from there

6.7 Function in Detail Business Partner Data

6.7.1 Data Model Business Partner

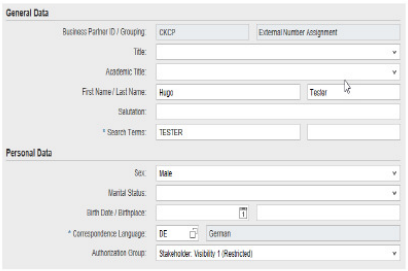


6.7.2 Central Data



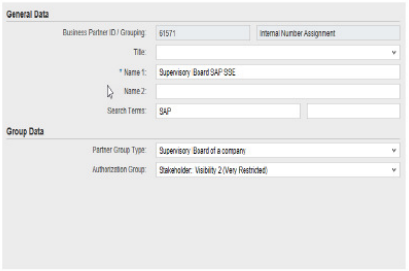
Organization

- 4 Name fields plus Search Terms
- Organizational data
- Selected fields visible only, more fields can be added via UI Configuration



Persons

- Several name fields like first name and last name plus search terms
- Personal data
- Selected fields visible only, more fields can be added via UI Configuration



Groups (of persons)

- 4 Name fields plus Search Terms
- Group data including Group Type
- Selected fields visible only, more fields can be added via UI Configuration

6.7.3 Addresses

Multiple Addresses can be maintained per BP

Address Data

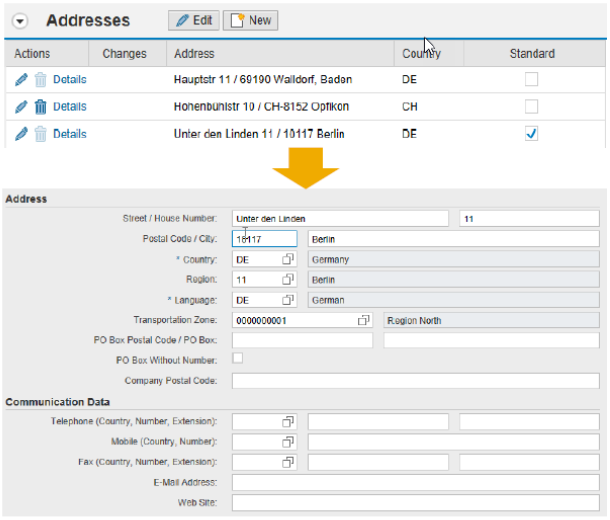
- Postal address
- PO Box address

Communication Data: Multiple …

- Telephone (landline and mobile) and Fax numbers
- Email addresses and Homepages

UI Layouts for Address data

- Layouts for Europe, USA, Canada, and Japan
- Layout for all users in IMG
  SAP NetWeaver → Application Server → Basis Services → Address Management → International Settings → Choose Address Screen Layout
- Layout per user via user parameter ADDRESS_SCREEN



6.7.4 Address Usages

Configure Address Types / Transactions

- Address Types: Visible in UI
- Address Transactions: Code against ID
- Configuration in IMG
  Cross-Application Components → SAP Business Partner → Business Partner → Basic Settings → Address Determination

Synchronization via CVI to ERP Customer / ERP Vendor

- Synchronize assigned address per ERP Customer / ERP Vendor
  ➔ See slides about 'Usage of Multiple Addresses')

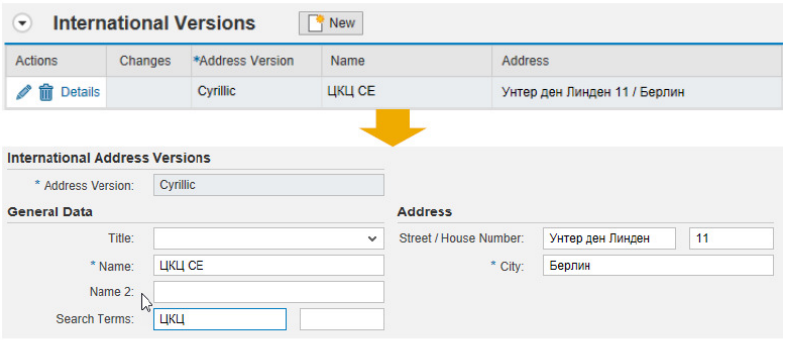
Address Usages Edit New

Actions	Changes	*Address Type	Description	*Address	Standard Usage
		0002	Home Address	Hauptstr 11 / 69190 Walldorf, Baden	☑
		0004	Holiday Home	Hohenbühlstr 10 / CH-8152 Opfikon	☑
		XXDEFAULT	Standard Address	Unter den Linden 11 / 10117 Berlin	☐

6.7.5 Address Versions

Address versions

- Maintain address data per character set
- Latin, Cyrillic, Greek, Kanji, Hebrew, etc.
- Detailed documentation see note 316331, attached document



Maintain multiple versions per address

Configuration of Address Versions
IMG: SAP NetWeaver → Application Server
→ Basis Services → Address Management
→ International Settings → Activate International Address Versions

Synchronization with ERP Customer / ERP Vendor

- Synchronize all address versions per address

6.7.6 Business Partner (BP) Roles

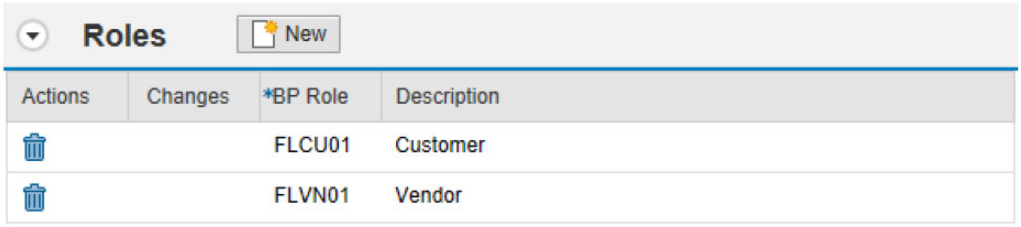
Assign multiple roles per Business Partner

Time-validity of BP Roles supported

- Fields can be added to the UI via UI Configuration

Synchronization with ERP Customer / ERP Vendor during activation

- Define roles to enable Customer Vendor Integration (CVI)
  IMG: Cross-Application Components → Master Data Synchronization → Customer/Vendor Integration → Business Partner Settings
  → Settings for Customer Integration → Set BP Role Category for Direction BP to Customer
  → Settings for Vendor Integration → Set BP Role Category for Direction BP to Vendor



6.7.7 BP - Bank Accounts

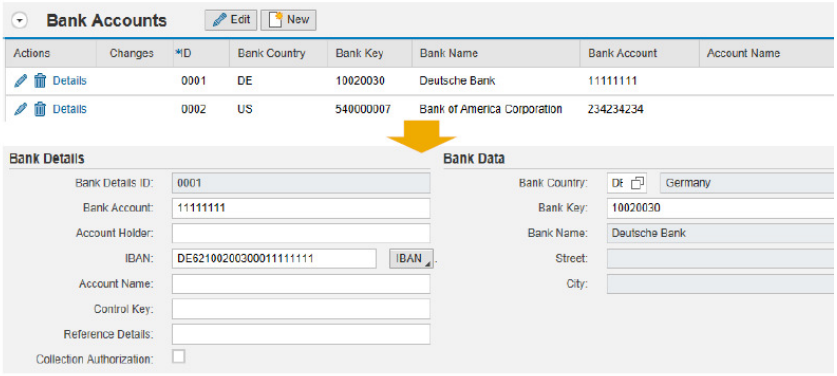
Multiple bank accounts per BP

Country-specific checks

- Bank ID and Bank Account Number
  SAP NetWeaver → General settings
  → Set Countries → Set Country-Specific Checks

Time-validity of Bank Accounts

- Fields can be added via Screen Configuration



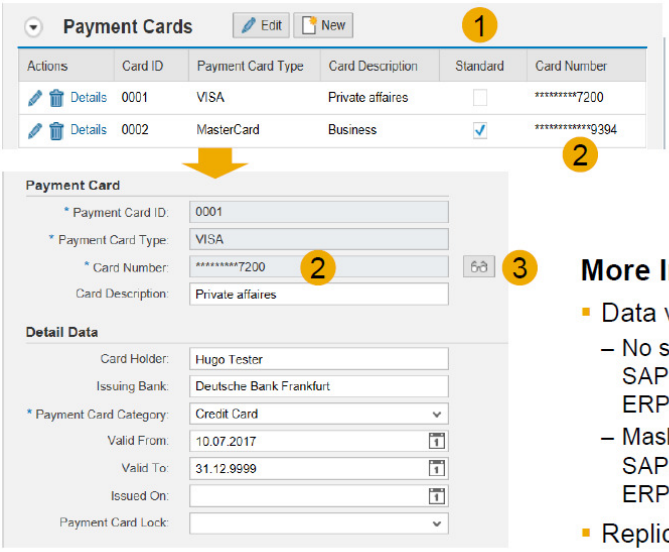
Legal requirements from Single European Payment Area (SEPA)

- IBAN: Maintain bank accounts with IBAN only (w/o Bank Identification Code)
- Mandates: Maintenance per local system (transactions BP, XD*, XK*), not supported in MDG

Synchronization with ERP Customer / ERP Vendor

- Synchronize all Bank Accounts to standard assigned ERP Customer / ERP Vendor

6.7.8 BP - Payment Cards



User Interface

1. Multiple payment cards per BP can be maintained in MDG-C. One can be marked as standard.
2. Card numbers can be masked on UI or even encrypted on database level (SAP-BP).
3. Only users with special authorization are allowed to see the full card number on request.

More Information

- Data will be stored during activation
  - No security measure or Masked display
    SAP-BP: Table BUT0CC, CCARD
    ERP Customer: tables VCKUN / VCNUM
  - Masked display & Encrypted when saved
    SAP-BP: Tables BUT0CC, PCA_*
    ERP Customer: Tables VCKUN / VCNUM
- Replication
  - via ALE (Idoc DEBMAS) or

6.7.9 BP - Identification Numbers

Maintain multiple Identification Numbers BP

Configuration of ID Type and ID Category

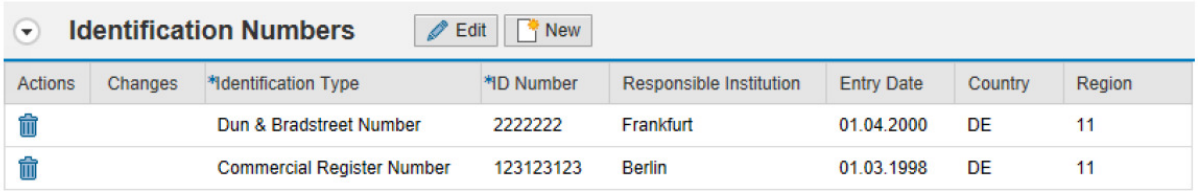
- Pre-defined values by SAP
- Customer can add values
- One or multiple numbers per ID Type

SEPA (Slides in What's new for MDG 7.0

- IBAN
- Mandates (local use only, not supported in MDG)

Synchronization with ERP Customer / ERP Vendor during activation

- No



Identification Numbers Edit New

Actions	Changes	*Identification Type	*ID Number	Responsible Institution	Entry Date	Country	Region
		Dun & Bradstreet Number	2222222	Frankfurt	01.04.2000	DE	11
		Commercial Register Number	123123123	Berlin	01.03.1998	DE	11

6.7.10 BP - Tax Numbers

Maintain multiple Identification Numbers BP

Configuration of ID Type and ID Category

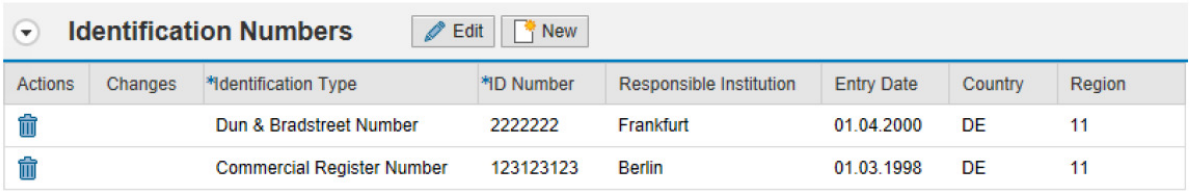
- Pre-defined values by SAP
- Customer can add values
- One or multiple numbers per ID Type

SEPA (Slides in What's new for MDG 7.0

- IBAN
- Mandates (local use only, not supported in MDG)

Synchronization with ERP Customer / ERP Vendor during activation

- No



Identification Numbers Edit New

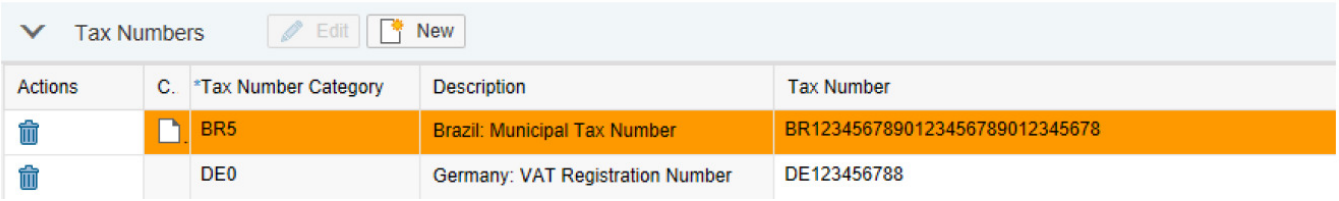
Actions	Changes	*Identification Type	*ID Number	Responsible Institution	Entry Date	Country	Region
		Dun & Bradstreet Number	2222222	Frankfurt	01.04.2000	DE	11
		Commercial Register Number	123123123	Berlin	01.03.1998	DE	11

6.7.11 Tax Numbers, Support of numbers with more than 20 characters

Tax numbers were limited to a length of 20 characters, now extended to up to 60 characters in the UI

Relevant for some countries like Brazil with tax numbers with more than 20 characters

Data base for long tax numbers
- SAP Business Partner: Tax Number Category ending with '5', data stored in DFKKBPTAXNUM-TAXNUMXL
- ERP Customer / ERP Vendor: data stored in new fields KNA1-STCD5 / LFA1-STCD5



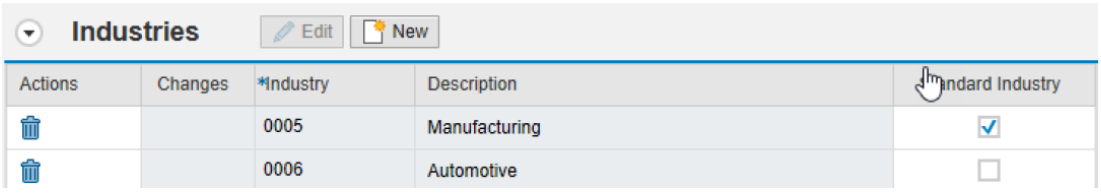
6.7.12 Industries

Maintain multiple Industries per BP

Optional: Maintain multiple industries for multiple Industry Systems (Catalogs)
- Add field Industry System to UI Configuration if multiple Industry Systems are relevant

Configure Industry Systems and Industries
Cross-Application Components → SAP Business Partner → Business Partner → Organizations → Maintain Industry Systems and Industries

Synchronization with ERP Customer / ERP Vendor
- Standard industry of standard industry system synchronized to Industry field (KNA1 / LFA1-BRSCH)
- Precondition: Define industry mapping for CVI
  IMG: Cross-Application Components → Master Data Synchronization → Customer/Vendor Integration → Business Partner Settings
  → Settings for Customer Integration → Field Assignment for Customer Integration → Assign Attributes → Assign Industries
  → Settings for Vendor Integration → Field Assignment for Vendor Integration → Assign Attributes → Assign Industries



6.7.13 BP Relationships

Maintain multiple relationships per relationship category

Relationship Categories

- SAP-delivered categories: Contact Person, Shareholder, Marriage, Parent / Child, etc.
- Customers can add further Categories (transaction BUBA)
- Cardinality and Time Constraint can be defined

Synchronization with ERP Customer / ERP Vendor

- Contact Persons with Workplace Address are synchronized

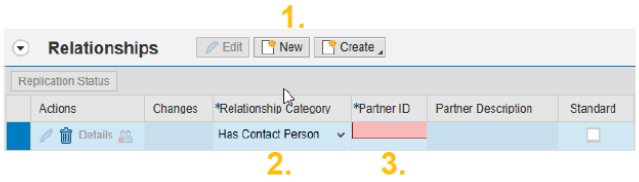
6.7.14 BP Relationships, Add Relationship in UI

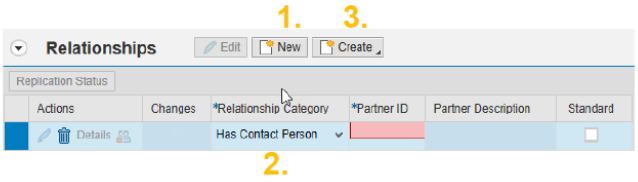
Add Relationship with **existing** BP

1. Press ‚New' to add a new relationship
2. Select the relationship category
3. Select related Business Partner via Search
4. Select relationship and maintain relationship data

Add Relationship with **new** BP

1. Press ‚New' to add a new relationship
2. Select the relationship category
3. Press ‚Create' and select category of related Business Partner
4. Enter data of related Business Partner and press ‚Done'
5. Select relationship and maintain relationship data





6.7.15 BP Relationships, Time Dependency

Validity dates can be maintained for BP Relationships

Time constraint and cardinality can be defined for each relationship category (see transaction BUBA)

Time dependency to be activated by customer

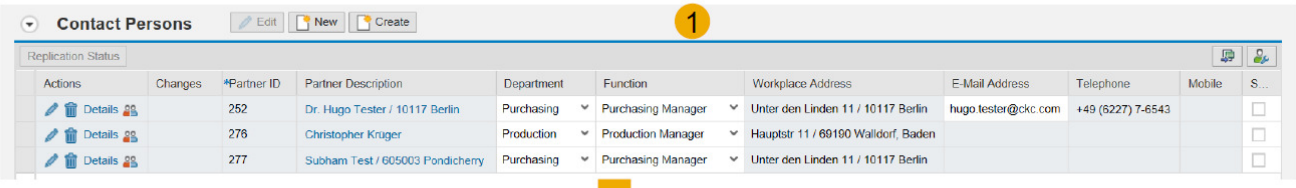
Manual action required, for more information see SAP Note 2249877

Business Background

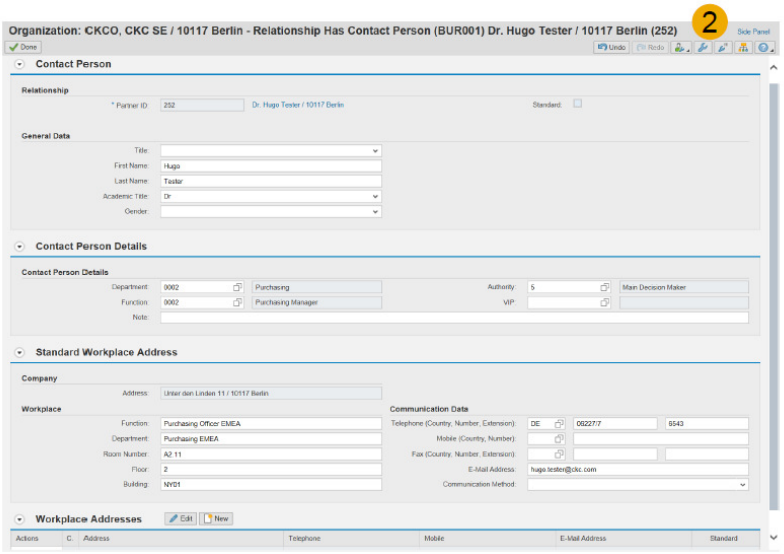
Relevant contact person information distributed across several pages (person data, relationship data, workplace address). The requirement was to merge most relevant data on one page, decrease the navigation, and therewith optimize the usability.

User Interface

1 ▪ New List UIBB for contact persons containing specific relationship information.
Note: Generic Relationship List UIBB is still available.



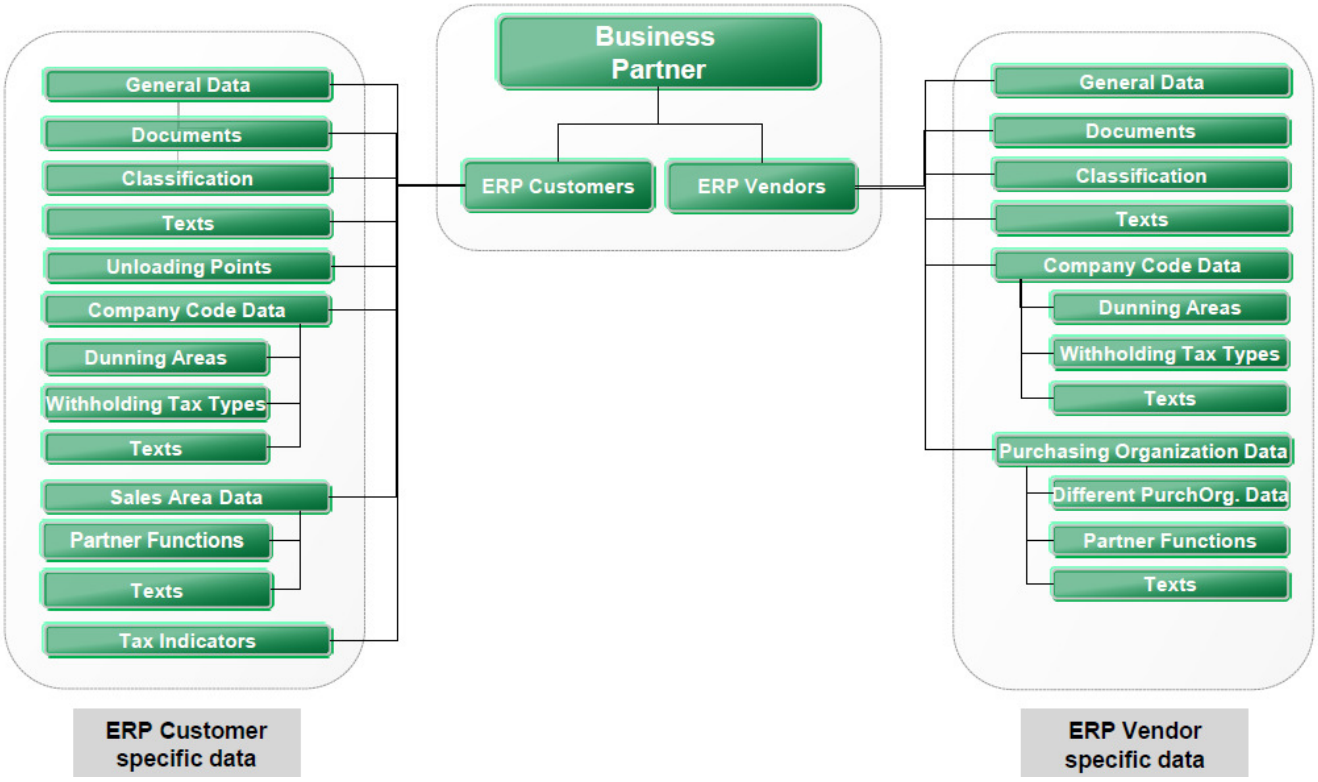
Navigation to details UI, see next slide



User Interface

2 ▪ One UI merging the most relevant
- Contact Person data
  (e.g. First Name, Last Name)
- Contact Person (Relationship) details
  (e.g. Department, Function)
- Standard Workplace Address
  (e.g. Building, Room Number, Communication data)
- Navigation to further Workplace Addresses

6.7.16 BP - Data Model ERP Customer / ERP Vendor



6.7.17 Assign ERP Customers / ERP Vendors, Use Cases & Concept

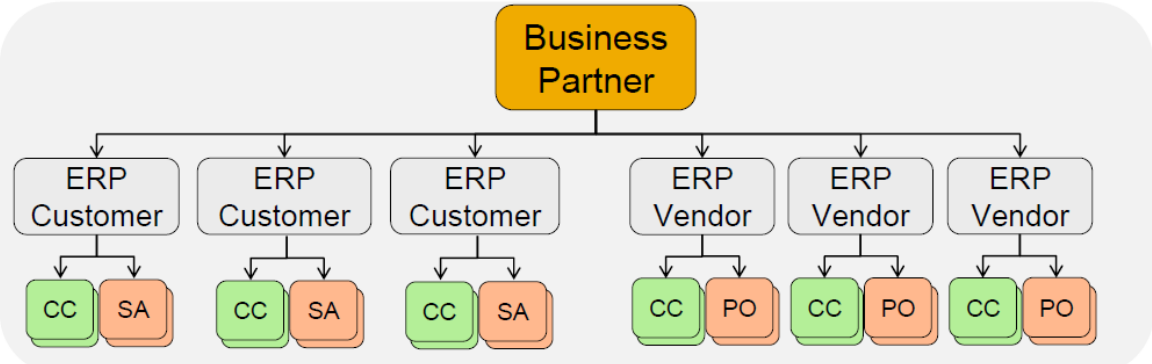
Multiple ERP Customers and ERP Vendors can be assigned to one Business Partner.

Use Case for assigning multiple ERP Customers with **same** address:

- Customer is treated differently within several business units of the same company code

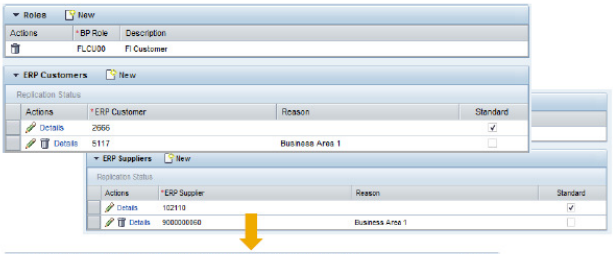
Use Case for assigning multiple ERP Customers with **different** addresses:

- Company includes main location as well as deviating bill-to, ship-to etc. addresses

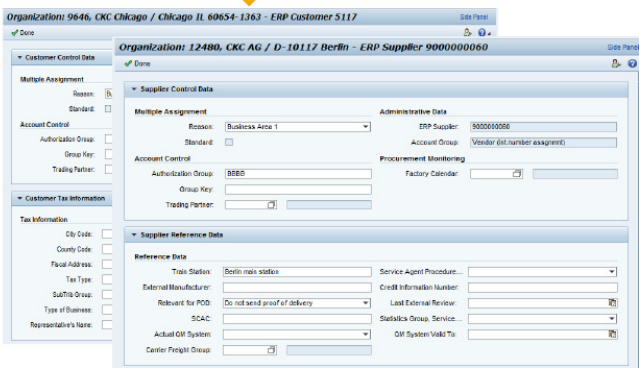


6.7.18 Assign ERP Customers / ERP Vendors, Implementation

Multiple ERP Customers and ERP Vendors can be assigned to one Business Partner and share the same Business Partner data.



ERP Customer and ERP Vendor specific data including sales / purchasing organization and company code data; can be maintained per assigned ERP Customer or ERP Vendor

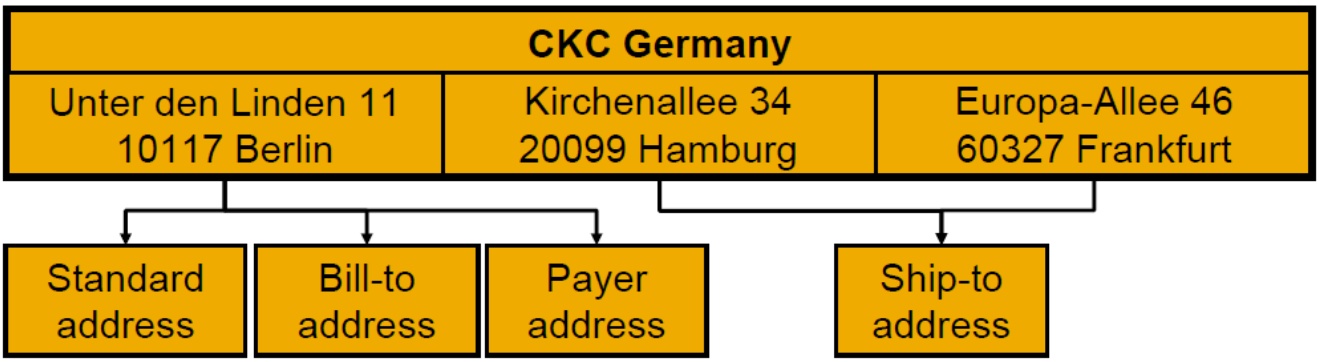


6.7.19 Assign ERP Customers / ERP Vendors, Usage of Multiple BP addresses (UseCase)

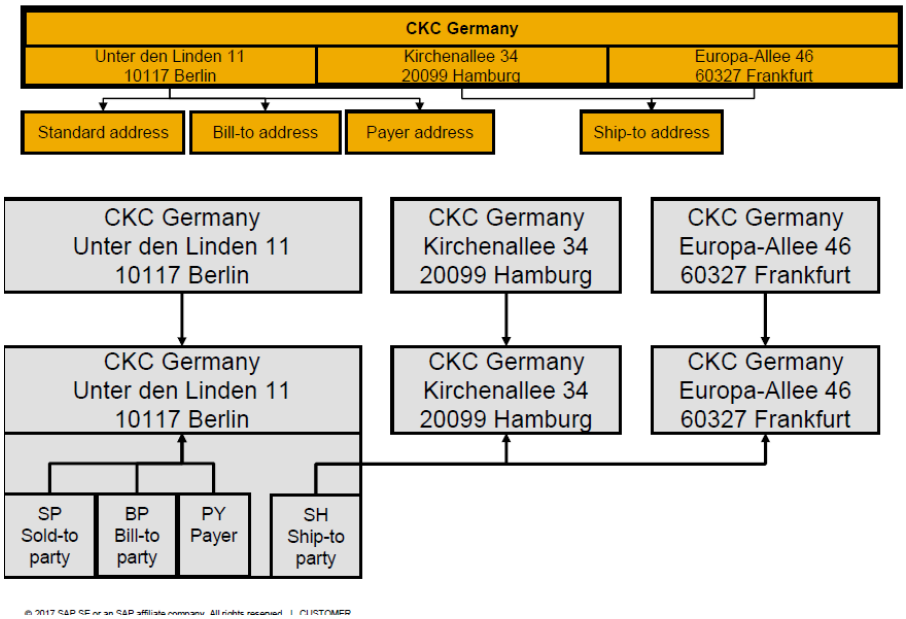
One Business Partner with 3 Addresses

Main address in Berlin is Sold-to, Bill-to, and Payer Address

Additional addresses in Hamburg and Frankfurt are ship-to addresses



6.7.20 Assign ERP Customers / ERP Vendors, One BP perAddress

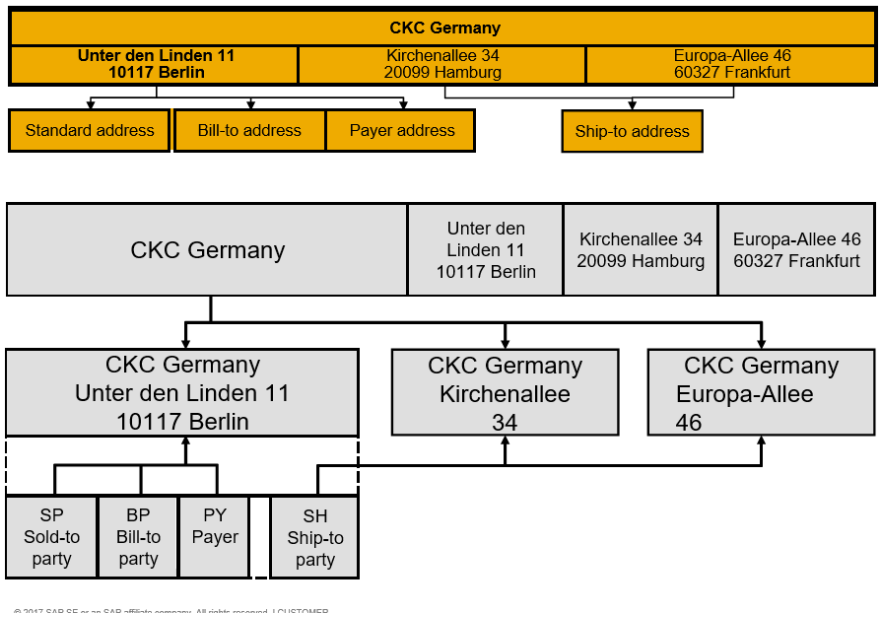


Business Partner
Three Business Partners required, or per address

ERP Customers
Relevant location for each process

Partner Functions (per Sales Area)
SP, BP, and PY by the same ERP Customer
For SH: link to other ERP Customer under deviating Business Partner

6.7.21 Assign ERP Customers / ERP Vendors, One BP per LegalEntity

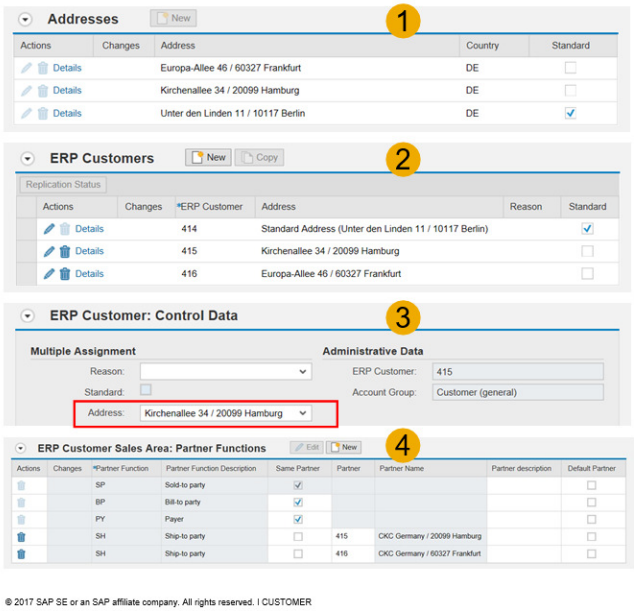


Business Partner
One Business Partner for sold-to party with three addresses

ERP Customers
Relevant location for each process

Partner Functions (per Sales Area)
SP, BP, and PY by the same ERP Customer
For SH: links to other ERP Customers under the same Business Partner

6.7.22 Assign ERP Customers / ERP Vendors, One BP per Legal Entity, UserInterface



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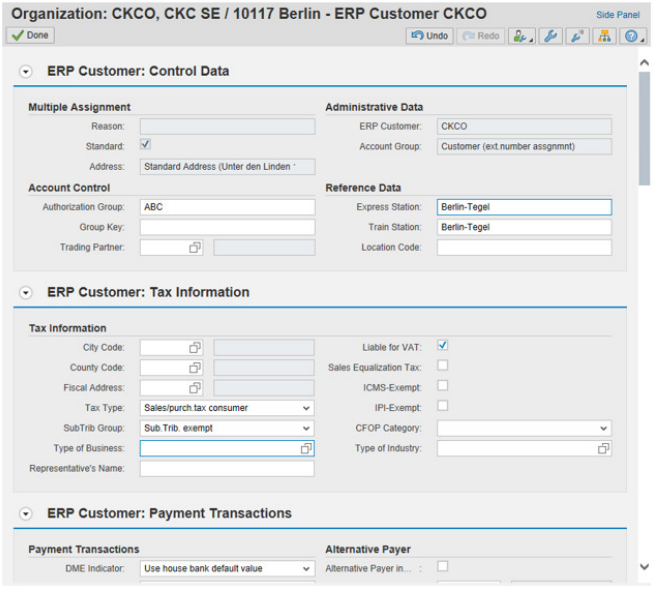
1 **Business Partner with Addresses**
Add new Business Partner for sold-to party with three addresses, one address marked as standard

2 **ERP Customers with Sales Area Data**
Assign three ERP Customers including Sales Area data to the Business Partner. One ERP Customer is marked as standard which has to use the standard address.

3 For additional assigned ERP Customers, users can decide whether to use the standard address or one of the BP addresses being assigned explicitly

4 **Partner Functions (per Sales Area)**
Maintain partner functions on sales area level for the standard ERP Customer. Partner Functions SP, BP, and PY are taken over by the same ERP Customer, SH (ship-to) is taken over by the other ERP Customers.
Note: In MDG on S/4HANA 1709 / MDG 9.1 activated ERP Customers only can be referred within Partner Functions. Eventually process will have to be split into two consecutive Change Requests.

6.7.23 General Data (Customer and Vendor)



General data can be maintained per ERP Customer / ERP Vendor

Account Group for

- Standard Assignment derived from BP Grouping according to CVI Configuration
- Additional Assignments selected by the user

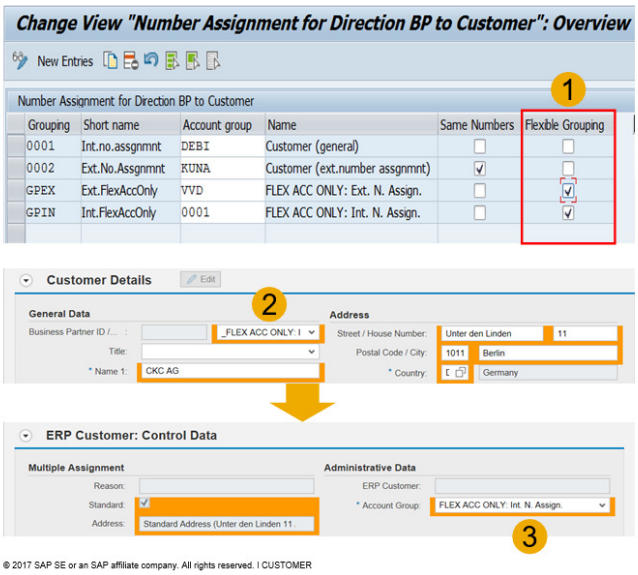
Authorization checks supported

- General Data (F_KNA1_GEN, F_LFA1_GEN)
- Account Group (F_KNA1_GRP, F_LFA1_GRP)

Data will be stored in tables KNA1 / LFA1 during activation

Replication via ALE and SOA Service

6.7.24 General Data (Customer and Vendor), Select account group during creation



Business Background

When first assigning an ERP Customer / ERP Vendor to a Business Partner, the account group was automatically derived according to CVI Configuration. Requirement to enable users to select another account group manually.

CVI Configuration

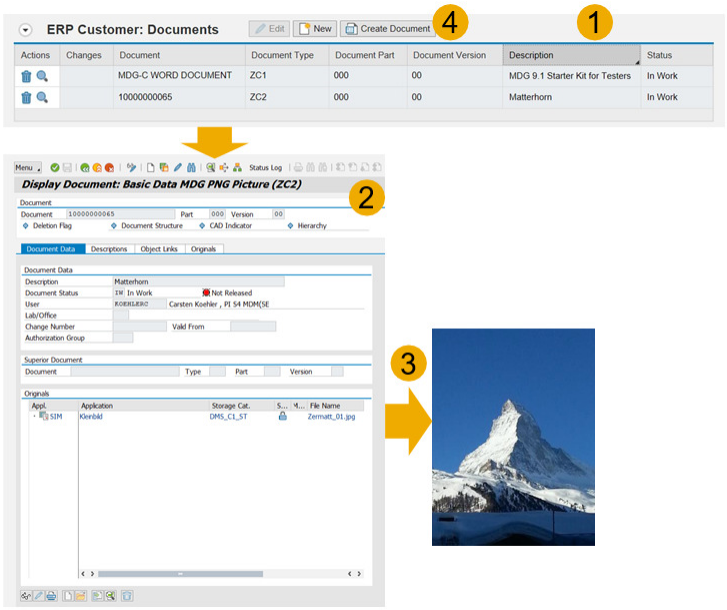
1 - Assignment BP Grouping to Account Group can now be marked as flexible. This means the user can overwrite the account group for standard assignment.

User Interface

2 - When creating a new Business Partner, select flexible BP Grouping from the value list.

3 - When adding a new ERP Customer, the account group is defaulted according to the CVI Configuration, but can be overwritten by the user.

6.7.25 Documents (Customer and Vendor)



User Interface

1 - Multiple documents can be assigned per ERP Customer / ERP Vendor.

2 - Documents as well as links are stored in Document Management System (DMS); this U can be called from MDG UI.

3 - From DMS UI, document properties can be managed, and document can be displayed.

4 - Documents can be uploaded to DMS from MDG UI.

Replication

- Document as well as link to ERP Customer / ERP Vendor via ALE (Idoc DOCMAS) or
- Link to ERP Customer / ERP Vendor via SOA Service BusinessPartnerSUITEBulkReplicateRequest

6.7.26 Lean Classification (Customer and Vendor),

Feedback for MDG-M Classification (up to MDG 9.0 and S/4HANA1610)

- UI-related feedback
  - Classes and characteristics are not on the same page
  - Users should not see a difference between normal attributes and classification

- More usable UI, possibility to adapt the UI
- Function-related feedback
  - Integration of derivations is missing
  - Simultaneous display and change of classification is not supported (authorizations for display not possible)
  - Simultaneous processing of different class types or classes using a parallel change request is not possible

Missing support in other MDG applications (up to MDG 9.0 and MDG on S/4HANA 1610)

- No support for classification in MDG Consolidation
- No support for classification for supplier and customer in MDG Central Governance

6.7.26.1 Lean Classification (Customer and Vendor), Concept

For Customer, Supplier and Material:
- MDG Single-Object Maintenance UI
- MDG Consolidation

Simplified UIs

Mass-enabled API

Simplified backend features to reduce complexity

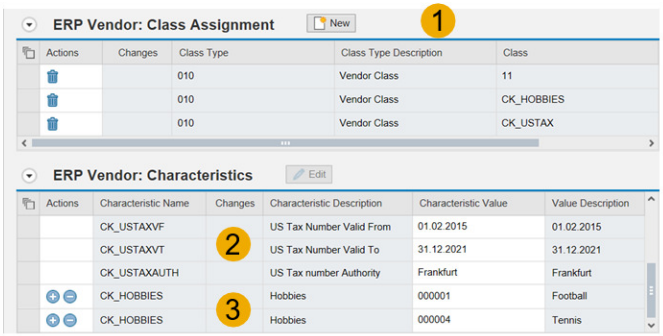
Class assignment and characteristics valuation work like normal entities in data model

Supported Features
- Derivations using BAdI CROSS_ENTITY_BADI (incl. defaulting for class types and classes)
- Checks using BRFplus
- Authorizations (incl. display authorization)
- Change Documents
- Multiple class types and class assignments
- Characteristic values: Single or multiple values, additional values, intervals, mandatory checks (required entry)

Restrictions (not supported)
- Object dependencies (preconditions, actions, selection condition, procedures)
- Overwriting of characteristic values and overwriting of inherited values
- Engineering Change Number

You can find detailed information about included/excluded features in SAP Note 2479869.

6.7.26.2 Lean Classification (Customer and Vendor), User Interface



1 **Assign Classes**
List UIBB to assign classes. Classes of class types assigned to object table LFA1 / KNA1 can be assigned.

2 **Enter Characteristic Values (Single Value)**
Single value characteristics assigned to selected classes are visible in one row and can be filled with a value.

3 **Enter Characteristic Values (Multiple Values)**
Multiple value characteristics are also visible in one row and can be recognized by '+' and '-' buttons on the left. You can use these buttons to insert or delete rows.

Configuration / Master Data

- based on Customizing configuration of classification system to e.g. maintain object types and class types (see Customizing under Cross-Application Components ☐ Classification System)
- based on master data of classification system to e.g. maintain characteristics and classes (see SAP Menu under Cross-Application Components ☐ Master Data)

Consolidation

- can be consolidated like other entities
- See separate presentation about Consolidation

Data Activation / Replication

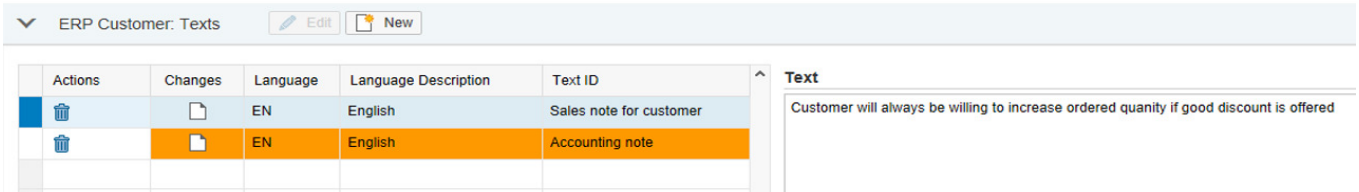
- Classification data will be stored in classification system during activation
  – Object Type KNA1 for ERP Customer
  – Object Type LFA1 for ERP Vendor
- ALE replication via standard IDOC *CLFMAS*
- SOA Service replication via BP service *BusinessPartnerSUITEBulkReplicateRequest*

6.7.26.3 Texts (Customer and Vendor)

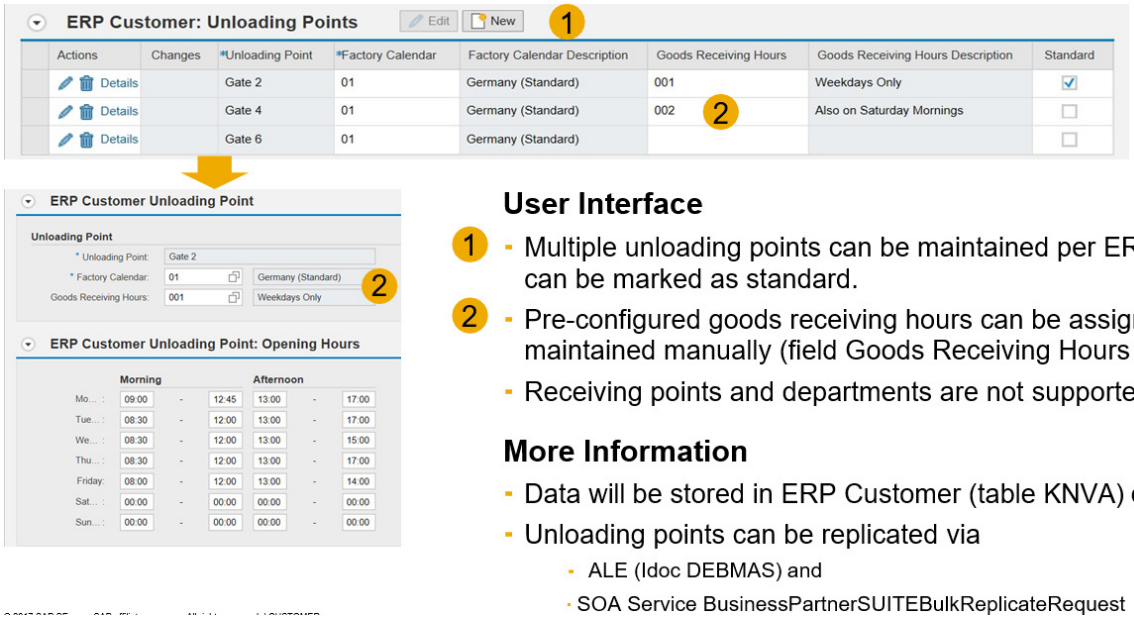
You can now enter explanatory texts for ERP customer and ERP vendors for different text types and for different languages.

Texts are available for
– ERP Customers on general level as well as on company code and sales area level
– ERP Vendors on general level as well as on company code and purchasing organization level

Configuration of Text IDs within IMG activities known from ERP Customer / ERP Vendor



6.7.27 Unloading Points (Customer)



User Interface

1. Multiple unloading points can be maintained per ERP Customer. One can be marked as standard.
2. Pre-configured goods receiving hours can be assigned or can be maintained manually (field Goods Receiving Hours is initial).
   - Receiving points and departments are not supported.

More Information

- Data will be stored in ERP Customer (table KNVA) during activation
- Unloading points can be replicated via
  - ALE (Idoc DEBMAS) and
  - SOA Service BusinessPartnerSUITEBulkReplicateRequest

6.7.28 Company Code data (Customer and Vendor)

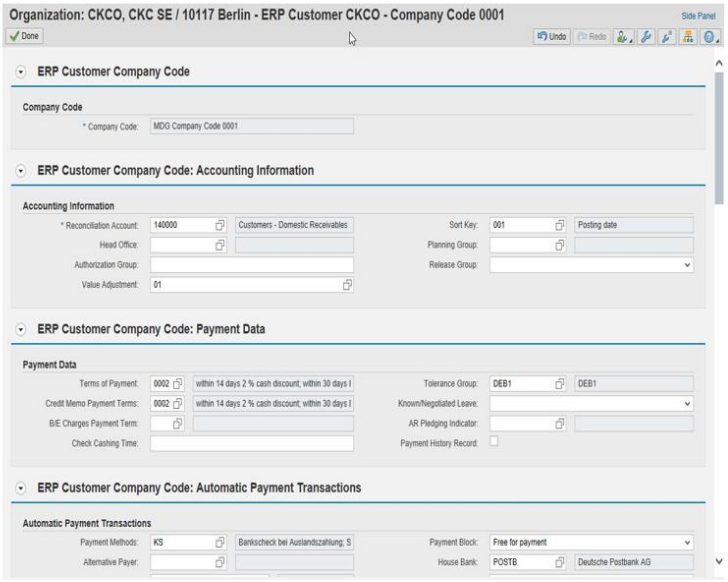
Multiple Company Codes can be assigned per ERP Customer / ERP Vendor

Screen Layout per Company Code supported
IMG: Financial Accounting (New) ☐ Accounts Receivable and Accounts Payable ☐ Customer Accounts ☐ Master Data ☐ Preparations for Creating Customer Master Data ☐ Define Screen Layout per Company Code (Customers)
Similar path for ERP Vendor

Authorization check per company code (F_KNA1_BUK, F_LFA1_BUK) supported

Data will be stored in tables KNB1 / LFB1 during activation

Replication via ALE and SOA Service



6.7.29 Dunning Areas (Customer and Vendor)

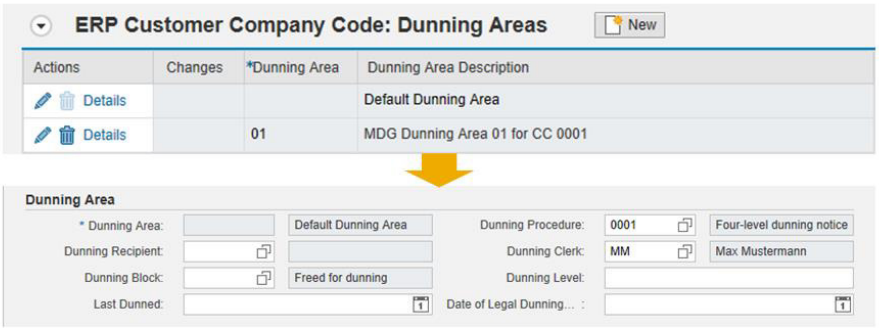
Multiple Dunning Areas per company code assigned to ERP Customer / ERP Vendor

Configure Dunning Areas and values for other fields in IMG for ERP Customer / ERP Vendor

Dunning Area ‘ ‘ (Default) always first entry

Data will be stored in tables KNB5 / LFB5 during activation

Replication via ALE and SOA Service



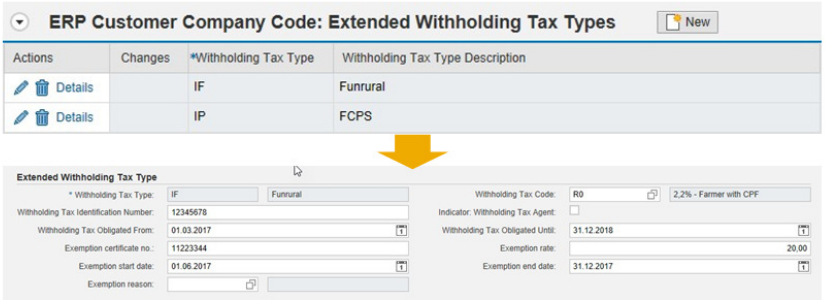
6.7.30 Withholding Tax Types (Customer and Vendor)

Multiple Withholding Tax Types per company code assigned to ERP Customer / ERP Vendor

Activate Extended Withholding Tax per Company Code
IMG: Financial Accounting (New) ☐ Financial Accounting Global Settings (New) ☐ Withholding Tax ☐ Extended Withholding Tax ☐ Company Code ☐ Activate Extended WithholdingTax

Data will be stored in tables KNBW / LFBW during activation

Replication via ALE and SOA Service



6.7.31 Sales Area Data (Customer)

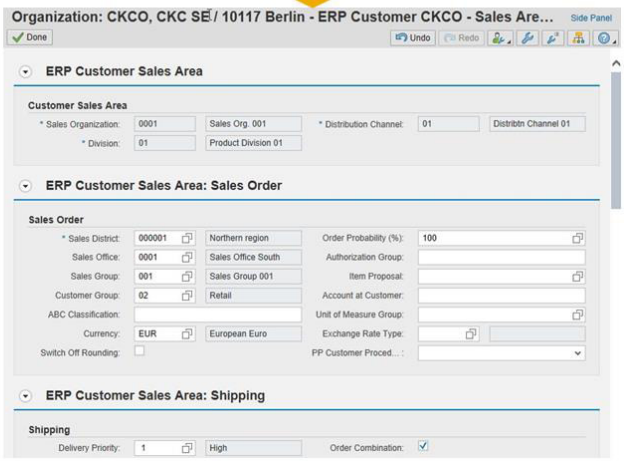
Multiple Sales Areas per ERP Customer

Screen layout per sales area supported (transaction SM30, view V_T079V)

Authorization check for ERP Customer per Sales Area (V_KNA1_VKO) supported

Data will be stored in tables KNVV during activation

Replication via ALE and SOA Service



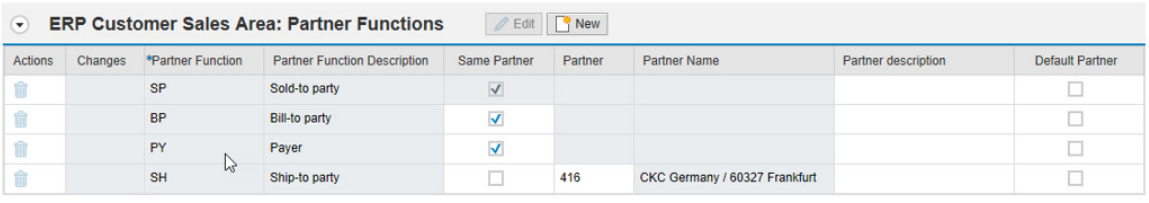
6.7.32 Partner Functions (Customer)

Multiple Partner Functions per Sales Area can be assigned to ERP Customer

Partner Determination for ERP Customer can be configured in IMG
IMG: Sales and Distribution ☐ Basic Functions ☐ Partner Determination ☐ Set Up Partner Determination

Data will be stored in tables KNVP during activation

Replication via ALE and SOA Service



6.7.33 Tax Indicators (Customer)

Tax Indicators to be maintained per relevant countries

Collect company code's countries of Sales Organizations assigned to ERP Customer

Data will be stored in table KNVI during activation

Tax Indicators can be replicated via ALE and SOA Service

ERP Customer: Tax Indicators Edit

Actions	Changes	*Country	Country Description	*Tax Category	Tax Category Description	Tax Classification	Tax Classification Description
		DE	Germany	MWST	Output Tax	1	Liable for Taxes
		GB	United Kingdom	MWST	Output Tax	0	Tax Exempt

6.7.34 Purchasing Organization Data (Vendor)

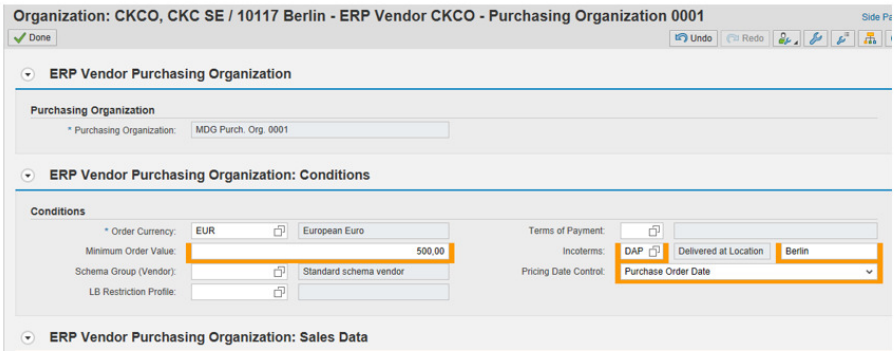
Multiple Purchasing Organizations can be assigned per ERP Vendor

Field modification per Purchasing Organization supported (transaction SM30, view V_T079M)

Authorization check per purchasing organization (V_KNA1_VKO) supported

Data will be stored in tables LFM1 during activation

Replication via ALE and SOA Service



6.7.35 Different Purchasing Organization Data (Vendor)

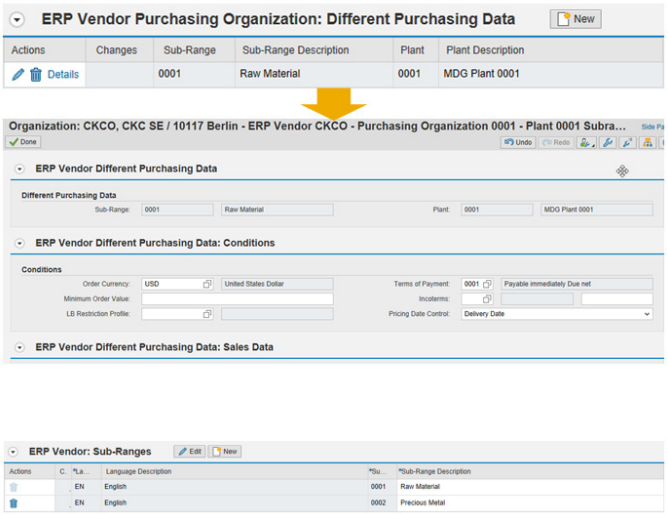
Multiple records per assigned ERP Vendor / Purchasing Organizations

Record can be dependent of a
- Plant
- Vendor Subrange
- Combination of Plant / Vendor Subrange

Data will be stored in table LFM2 during activation

Replication via ALE and SOA Service

Plants can be configured in IMG
Vendor Subranges are maintained per ERP Vendor on general data level (table WYT1)



6.7.36 Partner Functions (Vendor)

Multiple Partner Functions can be assigned per ERP Vendor / Purchasing Organization.

Partner Functions can be additionally dependent of 'Different Data'
- Plant
- Vendor Subrange
- Combination of Plant / Vendor Subrange

Partner Determination can be configured in IMG
IMG: Materials Management ☐ Purchasing ☐ Partner Determination

Data will be stored in table WYT3 during activation

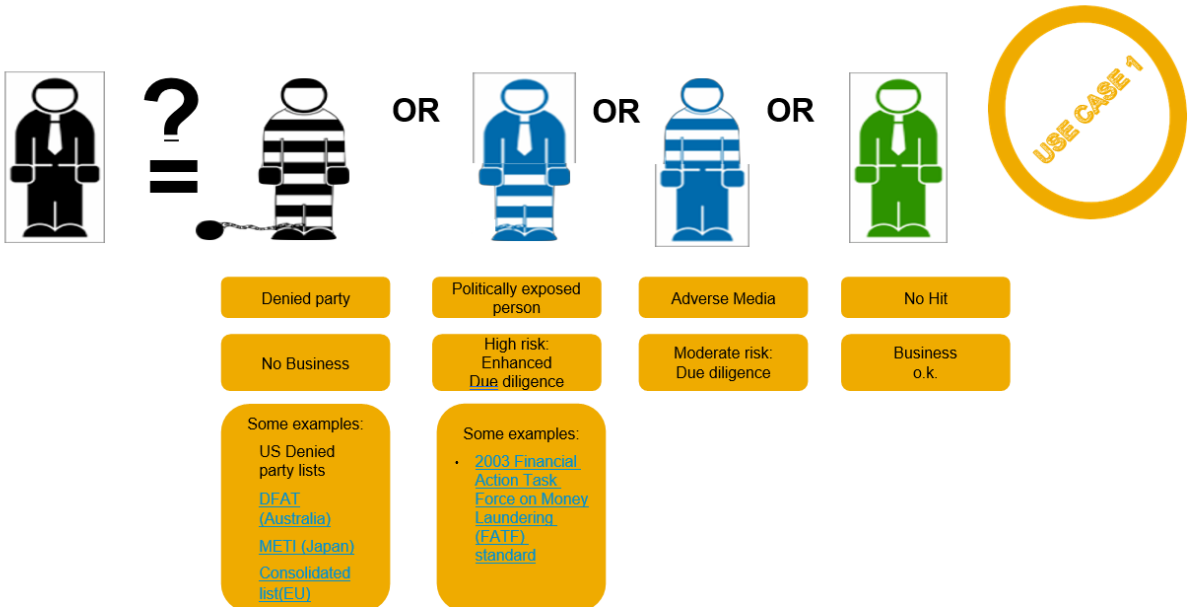
Replication via ALE and SOA Service

ERP Vendor Purchasing Organization: Partner Functions Edit | New | New - Different Data | Delete - Different Data

Actions	Changes	*Partner Function	Partner Function Description	Same Partner	Partner	Partner Name	Default Partner	Sub-Range	Sub-Range Description	Plant	Plant Description
		OA	Ordering address	☐	CKCOFLEXC	CKC Flex Account C / 69190 Walldorf	☐	0001	Raw Material	0001	MDG Plant 0001
		PI	Invoicing Party	☐	CKCOFLEXC	CKC Flex Account C / 69190 Walldorf	☐	0001	Raw Material	0001	MDG Plant 0001
		OA	Ordering address	☑			☐				
		VN	Vendor	☑			☑				
		PI	Invoicing Party	☑			☐				

6.7.37 Business Partner Screening

Manage different type of content; is my business partner on a published list?



Why to care about these parties?

$1.45 Billion in Penalties for Bank that Concealed Transactions with Sanctioned Parties (March 2015)

The Justice Department announced March 12 that a global financial institution headquartered in Germany and its New York branch have agreed a total of $1.45 billion in penalties for concealing hundreds of millions of dollars in transactions prohibited by U.S. sanctions laws on behalf of Iranian and Sudanese businesses.

Source: http://www.strtrade.com/news-publications-economic-sanctions-penalties-German-bank-031715.html

Diebold agrees to Pay $25.2 Million Penalty (October 2013)

The SEC alleges that Diebold spent approximately $1.6 million on leisure trips, entertainment, and other improper gifts for government bank officials in China. During this same time period, the SEC alleges, Diebold spent over $147,000 on leisure trips and entertainment for officials of government banks in Indonesia. As alleged in the complaint, Diebold executives in charge of the company's operations in Asia knew of these improper payments, which were falsely recorded in Diebold's books and records as training or other legitimate business expenses

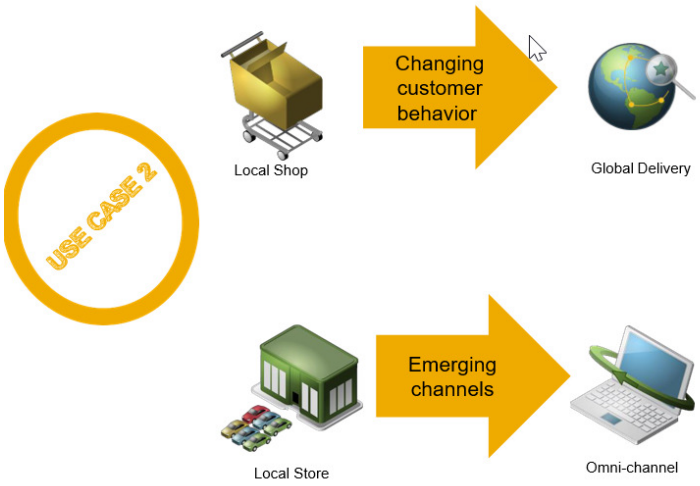
Source: http://www.justice.gov/opa/pr/diebold-incorporated-resolves-foreign-corrupt-practices-act-investigation-and-agrees-pay-252

MoneyGram, one of the world's largest remitters, fined $122,400 by AUSTRAC (Jan 2015)

MoneyGram Payment Systems, has been fined A$122,400 for contraventions of Australia's anti-money laundering and counter-terrorism financing laws.

Source: http://www.austrac.gov.au/media/media-releases/moneygram-one-world%E2%80%99s-largest-remitters-fined-122400-austrac

6.7.37.1 Digitized world increasing the risk of partner compliance

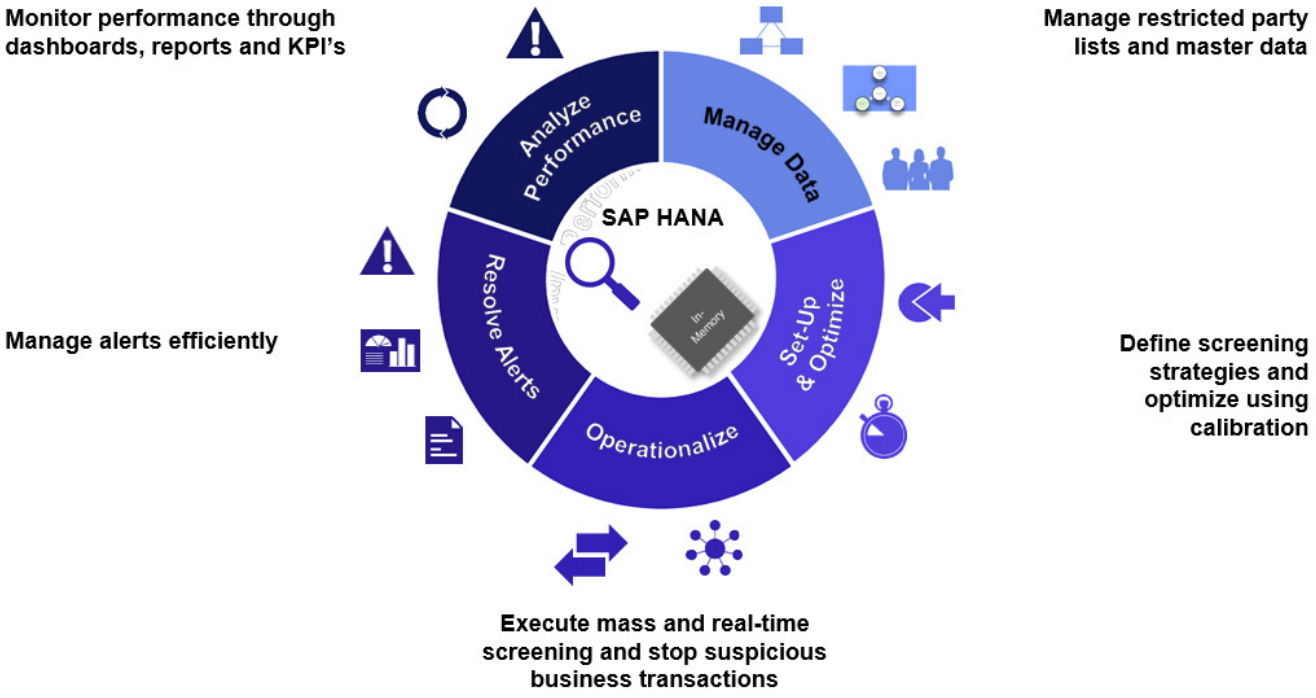


- Web shops leading to massive increase in number of Business partners
- Web shops ships globally and they need to know whom they are dealing with
- Businesses need unprecedented agility and flexibility from systems
- Technology should enable integrating new systems much quicker

6.7.37.2 Address Screening Solutions today

6.7.37.3 SAP Business Partner Screening - Vision

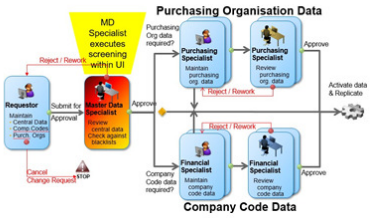
6.7.37.4 Achieve effective and efficient partner screening!



6.7.37.5 Three Scenarios

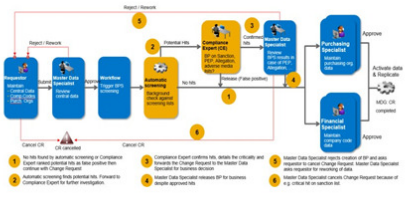
1) Decision in MDG

- Screening results showed as pop-up in MDG
- No involvement from BPS organization



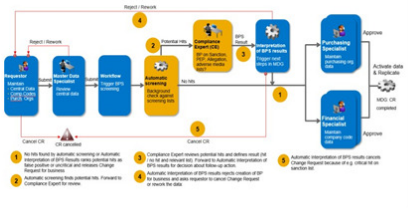
2) Consulting by Compliance Expert, Decision in MDG

- Results from screening made available in MDG in the notes
- Final decision in MDG

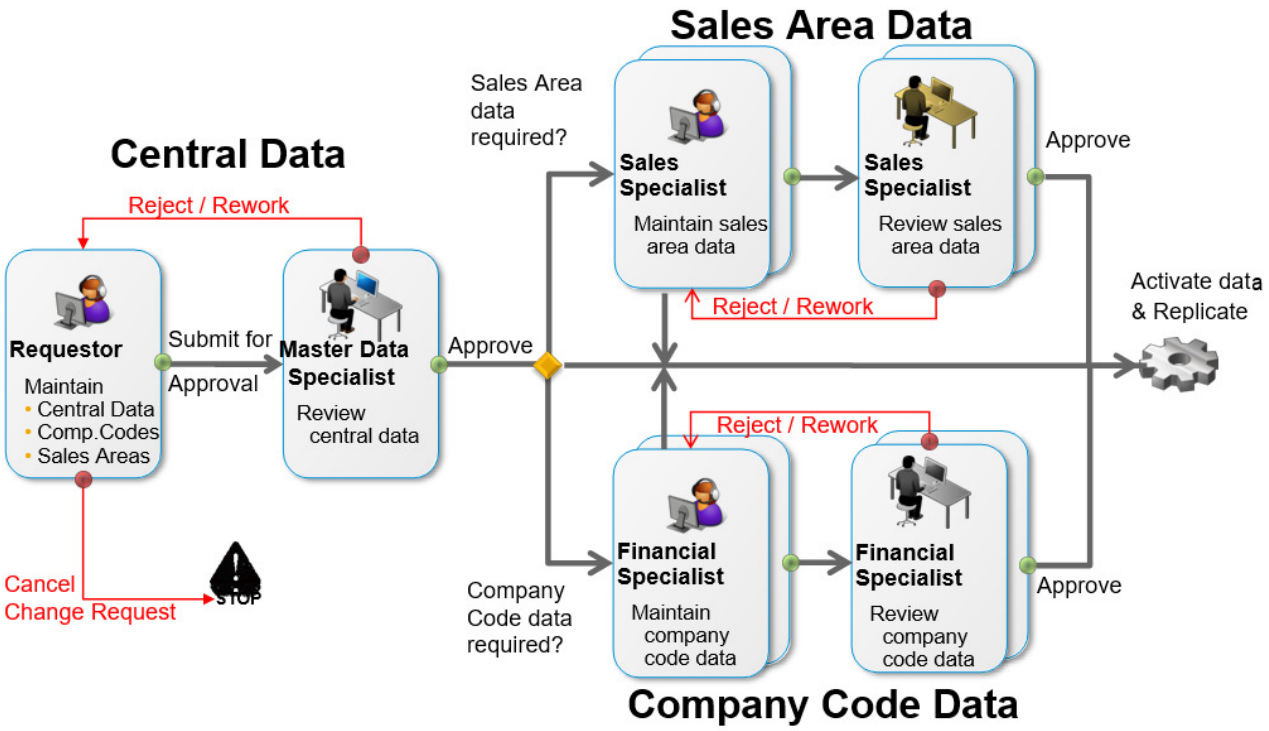


3) Decision in BPS

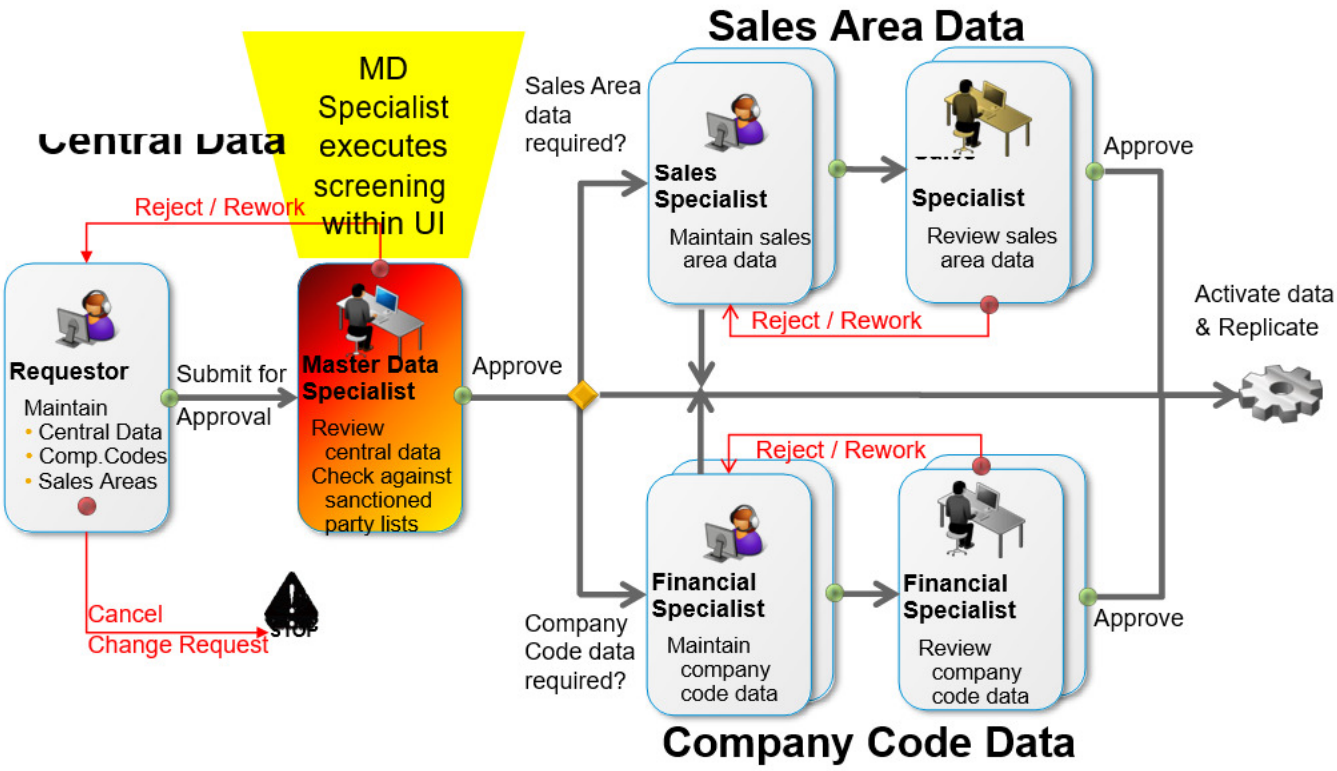
- Results from screening interpreted in MDG via customizing
- No further actions by a MDG specialist



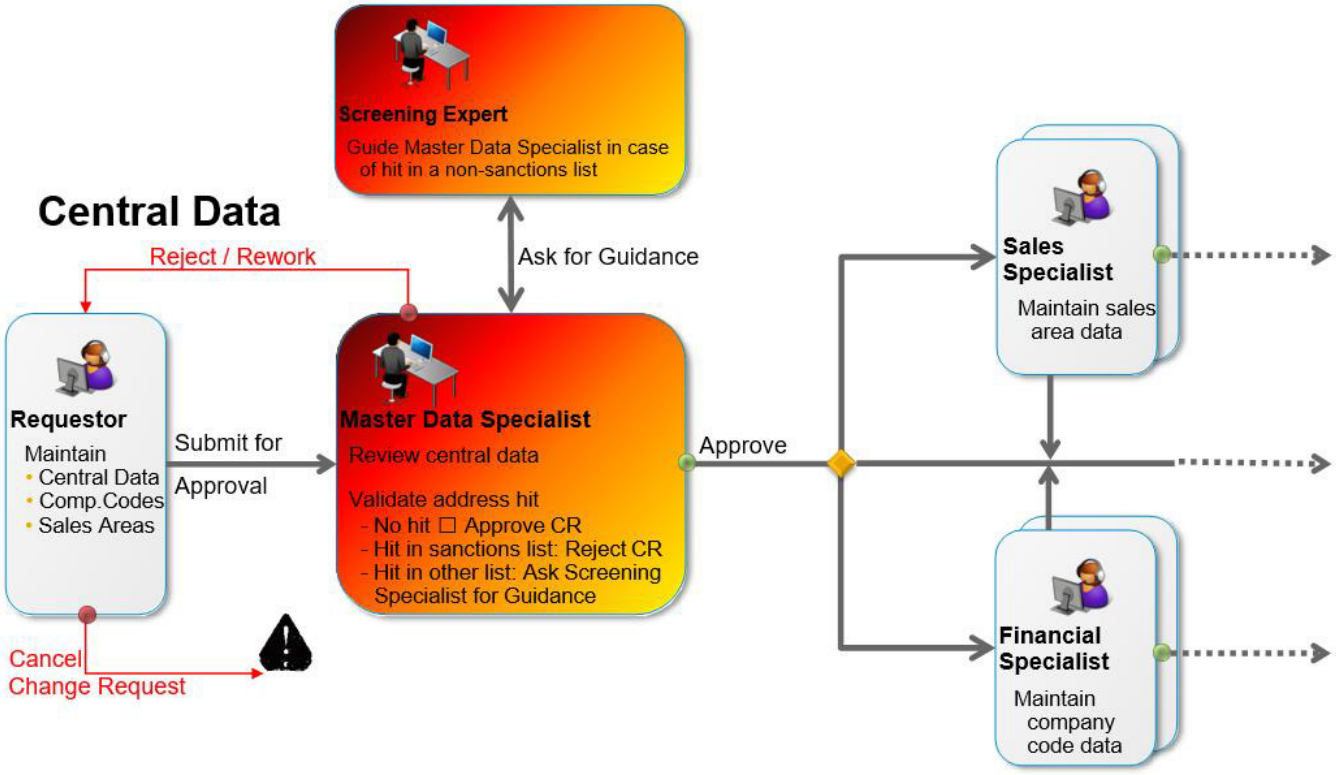
6.7.37.6 MDG-C process to Create Customer as is w/o SAP-BP Screening



6.7.37.7 Decision in MDG (Screening by Master Data Specialist)



6.7.37.8 Decision in MDG (Screening by Master Data Specialist, guidance by Screening Expert)



6.7.37.9 Activate screening per Change Request Step

Enrichment

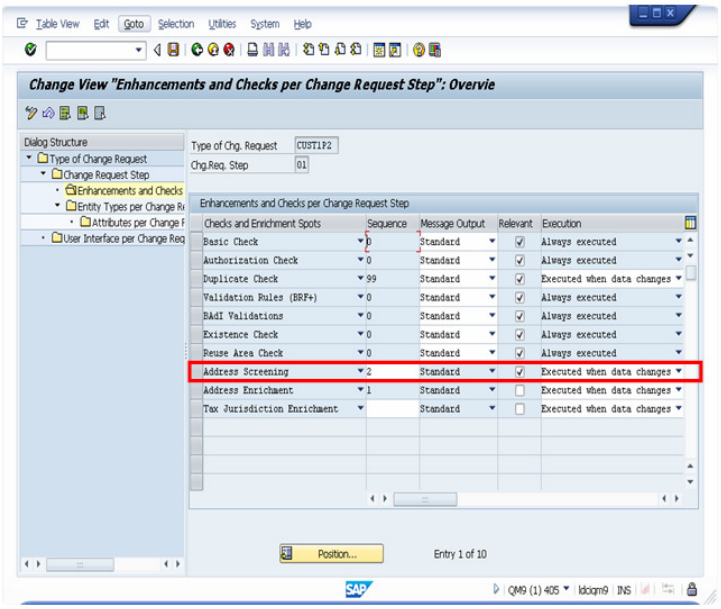
Implemented as Enrichment Spot ‘Address Screening’ with corresponding adapter in MDG

Activate per CR-Step

For each Change Request Type Screening can be activated per Change Request Step

Deployment Options

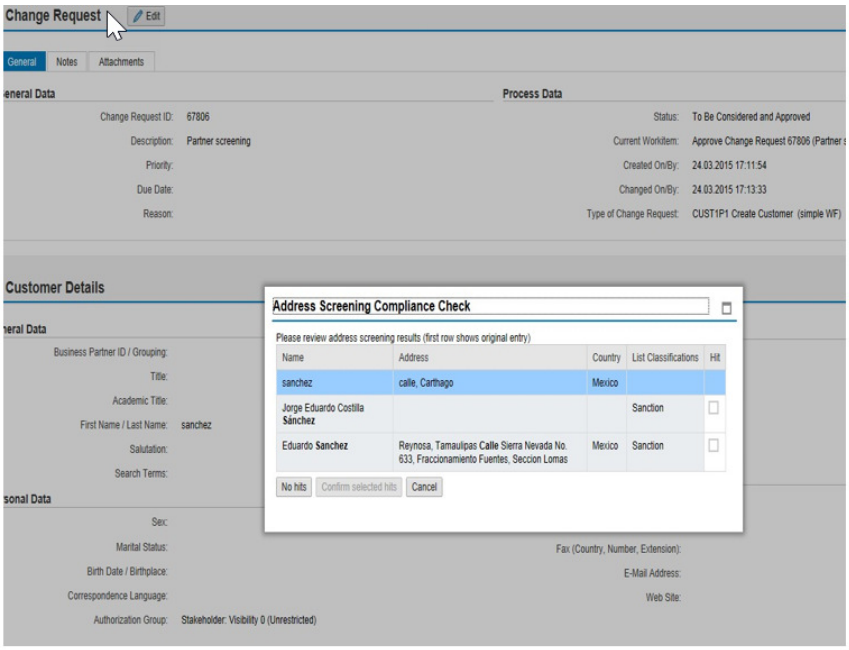
BP Screening can run on MDG instance (same HANA DB, different NW AS) or on different instance



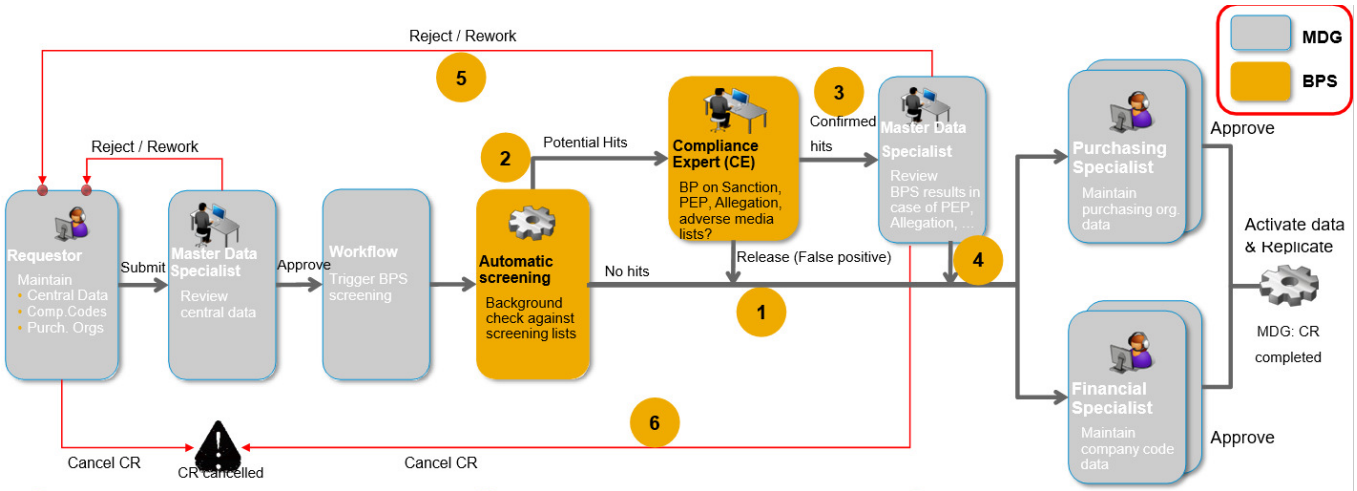
6.7.37.10 User Interaction

Compliance Check

- Screening automatically executed in process steps being activated.
- Popup appears in case of suspicions; user can decide about hit / no hit
- Hits are protocolled within BP Screening.
- User is free how to continue CR process independent of decision about hit / no hit.



6.7.37.11 Automatic screening: Consulting by Compliance Expert, Decision in MDG



1 No hits found by automatic screening or Compliance Expert ranked potential hits as false positive then continue with Change Request

2 Automatic screening finds potential hits. Forward to Compliance Expert for further investigation.

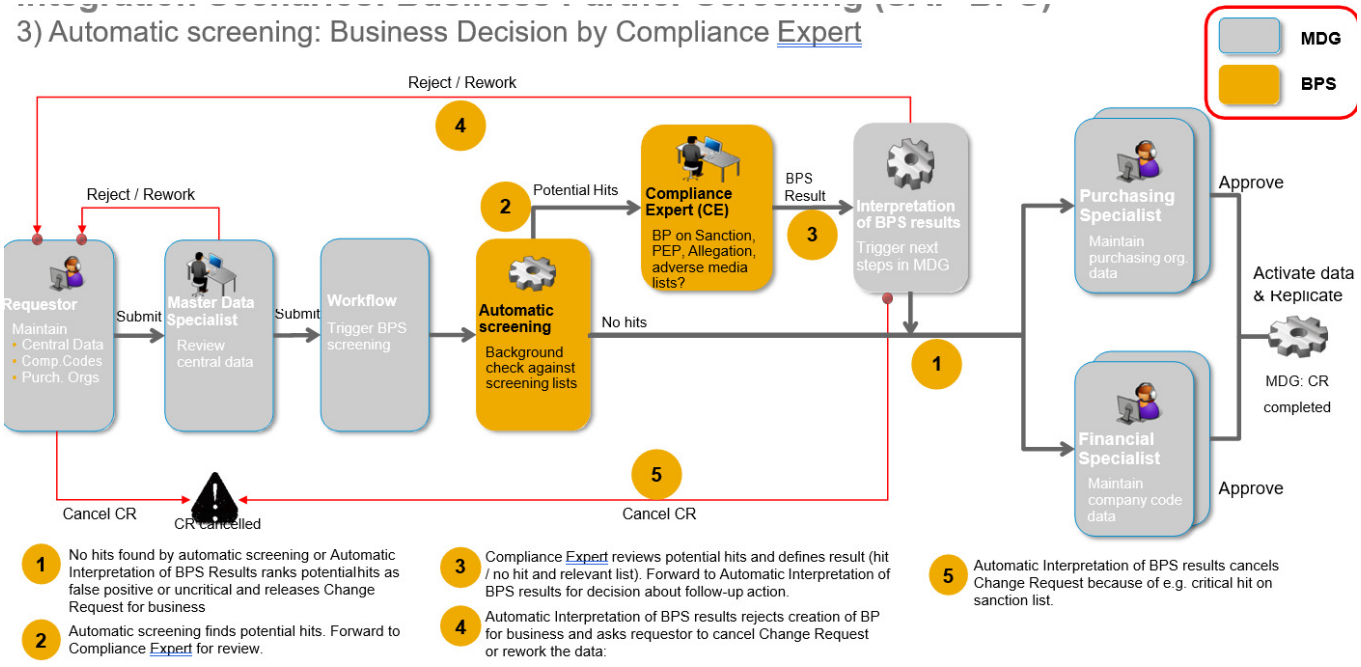
3 Compliance Expert confirms hits, details the criticality and forwards the Change Request to the Master Data Specialist for business decision

4 Master Data Specialist releases BP for business despite approved hits

5 Master Data Specialist rejects creation of BP and asks requestor to cancel Change Request. Master Data Specialist asks requestor for reworking of data.

6 Master Data Specialist cancels Change Request because of e.g. critical hit on sanction list.

6.7.37.12 Automatic screening: Business Decision by Compliance Expert[17]



17 SAP SE

6.8 Business Partner – Customer-Vendor Integration S/4 HANA[18]

6.8.1 Setting up Business Partner S/4 HANA

There are redundant object models in the traditional ERP system. Here the vendor master and customer

Master is used. The (mandatory) target approach in S/4HANA is the Business Partner approach.

Business Partner is now capable of centrally managing master data for business partners, customers, and vendors. With current development, BP is the single point of entry to create, edit, and display master data for business partners, customers, and vendors

6.8.1.1 Customer/Vendor Object Model Limitations

- Only one single address
- No relation between a vendor and a customer for the same real world entity (no role concept)
- No time-dependency

6.8.1.2 Business Partner – SAP Strategic Object Model

- General data are shared across different roles
- One business partner could perform multiple roles, e.g., customer and vendor (supplier)
- One business partner could have multiple addresses
- Time-dependency object attributes and relationships
- Provide harmonized architecture
- Customer/Vendor Integration (CVI) component ensure the synchronization between the Business Partner object and the customer/vendor objects

6.8.1.3 Dependencies/Pre-requisites

Business partner has been mandatory with S4 Hana Enterprise Management version i.e 1511 and 1610 & old transactions related to Vendor and customers have become obsolete.

It´s recommended but not mandatory that BuPa ID and Customer-ID / Vendor ID are the same.

The Business Partner Approach is not mandatory for the SAP Simple Finance 1503

6.8.1.4 Check and activate Business Functions

CA_SUPPLIER_SOA

CA_BP_SOA Switch ‘VENDOR_SFWS_SC1’ and switch ‘VENDOR_SFWS_SC2’ have to be active for the vendor contact person’s data to be synchronized with BP contact person’s data

[18] SAP SE

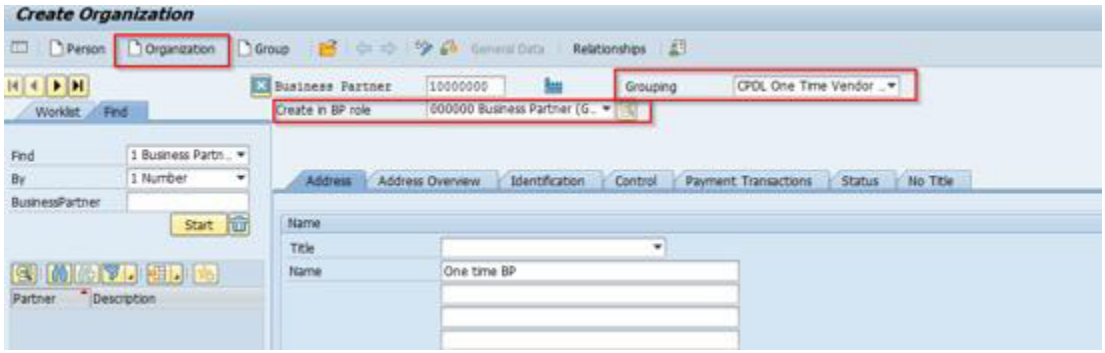
The appropriate specific transaction like XD01,XD02, XD03 or VD01, VD02, VD03 / XK01, XK02, XK03 or MK01, MK02, MK03 etc. are not available in SAP S/4HANA Enterprise management Version i.e. 1511 and 1610

Currently mass maintenance for customers and vendors via transaction MASS is not available.

Before you begin the BP conversion from an SAP ERP system to an SAP S/4 HANA system, you have to answer the questions Whether the Business Partner ID and Customer-ID /Vendor ID should be the same in the S/4 HANA System?

6.8.1.5 Business Partner Master Data Setup Process

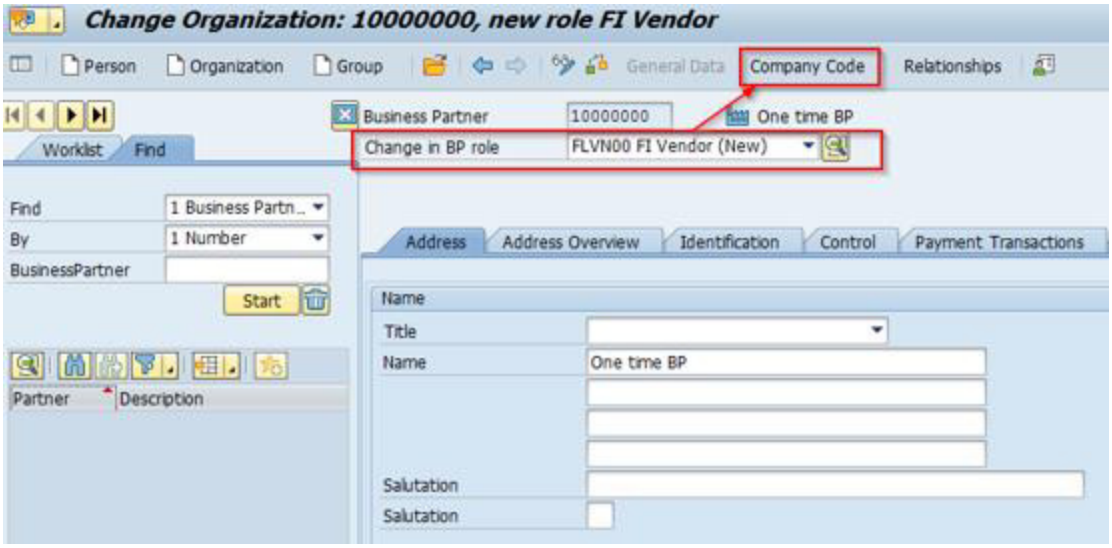
1. Setup BP General Data



Select General Business Partner Role and Grouping, here this grouping is linked with customer/ vendor Number range and this Grouping is key in linking the Vendor/Customer & BP groups as well as number ranges which is explained below in detail.

Table BUT000 will be updated once General BP role is completed.

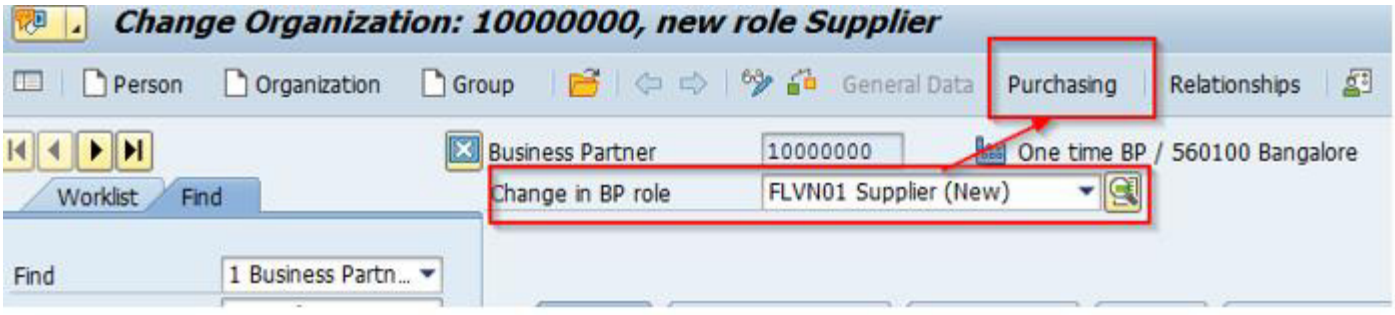
2. Setup FI Vendor



BP Role FLVN00 FI Vendor extension enables Invoice to be posted directly in FI.

Here vendor master table LFA1 and LFB1 will be updated.

FI document can be posted but cannot create PO as purchasing data is not yet extended so for extending purchasing data use ROLE FLVN01



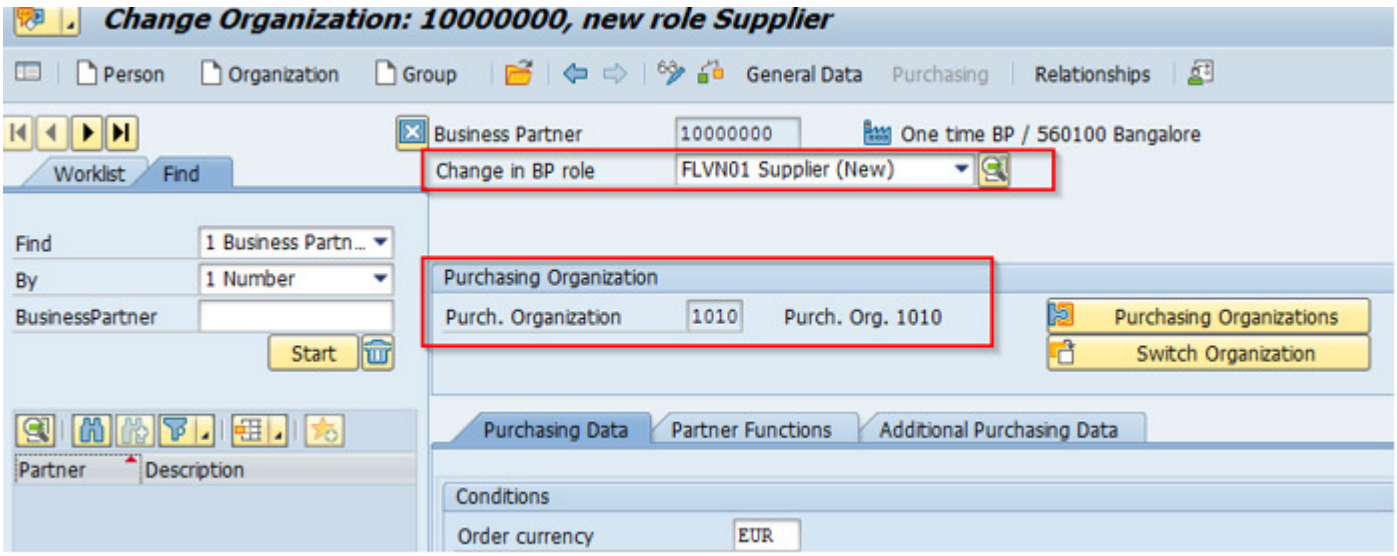


Table LFM1 will be updated once the BP Role is extended to Supplier BP Role

Once BP is extended to Purchase Organization using BP Role FLVN01 the vendor can be used PO can be created.

6.8.1.5.1 Setting up Customer

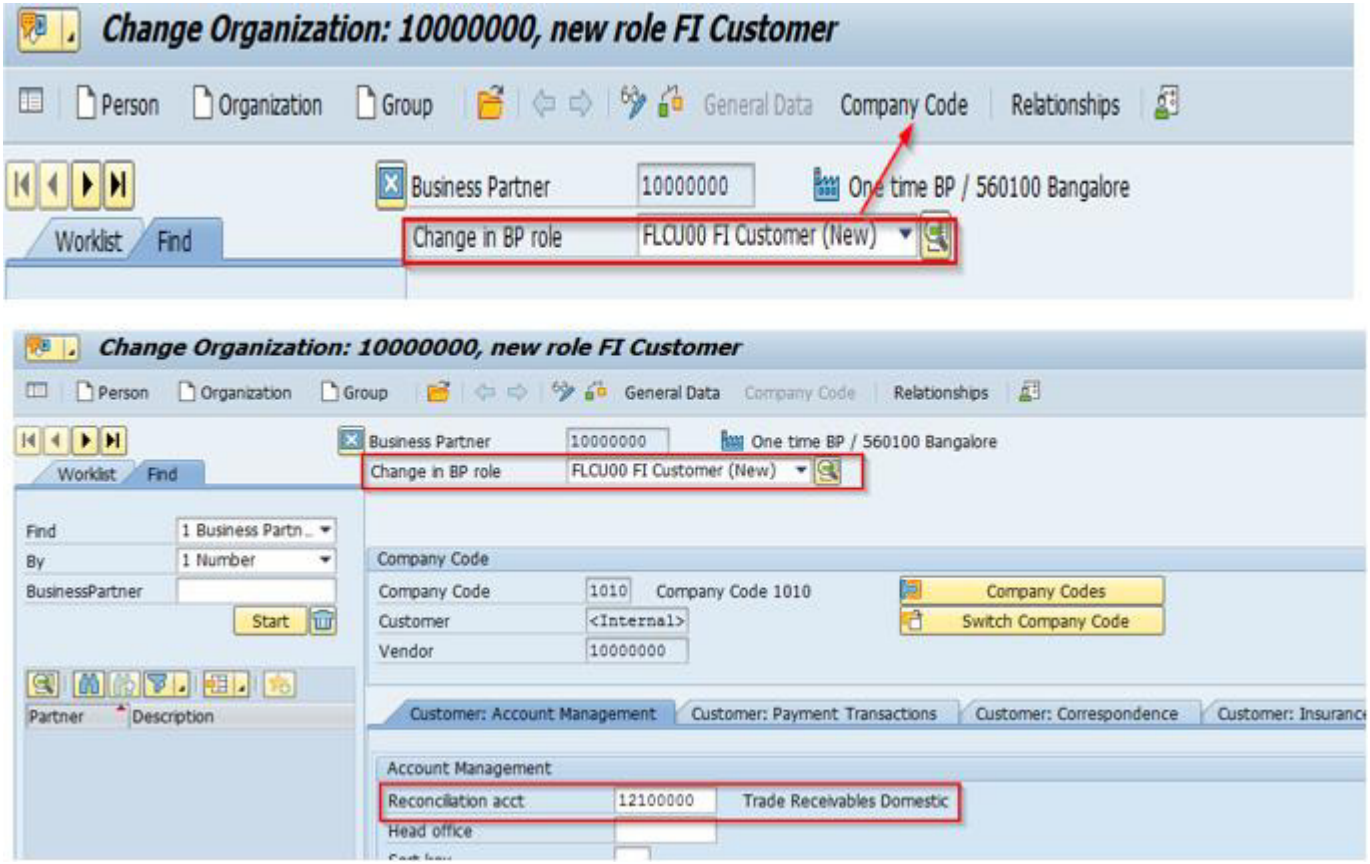
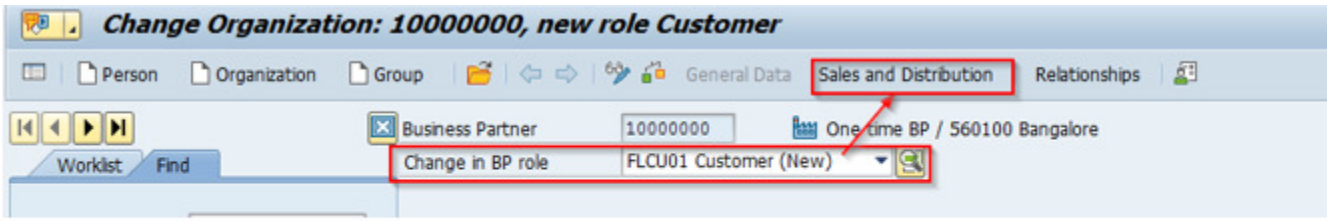


Table KNA1 will be updated once BP Role is extended to FI Customer and Finance can start posting directly in FI for example FB70 etc.

6.8.1.6 Extend Customer to Sales Area

BP Role FLCU01 is extended to sales area.



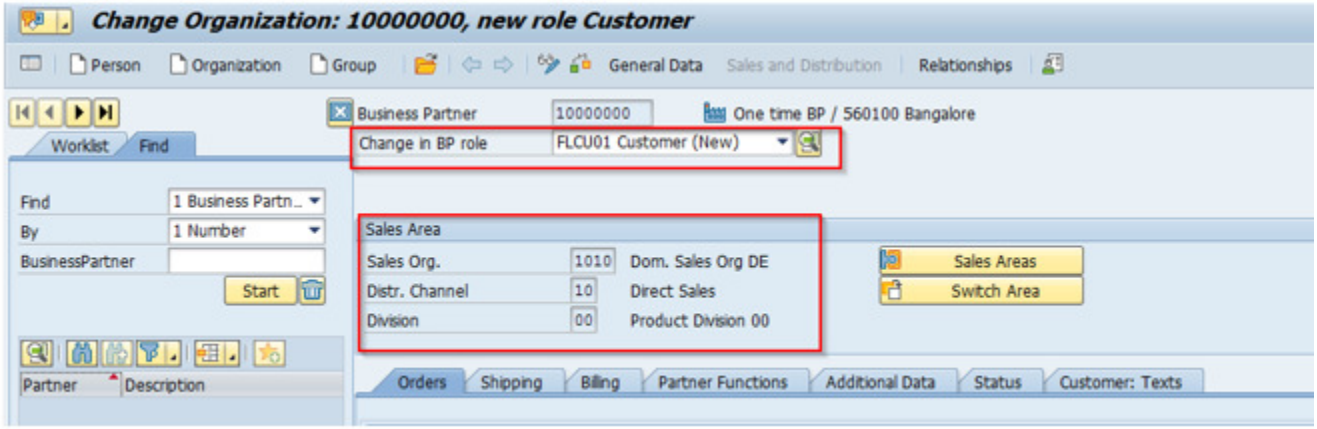
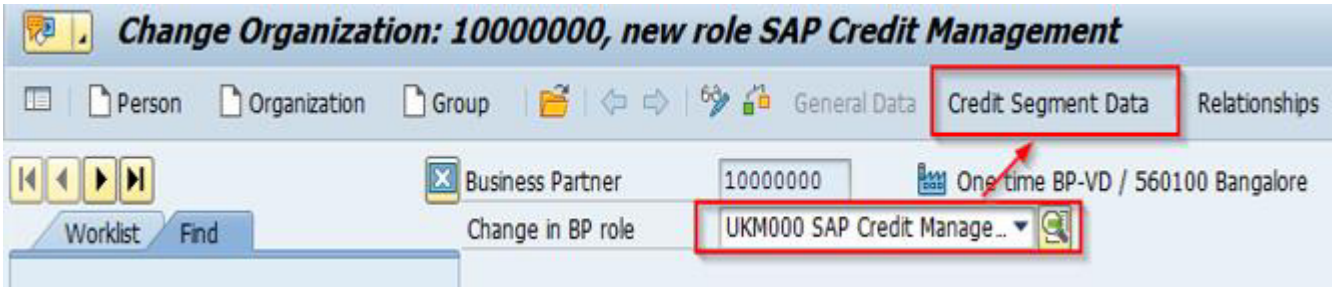


Table KNVV updated with Sales view details and henceforth Customer can be used in sales area.

6.8.1.7 Extend Customer to Credit Segment



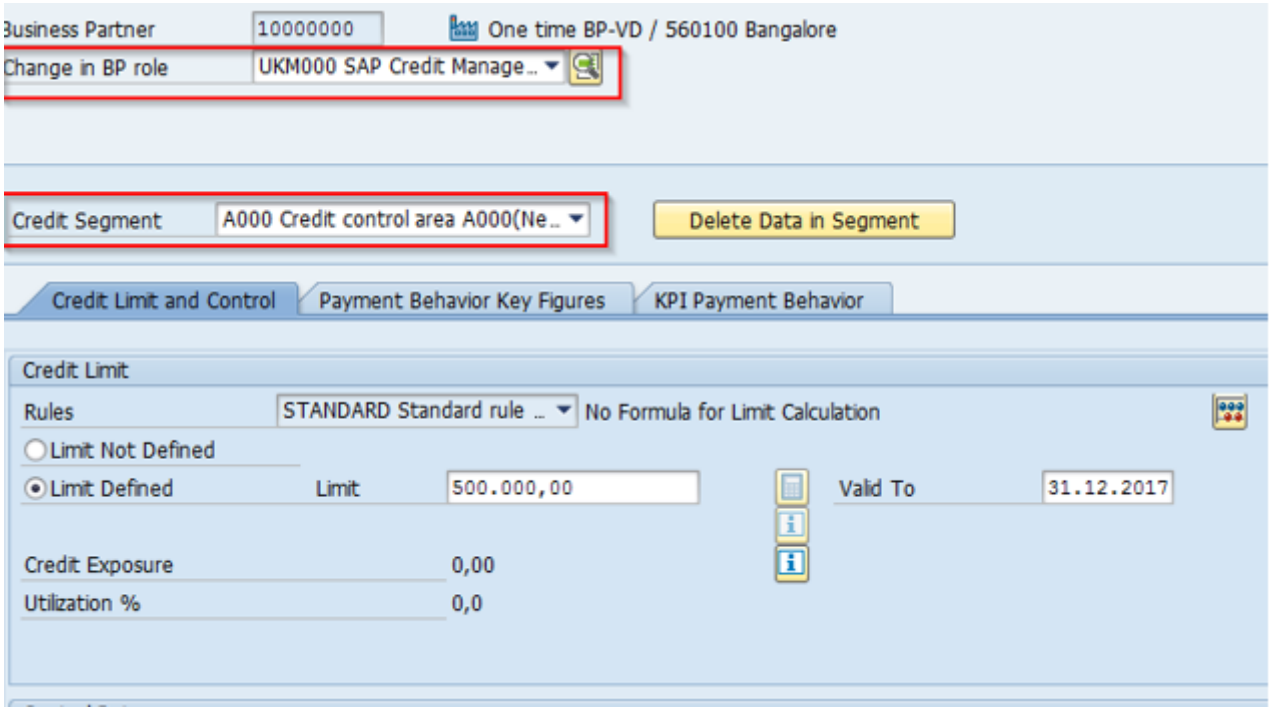


Table UKMBP_CMS_SGM updated

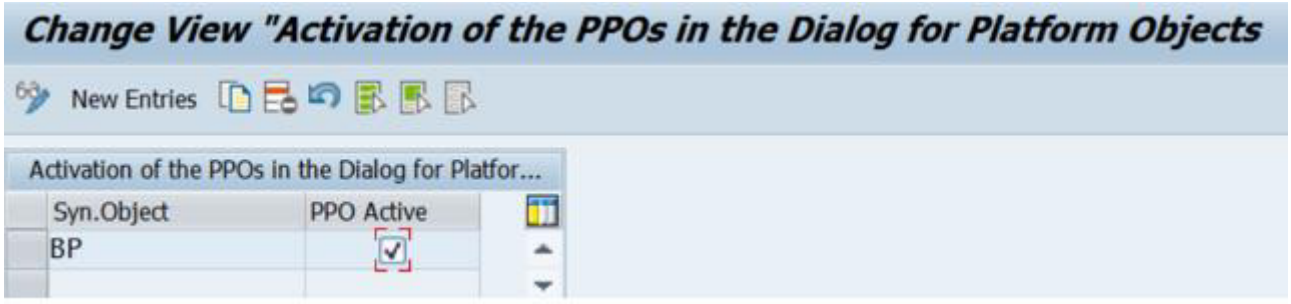
Note- S4 Hana credit Management setting should be enabled to check the credit limits for customers

6.8.2 Customizing

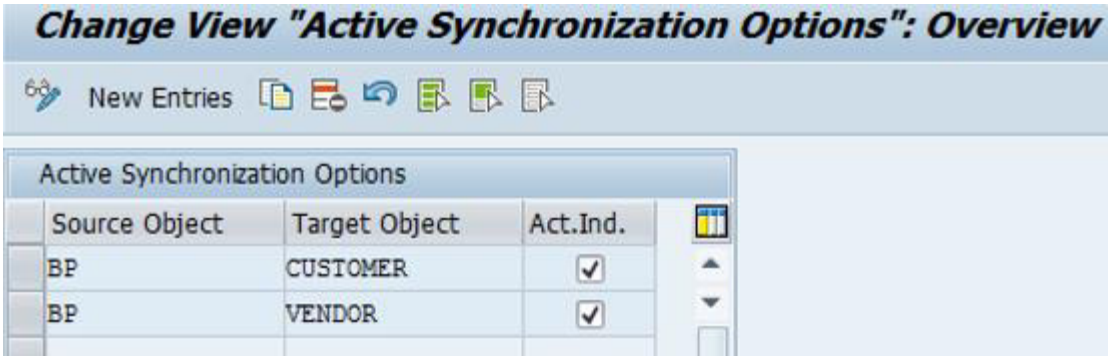
6.8.2.1 Business Partner Customizing

Activate the PPO Request for the synchronization object Business Partner

Cross-Application Components–> Master Data Synchronization –> Synchronization Control–> Synchronization



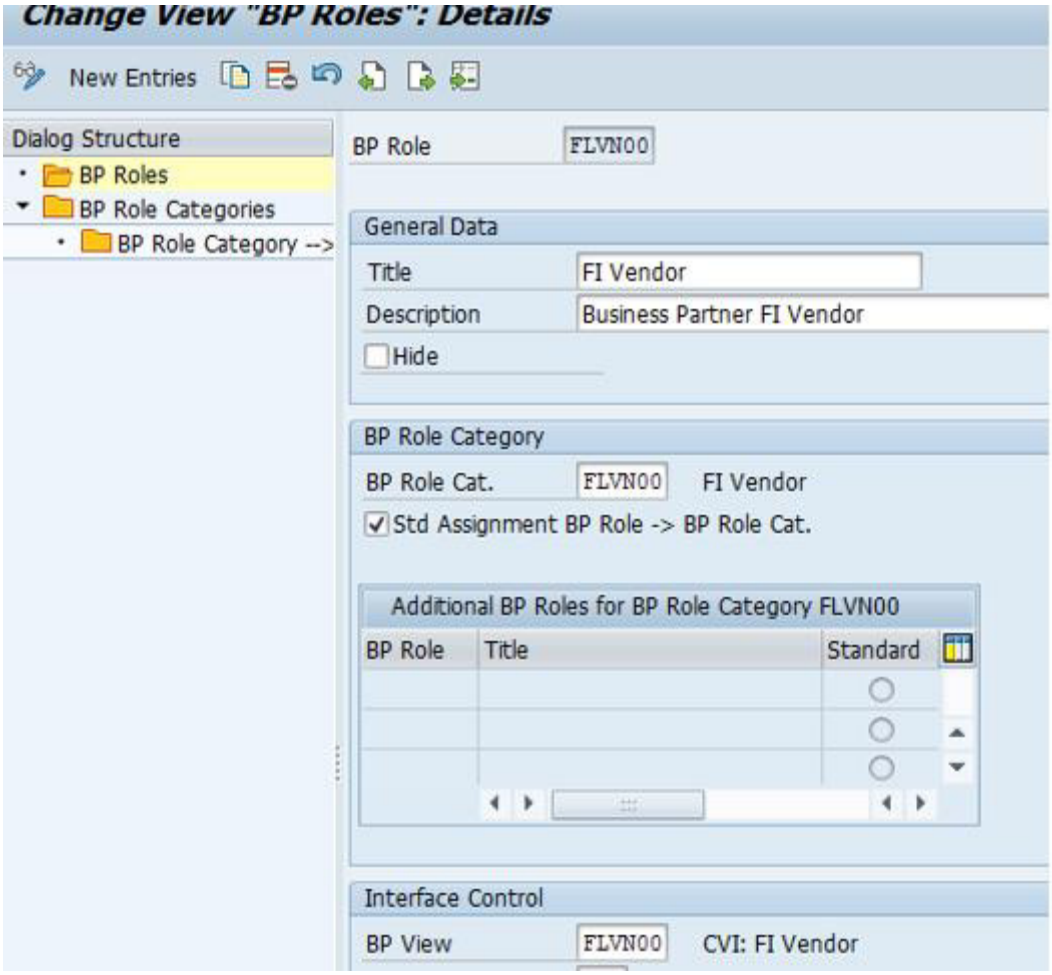
Active Synchronization between Customer/Vendor and BP is active



6.8.2.2 Define BP Roles

Assign BP Role to BP Role Category

Cross-Application Components–> SAP Business Partner–>Business Partner–> Basic Settings –> Business Partner Roles–> Define BP Roles



This setting is already available in standard delivered system for standard roles.

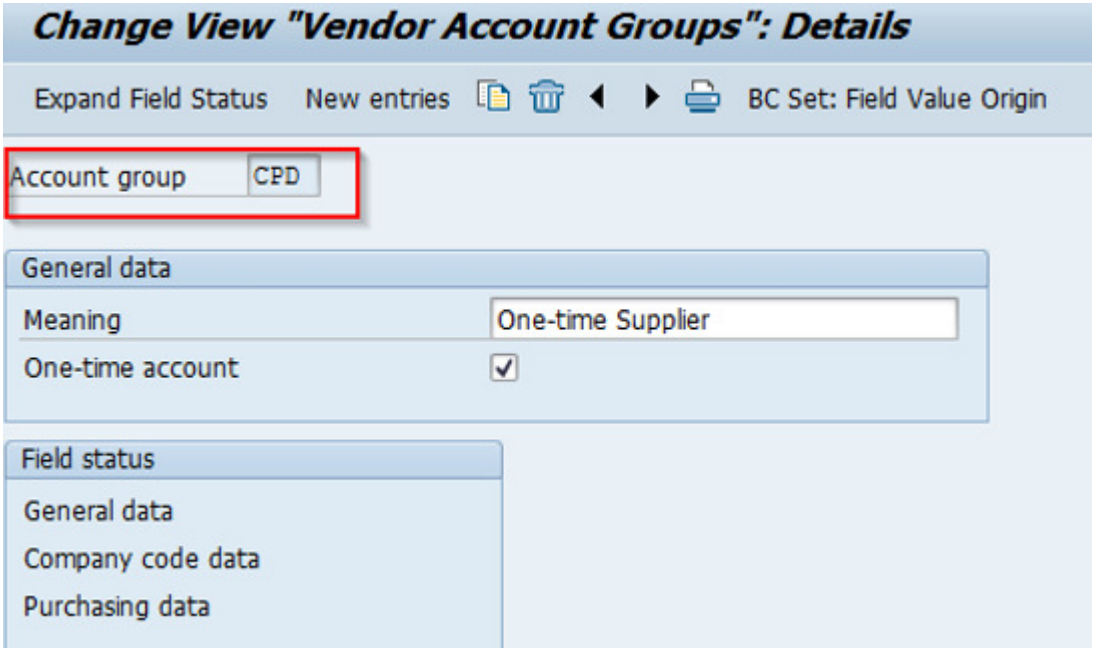
Account Group /Number Range Mapping

I am explaining below with an example of requirement for setting up One Time Vendor and One Time Customer Account and as a summary we need to setup Vendor/Customer Account group & assign number ranges to respective Vendor/Customer account group as it was required in classic ERP environment.

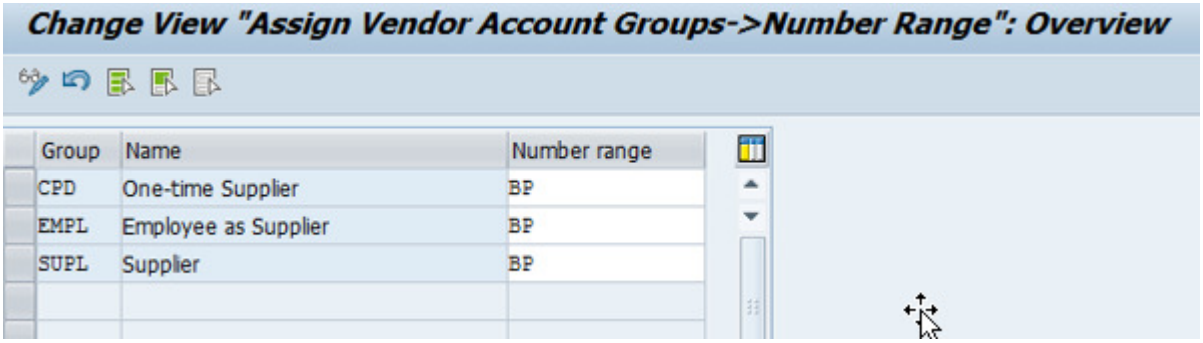
Additionally BP grouping is mapped with Vendor/Customer account group

6.8.2.3 Define Account Groups Vendor

Financial Accounting (New)–> Accounts Receivable and Accounts Payable–> Vendor Accounts–> Master Data–> Preparations for Creating Vendor Master Data



Assign Number Ranges to Vendor Account Groups



Similarly for Customers we need to define Account Group and assign number range to Customer account groups.

6.8.2.4 Define Number Assignment for Direction BP to Vendor/Customer

Cross-Application Components–> Master Data Synchronization–>Customer/Vendor Integration –> Business Partner Settings –> Settings for Vendor Integration –> Field Assignment for Vendor Integration–> Assign Keys

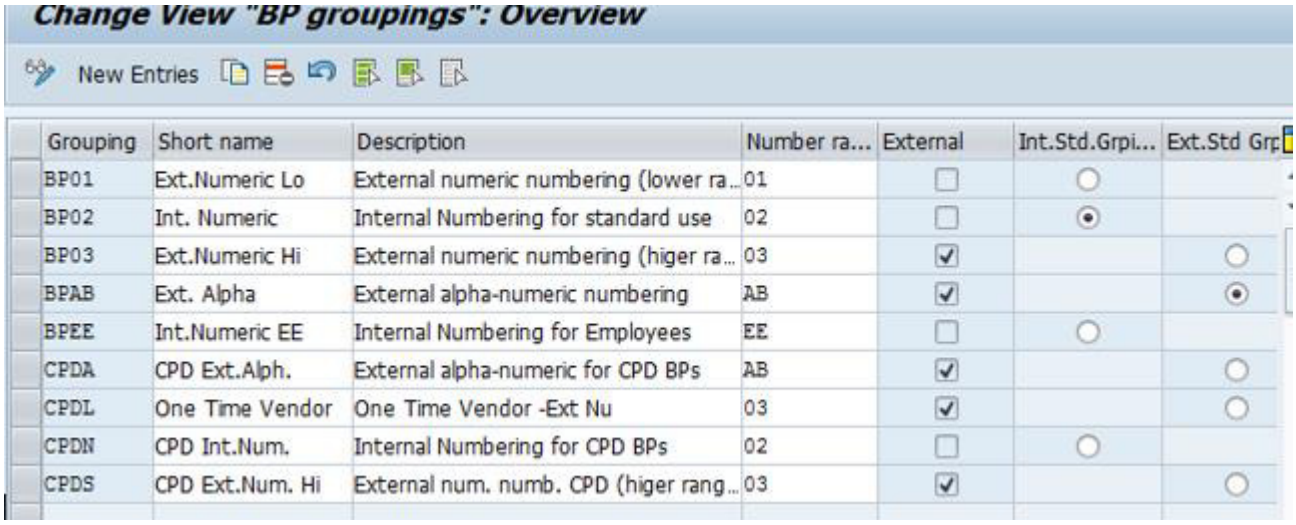
Define Number Assignment for Direction BP to Vendor

6.8.2.5 Define BP Number Range and assigning to BP Grouping

Cross-Application Components–> SAP Business Partner–> Business Partner–> Basic Settings –>Number Ranges and Groupings

Define Number Ranges-BP

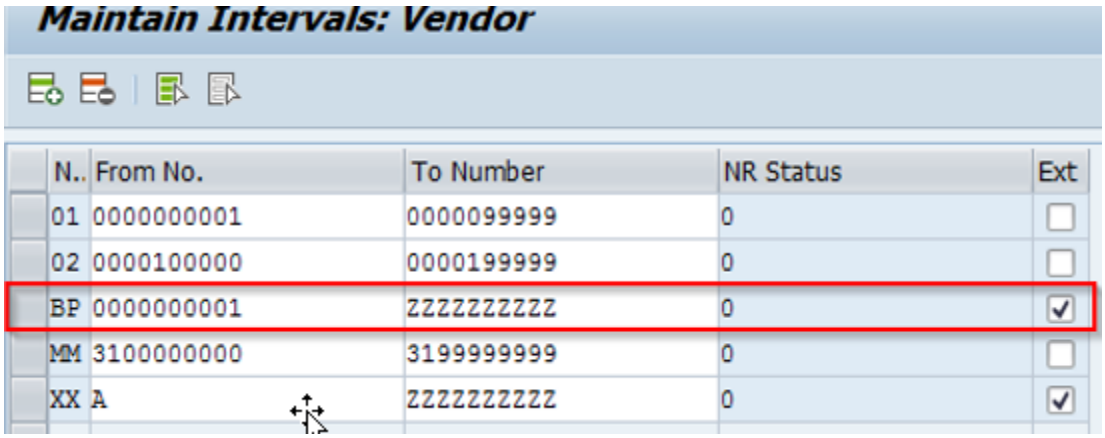
Define Groupings and Assign Number Ranges



6.8.2.6 Create Number Ranges for Vendor/Customer

Financial Accounting (New)–> Accounts Receivable and Accounts Payable–> Vendor Accounts–> Master Data –> Preparations for Creating Vendor Master Data

Create Number Ranges for Vendor Accounts



Similarly create Number Ranges for Customer Accounts.

This will enable to start work with new Business Partner in S4 Hana Enterprise Management Environment.

6.9 Excel Upload for Business Partner[19]

6.9.1 OVERVIEW

You can upload business Partner data in batch to MDG-S from a .csv file. This document provides the installation and configuration instructions, a link to the relevant source code and excel file, and instructions on how to use the excel upload.

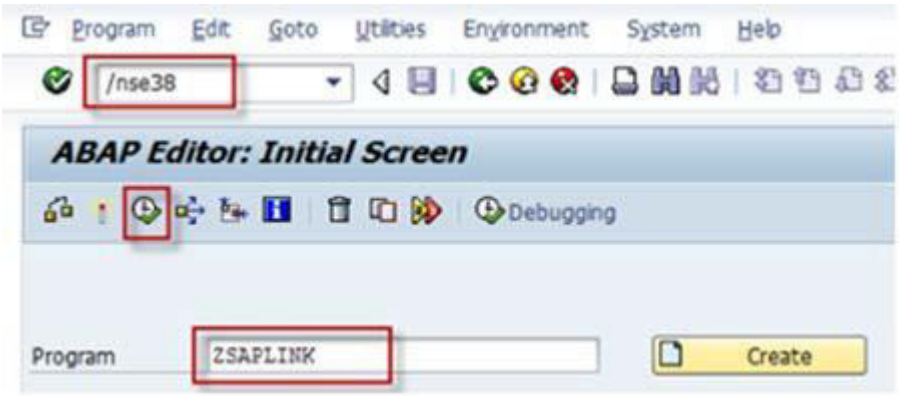
6.9.2 INSTALLATION AND CONFIGURATION OF THE EXCEL UPLOAD

6.9.2.1 Prerequisites

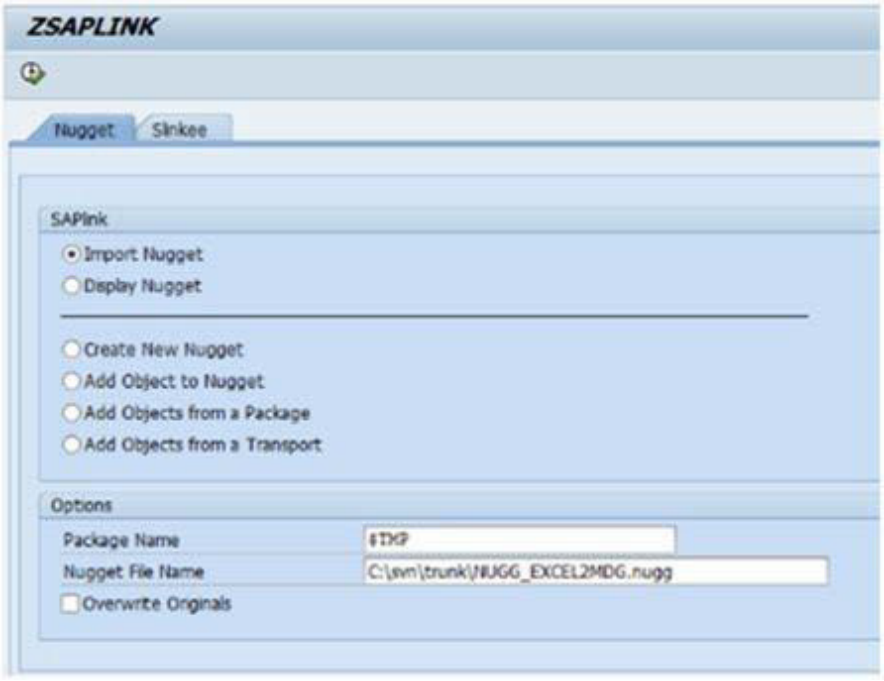
- You have an ERP system with the MDG-S business functions enabled
- You have installed SAPLINK.
- You have downloaded the Nugget with the ABAP Source code for an Excel to MDG-S upload, and a sample CSV file, from the following page.

6.9.2.2 Installation

1. Run transaction SE38 and execute program ZSAPLINK.



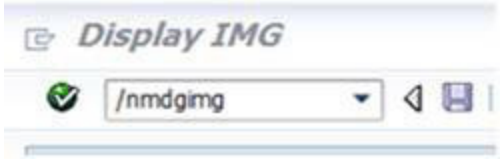
2. In the *Nugget* tab page, select *Import Nugget*. Specify a *Package Name* and a *Nugget File Name*.

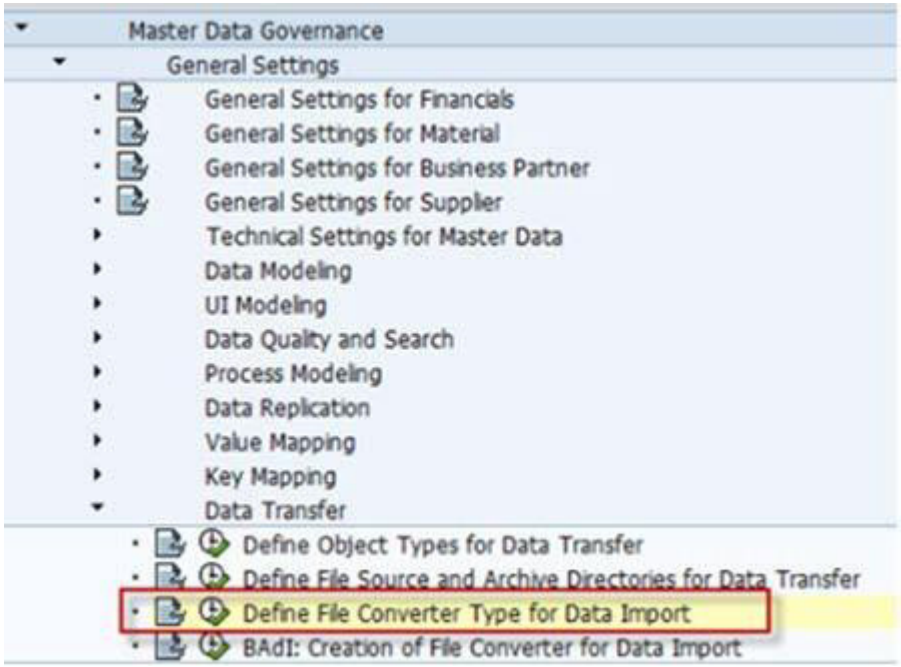


[19] SAP

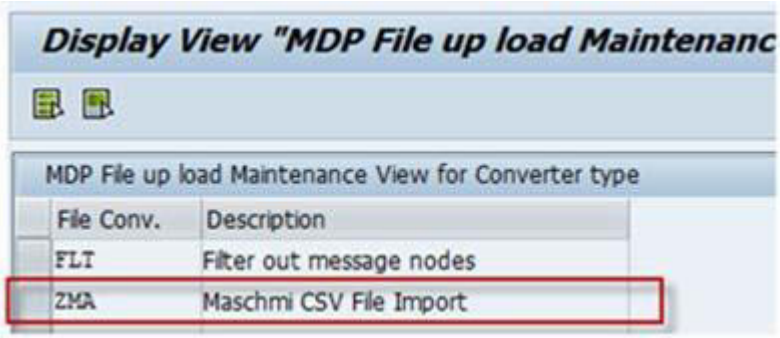
6.9.2.3 Configuration

1. In Customizing for Master Data Governance, open activity General Settings -> Data Transfer -> Define File Converter Type for Data Import.

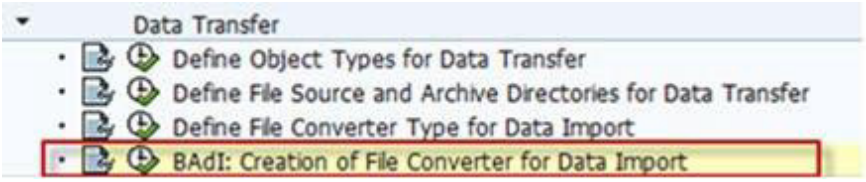




2. Identify the file converter type to be used.



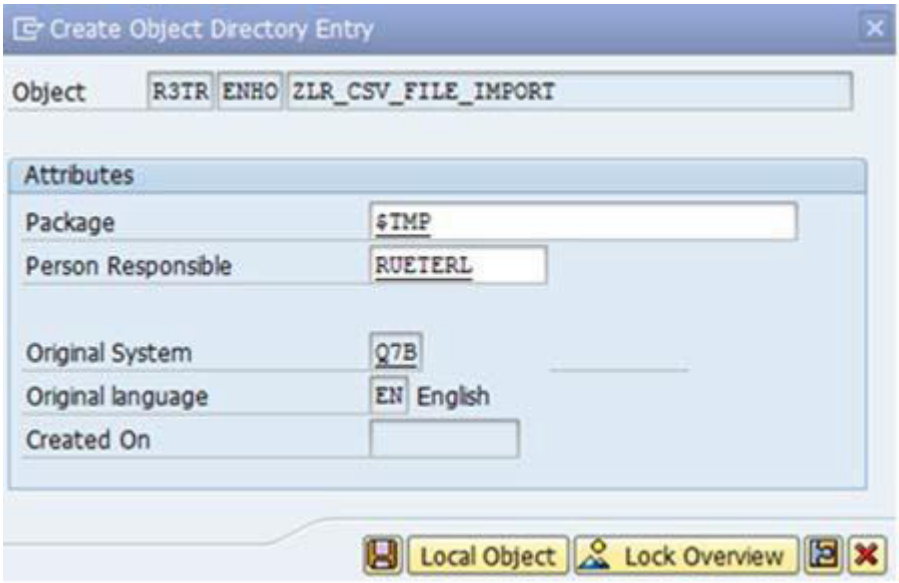
3. In Customizing for Master Data Governance, open activity *General Settings -> Data Transfer -> BAdI: Creation of File Converter for Data Import*.



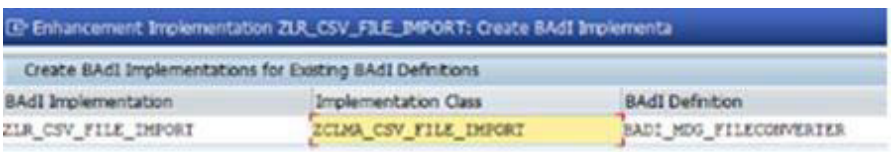
The system creates a BAdI implementation.

4. Specify a technical name and a short text for the enhancement implementation.

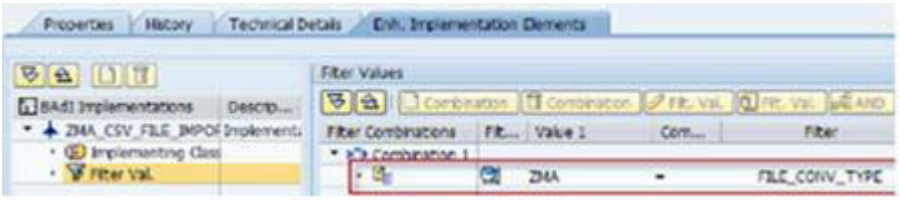
5. Create an Object Directory Entry.

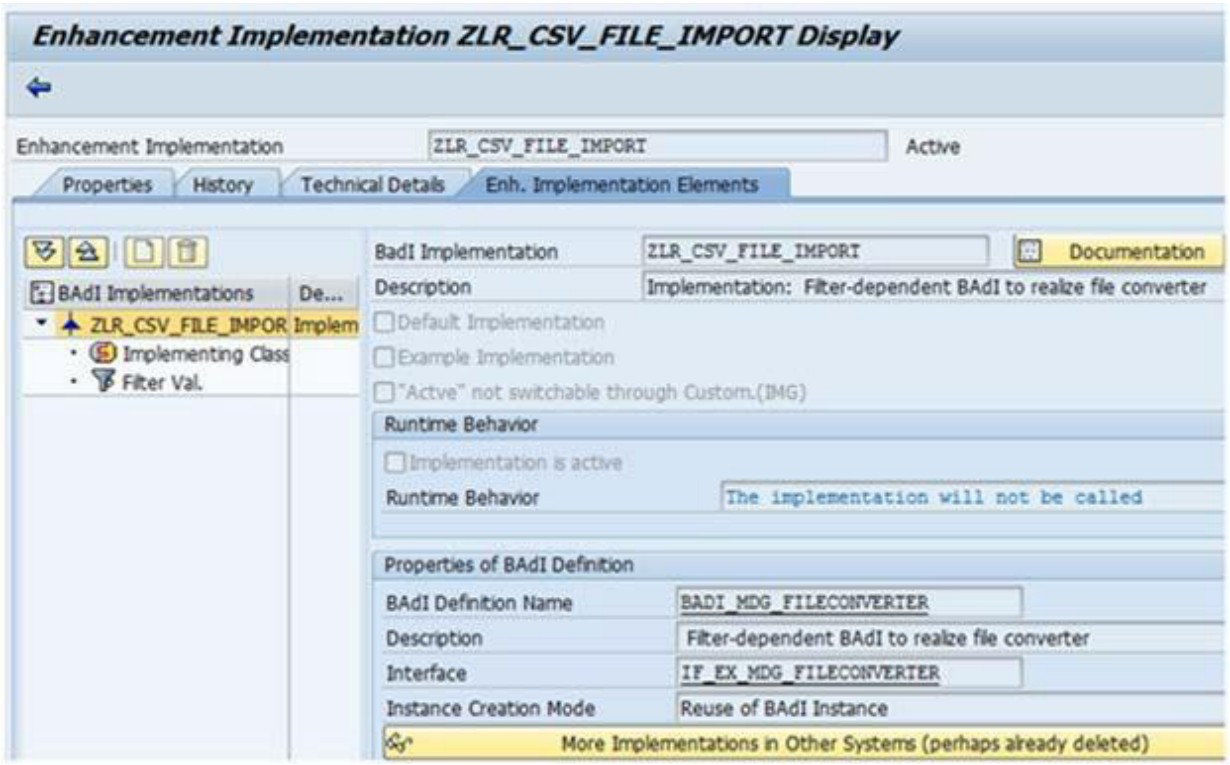


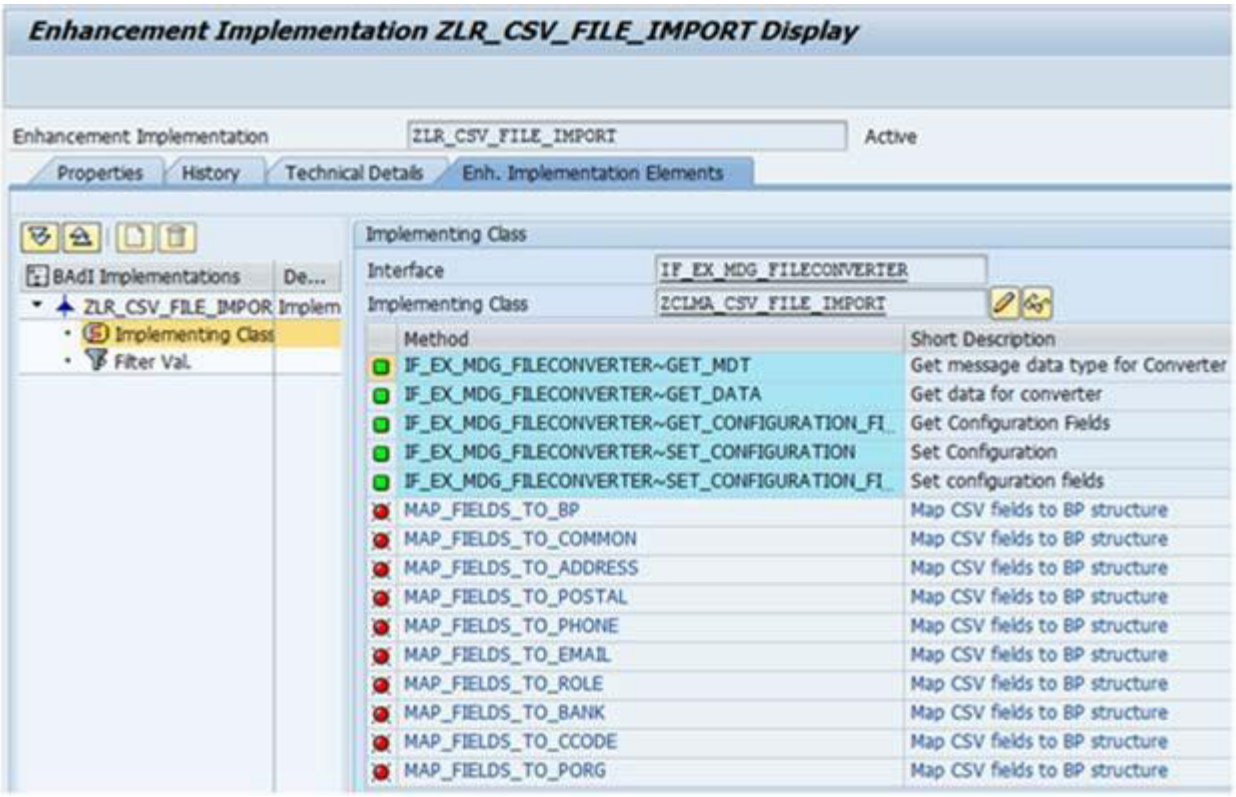
6. Specify an implementation class.



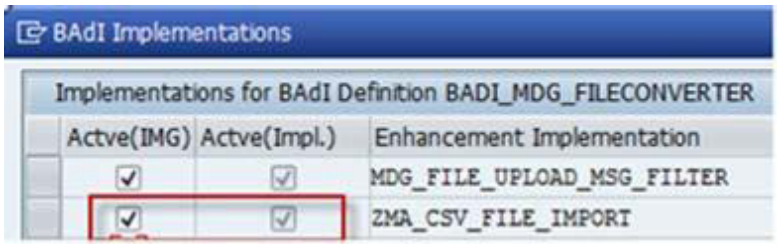
7. In the *Enhancement Implementation Elements* tab page, specify filter values and other fields.





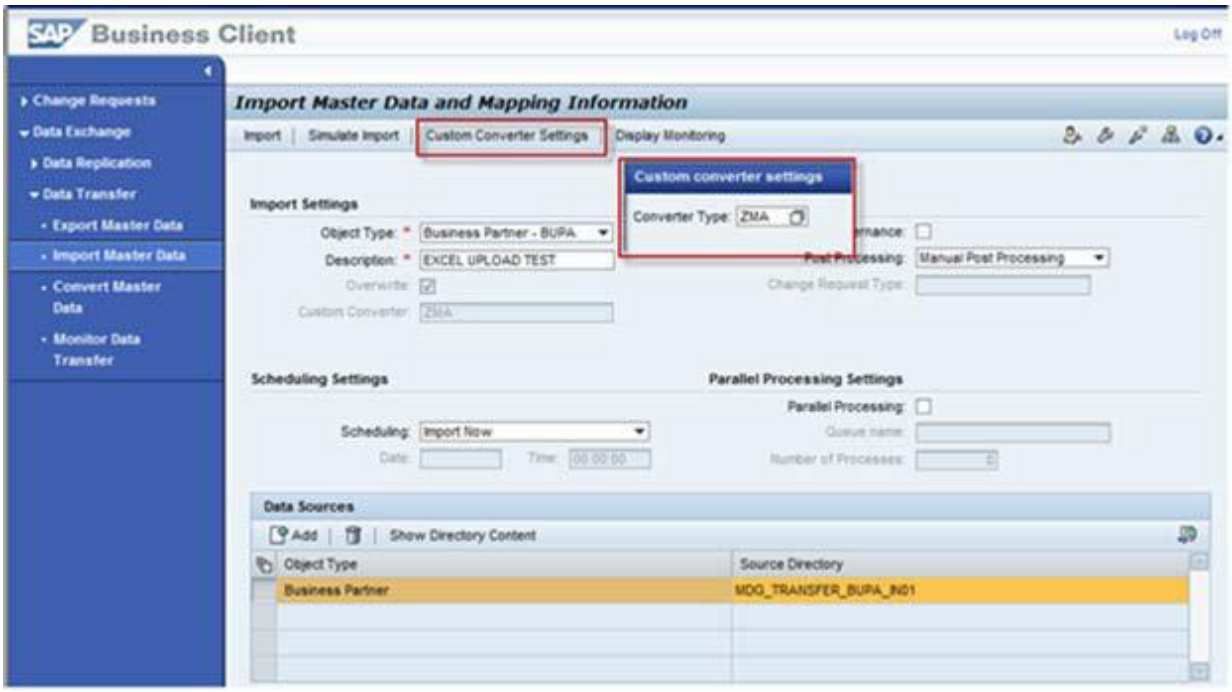


8. Make sure that your implementation is active before you proceed with testing the file upload.



USING THE EXCEL UPLOAD

To upload a .csv file, you use the standard UI for importing Master Data and select the custom converter to handle the CSV file. For testing you can use the .csv file provided here.

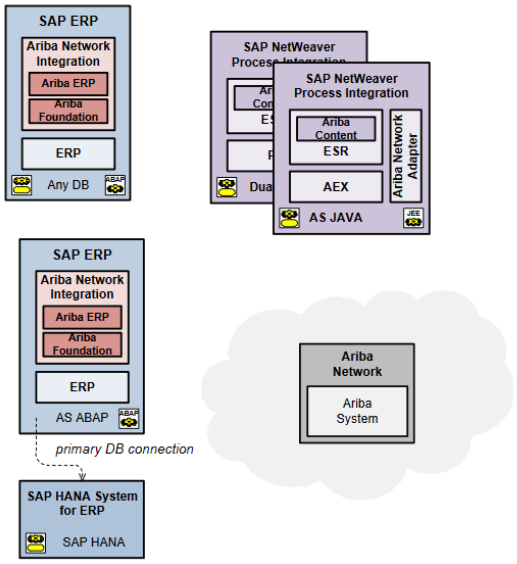

Business Client
Log Off
Change Requests
Data Exchange
Data Replication
Data Transfer
Export Master Data
Import Master Data
Convert Master Data
Monitor Data Transfer
Import Master Data and Mapping Information
Import
Simulate Import
Custom Converter Settings
Display Monitoring
Custom converter settings
Converter Type: ZMA
Import Settings
Object Type: * Business Partner - BUPA
Description: * EXCEL UPLOAD TEST
Overwrite:
Custom Converter: ZMA
Post Processing: Manual Post Processing
Change Request Type:
Scheduling Settings
Scheduling: Import Now
Date:
Time: 00:00:00
Parallel Processing Settings
Parallel Processing:
Queue name:
Number of Processes:
Data Sources
Add
Show Directory Content
Object Type
Source Directory
Business Partner
MDG_TRANSFER_BUPA_IN01

7 Integration MDG and ARIBA

Ariba is a full procurement suite on demand and at the same time the world largest and leading Supplier Network

With the Ariba Network Integration 1.0, SAP Business Suite becomes "Ariba-enabled". This means, SAP provides the possibility for customers (in particular buyers) to connect the following SAP Business Suite systems to the Ariba Network:

- **SAP ERP**
- **SAP Supplier Relationship Management (SAP SRM) in extended classic scenario**
  - Note: SAP SRM in classic scenario is covered by the integration of SAP ERP with Ariba Network
- **SAP Supply Network Collaboration (SAP SNC)**



Main usage scenario

- Integration from SAP ERP MM to Ariba Network

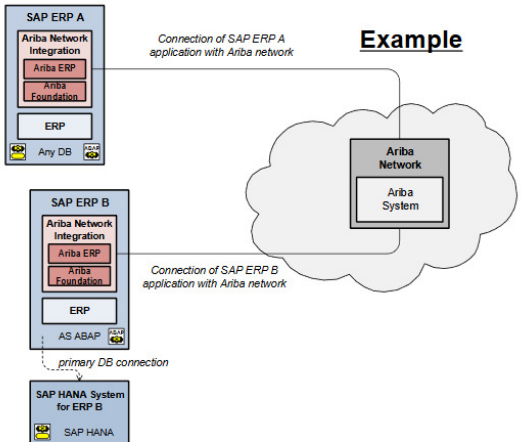
Main required product artifacts

- SAP ERP
  - SAP ERP system starting with SAP ERP 6.0
  - So far, only Ariba integration scenario with SAP ERP **core** functionality supported (see Administrator's Guide)
- Ariba Network Integration Add-Ons
  - Ariba Foundation
    - Add-on "ARBFNDI1" starting with AS ABAP 7.00
    - Add-on "ARBFNDI2" starting with AS ABAP 7.01
  - Ariba ERP
    - Add-on "ARBERPI1" based on add-on "ARBFNDI1"
- Ariba Network (in cloud environment)
- SAP NetWeaver PI (optional)
  - Middleware to connect SAP ERP with Ariba Network
  - Starting with SAP NetWeaver PI 7.1
  - Java stack sufficient, therefore SAP NetWeaver PI - JAVA only (AEX) supported
  - ESR Content with Ariba cXML interfaces
  - Ariba Network Adapter for SAP NetWeaver (starting with Release Cloud Integration 1.0) – released for SAP NetWeaver PI 7.1 up to EHP1 for SAP NetWeaver PI 7.4

7.1 Add-On deployment in SAP ERP system[20]

General Recommendation

- Ariba Network Integration add-ons are deployed in every SAP ERP system that you want to integrate with Ariba Network
- Use direct connectivity between SAP ERP and Ariba Network



Benefits

- No additional system or infrastructure component required
- Minimum release is SAP ERP 6.0
- Low landscape complexity, modification-free add-ons only
- No disruption, no update or upgrade required

Considerations

- SAP ERP system must be at least based on SAP ERP 6.0 SP Stack 15
- Starting with EHP4 FOR SAP ERP 6.0 “Forward Error Handling” (FEH) is supported
- Connectivity needs to be configured for each SAP ERP system
- SAP ERP systems are connected directly with external cloud environment but communication is always initiated from ERP system

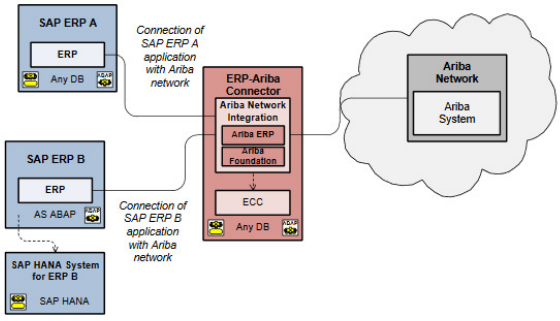
7.2 Hub deployment of Ariba Network Integration

Not Supported

- Ariba Network Integration add-ons are deployed in a separate system
- SAP ERP systems are integrated with Ariba via a central SAP ERP-Ariba-Connector system

Not Supported

Example

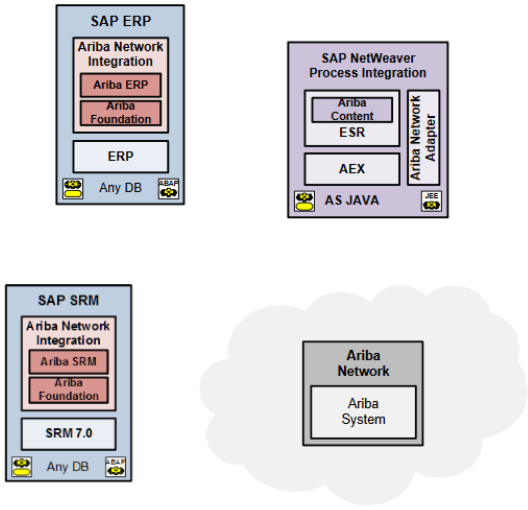


Implications

- No deployment separate from consuming SAP application possible
- Ariba Network Integration add-ons must be technically deployed in the SAP ERP system
- No support of old SAP ERP releases below SAP ERP 6.0

[20] archive.sap.com/kmuuid2/d060ac35-a269-3110-34b7-c7d3a1f54175/System Landscape Recommendations for Ariba Network Integration 1.0 for SAP Business Suite

7.3 Ariba Integration for SAP SRM



Main usage scenario: Extended Classic Scenario of SAP SRM

- Integration from SAP ERP to Ariba Network
- Integration from SAP SRM to Ariba Network

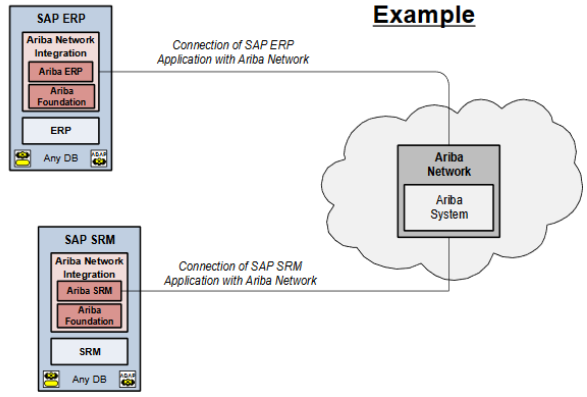
Main required product artifacts

- SAP ERP
  - SAP ERP system starting with SAP ERP 6.0 (see SAP ERP Administrator's Guide)
- SAP SRM
  - SAP SRM system starting with SAP SRM 7.0 (see SAP SRM Administrator's Guide)
- Ariba Network Integration Add-Ons
  - Ariba Foundation
    - Add-on "ARBFNDI1" starting with AS ABAP 7.00
    - Add-on "ARBFNDI2" starting with AS ABAP 7.01
  - Ariba ERP
    - Add-on "ARBERPI1"
  - Ariba SRM
    - Add-on "ARBSRMI1"
- Ariba Network (in cloud environment)
- SAP NetWeaver PI (optional)
  - Middleware to connect SAP ERP and SAP SRM with Ariba Network
  - Starting with SAP NetWeaver PI 7.1
  - Java stack sufficient, therefore SAP NetWeaver PI - JAVA only (AEX) supported
  - ESR Content with Ariba cXML interfaces
  - Ariba Network Adapter for SAP NetWeaver (starting with Release Cloud Integration 1.0) – released for SAP NetWeaver PI 7.1 up to EHP1 for SAP NetWeaver PI 7.3 (7.4 currently in certification state)

7.4 Add-On Deployment in SAP SRM System Connected to SAP ERP[21]

General Recommendation

- Ariba Network Integration add-ons have to be deployed in the SAP SRM system and in every connected SAP ERP back-end system that you want to integrate with Ariba Network
- Use direct connectivity between SAP ERP, SAP SRM, and Ariba Network



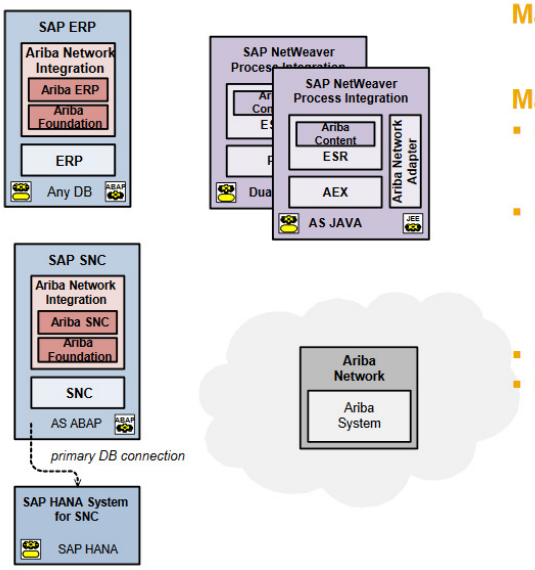
Benefits

- Minimum release is SAP ERP 6.0 and SAP SRM 7.0
- Low landscape complexity, modification-free add-ons only
- No disruption, no update or upgrade required

Considerations

- SAP ERP system must be at least based on SAP ERP 6.0 SP Stack 15
- Starting with EHP4 for SAP ERP 6.0 "Forward Error Handling" (FEH) is supported
- SAP SRM system must be at least based on SAP SRM 7.0 SP Stack 03
- Connectivity needs to be configured for each SAP ERP and SAP SRM system
- Although SAP ERP and SAP SRM are connected directly with Ariba Network, communication is always initiated from SAP ERP or SAP SRM

[21] sap.com/System Landscape Recommendations for Ariba Network Integration 1.0 for SAP Business Suite



Main usage scenario

- Integration from SAP SNC to Ariba Network
- SAP SNC is integrated with SAP ERP via PI

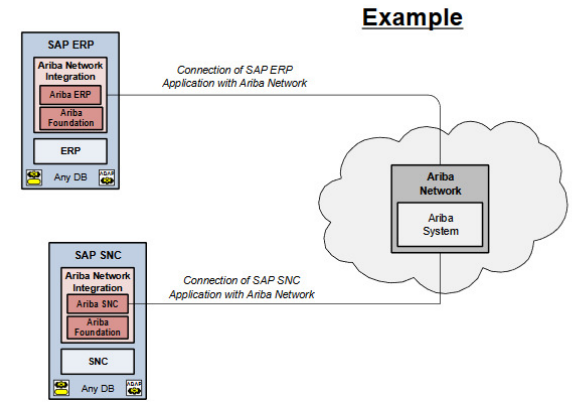
Main required product artifacts

- SAP SNC
  - SAP SNC system starting with SAP SNC 7.0 EHP1
  - See Administrator's Guide for Integration of SAP SNC
- Ariba Network Integration Add-Ons
  - Ariba Foundation
    - Add-on "ARBFNDI1" starting with AS ABAP 7.00
    - Add-on "ARBFNDI2" starting with AS ABAP 7.01
  - Ariba SNC
    - Add-on "ARBSNCI1" based on add-on "ARBFNDI1"
- Ariba Network (in cloud environment)
- SAP NetWeaver PI (optional)
  - Middleware to connect SAP ERP and SAP SNC with Ariba Network
  - Starting with SAP NetWeaver PI 7.1
  - Java stack sufficient, therefore SAP NetWeaver PI - JAVA only (AEX) supported
  - ESR Content with Ariba cXML interfaces
  - Ariba Network Adapter for SAP NetWeaver (starting with Release Cloud Integration 1.0) – released for SAP NetWeaver PI 7.1 up to EHP1 for SAP NetWeaver PI 7.4

7.5 SAP SNC System Connected to SAP ERP[22]

General Recommendation

- Ariba Network Integration Add-Ons are deployed in every SAP SNC system and optionally in every SAP ERP system (dependent if needed for the business scenario) that is to be integrated with Ariba Network
- Direct connectivity between SAP ERP, SAP SNC, and Ariba Network



Example

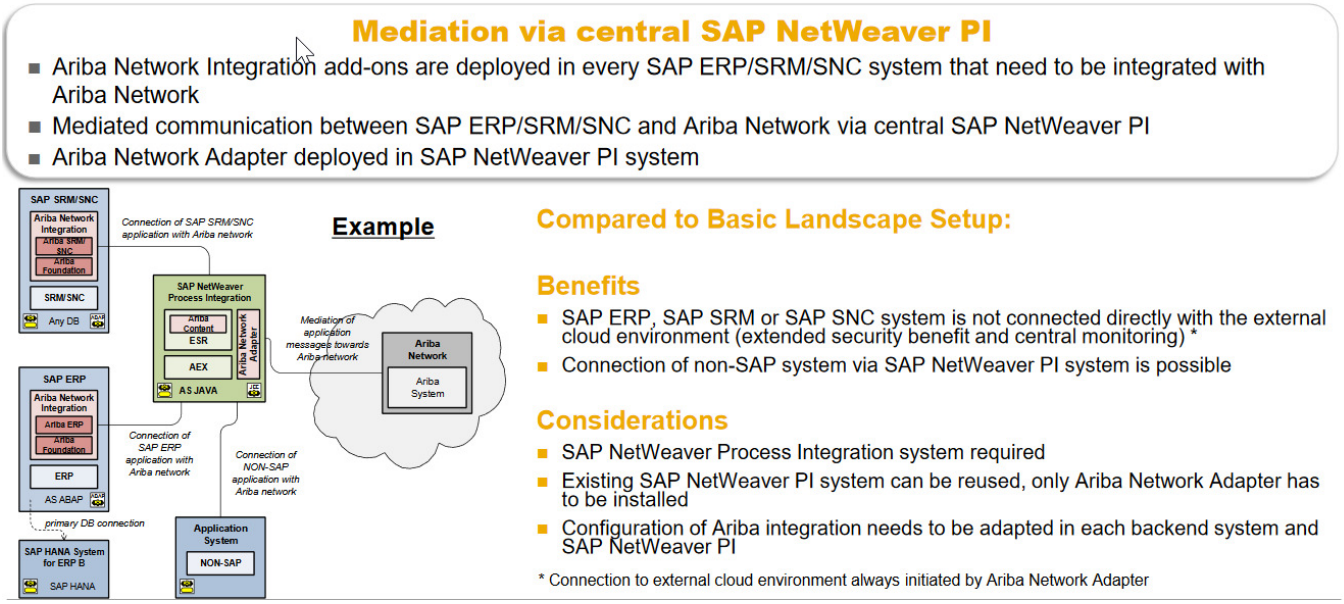
Benefits

- Minimum release is SAP ERP 6.0 and SAP SNC 7.01
- Low landscape complexity, modification-free add-ons only
- No disruption, no update or upgrade required

Considerations

- SAP SNC system must be at least based on SAP SNC 7.01 SP Stack 10
- SAP ERP system must be at least based on SAP ERP 6.0 SP Stack 15
- Starting with EHP4 for SAP ERP 6.0 "Forward Error Handling" (FEH) is supported
- Connectivity needs to be configured for each SAP ERP and SAP SNC system
- Although SAP ERP and SAP SNC are connected directly with Ariba Network, communication is always initiated from SAP ERP or SAP SNC

22 SAP SE or an SAP affiliate company.

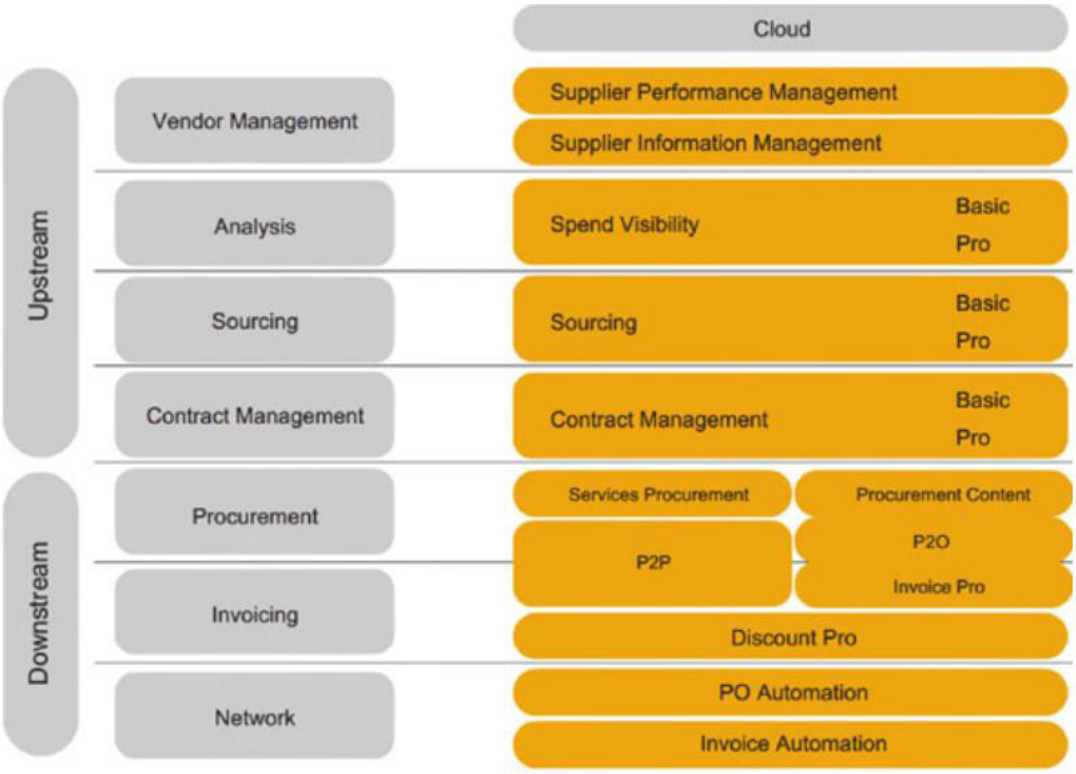


Ariba is the world's foremost business commerce network and leading cloud based procurement and invoicing application. It has the world's largest web-based trading community to help companies DISCOVER, CONNECT and COLLABORATE with the global network of partners. SAP bought Ariba few years back, and now it is known as SAP Ariba. SAP Ariba provides innovative business models to help you modernize and grow your business.

Most of the businesses suffer from a poor collaboration between buyer and trading partners. Because of this lot of time is wasted, sales opportunities are missed, payments are delayed and hence creates limited visibility to decision makers. SAP Ariba has its own network called Ariba network with more than 1 – 1.5 million companies registered on it. This enormous database provides better collaboration and is also a central point platform for buyer and trading partners. This platform provides a user-friendly interface that seems native for anyone who shops online.

SAP Ariba supports both sellers and buyers. Ariba platform provides sellers, the ability to manage catalogs, bids, purchases, and invoices and it provides buyers with the ability to search for suppliers, negotiate savings, purchase goods and services and track expenditure. **It is a single solution which offers better features and functionality in sourcing, contract life cycle management, invoice management, supplier life cycle management, and MDM catalog.** These extended features and functionality of Ariba helps both do better business, which existing ERP's are not capable of offering. The business process is divided into two different processes – upstream and downstream. Below picture displays the available solution components of SAP Ariba.

7.6 SAP Ariba Solution Components[23]



[23] *Karthik Ayyala SAP Solution Architect @YASH Technologies*

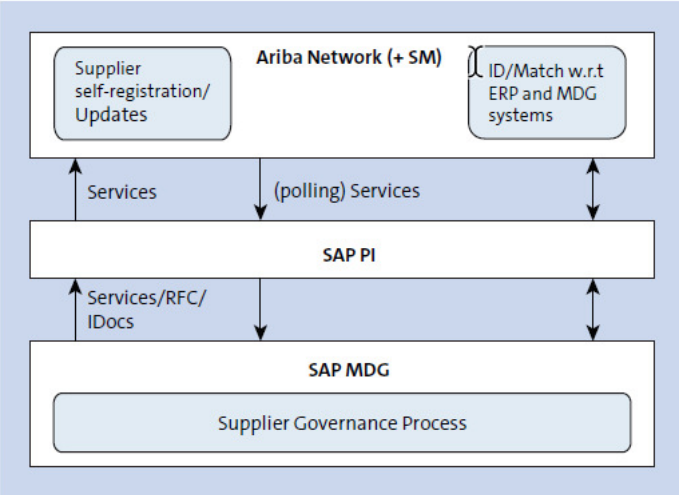
7.7 SAP Ariba Integration

SAP Ariba can be seamlessly integrated with any procurement ERP solution like SAP, PeopleSoft, Oracle, IBM Maximo and Others.

In the current blog, we will be discussing details of Ariba procurement solution integration with SAP ERP. Following are the features-

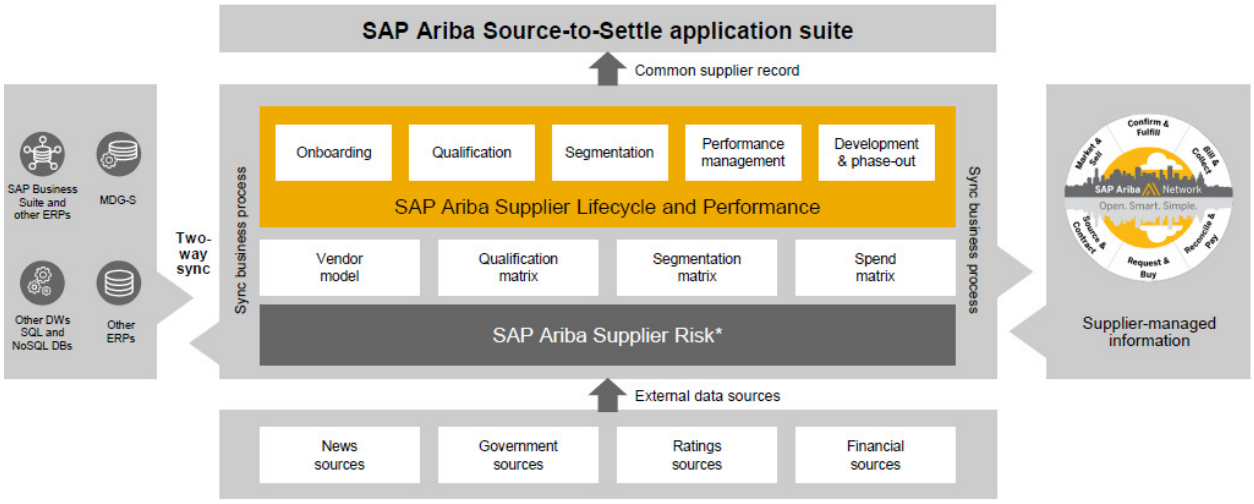
1. The administrators can export master data from SAP ERP to Ariba seamlessly.
2. The transactional data can be updated bi-directionally or in real time.
3. Ariba integration toolkit – A Java-based tool that Ariba provides to upload master data or download transactional data.
4. Ariba integration toolkit reads the CSV files in SAP ERP – zips them, sends them as MIME messages using the HTTP post – to the Ariba procurement solution.

7.7.1 ARIBA NETWORK



documents into a Central Finance instance (SAP S/4HANA Finance). In a multisystem landscape, Central Finance adds value through a nondisruptive implementation approach to consolidate the systems over time. The Central Finance deployment approach delivers harmonized data for consolidated financial and management reporting. Data from multiple source systems need to be mapped to the Central Finance system. SAP MDG can be leveraged to maintain key master data mappings (e.g., chart of accounts, controlling areas, plant, material, cost center, etc.) across various system to have a harmonized data set for reporting. Hence, when an SAP ERP Financials (SAP ERP FI) posting is pushed from the source system to the Central Finance instance, cor- responding master data key values are retrieved with respect to the systems (source system and Central Finance instance mapping in SAP MDG). Additionally, SAP Master Data Governance, Financials functionalities can be used to govern the SAP ERP FI master data objects such as SAP General Ledger (G/L) accounts, hierarchies, and so on. Thus, SAP MDG adds value to the Central Finance deployment scenario, along with the management of other master data domains and their respective mappings. For use of SAP Credit Management with SAP S/4HANA in the Central Finance sce- nario, refer to SAP Note 2337298 for a sample implementation for the BAdI UKM_CONV_ TO_PARTNER to convert customer numbers between systems.

7.7.2 ARIBA Management solution portfolio overview

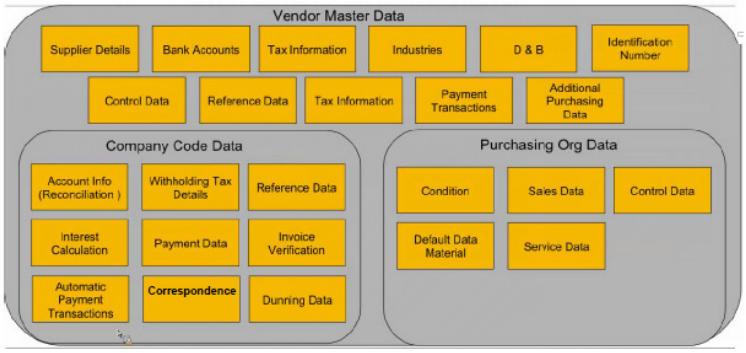


Product capabilities

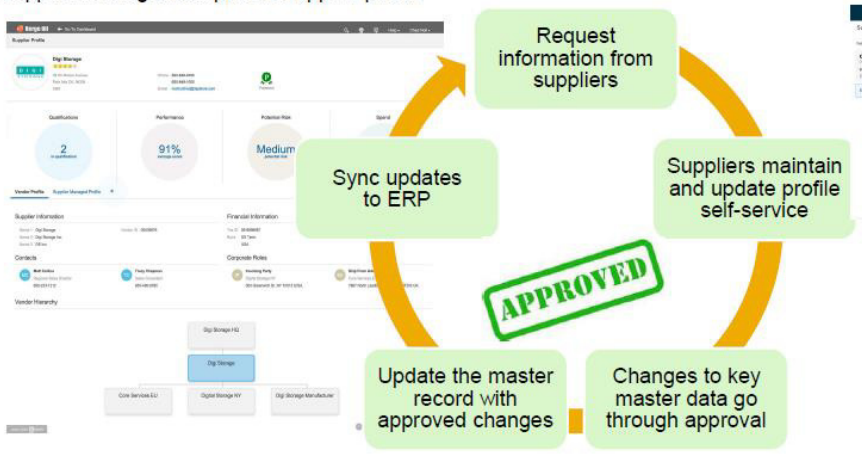
- Vendor model alignment with ERP: In the SAP Ariba supplier management system we are aligning the object model to have all details of the ERP. Some of the high level data are:
  - Vendor general data
  - Vendor purchasing data
  - Vendor company code data
  - Vendor bank details
  - Vendor partner functions
  - Vendor hierarchy
  - Vendor dunning data
  - Vendor attachments

Key benefits

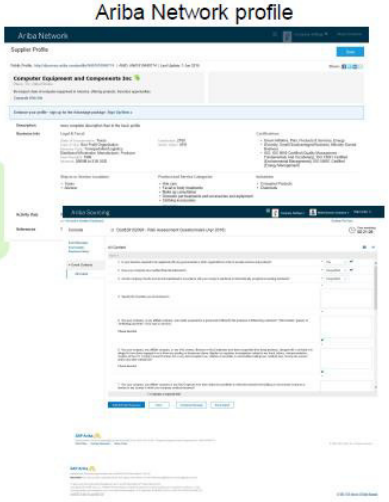
- Support vendor master in the cloud as an extension to the ERP
- Easier integration within SAP Ariba solutions and with SAP ERP



Supplier management private supplier profile



Ariba Network profile



7.7.3 Enhanced SAP ERP integration with access to SAP ERP master data including material master

Product capabilities

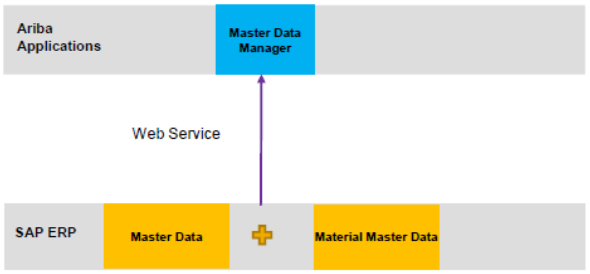
- Access to the replicated SAP ERP master data including material master in Ariba Sourcing and Ariba Contract Management
- Creation of integrated sourcing events and contract workspaces in Ariba Sourcing with replicated master data from SAP ERP and SAP ERP validation
- Creation of an SAP ERP Outline Agreement out of an Ariba Contract Workspace

Key benefits

- Build in operational compliance to ensure that the results of a strategic sourcing event is leveraged for the operational procurement process in SAP ERP
- Enhanced support for direct material sourcing events in industries like CPG, ENR, Automotive, Hi-Tech, Discrete Manufacturing etc.

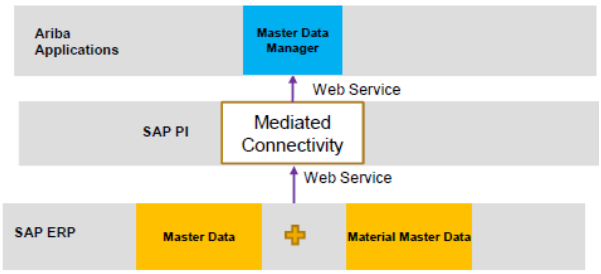
7.7.4 Master Data Integration

Direct Connectivity



Data is extracted as CSV files in ERP, Compressed to a ZIP file and sent with in a SOAP Message Envelope through Web Service calls directly to the application. No PI involved.

Mediated Connectivity

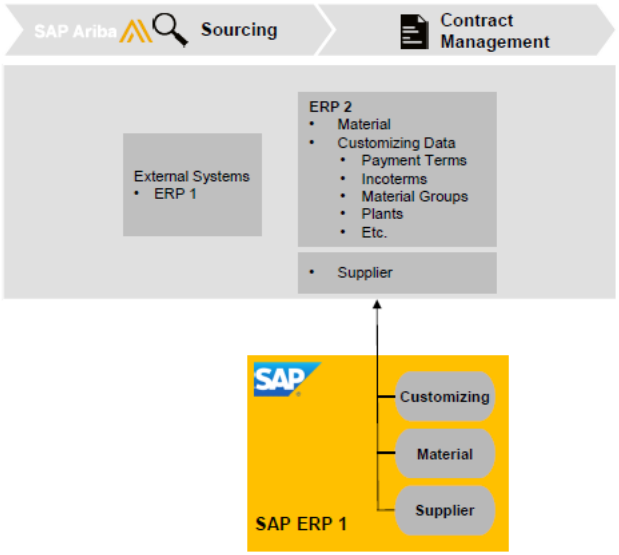


Data is extracted as CSV files in ERP, compressed to a ZIP file and sent with in a SOAP Message envelope; it is sent via PI as a 'Pass Through'. i.e. there will not be any mapping done in PI.

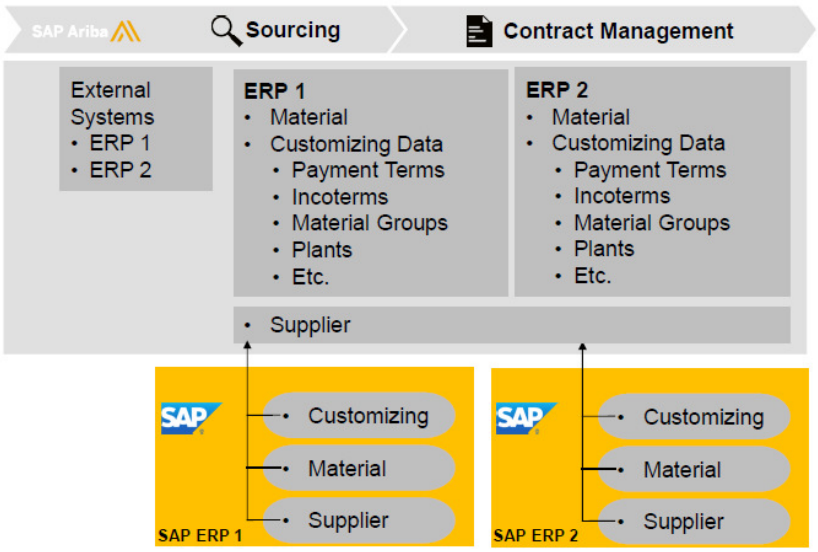
Master Data Extracted

- Company Code
- Material Group
- Inco Terms
- Item Category
- Payment Terms
- Plant
- Purchase Group
- Purchase Org
- Plant & Purchase Org Combo
- POrg & Company Code Combo
- Item Master & Plant Combo
- Plant & Company Code Combo

7.7.5 Master Data replication (single ERP case)

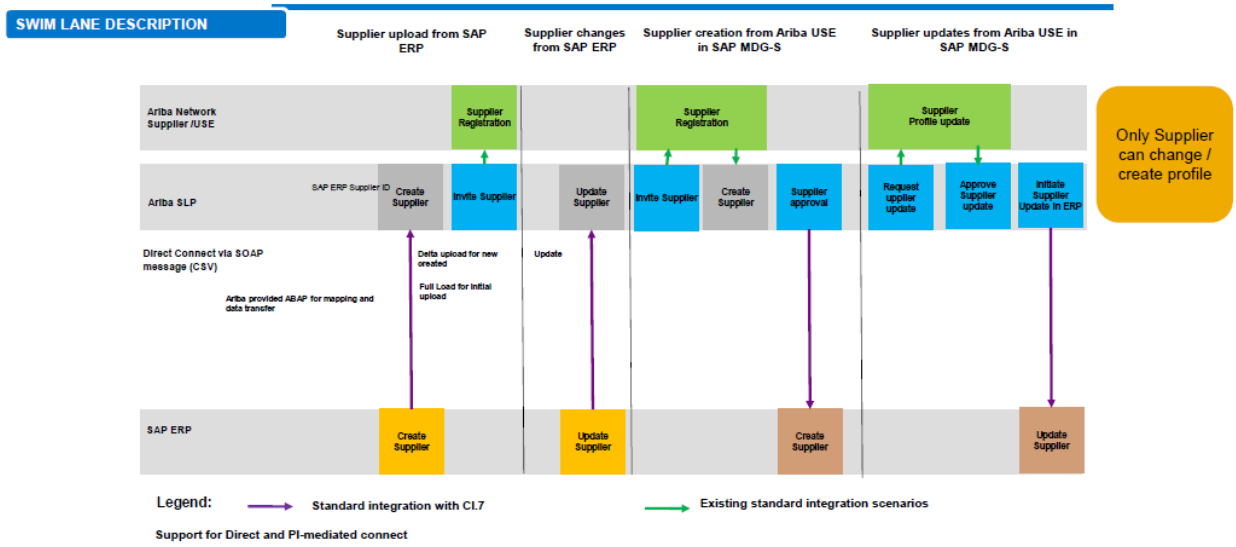


- SAP ERP extractors to replicate master data from SAP ERP to Ariba Upstream
- HTTPS web service Direct Connect (PI optional)
- SOAP Message Envelope (CSV)

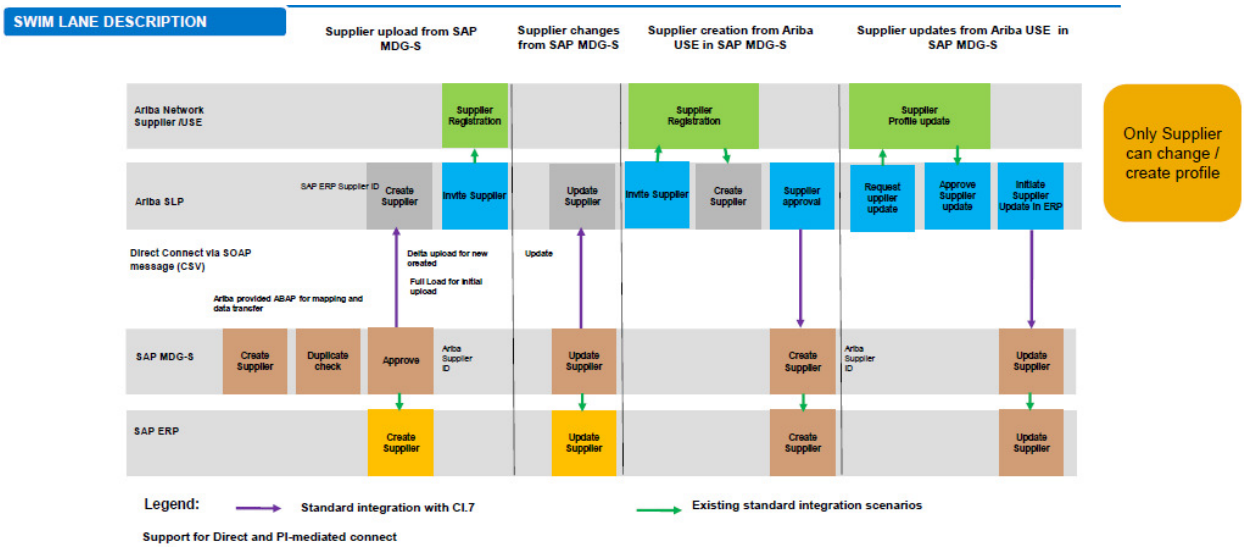


- SAP ERP extractors to replicate master data from SAP ERP to Ariba Upstream
- Support for multi ERP master data replictation
- Replicated master data stored in Ariba with source system reference (exception: supplier)

7.7.6 Option 1: SLP – SAP ECC Integration (CI-7) Process Flow



7.7.7 Option 2: SLP – SAP ECC / MDG-S Integration (CI-7) Process Flow



SAP Ariba can be seamlessly integrated with any procurement ERP solution like SAP, PeopleSoft, Oracle, IBM Maximo and Others.

7.8 Different methods to integrate Ariba and SAP

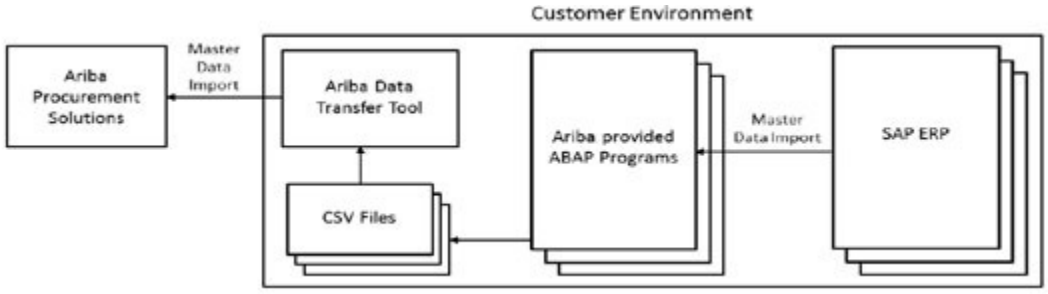
SAP and Ariba can talk to each other using various methods. The buyer administrator can enable the integration of data. It can be integrated through the following methods:

- File-based integration
- Web services
- Direct connectivity
- Mediated connectivity

7.8.1 1. File-Based integration:

The data transfer tool is an integral part of Ariba integration toolkit to import master data and transactional data. Recent SAP transports should be installed in SAP which consists of Ariba specific ABAP programs. Steps on how the master data information flows from SAP to Ariba using the file based integration method are shown below.

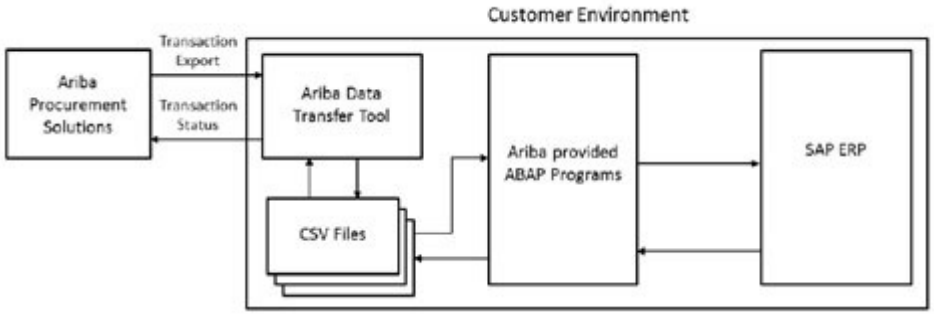
- It extracts master data from SAP and converts to a *.csv file.
- The *.csv files are then transferred to the Ariba using the Ariba integration toolkit.



The above figure illustrates the integration of master data flow using file-based integration method. Steps shown below indicate the flow of transactional data information from SAP to Ariba using the file based integration method.

- The data transfer tool exports data from Ariba procurement solution in the form of *.csv files.
- ABAP program then read these *.csv files and transfers data to SAP ERP.
- ABAP program then retrieves the status of each export transaction from SAP ERP.
- The data transfer tool reads these *.csv files and updates them into Ariba.

The following graphic illustrates the integration of transactional data –



7.8.2 2. Web services based integration:

The web services are used for real-time integration of Ariba with SAP ERP using SAP PI (Process integration). The integration setup using SAP PI is available, by default. If any other application system is being used, through an SOA supporting middleware, then it requires the complete set up to be done manually. The web services generate a SOAP message based on the WSDL. It dispatches it to a web services server, through SAP PI, using the configured URL.

7.8.3 3. Direct connectivity integration:

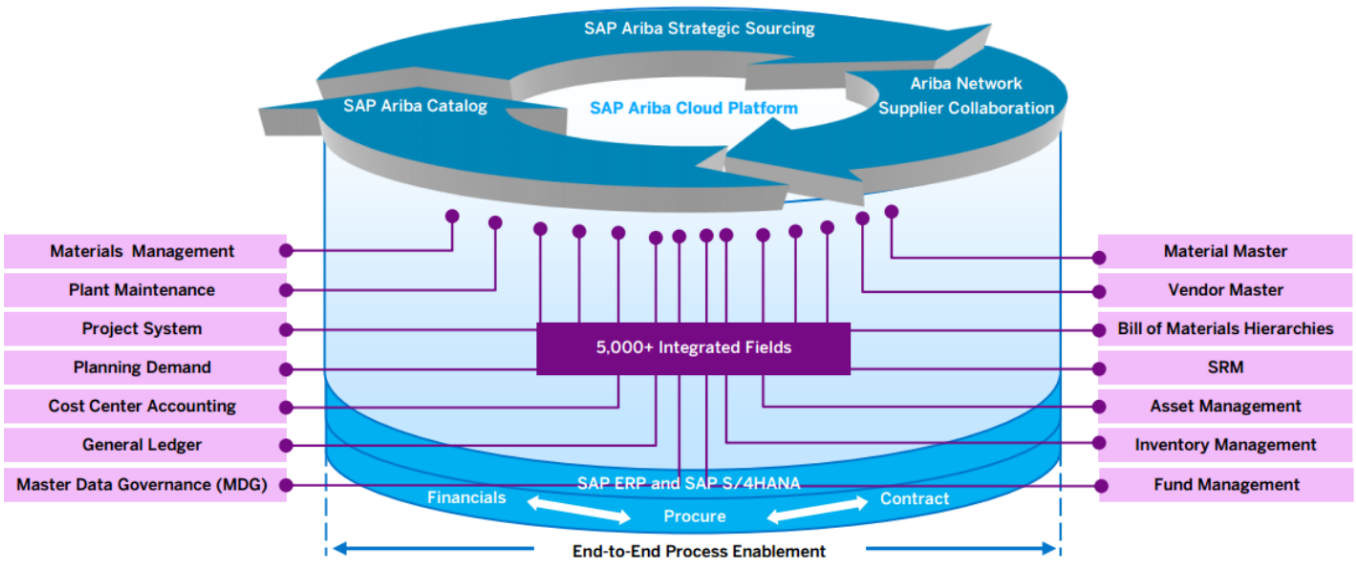
The direct connectivity method uses the *.csv files. The *.csv files containing the master data is sent to the Ariba system through SOAP messages to update. The direct connectivity integration method for master data is available from Ariba cloud integration release 4.0 and later whereas importing transactional data from release 6.0 and above.

7.8.4 4. Mediated connectivity integration:

There are two different ways through which the mediated connectivity method can be used. First one is by providing a SAP PI layer to exchange the data and the second one is by sending the information as SOAP messages. The mediated connectivity integration method for master data is available from Ariba cloud integration release 5.0 and later and transactional data from release 6.0 and later.

7.9 SAP ARIBA integration into SAP® ERP and SAP S/4HANA®

SAP Ariba is the only cloud provider with true bidirectional integration into SAP® ERP and SAP S/4HANA®, including more than 150 integration processes and over 5,000 fields integrated to the SAP core. Integration with SAP Ariba® solutions is achieved via the SAP Ariba cloud integration framework. The SAP Ariba cloud integration framework is a suite of components and complementary technologies that can be optionally deployed depending on a customer's specific landscape.



7.10 SAP ARIBA CLOUD SOLUTIONS INTEGRATION WITH SAP ERP and SAP S/4HANA[24]

SAP Ariba's products are focused on two key areas:

- Ariba Network
- SAP Ariba applications – SAP Ariba Strategic Sourcing, SAP Ariba Supplier Management, SAP Ariba Procurement, SAP Ariba Supply Chain, and SAP Ariba Financial Supply Chain solutions

- Ariba Network

  Ariba Network is a cloud-based network where buyers and suppliers collaborate by managing different business processes, either via the Ariba Network portal or by exchanging commerce extensible markup language (cXML) messages. Buying organizations use Ariba Network to find suppliers and to run automated business transactions from ordering to issuing payments. Suppliers use Ariba Network to find new business opportunities and to work more efficiently and effectively with their customers on all the shared aspects of business commerce: proposals, contracts, orders, invoices, and payments.

- SAP Ariba applications

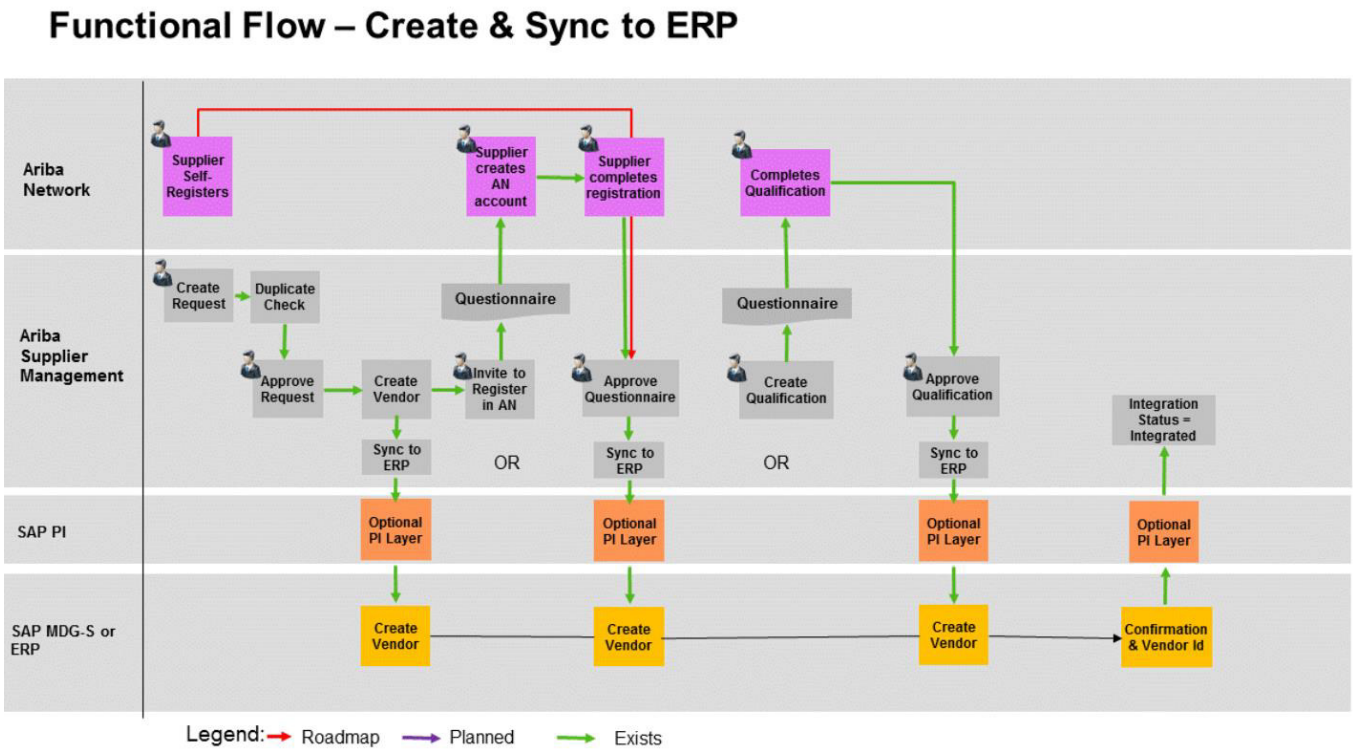
[24] Quelle https://www.forcebin.com/ariba-integration

The SAP Ariba Cloud Integration Gateway, enabled by SAP Cloud Platform Integration redefines how our customers integrate with the world's largest B2B network. It gives our suppliers a simple and efficient way to integrate with Ariba Network using standards other than cXML. It reduces the deployment time for suppliers to integrate with Ariba Network. This is currently only available for Ariba Network supplier-side integration.

7.11 Ariba Network and SAP ERP or SAP S/4HANA Integration Scenarios

7.11.1 Document Automation:

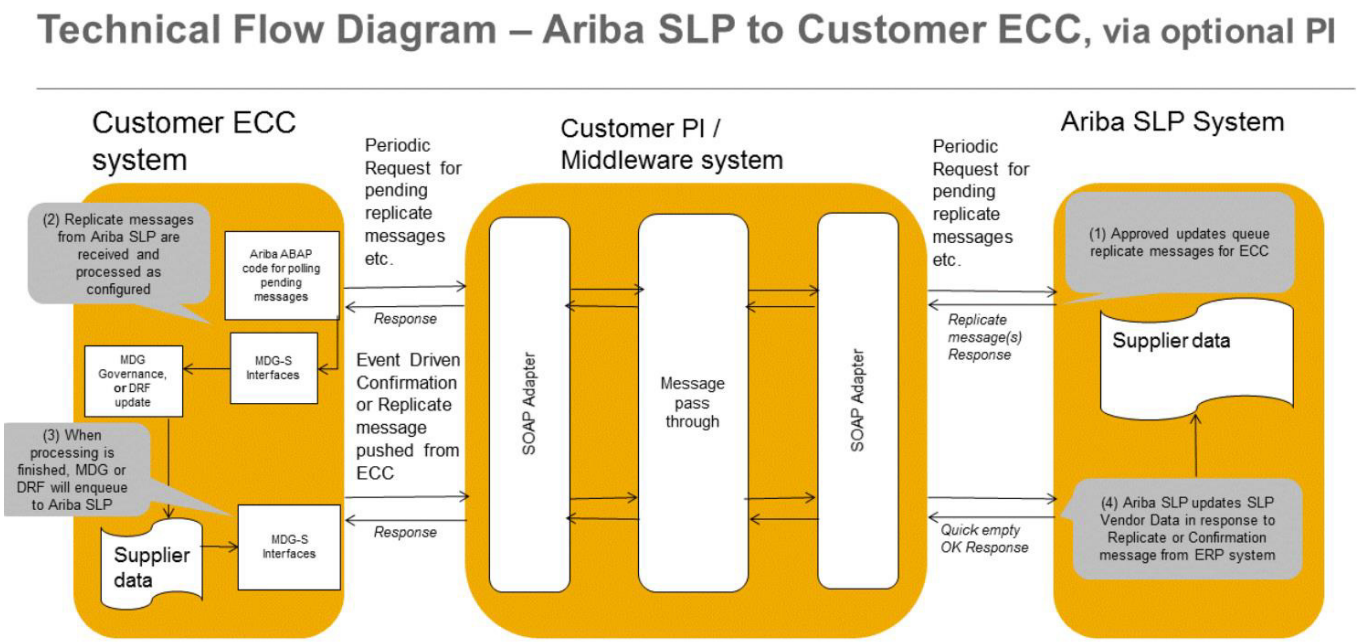
SAP Ariba offers many different integration scenarios to support many business processes, but the most typical process is document automation for purchase order (PO) and invoice automation. In this scenario, users create POs in their SAP ERP or SAP S/4HANA system, which sends them to the suppliers on Ariba Network for processing. Suppliers confirm orders or let their customers know if items are backordered; they can notify them that items have shipped and send a PO invoice for payment. Business rules enforce process and document compliance. For example, you can require that a supplier must confirm an order, create a ship notice, or wait for a goods receipt notice before they can send an invoice.



7.11.2 SAP Ariba Supply Chain Collaboration:

Ariba Network provides enhanced supply chain visibility for direct materials for discrete and process manufacturing, consumer products and life sciences, and merchandise spend in the retail industry.

To improve poor supply chain visibility, communication, and collaboration among supply chain partners, SAP Ariba Supply Chain Collaboration was developed to provide comprehensive direct material and supply chain collaboration tools with industry-specific aspects and a single access point and collaboration platform for all partners across different supply chains. These scenarios include processes for subcontracting, consignment, forecasting, and supplier-managed inventory as well as returns.



7.11.3 Contract Manufacturing Collaboration (Subcontract PO):

There are several steps in the subcontracting process.The first is the basic subcontracting purchase orders with components, purchase order confirmation, and ship notice of finished goods to the buyer.The buyer sends a goods receipt and invoice to the supplier.

In addition, basic subcontracting allows buyers/sellers to monitor component shipments/deliveries.Component shipments are supported as well as visibility of component inventory and component acknowledgement. In the flow below, there are two different options for component shipments.

• One-step process: In this process both the ship notice request and product activity message are sent in a single step. The proof of delivery is triggered at the seller side once the components have been received.

• Two-step process: Although the overall steps are the same as the one-step process, the main difference is when these steps are taken.The two-step process waits for the product to be received before the product is moved into the buyer system, and a product activity message is triggered.
• Finished production: Integrated suppliers have the option to notify the buyer of production of the finished goods via manufacturing execution system (MES) visibility.These messages are usually triggered from completed production orders in the supplier system and sent to inform the buyer of how much finished goods has been produced.

• Real-time consumption:Real-time consumption is when the supplier uses up the components originally shipped by the buyer, and notifies them. Once the buyer receives the notification, a product activity message is sent to the supplier from SAP with current component quantities as SAP needs to be the system of record.If real-time computation is not used, then the product activity message for component overview is sent when the goods receipts from the ship notice of finished goods is sent.

7.11.4 Consignment Collaboration (Consignment PO):

The nature of consignment inventory is that “change of ownership” is unrelated to the shipment/receipt processes. This is contrary to the basic design of most inventory/accounting systems’ transactional processes. Because of this, most inventory systems do not handle consignment inventory very well. This feature enables buyers to send consignment line items on POs to suppliers and receive consignment goods that are still owned by the suppliers. The feature provides visibility of the consignment PO items, visibility of the consignment material movements, transfer of ownership of stock, and invoice settlement based on the material movements.

• Consignment Movement: After the goods receipt is generated in SAP, the consignment movement using transaction MB1B triggers the product activity message to Ariba Network to transfer consignment stock to own stock.

• Consignment Settlement and Invoice: Real-time consumption is when the supplier uses up the components originally shipped by the buyer, and notifies them.Once the buyer receives the notification, a product activity message is sent to the supplier.

Forecast Collaboration Forecast collaboration provides several benefits to the supplier. First, by providing suppliers with near real-time visibility into demand forecast, they are able to more effectively schedule operations and plan capacity usage in order to meet the customer's requirements. It also allows suppliers to provide commitment against the buyer-requested forecast quantities, which can be mapped to process into the material requirements planning (MRP) software the buyer is using.

Supplier-Managed Inventory Collaboration In addition to sending a long-term plan, supplier-managed inventory (SMI) functionality will allow for suppliers to manage the stock levels for a buyer and send product based on scheduled agreements. This process would allow the supplier to see the gross demand and current inventory levels and to make any necessary adjustments based on capacity to satisfy the needs of the buyers. This process allows suppliers to manage the more day-to-day execution of the forecast plan and send back SMI agreement schedule lines to the buyer. They can then ship against those schedule lines and can be measured to the original forecast commitment agreed upon.

SAP Ariba Sourcing and SAP Ariba Contracts Integration with SAP ERP and SAP S/4HANA

SAP Ariba Sourcing The standard integration between SAP and SAP Ariba Sourcing supports integrated sourcing events initiated from SAP ERP or SAP S/4HANA:

• Manually created purchase requisitions in SAP ERP materials management (MM) that then create a request for quotation (RFQ)

• Automatically created purchase requisitions during MRP runs that then create an RFQ

• Manual creation of SAP ERP RFQs without purchase requisitions

An RFQ is sent through the Ariba Network adapter for SAP Process Integration to Ariba Network, and from there to SAP Ariba Sourcing as a sourcing request. The sourcing request is routed to the appropriate sourcing manager based upon the header data sent from SAP ERP or SAP S/4HANA. The sourcing manager is notified to review the sourcing request and edit and send it for approval if required, and then creates a sourcing
project based upon the most appropriate template.

The sourcing project allows the sourcing manager to manage every aspect of the sourcing event, including creation, collaboration with suppliers, monitoring the execution, scoring and grading of supplier

responses, and awarding suppliers. A sourcing event can be a request for information (RFI), a request for proposal (RFP), or an auction. When the sourcing manager creates awards for the event, SAP Ariba Sourcing sends a cXML document based on the award to the ERP system. The ERP creates a purchase order or contract from the cXML document.

7.11.5 SAP Ariba Contracts

hen a buyer publishes a contract, SAP Ariba Contracts sends contract information such as the contract header data and lineitem data to SAP ERP or SAP S/4HANA. SAP ERP or SAP S/4HANA then creates an outline agreement based on the contract information received from SAP Ariba Contracts.

7.11.6 Master Data Integration

Buyers using SAP Ariba Sourcing or SAP Ariba Contracts can import organizational data, cross-application data such as incoterms and payment terms, and material master data from SAP ERP systems. This allows buyers to initiate the sourcing process in SAP Ariba Sourcing and send the award to SAP ERP or SAP S/4HANA.

7.11.7 Direct Material Sourcing

Buyers using the SAP Ariba Strategic Sourcing Suite will be able to integrate additional master data elements such as bill of material (BOM) and purchasing information records (PIR) with SAP ERP or SAP S/4HANA.

7.11.8 Service Items Integration

External services procurement containing a flat list of service line items is supported in the integration flow from RFQ in SAP ERP or SAP S/4HANA to sourcing request, award to PO or outline agreement in SAP ERP, and contract to SAP ERP outline agreement.

SAP Ariba Supplier Management Solutions Integration with SAP ERP, SAP S/4HANA, and SAP Master Data Governance

The integration of SAP Ariba Supplier Management solutions with SAP ERP, SAP S/4HANA, and SAP Master Data Governance (SAP MDG) now enables seamless sharing and reuse of supplier information between the systems. This bidirectional integration is available in both the direct connect and mediated connectivity options using SAP NetWeaver Process Integration, and it includes the initial load as well as incremental updates. The two main use cases supported are:

• Upload existing supplier master data from SAP ERP, SAP S/4HANA, or SAP MDG with SAP Ariba Supplier Management followed by the invitation to suppliers to register via Ariba Network.

• Download new supplier data from SAP Ariba Supplier Management with SAP ERP, SAP S/4HANA, or SAP MDG for suppliers who self-register and are successfully qualified.

SAP Ariba Procurement Solution Integration with SAP ERP and SAP S/4HANA

The data managed by SAP Ariba Procurement solutions includes the buyer catalogs, transaction documents (such as requisitions, POs, receipts, invoices), and master data such as user, accounting, and supplier data. The SAP Ariba cloud integration framework enables the integration between SAP Ariba Procurement solutions and SAP ERP or SAP S/4HANA for the exchange of both transactional data and master data.

Data can be extracted and transferred directly between SAP ERP or SAP S/4HANA systems and SAP Ariba Procurement solutions or through a mediated connectivity using SAP NetWeaver Process Integration.

SAP Ariba Procurement Solutions

SAP Ariba Procurement solutions cover the operational procurement processes, which include:

• Managing catalogs

• Creating and approving purchase requisitions

• Creating and sending POs

• Recording the receipt of goods and services

• Receiving invoices (sent from suppliers electronically through Ariba Network, or as paper invoices)

• Reconciling invoices against contracts, POs, and/or receipts

• Real-time budget check with fund reservation in SAP ERP

• Taxes and charges in requisitions and purchase orders

7.12 THE FUTURE OF SAP: SAP S/4HANA AND THE INTEGRATED BUSINESS NETWORK

SAP S/4HANA is the next-generation business suite product that provides innovative, tightly coupled solutions that support endto-end business process integration between SAP S/4HANA core components and SAP software-as-a-service (SaaS) cloud solutions, such as:

- SAP Ariba procurement and sourcing solutions –Ariba Network connects more than 2.5 million buyers and sellers, generating over U.S. $1 trillion in commerce annually.
- Concur travel and expense (T&E) –Concur'stravel and expense ecosystem consists of 20,000 customers, 30 million end users, and hundreds of developer partners, all processing more than U.S. $50 billion worth of T&E spend a year.
- SAP Fieldglass contingent labor and services –More than 7 million users and 40,000 suppliers are managing over U.S. $25 billion in global spend annually using SAP Fieldglass solutions, placing 1.4 million contingent workers in over 100 countries.

SAP S/4HANA can be deployed on premise or in the cloud. It is built on the SAP HANA core, where the basic setup, master data, and core processes are defined and applied, and around which everything else can be built in a sustainable way.

Today, SAP ERP integration to SAP Ariba cloud solutions is made possible via SAP ERP add-ons and the Ariba Network adapter for SAP NetWeaver. With SAP S/4HANA, the add-ons become the basis for native integration support. Currently, the SAP S/4HANA 1511 On Premise and 1610 On Premise versions are supported.

SAP Ariba will continue to provide integration support through the SAP NetWeaver Process Integration adapter and upcoming SAP HANA Cloud integration releases.

Upcoming SAP Ariba Cloud Integration Enhancements
SAP Ariba will deliver major innovations in its next cloud integration release to help customers integrate

their back-end SAP ERP and SAP S/4HANA systems and reduce the time needed to integrate SAP Ariba solutions.

On the Ariba Network buy-side, the key planned integration enhancements are:

- SAP Ariba Cloud Integration Gateway for Ariba Network buyside integration
- Collaborative supply chain -Quality management
- SAP S/4HANA 1709

8 Literature sources

SAP Master Data Governance product page: *http://help.sap.com/mdg*
SAP Master Data Governance landing page: *www.sap.com/mdg*
SAP Master Data Governance Community page: *www.sap.com/community/topic/ master-data-governance.html*
Stammdatenmanagement mit SAP Master Data Governance (SAP PRESS)
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Logistics with SAP S/4HANA: An Introduction (SAP PRESS)
By Deb Bhattacharjee and Eric Monti
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SAP S/4HANA Cloud: Use Cases, Functionality, and Extensibility (SAP PRESS)
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(SAP PRESS)
29. September 2017 by Susanne Wagner and Marcus Mock
SAP Fiori: Implementation and Development (SAP PRESS)
31. July 2017 by Michael Englbrecht and Michael Wegelin
SAP Master Data Governance on SAP S/4HANA 1709 / SAP Master Data Governance 9.1
March 2018
Rheinwerk Verlag SAP Master Data Governance (SAP MDG) on SAP S/4HANA

9 Additional Information

9.1 Links

Customer Vendor Integration on SAP Help
FPM on SCN
• FPM Cookbook
MDG Guides on Service Market Place
MDG Extensibility Options on SDN

9.2 How-to Guides

Extensibility Options for SAP Master Data Governance • Business Partner / Customer / Supplier Data

- How-to Guide: Create and Register a Custom Handler Class
- How-to Guide: Create or Redefine a UI Feeder Class
- How-to Guide: Extend MDG-S / MDG-C Data Model by a New Entity Type (Flex Option)
- How-to Guide: Extend MDG-S / MDG-C Data Model by a New Entity Type (Reuse Option)
- How-to Guide: Extend MDG-S / MDG-C Data Model by a New Field (Reuse Option)
- How-to Guide: Default the ERP Vendor Creation with MDG-S

- How-to Guide: Filter an Input Help for a Business System

9.3 SAP Notes

- 1637249 specifying required information for OSS support
- 2105467 specifying required information for Performance Issues
- 2221398 providing a detailed overview of all fields in MDG-BP.